

USAJMO

§1 USAJMO 2010

1. (*USAJMO 2010*) A permutation of the set of positive integers $[n] = \{1, 2, \dots, n\}$ is a sequence $(a_1, a_2, \dots, a_n)$ such that each element of $[n]$ appears precisely one time as a term of the sequence. For example, $(3, 5, 1, 2, 4)$ is a permutation of $[5]$. Let $P(n)$ be the number of permutations of $[n]$ for which ka_k is a perfect square for all $1 \leq k \leq n$. Find with proof the smallest n such that $P(n)$ is a multiple of 2010.

Solution 1

We claim that the smallest n is $67^2 = \boxed{4489}$.

Let $S = \{1, 4, 9, \dots\}$ be the set of positive perfect squares. We claim that the relation $R = \{(j, k) \in [n] \times [n] \mid jk \in S\}$ is an equivalence relation on $[n]$.

- It is reflexive because $k^2 \in S$ for all $k \in [n]$.
- It is symmetric because $jk \in S \implies kj = jk \in S$.
- It is transitive because if $jk \in S$ and $kl \in S$, then $jk \cdot kl = jlk^2 \in S \implies jl \in S$, since S is closed under multiplication and a non-square times a square is always a non-square.

We are restricted to permutations for which $ka_k \in S$, in other words to permutations that send each element of $[n]$ into its equivalence class. Suppose there are N equivalence classes: $C_1, \dots, C_N$. Let n_i be the number of elements of C_i , then $P(n) = \prod_{i=1}^N n_i!$.

Now $2010 = 2 \cdot 3 \cdot 5 \cdot 67$. In order that $2010 \mid P(n)$, we must have $67 \mid n_m!$ for the class C_m with the most elements. This means $n_m \geq 67$, since no smaller factorial will have 67 as a factor. This condition is sufficient, since $n_m!$ will be divisible by 30 for $n_m \geq 5$, and even more so $n_m \geq 67$.

The smallest element g_m of the equivalence class C_m is square-free, since if it were divisible by the square of a prime, the quotient would be a smaller element of C_m . Also, each prime p that divides g_m divides all the other elements k of C_m , since $p^2 \mid kg_m$ and thus $p \mid k$. Therefore $g_m \mid k$ for all $k \in C_m$. The primes that are not in g_m occur an even number of times in each $k \in C_m$.

Thus the equivalence class $C_m = \{g_m k^2 \leq n\}$. With $g_m = 1$, we get the largest possible n_m . This is just the set of squares in $[n]$, of which we need at least 67, so $n \geq 67^2$. This condition is necessary and sufficient.

Solution 2

Solution 1 can also be rephrased as follows, in a longer way, but with fewer highly technical words such as "equivalence relation":

It is possible to write all positive integers n in the form $p \cdot m^2$, where m^2 is the largest perfect square dividing n , so p is not divisible by the square of any prime. Obviously, one working permutation of $[n]$ is simply $(1, 2, \dots, n)$; this is acceptable, as ka_k is always k^2 in this sequence.

Lemma 1. We can permute any numbers that, when each divided by the largest perfect square that divides it, yield equal quantities p .

Proof. Let p_k and m_k be the values of p and m , respectively, for a given k as defined above, such that p is not divisible by the square of any prime. We can obviously permute two numbers which have the same p , since if $p_j = p_w$ where j and w are 2 values of k , then $j \cdot w = p_j^2 \cdot m_j^2 \cdot m_w^2$, which is a perfect square. This proves that we can permute any numbers with the same value of p .

End Lemma

Lemma 2. We will prove the converse of Lemma 1: Let one number have a p value of ϕ and another, γ . $\phi \cdot f$ and $\gamma \cdot g$ are both perfect squares.

Proof. $\phi \cdot f$ and $\gamma \cdot g$ are both perfect squares, so for $\phi \cdot \gamma$ to be a perfect square, if g is greater than or equal to f , g/f must be a perfect square, too. Thus g is f times a square, but g cannot divide any squares besides 1, so $g = 1f$; $g = f$. Similarly, if $f \geq g$, then $f = g$ for our rules to keep working.

End Lemma

We can permute l numbers with the same p in $l!$ ways. We must have at least 67 numbers with a certain p so our product will be divisible by 67. Obviously, then it will also be divisible by 2, 3, and 5, and thus 210, as well. Toms as h , in general, we need numbers all the way up to $h \cdot 67^2$, so obviously, 67^2 is the smallest such number such that we can get a $67!$ term; here 67 p terms are 1. Thus we need the integers $1, 2, \dots, 67^2$, so 67^2 , or 4489, is the answer.

Solution 3 (Not a formal proof but understandable)

We claim the smallest n is $67^2 = \text{span style="border: 1px solid black; padding: 0 2px;">4489 Looking at small cases we see that $P(n)$ changes every time n increases to a perfect square. We find that we can not permute the non squares around because n (not a perfect square) $\cdot a_w$ where $w \neq n$ will not give a perfect square. But we can permute the perfect squares around to other perfect squares since a square times a square makes another square we can do this $(\# \text{ of squares below } n)!$ times. So now we need to find $n = a^2$ such that $a!$ is divisible by 210 which largest prime factor is 67 so, n is n is $67^2 = \text{span style="border: 1px solid black; padding: 0 2px;">4489 bjump$$

Solution Number Sense

We have to somehow calculate the number of permutations for a given n . How in the world do we do this? Because we want squares, why not call a number $k = m \cdot s^2$, where s is the largest square that allows m to be non-square? $m = 1$ is the only square m can be, which only happens if k is a perfect square.

For example, $126 = 14 \cdot 3^2$, therefore in this case $k = 126, m = 14, s = 3$.

I will call a permutation of the numbers P , while the original $1, 2, 3, 4, \dots$ I will call S .

Note that essentially we are looking at "pairing up" elements between P and S such that the product of P_k and S_k is a perfect square. How do we do this? Using the representation above.

Each square has to have an even exponent of every prime represented in its prime factorization. Therefore, we can just take all exponents of the primes $\text{mod } 2$ and if

there are any odd numbers, those are the ones we have to match- in effect, they are the m numbers mentioned at the beginning.

By listing the m values, in my search for "dumb" or "obvious" ideas I am pretty confident that only values with identical ms can be matched together. With such a solid idea let me prove it.

If we were to "pair up" numbers with different ms , take for example S_{18} with an m of 2 and $P_{18} = 26$ with an m of 26, note that their product gives a supposed m of 13 because the 2 values cancel out. But then, what happens to the extra 13 left? It doesn't make a square, contradiction. To finish up this easy proof, note that if a "pair" has different m values, and the smaller one is m_1 , in order for the product to leave a square, the larger m value has to have not just m_1 but another square inside it, which is absurd because we stipulated at the beginning that m was square-free except for the trivial multiplication identity, 1.

Now, how many ways are there to do this? If there are c_1 numbers with $m = 1$, there are clearly $(c_1)!$ ways of sorting them. The same goes for $m = 2, 3, etc.$ by this logic. Note that the $P(n)$ as stated by the problem requires a 67 thrown in there because $2010 = 2 * 3 * 5 * 67$, so there has to be a S_n with 67 elements with the same m . It is evident that the smallest n will occur when $m = 1$, because if m is bigger we would have to expand n to get the same number of m values. Finally, realize that the only numbers with $m = 1$ are square numbers! So our smallest $n = 67^2 = 4489$, and we are done.

I relied on looking for patterns a lot in this problem. When faced with combo/number theory, it is always good to draw a sketch. Never be scared to try a problem on the USAJMO. It takes about 45 minutes. Well, it's 2010 and a number 1. Cheers!

-expiLnCalc

Solution 4

It's well known that there exists $f(n)$ and $g(n)$ such that $n = f(n) \cdot g(n)$, no square divides $f(n)$ other than 1, and $g(n)$ is a perfect square.

Lemma: $k \cdot a_k$ is a perfect square if and only if $f(k) = f(a_k)$

We prove first: If $f(k) = f(a_k)$, $k \cdot a_k$ is a perfect square.

$k \cdot a_k = f(k) \cdot g(k) \cdot f(a_k) \cdot g(a_k) = f(k)^2 \cdot g(k) \cdot g(a_k)$, which is a perfect square.

We will now prove: If $k \cdot a_k$ is a perfect square, $f(k) = f(a_k)$.

We do proof by contrapositive: If $f(k) \neq f(a_k)$, $k \cdot a_k$ is not a perfect square.

$v_p(k)$ is the p -adic valuation of k . (Basically how many factors of p you can take out of k)

Note that if $f(k) \neq f(a_k)$, By the Fundamental Theorem of Arithmetic, $f(k)$ and $f(a_k)$'s prime factorization are different, and thus there exists a prime p , such that $v_p(f(k)) \neq v_p(f(a_k))$. Also, since $f(k)$ and $f(a_k)$ is squarefree, $v_p(k), v_p(a_k) \leq 1$. Thus, $v_p(k \cdot a_k) = 1$, making $k \cdot a_k$ not a square.

End Lemma

Thus, we can only match k with a_k if they have the same f value. Thus, to find $P(k)$, we can do it by f value, permuting the a_k with f value 1, then 2, ... Thus, our answer is:

$$P(n) = \prod_{1 \leq i \leq n, g(i)=1} \left\lfloor \sqrt{\frac{n}{i}} \right\rfloor !$$

For all $n < 67^2$, $P(n)$ doesn't have a factor of 67. However, if $n = 67^2$, the first term will be a multiple of 2010, and thus the answer is $67^2 = \boxed{4489}$

-Alexlikemath

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

Solution 5(stupid but simple)

The answer is 4489.(67 squared)

Proof

We start with trying to fulfill ways to do $P(n)$. When n is 1 there is one way to do it and fulfill ka_k as a perfect square. Same with the numbers 2 and 3. The order for both of these number are all in increasing order

$$1, 2$$

$$1, 2, 3$$

Now when it comes to 4 we could do

$$1, 2, 3, 4$$

and

$$4, 2, 3, 1$$

This means when n is a perfect square it increases where k and a_k are perfect squares. The prime factorization of 2010 is:

$$2010 = 2 * 5 * 3 * 67$$

Because we only care about 67, the biggest divisor we do 67 squared because n has to be a square.

$$67^2 = 4489$$

-Multpil2

Solution 6

We start by realizing that we could change this confusing set theory problem into a number theory problem. Notice that we have $(a_1, a_2, \dots, a_n)$ as our permutation of the set $(1, 2, \dots, n)$. Essentially, the problem is asking us to find the number of permutations of this original set such that $a_1, 2a_2, \dots, na_n$ are all perfect squares. From here, we realize that we could write the following:

$$a_1 = (p_1)^2$$

$$a_2 = 2(p_2)^2$$

.

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$$a_n = n(p_n)^2$$

Thus, we just transformed this problem into the question that how many ways are there such that the elements of the form $n(p_n)^2$ are elements of the original set $[n]$. We could easily solve this by looking at ranges. First, we realize $1 \leq (p_1)^2 \leq n$.

Solving, we get $1 \leq p_1 \leq \lfloor \sqrt{n} \rfloor$. Thus, there are $\lfloor \sqrt{n} \rfloor$ possibilities for p_1 . We can now see the pattern by solving the following inequalities and find the total number of permutations are: $\prod_{1 \leq i \leq n} \lfloor \sqrt{\frac{n}{i}} \rfloor$. Thus, we need $2010 = 2 * 3 * 5 * 67$ to divide that product. Notice this means we need the prime factors 2, 3, 5, 67 to divide that product. Base cases tells us that the prime factors of 2, 3, 5 are trivial but 67 needs to be focused on. We need 67 to divide that product. This tells us that $\lfloor \sqrt{\frac{n}{i}} \rfloor$ must be divisible by 67. Already, we can tell that n has to be at least $67^2 = 4489$ for this to even hold. This would be the minimum n for 67 to divide that product. We can test out 4489 and see that we get factors of 2, 3, 5 in the product along with our intention of including the prime factor of 67. Therefore, what we thought was the minimum actually holds and thus our answer should be 4489.

Solution 7

For a given sequence $a_1, a_2, a_3, \dots, a_n$ to obey all the requirements, we must have $a_i \cdot i$ equal to a perfect square for all i from 1 to n inclusive.

We first consider a simple permutation $1, 2, 3, \dots, n$ of $1, 2, 3, \dots, n$, which satisfies the condition.

Lemma 1. Starting from the sequence $1, 2, 3, \dots, n$, we may swap any two elements whose indices multiply to equal a perfect square, and the resulting sequence will still satisfy all requirements.

Proof.

Suppose we swap the elements in positions i and j . We are given that $i \cdot j$ is a perfect square. After the swap, the elements at position $i = j$ and the element at position $j = i$, so the products are $i \cdot j$ and $j \cdot i$, which are both perfect squares. For all other indices $k \neq i, j$, the product $k \cdot a_k$ remains unchanged (where a_k represents the k th element in the permutation). Hence, the sequence remains valid.

Lemma 2. If two positive integers a and b satisfy that ab is a perfect square, then there exists a positive integer n and integers x, y such that $a = nx^2$ and $b = ny^2$.

Proof.

Let $ab = \prod_{i=1}^k (p_i)^{2e_i}$ be the prime factorization of ab . Suppose the exponent of p_i in a is m_i and in b is n_i . Since $m_i + n_i = 2e_i$, both m_i and n_i have the same parity. Let $r_i = \min(m_i, n_i)$; then we can factor out n and let it equal $(p_i)^{r_i}$. Hence, $m_i - r_i$ and $n_i - r_i$ are both even, so it is possible to write $a = nx^2$ and $b = ny^2$ for some integers x, y .

Let G_i be the set of all numbers of the form $i \cdot a^2$ contained in $1, 2, \dots, n$, where i is a fixed positive integer.

Lemma 3. For each G_i , the elements can be permuted in any order while preserving validity of the sequence.

Proof.

The elements of G_i are $i \cdot 1^2, i \cdot 2^2, i \cdot 3^2, \dots, i \cdot t^2$, where $i \cdot t^2 \leq n$. If we place $i \cdot x^2$ in the position originally occupied by $i \cdot y^2$, then the corresponding product is

$$(i \cdot y^2)(i \cdot x^2) = i^2(xy)^2$$

, which is a perfect square. Hence, any permutation within g_i still satisfies the given condition.

Therefore, each G_i may be permuted independently, and the total number of valid sequences is

$$p(n) = (\lfloor \sqrt{n} \rfloor)! \cdot (\lfloor \sqrt{n/2} \rfloor)! \cdot (\lfloor \sqrt{n/3} \rfloor)! \cdots (\lfloor \sqrt{n/n} \rfloor)!$$

We require $p(n)$ to be divisible by $2010 = 2 \cdot 3 \cdot 5 \cdot 67$. Since 67 is the largest prime factor of 2010, it suffices that $(\lfloor \sqrt{n} \rfloor)! \geq 67!$, because then $67!$ contains all the prime factors of 2010. To minimize n , we take $\sqrt{n} = 67$, giving

$$n = 67^2 = \boxed{4489}.$$

Irfans123

2. (*USAJMO 2010*) Let $n > 1$ be an integer. Find, with proof, all sequences $x_1, x_2, \dots, x_{n-1}$ of positive integers with the following three properties:

- a) $x_1 < x_2 < \cdots < x_{n-1}$;
- b) $x_i + x_{n-i} = 2n$ for all $i = 1, 2, \dots, n-1$;
- c) given any two indices i and j (not necessarily distinct) for which $x_i + x_j < 2n$, there is an index k such that $x_i + x_j = x_k$.

Solution

The sequence is $2, 4, 6, \dots, 2n-2$.

Proof 1 We will prove that any sequence $x_1, \dots, x_{n-1}$, that satisfies the given conditions, is an arithmetic progression with x_1 as both the first term and the increment. Once this is proved, condition (b) implies that $x_1 + x_{n-1} = x_1 + (n-1)x_1 = nx_1 = 2n$. Therefore $x_1 = 2$, and the sequence is just the even numbers from 2 to $2n-2$. The sequence of successive even numbers clearly satisfies all three conditions, and we are done.

First a degenerate case. If $n = 2$, there is only one element x_1 , and condition (b) gives $x_1 + x_1 = 4$ or $x_1 = 2$. Conditions (a) and (c) are vacuously true.

Otherwise, for $n > 2$, we will prove by induction on m that the difference $x_{n-m} - x_{n-1-m} = x_1$ for all $m \in [1, n-2]$, which makes all the differences $x_{n-1} - x_{n-2} = \dots = x_2 - x_1 = x_1$, i.e. the sequence is an arithmetic progression with x_1 as the first term and increment as promised.

So first the $m = 1$ case. With $n > 2$, x_{n-2} exists and is less than x_{n-1} by condition (a). Now since by condition (b) $x_1 + x_{n-1} = 2n$, we conclude that $x_1 + x_{n-2} < 2n$, and therefore by condition (c) $x_1 + x_{n-2} = x_k$ for some k . Now, since $x_1 > 0$, $x_k > x_{n-2}$ and can only be x_{n-1} . So $x_1 + x_{n-2} = x_{n-1}$.

Now for the induction step on all values of m . Suppose we have shown that for all $i \leq m$, $x_1 + x_{n-1-i} = x_{n-i}$. If $m = n-2$ we are done, otherwise $m < n-2$, and by condition (c) $x_1 + x_{n-2-m} = x_k$ for some k . This x_k is larger than x_{n-2-m} , but smaller than $x_1 + x_{n-1-m} = x_{n-m}$ by the inductive hypothesis. It then follows that $x_1 + x_{n-2-m} = x_{n-1-m}$, the only element of the sequence between x_{n-2-m} and x_{n-m} . This establishes the result for $i = m+1$.

So, by induction $x_1 + x_{n-1-m} = x_{n-m}$ for all $m \in [1, n-2]$, which completes the proof.

Proof 2 Let $S = \{x_1, x_2, \dots, x_{n-1}\}$. Notice that

$$x_1 < x_1 + x_1 < x_1 + x_2 < \dots < x_1 + x_{n-2} < 2n.$$

Then by condition (c), we must have $x_1, x_1 + x_1, \dots, x_1 + x_{n-2} \in S$. This implies that $x_1 = x_1, x_1 + x_1 = x_2, \dots, x_1 + x_{n-2} = x_{n-1}$, or that $x_k = kx_1$. Then we have $x_1 + x_{n-1} = n(x_1) = 2n \rightarrow x_1 = 2$, and the rest is trivial.

Solution 2

The claim is that in this sequence, if there are 2 elements a, b where $a, b < n$, such that $\gcd(a, b) = 1$, then the sequence contains every number less than $2n$.

Proof: Let a and b be the numbers less than n such that $\gcd(a, b) = 1$. We take this sequence modulo n . This means that if x_i is an element in this sequence then $-x_i$ is as well. $a, b, 2n - a, 2n - b$ are all elements in the sequence. Clearly, one of $2n - a + b$ and $2n - b + a$ is less than $2n$, which means that $\pm(a - b)$ are in this sequence modulo n . Now we want to show every number is achievable. We have already established that a and b are relatively prime, so by euclidean algorithm, if we take the positive difference of a' and b' every time, we will get that 1 is in our sequence. Then, we can simply add or subtract 1 as many times from a as desired to get every single number.

We have proved that there are no two numbers that can be relatively prime in our sequence, implying that no two consecutive numbers can be in this sequence. Because our sequence has $n - 1$ terms, our sequence must be one of $2, 4, 6, \dots, 2(n - 1)$ or $1, 3, 5, \dots$, the latter obviously fails, so $2, 4, 6, \dots, 2(n - 1)$ is our only possible sequence.

Solution 3

We can add x_1 to every expression in property (a) to get

$$2x_1 < x_1 + x_2 < \dots < x_1 + x_{n-1} = 2n.$$

Therefore, we have $n - 2$ distinct (because all the x_i are distinct) expressions of the form $x_1 + x_i$ for $i = 1, 2, \dots, n - 1$ that are all less than $2n$, which means these $n - 2$ expressions are also equal to some x_k .

Now, we have that the x_i are all positive, so $x_1 > 0$. Adding x_1 to both sides, $2x_1 > x_1$. Therefore, since the $n - 2$ expressions are all at least $2x_1$, they have to be equal to $x_2, x_3, \dots, x_{n-1}$.

Since there are $n - 2$ distinct expressions $x_1 + x_i$ equal to $n - 2$ distinct expressions x_j , we have that each $x_1 + x_i$ is equal to one x_j . Using the orders of the $x_1 + x_i$ and the x_j , we find that

$$\begin{aligned} x_1 + x_1 &= x_2 \\ x_1 + x_2 &= x_3 \\ &\dots \\ x_1 + x_{n-2} &= x_{n-1}. \end{aligned}$$

This gives us

$$\begin{aligned} x_1 &= x_1 \\ x_2 &= 2x_1 \end{aligned}$$

$$\begin{aligned} x_3 &= 3x_1 \\ &\dots \\ x_{n-1} &= (n-1)x_1 \end{aligned}$$

Now, we can use property (b) with $i = 1$, so $x_1 + x_{n-1} = 2n$, which means $x_1 = 2$. This gives the sequence

$$\boxed{2, 4, 6, \dots, 2(n-1)}$$

-sixeoneeight

3. (*USAJMO 2010*) Let $AXYZB$ be a convex pentagon inscribed in a semicircle of diameter AB . Denote by P, Q, R, S the feet of the perpendiculars from Y onto lines AX, BX, AZ, BZ , respectively. Prove that the acute angle formed by lines PQ and RS is half the size of $\angle XOZ$, where O is the midpoint of segment AB .

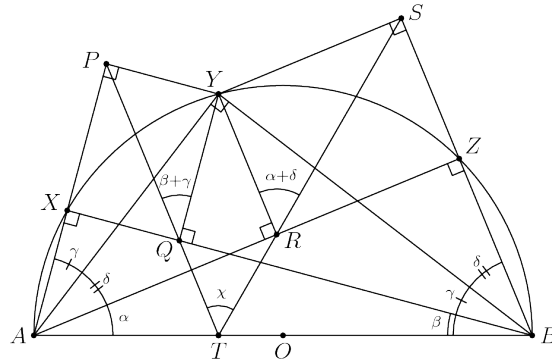
Solution 0 (play like a God)

Angles are directed mod 180 don't worry (take so that the solution can be applied to any configuration) Let the intersection of pq and rs is M .

$m\angle ARM = m\angle ZRS = m\angle ZYS = m\angle YZR = m\angle YZA = m\angle YXP = m\angle XYQ = m\angle XPQ = m\angle APM$ This implies $APRM$ is cyclic So: $m\angle PMR = m\angle PAR = m\angle XAZ = \frac{m\angle XOZ}{2}$ and we are done. For any concern mail on harshsahu1098@gmail.com

Solution 1

Let $\alpha = \angle BAZ$, $\beta = \angle ABX$. Since XY is a chord of the circle with diameter AB , $\angle XAY = \angle XBY = \gamma$. From the chord YZ , we conclude $\angle YAZ = \angle YBZ = \delta$.



Triangles BQY and APY are both right-triangles, and share the angle γ , therefore they are similar, and so the ratio $PY : YQ = AY : YB$. Now by **Thales' theorem** the angles $\angle AXB = \angle AYB = \angle AZB$ are all right-angles. Also, $\angle PYQ$, being the fourth angle in a quadrilateral with 3 right-angles is again a right-angle. Therefore $\triangle PYQ \sim \triangle AYB$ and $\angle YQP = \angle YBA = \gamma + \beta$. Similarly, $RY : YS = AY : YB$, and so $\angle YRS = \angle YAB = \alpha + \delta$.

Now RY is perpendicular to AZ so the direction RY is α counterclockwise from the vertical, and since $\angle YRS = \alpha + \delta$ we see that SR is δ clockwise from the vertical. (Draw an actual vertical line segment if necessary.)

Similarly, QY is perpendicular to BX so the direction QY is β clockwise from the vertical, and since $\angle YQP$ is $\gamma + \beta$ we see that QY is γ counterclockwise from the vertical.

Therefore the lines PQ and RS intersect at an angle $\chi = \gamma + \delta$. Now by the central angle theorem $2\gamma = \angle XOY$ and $2\delta = \angle YOZ$, and so $2(\gamma + \delta) = \angle XOZ$, and we are done.

Note that $RTQY$ is a quadrilateral whose angles sum to 360° ; can you find a faster approach using this fact?

Footnote We can prove a bit more. Namely, the extensions of the segments RS and PQ meet at a point on the diameter AB that is vertically below the point Y .

Since $YR = AY \sin(\delta)$ and is inclined α counterclockwise from the vertical, the point S is $AY \sin(\delta) \sin(\alpha)$ horizontally to the right of Y .

Now $AS = AY \cos(\delta)$, so S is $AS \sin(\alpha) = AY \cos(\delta) \sin(\alpha)$ vertically above the diameter AB . Also, the segment SR is inclined δ clockwise from the vertical, so if we extend it down from S towards the diameter AB it will meet the diameter at a point which is $AY \cos(\delta) \sin(\alpha) \tan(\delta) = AY \sin(\delta) \sin(\alpha)$ horizontally to the left of S . This places the intersection point of RS and AB vertically below Y .

Similarly, and by symmetry the intersection point of PQ and AB is directly below Y on AB , so the lines through PQ and RS meet at a point T on the diameter that is vertically below Y .

Footnote to the Footnote The Footnote's claim is more easily proved as follows.

Note that because $\angle QPY$ and $\angle YAB$ are both complementary to $\beta + \gamma$, they must be equal. Now, let PQ intersect diameter AB at T' . Then $PYT'A$ is cyclic and so $\angle YT'A = 180^\circ - \angle APY = 90^\circ$. Hence $T'YSB$ is cyclic as well, and so we deduce that $\angle YST' = \angle YBT' = 90^\circ - \alpha - \delta = \angle YSR$. Hence S, R, T' are collinear and so $T = T'$. This proves the Footnote.

Footnote to the Footnote to the Footnote The Footnote's claim can be proved even more easily as follows.

Drop an altitude from Y to AB at point T . Notice that P, Q, T are collinear because they form the Simson line of $\triangle AXB$ from Y . Also notice that P, Q, T are collinear because they form the Simson line of $\triangle AZB$ from Y . Since T is at the diameter AB , lines PQ and SR must intersect at the diameter.

Footnote to the Fourth Power There is another, simpler solution using Simson lines. Can you find it?

Operation Diagram Of course, as with any geometry problem, DRAW A HUGE DIAGRAM spanning at least one page. And label all your right angles, noting rectangles $PXQY$ and $YSZR$. It looks like there are a couple of key angles we need to diagram. Let's take $\angle ZAB = \alpha$, $\angle XBA = \beta$, $\angle YAZ = \angle YBZ = \delta$. From there $\angle XOZ = 180^\circ - \angle XOY - \angle ZOY = 180 - 2(\beta + \alpha)$.

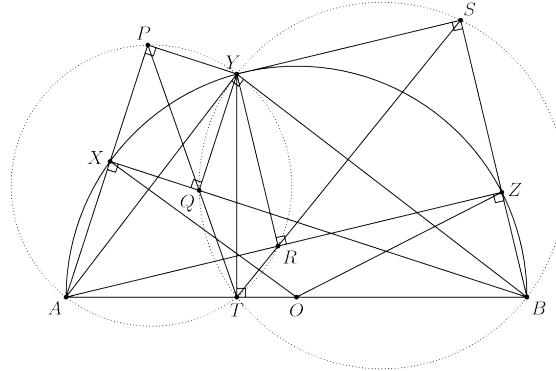
Move on to the part about the intersection of PQ and RS . Call the intersection J . Note that by Simson Lines from point Y to $\triangle ABX$ and $\triangle AZB$, YJ is perpendicular to AB and J lies on AB . Immediately note that we are trying to show that $\angle PJS = 90 - \beta - \alpha$.

It suffices to show that referencing quadrilateral QRJ , where J represents the intersection of XB, AZ , we have $\text{reflex } \angle QJR + \angle BQJ + \angle ARJ = 270 + \alpha + \beta$. Note that the reflex angle is $180^\circ + \angle AXZ = 180^\circ + (90^\circ - \angle XAZ) = 270^\circ - ((90 - \beta) - \alpha) = 180^\circ + \alpha + \beta$, therefore it suffices to show that $\angle BQJ + \angle ARJ = 90^\circ$. To make this proof more accessible, note that via (cyclic) rectangles $PXQY$ and $YSZR$, it suffices to prove $\angle YPJ + \angle YSJ = 90^\circ$.

Note $\angle YPJ = \angle YPQ = \angle YXQ = \angle YXB = \angle YAB = \alpha + \delta$. Note $\angle YSJ = \angle YSR = \angle YZR = \angle YZA = \angle YBA = \angle YBX + \angle XBA = ((90^\circ - \alpha) - \delta - \beta) + \beta = 90^\circ - \alpha - \delta$, which completes the proof.

Footnote to Operation Diagram For reference/feasibility records: took expiLnCalc 56 minutes (consecutively). During the problem expiLnCalc realized that the inclusion of δ was necessary when trying to show that $\angle YSJ + \angle YPJ = 90^\circ$. Don't be afraid to attempt several different strategies, and always be humble!

Solution 2



Let T be the foot of the perpendicular from Y to $\overline{AB}$, let O be the center of the semi-circle.

Since we have a semi-circle, if we were to reflect it over $\overline{AB}$, we would have a full circle, with $\triangle AXB$ and $\triangle AZB$ inscribed in it. Now, notice that Y is a point on that full circle, so we can say that T lies on the Simson Line $\overline{PQ}$ from Y to $\triangle AXB$ and that it also lies on the Simson line $\overline{RS}$ from Y to $\triangle AZB$. Thus, T lies on two distinct lines in a plane, which means that $T = \overline{PQ} \cap \overline{RS}$. Therefore, it suffices to show that $\angle PTS = \frac{1}{2}\angle XOZ$.

Since $m\angle YTA + m\angle YPA = 90^\circ + 90^\circ = 180^\circ$ and $m\angle YTB + m\angle YSB = 90^\circ + 90^\circ = 180^\circ$, we know that $TAPY$ and $TBSY$ are cyclic quadrilaterals.

We use this fact to get

$$\angle PTS = \angle PTY + \angle YTS = \angle PAY + \angle YBS = \angle XAY + \angle YBZ. (1)$$

Now note that $\angle XAY$ is the inscribed angle of minor arc $\widehat{PY}$, and $\angle XOY$ is the central angle of minor arc $\widehat{AB}$, so $\angle XAY = \frac{\widehat{PY}}{2} = \frac{\angle XOY}{2}$. Similarly, $\angle YBZ = \frac{\widehat{YZ}}{2} = \frac{\angle YOZ}{2}$. Thus we can say

$$\angle XAY + \angle YBZ = \frac{\angle XOY}{2} + \frac{\angle YOZ}{2} = \frac{\angle XOY + \angle YOZ}{2} = \frac{\angle XOZ}{2}. (2)$$

Combining statements (1) and (2), we can say that $\angle PTS = \frac{\angle XOZ}{2}$, as desired. $\square$

4. (USAJMO 2010) A triangle is called a *parabolic triangle* if its vertices lie on a parabola $y = x^2$. Prove that for every nonnegative integer n , there is an odd number m and a parabolic triangle with vertices at three distinct points with integer coordinates with area $(2^n m)^2$.

Solution

Let the vertices of the triangle be $(a, a^2), (b, b^2), (c, c^2)$. The area of the triangle is the absolute value of A in the equation:

$$A = \frac{1}{2} \det \begin{vmatrix} b-a & c-a \\ b^2-a^2 & c^2-a^2 \end{vmatrix} = \frac{(b-a)(c-a)(c-b)}{2}$$

If we choose $a < b < c$, $A > 0$ and gives the actual area. Furthermore, we clearly see that the area does not change when we subtract the same constant value from each of a , b and c . Thus, all possible areas can be obtained with $a = 0$, in which case $A = \frac{1}{2}bc(c-b)$.

If a particular choice of b and c gives an area $A = (2^n m)^2$, with n a positive integer and m a positive odd integer, then setting $b' = 4b$, $c' = 4c$ gives an area $A' = 4^3 A = 8^2 A = (2^{n+3} m)^2$.

Therefore, if we can find solutions for $n = 0$, $n = 1$ and $n = 2$, all other solutions can be generated by repeated multiplication of b and c by a factor of 4.

Setting $b = 1$ and $c = 2$, we get $A = 1 = (2^0 \cdot 1)^2$, which yields the $n = 0$ case.

Setting $b = 1$ and $c = 9$, we get $A = 9 \cdot 4 = (2^1 \cdot 3)^2$, which yields the $n = 1$ case.

Setting $b = 1$ and $c = 5$, we get $A = 1 \cdot 2 \cdot 5 = 10$. Multiplying these values of b and c by 10, we get $b' = 10$, $c' = 50$, $A' = 10 \cdot 20 \cdot 50 = 100^2 = (2^2 \cdot 5^2)^2$, which yields the $n = 2$ case. This completes the construction.

Solution 2

We proceed via induction on n . Notice that we prove instead a stronger result: there exists a parabolic triangle with area $(2^n m)^2$ with two of the vertices sharing the same y -coordinate.

BASE CASE: If $n = 0$, consider the parabolic triangle ABC with $A(0, 0)$, $B(1, 1)$, $C(-1, 1)$ that has area $1/2 \cdot 1 \cdot 2 = 1$, so that $n = 0$ and $m = 1$. If $n = 1$, let $ABC = A(5, 25)$, $B(4, 16)$, $C(-4, 16)$. Because ABC has area $1/2 \cdot 8 \cdot 9 = 36$, we set $n = 1$ and $m = 3$. If $n = 2$, consider the triangle formed by $A(21, 441)$, $B(3, 9)$, $C(-3, 9)$. It is parabolic and has area $1/2 \cdot 6 \cdot 432 = 1296 = 36^2$, so $n = 2$ and $m = 9$.

INDUCTIVE STEP: If $n = k$ produces parabolic triangle ABC with $A(a, a^2)$, $B(b, b^2)$, and $C(-b, b^2)$, consider $A'B'C'$ with vertices $A(4a, 16a^2)$, $B(4b, 16b^2)$, and $C(-4b, 16b^2)$. If ABC has area $(2^k m)^2$, then $A'B'C'$ has area $(2^{k+3} m)^2$, which is easily verified using the $1/2 \cdot \text{base} \cdot \text{height}$ formula for triangle area. This completes the inductive step for $k \implies k + 3$.

Hence, for every nonnegative integer n , there exists an odd m and a parabolic triangle with area $(2^n m)^2$ with two vertices sharing the same ordinate. The problem statement is a direct result of this result. -MathGenius__

Solution 3 (without induction)

First, consider triangle with vertices $(0, 0)$, $(1, 1)$, $(-1, 1)$. This has area 1 so $n = 0$ case is satisfied.

Then, consider triangle with vertices (a, a^2) , $(-a, a^2)$, (b, b^2) , and set $a = 2^{2n}$ and $b = 2^{4n-2} + 1$. The area of this triangle is $\frac{1}{2} \cdot \text{base} \cdot \text{height} = a(b^2 - a^2)$. We have that $b^2 - a^2 = 2^{8n-4} + 2^{4n-1} + 1 - 2^{4n} = 2^{8n-4} - 2^{4n-1} + 1 = (2^{4n-2} - 1)^2$. We desire $A = a(b^2 - a^2) = 2^{2n} m^2$, or $2^{4n-2} - 1 = m$, and m is clearly always odd for positive n , completing the proof.

Solution 4

We simply need to provide an example for all n that satisfies the condition, and we do so.

Let $a = 2^{2n+1} + 1$. Then consider the triangle with coordinates $(0, 0), (a, a^2), (a^2, a^4) = (x_1, y_1), (x_2, y_2), (x_3, y_3)$.

By the shoelace formula, this triangle has area

$$\frac{1}{2}|x_1y_2 - y_1x_2 + x_2y_3 - y_2x_3| = \frac{1}{2}|a(a^4) - a^2(a^2)| = \frac{1}{2}|a^5 - a^4| = \frac{a^4(a-1)}{2} = 2^{2n}(2^{2n+1}+1)^4$$

which clearly can be written in the form $(2^n \times m)^2 = 2^{2n} \times m^2$, where $m^2 = (2^{2n+1} + 1)^4$ or $m = (2^{2n+1} + 1)^2$. Now, we just have to prove that $(2^{2n+1} + 1)^2$ is always odd. This is true because 2^{2n+1} is even (because it's a power of 2), so $2^{2n+1} + 1$ is odd, and since an odd number squared is odd, the whole thing is odd as well. And we are done, since we have proved that for all n , we can show that there exists such a triangle by merely providing an example. $\square$

5. (*USAJMO 2010*) Two permutations $a_1, a_2, \dots, a_{2010}$ and $b_1, b_2, \dots, b_{2010}$ of the numbers $1, 2, \dots, 2010$ are said to intersect if $a_k = b_k$ for some value of k in the range $1 \leq k \leq 2010$. Show that there exist 1006 permutations of the numbers $1, 2, \dots, 2010$ such that any other such permutation is guaranteed to intersect at least one of these 1006 permutations.

Solution 1

Let n be a positive integer. Let m be the smallest positive integer with $m > n - m$. Since $n > n - n = 0$, $m \leq n$. Let $N = \{1, \dots, n\}$ be the set of positive integers from 1 to n . Let $M \subset N$, be $M = \{1, \dots, m\}$.

Let P_n be the set of permutations of N .

Let C_m be the set of cyclic permutations of M , there are m cyclic permutations in all, and C_m acts transitively on M , i.e. for every pair of elements $a, b \in M$, there is an element of C_m that maps a to b .

Let $C'_m \subset P_n$ be the permutations in P_n that leave $N \setminus M$ fixed, and restricted to M yield one of the permutations in C_m . There is a natural one-to-one correspondence between C'_m and C_m .

We claim that the m permutations C'_m intersect every permutation in P_n .

Suppose, to the contrary, that there exists a permutation $p \in P_n$ that does not intersect any permutation in C'_m . Since C'_m act transitively on $M \subset N$ the permutation p cannot send any element of M to any other element of M , therefore it must send all the elements of M to $N \setminus M$, but since $N \setminus M$ has $n - m$ elements and $m > n - m$, this is impossible by the pigeonhole principle. Therefore such a permutation cannot exist, and the permutations in C'_m intersect every permutation in P_n .

For $n = 2010$ we get $m = 1006$, which is the required special case of the general result above.

Solution 2

I construct the following permutations by continuously rotating the first 1006 numbers):

(1, 2, 3 ... 1005, 1006, 1007 ... 2009, 2010)

(2, 3, 4 ... 1006, 1, 1007 ... 2009, 2010)

...

...

...

(1006, 1, 2 ... 1005, 1007, ... 2009, 2010)

I claim that these permutations above satisfy the property that any other permutation will intersect with at least one of them.

Proof: Assume for the sake of contradiction that there exists a permutation that won't intersect with these, say $(a_1, a_2, a_3 \dots a_{2010})$.

Lemma: If there exists a k such that $1 \leq k \leq 1006$ $a_k \leq 1006$, we will get an intersection with the permutations.

Note that for any 2 distinct permutations in our list, the numbers in k th index must be different, since each permutation is a rotation of an previous permutation. Also, the numbers can't exceed 1006, so, each number must occur exactly once in the k th index.

End Lemma

Using the lemma, in order to avoid intersections, we need $1007 \leq a_k \leq 2010$, for all $1 \leq k \leq 1006$.

But, there are 1004 numbers such that $1007 \leq n \leq 2010$, but we need to have 1004 numbers to fill in 1006 spots, and thus, by pigeonhole principle (the numbers are the holes, the spots are the pigeons), there must be 2 spots that have the same number! However, in a permutation, all numbers must be distinct, so we have a contradiction.

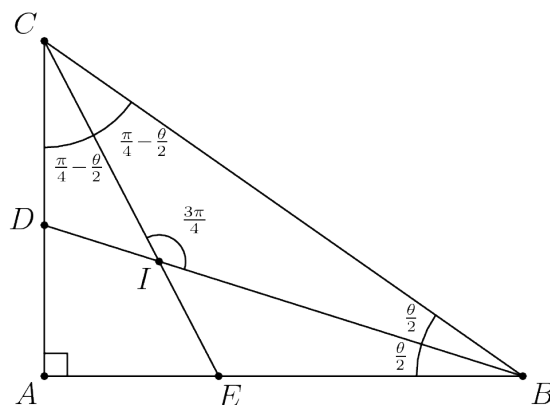
So, we have shown such a set of permutations exists satisfying that any permutation is guaranteed to intersect one of them, and the proof is complete. ■

-Alexlikemath

6. (USAJMO 2010) Let ABC be a triangle with $\angle A = 90^\circ$. Points D and E lie on sides AC and AB , respectively, such that $\angle ABD = \angle DBC$ and $\angle ACE = \angle ECB$. Segments BD and CE meet at I . Determine whether or not it is possible for segments AB, AC, BI, ID, CI, IE to all have integer lengths.

Solution

We know that angle $BIC = 135^\circ$, as the other two angles in triangle BIC add to 45° . Assume that only AB, AC, BI , and CI are integers. Using the [Law of Cosines](#) on triangle BIC ,



$BC^2 = BI^2 + CI^2 - 2BI \cdot CI \cdot \cos 135^\circ$. Observing that $BC^2 = AB^2 + AC^2$ is an integer and that $\cos 135^\circ = -\frac{\sqrt{2}}{2}$, we have

$$BC^2 - BI^2 - CI^2 = BI \cdot CI \cdot \sqrt{2}$$

and therefore,

$$\sqrt{2} = \frac{BC^2 - BI^2 - CI^2}{BI \cdot CI}$$

The LHS ($\sqrt{2}$) is irrational, while the RHS is the quotient of the division of two integers and thus is rational. Clearly, there is a contradiction. Therefore, it is impossible for AB, AC, BI , and CI to all be integers, which invalidates the original claim that all six lengths are integers, and we are done.

Solution 2

The answer is no.

Suppose otherwise. It is easy to see (through simple angle chasing) that $\angle DIC = 45^\circ$. Also, since I is the incenter, we have $\angle IAC = 45^\circ$. Using the Law of Cosines, we have

$$CD^2 = IC^2 + ID^2 - \sqrt{2}(IC)(ID),$$

so that CD is irrational. But $\triangle IAC \sim \triangle DIC$, thus $IC^2 = CD \cdot AC$, implying that CD is rational, contradiction. ■

Solution 3

The result can be also proved without direct appeal to trigonometry, via just the angle bisector theorem and the structure of Pythagorean triples. (This is a lot more work).

A triangle in which all the required lengths are integers exists if and only if there exists a triangle in which AB and AC are relatively-prime integers and the lengths of the segments BI, ID, CI, IE are all rational (we divide all the lengths by the $\gcd(AB, AC)$ or conversely multiply all the lengths by the least common multiple of the denominators of the rational lengths).

Suppose there exists a triangle in which the lengths AB and AC are relatively-prime integers and the lengths IB, ID, CI, IE are all rational.

Since CE is the bisector of $\angle ACB$, by the angle bisector theorem, the ratio $IB : ID = CB : CD$, and since BD is the bisector of $\angle ABC$, $CB : CD = (AB + BC) : AC$. Therefore, $IB : ID = (AB + BC) : AC$. Now $IB : ID$ is by assumption rational, so $(AB + BC) : AC$ is rational, but AB and AC are assumed integers so $BC : AC$ must also be rational. Since BC is the hypotenuse of a right-triangle, its length is the square root of an integer, and thus either an integer or irrational, so BC must be an integer.

With AB and AC relatively-prime, we conclude that the side lengths of $\triangle ABC$ must be a Pythagorean triple: $(2pq, p^2 - q^2, p^2 + q^2)$, with $p > q$ relatively-prime positive integers and $p + q$ odd.

Without loss of generality, $AC = 2pq, AB = p^2 - q^2, BC = p^2 + q^2$. By the angle bisector theorem,

$$AE = \frac{AB \cdot AC}{AC + CB} = \frac{2pq(p^2 - q^2)}{p^2 + q^2 + 2pq} = \frac{2pq(p - q)}{p + q}$$

Since $\triangle CAE$ is a right-triangle, we have:

$$\begin{aligned} CE^2 &= AC^2 + AE^2 = 4p^2q^2 + \left(\frac{2pq(p - q)}{p + q}\right)^2 = 4p^2q^2 \left[1 + \left(\frac{p - q}{p + q}\right)^2\right] \\ &= \frac{4p^2q^2}{(p + q)^2} [(p + q)^2 + (p - q)^2] = \frac{4p^2q^2}{(p + q)^2} (2p^2 + 2q^2) \end{aligned}$$

and so CE is rational if and only if $2p^2 + 2q^2$ is a perfect square.

Also by the angle bisector theorem,

$$AD = \frac{AB \cdot AC}{AB + BC} = \frac{2pq(p^2 - q^2)}{p^2 + q^2 + p^2 - q^2} = \frac{q(p^2 - q^2)}{p}$$

and therefore, since $\triangle DAB$ is a right-triangle, we have:

$$\begin{aligned} BD^2 &= AB^2 + AD^2 = (p^2 - q^2)^2 + \left(\frac{q(p^2 - q^2)}{p}\right)^2 \\ &= (p^2 - q^2)^2 \left[1 + \frac{q^2}{p^2}\right] = \frac{(p^2 - q^2)^2}{p^2} (p^2 + q^2) \end{aligned}$$

and so BD is rational if and only if $p^2 + q^2$ is a perfect square.

Combining the conditions on CE and BD , we see that $2p^2 + 2q^2$ and $p^2 + q^2$ must both be perfect squares. If it were so, their ratio, which is 2, would be the square of a rational number, but $\sqrt{2}$ is irrational, and so the assumed triangle cannot exist.

Solution 4

We proceed by contradiction.

FTSOC, let AB, BI, ID have integer lengths. Then $BD = BI + ID \in \mathbb{Z}$ as well. By trigonometry,

$$BD = \frac{AB}{\cos\left(\frac{\angle ABC}{2}\right)}.$$

Rearranging we find $\cos\left(\frac{\angle ABC}{2}\right) = \frac{AB}{BD} \in \mathbb{Q}$. As $0 < \angle ABC < \frac{\pi}{2}$ so $0 < \frac{\angle ABC}{2} < \frac{\pi}{4}$ but there are no possible angles in the interval that result in a rational cosine by Niven's theorem. So we have contradiction.

§2 USAJMO 2011

1. (*USAJMO 2011*) Find, with proof, all positive integers n for which $2^n + 12^n + 2011^n$ is a perfect square.

Solution 1

Let $2^n + 12^n + 2011^n = x^2$. Then $(-1)^n + 1 \equiv x^2 \pmod{3}$. Since all perfect squares are congruent to 0 or 1 modulo 3, this means that n must be odd. Proof by Contradiction: We wish to show that the only value of n that satisfies is $n = 1$. Assume that $n \geq 2$. Then consider the equation $2^n + 12^n = x^2 - 2011^n$. From modulo 2, we easily know x is odd. Let $x = 2a + 1$, where a is an integer. $2^n + 12^n = 4a^2 + 4a + 1 - 2011^n$. Dividing by 4, $2^{n-2} + 3^n = a^2 + a + \frac{1}{4}(1 - 2011^n)$. Since $n \geq 2$, $n - 2 \geq 0$, so 2^{n-2} similarly, the entire LHS is an integer, and so are a^2 and a . Thus, $\frac{1}{4}(1 - 2011^n)$ must be an integer. Let $\frac{1}{4}(1 - 2011^n) = k$. Then we have $1 - 2011^n = 4k$. $1 - 2011^n \equiv 0 \pmod{4}$. $(-1)^n \equiv 1 \pmod{4}$. Thus, n is even. However, it has already been shown that n must be odd. This is a contradiction. Therefore, n is not greater than or equal to 2, and must hence be less than 2. The only positive integer less than 2 is 1.

Solution 2

If $n = 1$, then $2^n + 12^n + 2011^n = 2025 = 45^2$, a perfect square.

If $n > 1$ is odd, then $2^n + 12^n + 2011^n \equiv 0 + 0 + (-1)^n \equiv 3 \pmod{4}$.

Since all perfect squares are congruent to 0, 1 $\pmod{4}$, we have that $2^n + 12^n + 2011^n$ is not a perfect square for odd $n > 1$.

If $n = 2k$ is even, then $(2011^k)^2 < 2^{2k} + 12^{2k} + 2011^{2k} = 4^k + 144^k + 2011^{2k} < 2011^k + 2011^k + 2011^{2k} < (2011^k + 1)^2$.

Since $(2011^k)^2 < 2^n + 12^n + 2011^n < (2011^k + 1)^2$, we have that $2^n + 12^n + 2011^n$ is not a perfect square for even n .

Thus, $n = 1$ is the only positive integer for which $2^n + 12^n + 2011^n$ is a perfect square.

Solution 3

Looking at residues mod 3, we see that n must be odd, since even values of n leads to $2^n + 12^n + 2011^n \equiv 2 \pmod{3}$. Also as shown in solution 2, for $n > 1$, n must be even. Hence, for $n > 1$, n can neither be odd nor even. The only possible solution is then $n = 1$, which indeed works.

Solution 4

Take the whole expression mod 12. Note that the perfect squares can only be of the form 0, 1, 4 or 9 $\pmod{12}$. Note that since the problem is asking for positive integers, 12^n is always divisible by 12, so this will be disregarded in this process. If n is even, then $2^n \equiv 4 \pmod{12}$ and $2011^n \equiv 7^n \equiv 1 \pmod{12}$. Therefore, the sum in the problem is congruent to 5 $\pmod{12}$, which cannot be a perfect square. Now we check the case for which n is an odd number greater than 1. Then $2^n \equiv 8 \pmod{12}$ and $2011^n \equiv 7^n \equiv 7 \pmod{12}$. Therefore, this sum would be congruent to 3 $\pmod{12}$, which cannot be a perfect square. The only case we have not checked is $n = 1$. If $n = 1$, then the sum in the problem is equal to $2 + 12 + 2011 = 2025 = 45^2$. Therefore the only possible value of n such that $2^n + 12^n + 2011^n$ is a perfect square is $n = 1$.

Solution 5

We will first take the expression modulo 3. We get $2^n + 12^n + 2011^n \equiv -1^n + 1^n \pmod{3}$.

Lemma 1: All perfect squares are equal to 0 or 1 modulo 3. We can prove this by testing the residues modulo 3. We have $0^2 \equiv 0 \pmod{3}$, $1^2 \equiv 1 \pmod{3}$, and $2^2 \equiv 1 \pmod{3}$, so the lemma is true.

We know that if n is odd, $-1^n + 1^n \equiv 0 \pmod{3}$, which satisfies the lemma's conditions. However, if n is even, we get $2 \pmod{3}$, which does not satisfy the lemma's conditions. So, we can conclude that n is odd.

Now, we take the original expression modulo 4. For right now, we will assume that $n > 1$, and test $n = 1$ later. For $n > 1$, $2^n \equiv 0 \pmod{4}$, so $2^n + 12^n + 2011^n \equiv -1^n \pmod{4}$.

Lemma 2: All perfect squares are equal to 0 or 1 modulo 4. We can prove this by testing the residues modulo 4. We have $0^2 \equiv 0 \pmod{4}$, $1^2 \equiv 1 \pmod{4}$, $2^2 \equiv 0 \pmod{4}$, and $3^2 \equiv 1 \pmod{4}$, so the lemma is true.

We know that if n is even, $-1^n \equiv 0 \pmod{4}$, which satisfies the lemma's conditions. However, if n is odd, $-1^n \equiv -1 \equiv 3 \pmod{4}$, which does not satisfy the lemma's conditions. Therefore, n must be even.

However, a number cannot be even and odd at the same time, so this is impossible. Now, we only have to test $n = 1$. We know that $2^1 + 12^1 + 2011^1 = 45^2$, so the only integer is $\boxed{n = 1}$.

Solution 6

The answer is $n = 1$, which is easily verified to be a valid integer n . Notice that

$$2^n + 12^n + 2011^n \equiv 2^n + 7^n \pmod{12}.$$

Then for $n \geq 2$, we have $2^n + 7^n \equiv 3, 5 \pmod{12}$ depending on the parity of n . But perfect squares can only be $0, 1, 4, 9 \pmod{12}$, contradiction. Therefore, we are done. ■

2. (USAJMO 2011) Let a, b, c be positive real numbers such that $a^2 + b^2 + c^2 + (a + b + c)^2 \leq 4$. Prove that

$$\frac{ab+1}{(a+b)^2} + \frac{bc+1}{(b+c)^2} + \frac{ca+1}{(c+a)^2} \geq 3.$$

Solution 1

Since

$$\begin{aligned} (a+b)^2 + (b+c)^2 + (c+a)^2 &= 2(a^2 + b^2 + c^2 + ab + bc + ca) \\ &= a^2 + b^2 + c^2 + (a+b+c)^2, \end{aligned}$$

it is natural to consider a change of variables:

$$\begin{aligned} \alpha &= b+c \\ \beta &= c+a \\ \gamma &= a+b \end{aligned}$$

with the inverse mapping given by:

$$a = \frac{\beta + \gamma - \alpha}{2}$$

$$b = \frac{\alpha + \gamma - \beta}{2}$$

$$c = \frac{\alpha + \beta - \gamma}{2}$$

With this change of variables, the constraint becomes

$$\alpha^2 + \beta^2 + \gamma^2 \leq 4,$$

while the left side of the inequality we need to prove is now

$$\begin{aligned} & \frac{\gamma^2 - (\alpha - \beta)^2 + 4}{4\gamma^2} + \frac{\alpha^2 - (\beta - \gamma)^2 + 4}{4\alpha^2} + \frac{\beta^2 - (\gamma - \alpha)^2 + 4}{4\beta^2} \geq \\ & \frac{\gamma^2 - (\alpha - \beta)^2 + \alpha^2 + \beta^2 + \gamma^2}{4\gamma^2} + \frac{\alpha^2 - (\beta - \gamma)^2 + \alpha^2 + \beta^2 + \gamma^2}{4\alpha^2} + \frac{\beta^2 - (\gamma - \alpha)^2 + \alpha^2 + \beta^2 + \gamma^2}{4\beta^2} = \\ & \frac{2\gamma^2 + 2\alpha\beta}{4\gamma^2} + \frac{2\alpha^2 + 2\beta\gamma}{4\alpha^2} + \frac{2\beta^2 + 2\gamma\alpha}{4\beta^2} = \\ & \frac{3}{2} + \frac{\alpha\beta}{2\gamma^2} + \frac{\beta\gamma}{2\alpha^2} + \frac{\gamma\alpha}{2\beta^2}. \end{aligned}$$

Therefore it remains to prove that

$$\frac{\alpha\beta}{2\gamma^2} + \frac{\beta\gamma}{2\alpha^2} + \frac{\gamma\alpha}{2\beta^2} \geq \frac{3}{2}.$$

We note that the product of the three (positive) terms is $1/8$, therefore by AM-GM their mean is at least $1/2$, and thus their sum is at least $3/2$ and we are done.

Solution 2

Rearranging the condition yields that

$$a^2 + b^2 + c^2 + ab + bc + ac \leq 2$$

Now note that

$$\frac{2ab + 2}{(a + b)^2} \geq \frac{2ab + a^2 + b^2 + c^2 + ab + bc + ac}{(a + b)^2} = \frac{(a + b)^2 + (c + a)(c + b)}{(a + b)^2}$$

Summing this for all pairs of $\{a, b, c\}$ gives that

$$\sum_{cyc} \frac{2ab + 2}{(a + b)^2} \geq 3 + \sum_{cyc} \frac{(c + a)(c + b)}{(a + b)^2} \geq 6$$

By AM-GM. Dividing by 2 gives the desired inequality.

Solution 3

Given $a^2 + b^2 + c^2 + (a + b + c)^2 \leq 4$, prove that

$$\frac{ab + 1}{(a + b)^2} + \frac{bc + 1}{(b + c)^2} + \frac{ac + 1}{(a + c)^2} \geq 3.$$

From the given constraint $a^2 + b^2 + c^2 + (a + b + c)^2 \leq 4$, we can expand to obtain:

$$a^2 + b^2 + c^2 + a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \leq 4$$

$$2(a^2 + b^2 + c^2) + 2(ab + bc + ca) \leq 4$$

$$\frac{a^2 + b^2 + c^2}{2} + \frac{ab + bc + ca}{2} \leq 1.$$

Now, observe that we can rewrite the left-hand side of our inequality as:

$$\begin{aligned} & \sum_{\text{cyc}} \frac{ab + 1}{(a + b)^2} \\ &= \sum_{\text{cyc}} \frac{ab + \frac{a^2 + b^2 + c^2}{2} + \frac{ab + bc + ca}{2}}{(a + b)^2} \\ &= \sum_{\text{cyc}} \left[\frac{ab + \frac{a^2 + b^2}{2}}{(a + b)^2} + \frac{\frac{c^2}{2} + \frac{bc + ca}{2}}{(a + b)^2} \right] \\ &= \sum_{\text{cyc}} \left[\frac{ab + \frac{a^2 + b^2}{2}}{(a + b)^2} \right] + \sum_{\text{cyc}} \left[\frac{\frac{a^2 + c^2}{2}}{(a + c)^2} + \frac{\frac{ac + bc}{2}}{(a + b)^2} \right]. \end{aligned}$$

We now evaluate each part separately. Note that

$$\frac{ab + \frac{a^2 + b^2}{2}}{(a + b)^2} = \frac{\frac{(a + b)^2}{2}}{(a + b)^2} = \frac{1}{2}.$$

Therefore,

$$\sum_{\text{cyc}} \frac{ab + \frac{a^2 + b^2}{2}}{(a + b)^2} = \frac{3}{2}.$$

By the RMS-AM inequality, we have $(a + c)^2 \leq 2(a^2 + c^2)$, which gives $\frac{1}{(a + c)^2} \geq \frac{1}{2(a^2 + c^2)}$. Thus,

$$\begin{aligned} \sum_{\text{cyc}} \frac{\frac{a^2 + c^2}{2}}{(a + c)^2} &\geq \sum_{\text{cyc}} \frac{\frac{a^2 + c^2}{2}}{2(a^2 + c^2)} \\ &= \sum_{\text{cyc}} \frac{1}{4} = \frac{3}{4}. \end{aligned}$$

The remaining terms can be written as:

$$\begin{aligned} \sum_{\text{cyc}} \frac{\frac{ac + bc}{2}}{(a + b)^2} &= \frac{1}{2} \sum_{\text{cyc}} \frac{c(a + b)}{(a + b)^2} \\ &= \frac{1}{2} \sum_{\text{cyc}} \frac{c}{a + b} \\ &= \frac{1}{2} \left(\frac{a}{b + c} + \frac{b}{a + c} + \frac{c}{a + b} \right). \end{aligned}$$

By Nesbitt's inequality,

$$\frac{a}{b + c} + \frac{b}{a + c} + \frac{c}{a + b} \geq \frac{3}{2},$$

so this part contributes at least $\frac{3}{4}$. Adding all three parts together:

$$\frac{ab + 1}{(a + b)^2} + \frac{bc + 1}{(b + c)^2} + \frac{ac + 1}{(a + c)^2} \geq \frac{3}{2} + \frac{3}{4} + \frac{3}{4} = 3.$$

Therefore, the inequality is proven. $\square$

3. (*USAJMO 2011*) For a point $P = (a, a^2)$ in the coordinate plane, let $\ell(P)$ denote the line passing through P with slope $2a$. Consider the set of triangles with vertices of the form $P_1 = (a_1, a_1^2)$, $P_2 = (a_2, a_2^2)$, $P_3 = (a_3, a_3^2)$, such that the intersections of the lines $\ell(P_1)$, $\ell(P_2)$, $\ell(P_3)$ form an equilateral triangle Δ . Find the locus of the center of Δ as $P_1P_2P_3$ ranges over all such triangles.

Solution 1

Note that the lines $\ell(P_1), \ell(P_2), \ell(P_3)$ are

$$y = 2a_1x - a_1^2, y = 2a_2x - a_2^2, y = 2a_3x - a_3^2,$$

respectively. It is easy to deduce that the three points of intersection are

$$\left(\frac{a_1 + a_2}{2}, a_1a_2\right), \left(\frac{a_2 + a_3}{2}, a_2a_3\right), \left(\frac{a_3 + a_1}{2}, a_3a_1\right).$$

The slopes of each side of this equilateral triangle are

$$2a_1, 2a_2, 2a_3,$$

and we want to find the locus of

$$\left(\frac{a_1 + a_2 + a_3}{3}, \frac{a_1a_2 + a_2a_3 + a_3a_1}{3}\right).$$

Define the three complex numbers $w_n = 1 + 2a_ni$ for $n = 1, 2, 3$. Then note that the slope - that is, the imaginary part divided by the real part - of all w_n^3 is constant, say it is k . Then for $n = 1, 2, 3$,

$$\begin{aligned} \frac{\Im(w_n^3)}{\Re(w_n^3)} &= \frac{\Im((1 + 2a_ni)^3)}{\Re((1 + 2a_ni)^3)} \\ &= \frac{\Im(1 + 6a_ni - 12a_n^2 - 8a_n^3i)}{\Re(1 + 6a_ni - 12a_n^2 - 8a_n^3i)} \\ &= \frac{6a_n - 8a_n^3}{1 - 12a_n^2} \\ &= k. \end{aligned}$$

Rearranging, we get that

$$8a_n^3 - 12ka_n^2 - 6a_n + k = 0,$$

or

$$a_n^3 - \frac{3ka_n^2}{2} - \frac{3a_n}{4} + \frac{k}{8} = 0.$$

Note that this is a cubic, and the roots are a_1, a_2 and a_3 which are all distinct, and so there are no other roots. Using Vieta's, we get that

$$a_1 + a_2 + a_3 = \frac{3k}{2},$$

and

$$a_1a_2 + a_2a_3 + a_3a_1 = -\frac{3}{4}.$$

Obviously all values of k are possible, and so our answer is the line

$$y = -\frac{1}{4}.$$

■

Solution 2

Note that all the points $P = (a, a^2)$ belong to the parabola $y = x^2$ which we will denote p . This parabola has a focus $F = (0, \frac{1}{4})$ and directrix $y = -\frac{1}{4}$ which we will denote d . We will prove that the desired locus is d .

First note that for any point P on p , the line $\ell(P)$ is the tangent line to p at P . This is because $\ell(P)$ contains P and because $[\frac{d}{dx}]x^2 = 2x$. If you don't like calculus, you can also verify that $\ell(P)$ has equation $y = 2a(x - a) + a^2$ and does not intersect $y = x^2$ at any point besides P . Now for any point P on p let P' be the foot of the perpendicular from P onto d . Then by the definition of parabolas, $PP' = PF$. Let q be the perpendicular bisector of $\overline{P'F}$. Since $PP' = PF$, q passes through P . Suppose K is any other point on q and let K' be the foot of the perpendicular from K to d . Then in right $\triangle KKK'P'$, KK' is a leg and so $KK' < KP' = KF$. Therefore K cannot be on p . This implies that q is exactly the tangent line to p at P , that is $q = \ell(P)$. So we have proved Lemma 1: If P is a point on p then $\ell(P)$ is the perpendicular bisector of $\overline{P'F}$.

We need another lemma before we proceed. Lemma 2: If F is on the circumcircle of $\triangle XYZ$ with orthocenter H , then the reflections of F across $\overleftrightarrow{XY}$, $\overleftrightarrow{XZ}$, and $\overleftrightarrow{YZ}$ are collinear with H .

Proof of Lemma 2: Say the reflections of F and H across $\overleftrightarrow{YZ}$ are C' and J , and the reflections of F and H across $\overleftrightarrow{XY}$ are A' and I . Then we angle chase $\angle JYZ = \angle HYZ = \angle HXZ = \angle JXZ = m(\widehat{JZ})/2$ where $m(\widehat{JZ})$ is the measure of minor arc JZ on the circumcircle of $\triangle XYZ$. This implies that J is on the circumcircle of $\triangle XYZ$, and similarly I is on the circumcircle of $\triangle XYZ$. Therefore $\angle C'HJ = \angle FJH = m(\widehat{XF})/2$, and $\angle A'HX = \angle FHX = m(\widehat{FX})/2$. So $\angle C'HJ = \angle A'HX$. Since J , H , and X are collinear it follows that C' , H and A' are collinear. Similarly, the reflection of F over $\overleftrightarrow{XZ}$ also lies on this line, and so the claim is proved.

Now suppose A , B , and C are three points of p and let $\ell(A) \cap \ell(B) = X$, $\ell(A) \cap \ell(C) = Y$, and $\ell(B) \cap \ell(C) = Z$. Also let A'' , B'' , and C'' be the midpoints of $\overline{A'F}$, $\overline{B'F}$, and $\overline{C'F}$ respectively. Then since $\overleftrightarrow{A''B''} \parallel \overleftrightarrow{A'B'} = d$ and $\overleftrightarrow{B''C''} \parallel \overleftrightarrow{B'C'} = d$, it follows that A'' , B'' , and C'' are collinear. By Lemma 1, we know that A'' , B'' , and C'' are the feet of the altitudes from F to $\overleftrightarrow{XY}$, $\overleftrightarrow{XZ}$, and $\overleftrightarrow{YZ}$. Therefore by the Simson Line Theorem, F is on the circumcircle of $\triangle XYZ$. If H is the orthocenter of $\triangle XYZ$, then by Lemma 2, it follows that H is on $\overleftrightarrow{A'C'} = d$. It follows that the locus described in the problem is a subset of d .

Since we claim that the locus described in the problem is d , we still need to show that for any choice of H on d there exists an equilateral triangle with center H such that the lines containing the sides of the triangle are tangent to p . So suppose H is any point on d and let the circle centered at H through F be O . Then suppose A is one of the intersections of d with O . Let $\angle HFA = 3\theta$, and construct the ray through F on the same halfplane of $\overleftrightarrow{HF}$ as A that makes an angle of 2θ with $\overleftrightarrow{HF}$. Say this ray intersects O in a point B besides F , and let q be the perpendicular

bisector of $\overline{HB}$. Since $\angle HFB = 2\theta$ and $\angle HFA = 3\theta$, we have $\angle BFA = \theta$. By the inscribed angles theorem, it follows that $\angle AHB = 2\theta$. Also since HF and HB are both radii, $\triangle HFB$ is isosceles and $\angle HBF = \angle HFB = 2\theta$. Let P'_1 be the reflection of F across q . Then $2\theta = \angle FBH = \angle C'HB$, and so $\angle C'HB = \angle AHB$. It follows that P'_1 is on $\overleftrightarrow{AH} = d$, which means q is the perpendicular bisector of $\overline{FP'_1}$.

Let q intersect O in points Y and Z and let X be the point diametrically opposite to B on O . Also let $\overline{HB}$ intersect q at M . Then $HM = HB/2 = HZ/2$. Therefore $\triangle HMZ$ is a $30 - 60 - 90$ right triangle and so $\angle ZHB = 60^\circ$. So $\angle ZHY = 120^\circ$ and by the inscribed angles theorem, $\angle ZXY = 60^\circ$. Since $ZX = ZY$ it follows that $\triangle ZXY$ is an equilateral triangle with center H .

By Lemma 2, it follows that the reflections of F across $\overleftrightarrow{XY}$ and $\overleftrightarrow{XZ}$, call them P'_2 and P'_3 , lie on d . Let the intersection of $\overleftrightarrow{YZ}$ and the perpendicular to d through P'_1 be P_1 , the intersection of $\overleftrightarrow{XY}$ and the perpendicular to d through P'_2 be P_2 , and the intersection of $\overleftrightarrow{XZ}$ and the perpendicular to d through P'_3 be P_3 . Then by the definitions of P'_1 , P'_2 , and P'_3 it follows that $FP_i = P_iP'_i$ for $i = 1, 2, 3$ and so P_1 , P_2 , and P_3 are on p . By lemma 1, $\ell(P_1) = \overleftrightarrow{YZ}$, $\ell(P_2) = \overleftrightarrow{XY}$, and $\ell(P_3) = \overleftrightarrow{XZ}$. Therefore the intersections of $\ell(P_1)$, $\ell(P_2)$, and $\ell(P_3)$ form an equilateral triangle with center H , which finishes the proof.—Killbilledtoucan

Solution 3

Note that the lines $\ell(P_1), \ell(P_2), \ell(P_3)$ are

$$y = 2a_1x - a_1^2, y = 2a_2x - a_2^2, y = 2a_3x - a_3^2,$$

respectively. It is easy to deduce that the three points of intersection are

$$\left(\frac{a_1 + a_2}{2}, a_1a_2\right), \left(\frac{a_2 + a_3}{2}, a_2a_3\right), \left(\frac{a_3 + a_1}{2}, a_3a_1\right).$$

The slopes of each side of this equilateral triangle are

$$2a_1, 2a_2, 2a_3,$$

and we want to find the locus of

$$\left(\frac{a_1 + a_2 + a_3}{3}, \frac{a_1a_2 + a_2a_3 + a_3a_1}{3}\right).$$

We know that

$$2a_1 = \tan(\theta), 2a_2 = \tan(\theta + 120), 2a_3 = \tan(\theta - 120)$$

for some θ . Therefore, we can use the tangent addition formula to deduce

$$\frac{a_1 + a_2 + a_3}{3} = \frac{\tan(\theta) + \tan(\theta + 120) + \tan(\theta - 120)}{6} = \frac{3\tan\theta - \tan^3\theta}{2 - 6\tan^2\theta}$$

and

$$\begin{aligned} \frac{a_1a_2 + a_2a_3 + a_3a_1}{3} &= \frac{\tan\theta(\tan(\theta - 120) + \tan(\theta + 120)) + \tan(\theta - 120)\tan(\theta + 120)}{12} \\ &= \frac{9\tan^2\theta - 3}{12(1 - 3\tan^2\theta)} \end{aligned}$$

$$= -\frac{1}{4}.$$

Now we show that $\frac{a_1+a_2+a_3}{3}$ can be any real number. Let's say

$$\frac{3 \tan \theta - \tan^3 \theta}{2 - 6 \tan^2 \theta} = k$$

for some real number k . Multiplying both sides by $2 - \tan^2 \theta$ and rearranging yields a cubic in $\tan \theta$. Clearly this cubic has at least one real solution. As $\tan \theta$ can take on any real number, all values of k are possible, and our answer is the line

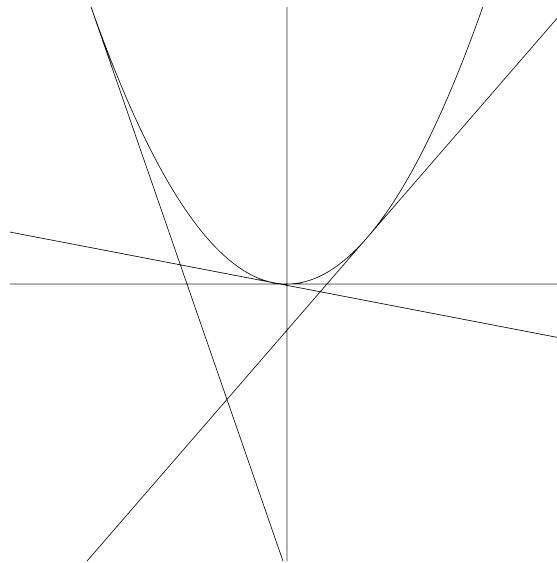
$$y = -\frac{1}{4}.$$

Of course, as the denominator could equal 0, we must check $\tan \theta = \pm \frac{1}{\sqrt{3}}$.

$$3 \tan \theta - \tan^3 \theta = k(2 - 6 \tan^2 \theta).$$

The left side is nonzero, while the right side is zero, so these values of θ do not contribute to any values of k . So, our answer remains the same. ■ Benq

Work in progress: Solution 4 (Clean algebra)



It can be easily shown that the center of $\triangle$ has coordinates $(\frac{a_1+a_2+a_3}{3}, \frac{a_1a_2+a_2a_3+a_3a_1}{3})$.

Without loss of generality, let $a_1 > a_2 > a_3$. Notice that $\ell(P_2)$ is a 60° clockwise rotation of $\ell(P_1)$, $\ell(P_3)$ is a 60° clockwise rotation of $\ell(P_2)$, and $\ell(P_1)$ is a 60° clockwise rotation of $\ell(P_3)$. By definition, $\arctan(2a_i)$ is the (directed) angle from the x-axis to $\ell(P_i)$. Remember that the range of $\arctan(x)$ is $(-90^\circ, 90^\circ)$. We have

$$\begin{aligned} \arctan(2a_1) - \arctan(2a_2) &= 60^\circ \\ \arctan(2a_2) - \arctan(2a_3) &= 60^\circ \\ \arctan(2a_3) - \arctan(2a_1) &= -120^\circ. \end{aligned}$$

Taking the tangent of both sides of each equation and rearranging, we get

$$2a_1 - 2a_2 = \sqrt{3}(1 + 4a_1a_2)$$

$$\begin{aligned} 2a_2 - 2a_3 &= \sqrt{3}(1 + 4a_2a_3) \\ 2a_3 - 2a_1 &= \sqrt{3}(1 + 4a_3a_1). \end{aligned}$$

We add these equations to get

$$\sqrt{3}(3 + 4(a_1a_2 + a_2a_3 + a_3a_1)) = 0.$$

We solve for $a_1a_2 + a_2a_3 + a_3a_1$ to get

$$a_1a_2 + a_2a_3 + a_3a_1 = -\frac{3}{4}.$$

So, the y-coordinate of $\triangle$ is $-\frac{1}{4}$.

We will prove that the x-coordinate of $\triangle$ can be any real number. If $2a_1$ tends to infinity, then $2a_2$ tends to $\frac{\sqrt{3}}{2}$ and $2a_3$ tends to $-\frac{\sqrt{3}}{2}$. So, $\frac{a_1+a_2+a_3}{3}$ can be arbitrarily large. Similarly, if we let $2a_3$ tend to negative infinity, then $2a_1$ tends to $\frac{\sqrt{3}}{2}$ and $2a_2$ tends to $-\frac{\sqrt{3}}{2}$. So, $\frac{a_1+a_2+a_3}{3}$ can be arbitrarily small. Since $\frac{a_1+a_2+a_3}{3}$ is continuous, it can take any real value. So, the locus is the line $y = -\frac{1}{4}$.

4. (*USAJMO 2011*) A *word* is defined as any finite string of letters. A word is a *palindrome* if it reads the same backwards as forwards. Let a sequence of words $W_0, W_1, W_2, \dots$ be defined as follows: $W_0 = a$, $W_1 = b$, and for $n \geq 2$, W_n is the word formed by writing W_{n-2} followed by W_{n-1} . Prove that for any $n \geq 1$, the word formed by writing $W_1, W_2, \dots, W_n$ in succession is a palindrome.

Solution

Let r be the reflection function on the set of words, namely $r(a_1 \dots a_n) = a_n \dots a_1$ for all words $a_1 \dots a_n$, $n \geq 1$. Then the following property is evident (e.g. by mathematical induction):

$$r(w_1 \dots w_k) = r(w_k) \dots r(w_1), \text{ for any words } w_1, \dots, w_k, k \geq 1.$$

We use mathematical induction to prove the statement of the problem. First, $W_1 = b$, $W_1W_2 = bab$, $W_1W_2W_3 = babbab$ are palindromes. Second, suppose $n \geq 3$, and that the words $W_1W_2 \dots W_k$ ($k = 1, 2, \dots, n$) are all palindromes, i.e. $r(W_1W_2 \dots W_k) = W_1W_2 \dots W_k$. Now, consider the word $W_1W_2 \dots W_{n+1}$:

$$\begin{aligned} r(W_1W_2 \dots W_{n+1}) &= r(W_{n+1})r(W_1W_2 \dots W_{n-2}W_{n-1}W_n) \\ &= r(W_{n-1}W_n)W_1W_2 \dots W_{n-2}W_{n-1}W_n \\ &= r(W_{n-1}W_n)r(W_1W_2 \dots W_{n-2})W_{n+1} \\ &= r(W_1W_2 \dots W_{n-2}W_{n-1}W_n)W_{n+1} \\ &= W_1W_2 \dots W_nW_{n+1}. \end{aligned}$$

By the principle of mathematical induction, the statement of the problem is proved.
Lightest 21:54, 1 April 2012 (EDT)

5. (USAJMO 2011) Points A, B, C, D, E lie on a circle ω and point P lies outside the circle. The given points are such that (i) lines PB and PD are tangent to ω , (ii) P, A, C are collinear, and (iii) $\overline{DE} \parallel \overline{AC}$. Prove that $\overline{BE}$ bisects $\overline{AC}$.

Solution

Since $\overline{DE} \parallel \overline{AC}$, we have that $AD = CE \rightarrow \angle ABD = \angle CBE$. But it is well known that $ABCD$ is a harmonic quadrilateral, thus $\overline{BD}$ is a symmedian of triangle ABC , from which it follows that $\overline{BE}$ is a median of $\triangle ABC$. ■

Solution 1

Connect segment PO , and name the intersection of PO and the circle as point M . Since PB and PD are tangent to the circle, it's easy to see that M is the midpoint of arc BD .

$$\angle BOA = 1/2 \text{ arc } AB + 1/2 \text{ arc } CE$$

Since $AC \parallel DE$, arc $AD = \text{arc } CE$,

$$\text{thus, } \angle BOA = 1/2 \text{ arc } AB + 1/2 \text{ arc } AD = 1/2 \text{ arc } BD = \text{arc } BM = \angle BOM$$

Therefore, $PBFO$ is cyclic, $\angle PFO = \angle OBP = 90^\circ$, $AF = AC$ (F is the intersection of BE and AC)

BE bisects AC , proof completed!

Solution 2

Let O be the center of the circle, and let X be the intersection of AC and BE . Let $\angle OPA$ be x and $\angle OPD$ be y .

$$\angle OPB = \angle OPD = y, \angle BED = \frac{\angle DOB}{2} = 90 - y, \angle ODE = \angle PDE - 90 = 90 - x - y$$

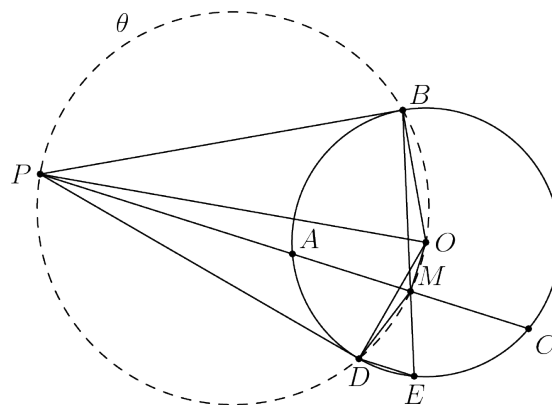
$$\angle OBE = \angle PBE - 90 = x = \angle OPA$$

Thus $PBXO$ is a cyclic quadrilateral and $\angle OXP = \angle OBP = 90$ and so X is the midpoint of chord AC .

Solution 3

This is the solution from EGMO Problem 1.43 page 242

Let O be the center of the circle, and let M be the midpoint of AC . Let θ denote the circle with diameter OP . Since $\angle OBP = \angle OMP = \angle ODP = 90^\circ$, B, D , and M all lie on θ .



Since quadrilateral $BOMP$ is cyclic, $\angle BMP = \angle BOP$. Triangles BOP and DOP are congruent, so $\angle BOP = \angle BOD/2 = \angle BED$, so $\angle BMP = \angle BED$. Because AC and DE are parallel, M lies on BE (using Euclid's Parallel Postulate).

-Evan Chen (vEnhance)

Solution 4

Note that by Lemma 9.9 of EGMO, $(A, C; B, D)$ is a harmonic bundle. We project through E onto $\overline{AC}$,

$$-1 = (A, C; B, D) \stackrel{E}{=} (A, C; M, P_\infty)$$

Where P_∞ is the point at infinity for parallel lines $\overline{DE}$ and $\overline{AC}$. Thus, we get $\frac{MA}{MC} = -1$, and M is the midpoint of AC . novus677

The power of angle chasing=

6. (*USAJMO 2011*) Consider the assertion that for each positive integer $n \geq 2$, the remainder upon dividing 2^{2^n} by $2^n - 1$ is a power of 4. Either prove the assertion or find (with proof) a counterexample.

Solution

We will show that $n = 25$ is a counter-example.

Since $2^n \equiv 1 \pmod{2^n - 1}$, we see that for any integer k , $2^{2^n} \equiv 2^{(2^n - kn)} \pmod{2^n - 1}$. Let $0 \leq m < n$ be the residue of $2^n \pmod{n}$. Note that since $m < n$ and $n \geq 2$, necessarily $2^m < 2^n - 1$, and thus the remainder in question is 2^m . We want to show that $2^m \pmod{2^n - 1}$ is an odd power of 2 for some n , and thus not a power of 4.

Let $n = p^2$ for some odd prime p . Then $\varphi(p^2) = p^2 - p$. Since 2 is co-prime to p^2 , we have

$$2^{\varphi(p^2)} \equiv 1 \pmod{p^2}$$

and thus

$$2^{p^2} \equiv 2^{(p^2 - p) + p} \equiv 2^p \pmod{p^2}.$$

Therefore, for a counter-example, it suffices that $2^p \pmod{p^2}$ be odd. Choosing $p = 5$, we have $2^5 = 32 \equiv 7 \pmod{25}$. Therefore, $2^{25} \equiv 7 \pmod{25}$ and thus

$$2^{2^{25}} \equiv 2^7 \pmod{2^{25} - 1}.$$

Since 2^7 is not a power of 4, we are done.

Solution 2

Lemma (useful for all situations): If x and y are positive integers such that $2^x - 1$ divides $2^y - 1$, then x divides y . Proof: $2^y \equiv 1 \pmod{2^x - 1}$. Replacing the 1 with a 2^x and dividing out the powers of two should create an easy induction proof which will be left to the reader as an Exercise.

Consider $n = 25$. We will prove that this case is a counterexample via contradiction.

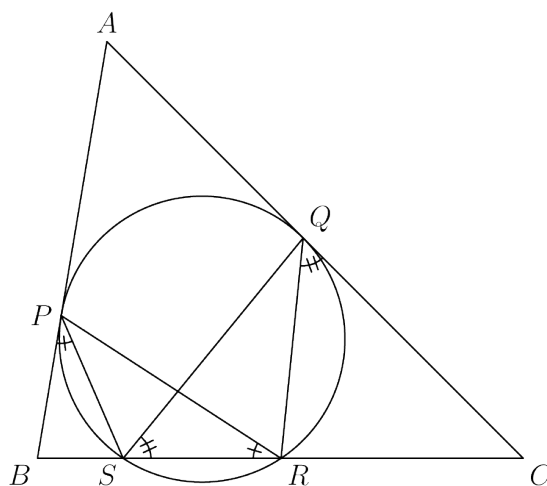
Because $4 = 2^2$, we will assume there exists a positive integer k such that $2^{2^n} - 2^{2k}$ divides $2^n - 1$ and $2^{2k} < 2^n - 1$. Dividing the powers of 2 from LHS gives $2^{2^n - 2k} - 1$ divides $2^n - 1$. Hence, $2^n - 2k$ divides n . Because $n = 25$ is odd, $2^{24} - k$ divides 25. Euler's theorem gives $2^{24} \equiv 2^4 \equiv 16 \pmod{25}$ and so $k \geq 16$. However, $2^{2k} \geq 2^{32} > 2^{25} - 1$, a contradiction. Thus, $n = 25$ is a valid counterexample.

§3 USAJMO 2012

1. (*USAJMO 2012*) Given a triangle ABC , let P and Q be points on segments $\overline{AB}$ and $\overline{AC}$, respectively, such that $AP = AQ$. Let S and R be distinct points on segment $\overline{BC}$ such that S lies between B and R , $\angle BPS = \angle PRS$, and $\angle CQR = \angle QSR$. Prove that P, Q, R, S are concyclic (in other words, these four points lie on a circle).

Solution

Since $\angle BPS = \angle PRS$, the circumcircle of triangle PRS is tangent to AB at P . Similarly, since $\angle CQR = \angle QSR$, the circumcircle of triangle QRS is tangent to AC at Q .



For the sake of contradiction, suppose that the circumcircles of triangles PRS and QRS are not the same circle. Since $AP = AQ$, $AP^2 = AQ^2$, so the power of A with respect to both circumcircles is the same. Thus, A lies on the radical axis of both circles. However, both circles pass through R and S , so the radical axis of both circles is RS . Hence, A lies on RS , which is a contradiction.

Therefore, the two circumcircles are the same circle. In other words, P, Q, R , and S all lie on the same circle.

Solution 2

Note that (as in the first solution) the circumcircle of triangle PRS is tangent to AB at P . Similarly, since $\angle CQR = \angle QSR$, the circumcircle of triangle QRS is tangent to AC at Q .

Now, suppose these circumcircles are not the same circle. They already intersect at R and S , so they cannot intersect anymore. Thus, AS must touch the two circumcircles at points M and N , with M on the circumcircle of triangle PRS . By Power of a Point, $AQ^2 = AM \cdot AS$ and $AP^2 = AN \cdot AS$. Hence, because $AP = AQ$, $AM = AN$, a contradiction because then, as they lie on the same line segment, M and N must be the same point! (Note line segment, not line.) Hence, the two circumcircles are the same circle.

2. (*USAJMO 2012*) Find all integers $n \geq 3$ such that among any n positive real numbers $a_1, a_2, \dots, a_n$ with

$$\max(a_1, a_2, \dots, a_n) \leq n \cdot \min(a_1, a_2, \dots, a_n),$$

there exist three that are the side lengths of an acute triangle.

Solution

Without loss of generality, assume that the set $\{a\}$ is ordered from least to greatest so that the bounding condition becomes $a_n \leq n \cdot a_1$. Now set $b_i \equiv \frac{a_i}{a_1}$, and since a triangle with sidelengths from $\{a\}$ will be similar to the corresponding triangle from $\{b\}$, we simply have to show the existence of acute triangles in $\{b\}$. Note that $b_1 = 1$ and for all i , $b_i \leq n$.

Now three arbitrary sidelengths x , y , and z , with $x \leq y \leq z$, will form a valid triangle if and only if $x + y > z$. Furthermore, this triangle will be acute if and only if $x^2 + y^2 > z^2$. However, the first inequality can actually be inferred from the second, since $x + y > z \iff x^2 + y^2 + 2xy > z^2$ and $2xy$ is trivially greater than 0. So we just need to find all n such that there is necessarily a triplet of b 's for which $b_i^2 + b_j^2 > b_k^2$ (where $b_i < b_j < b_k$).

We now make another substitution: $c_i \equiv b_i^2$. So $c_1 = 1$ and for all i , $c_i \leq n^2$. Now we examine the smallest possible sets $\{c\}$ for small n for which the conditions of the problem are not met. Note that by smallest, we mean the set whose greatest element is as small as possible. If $n = 3$, then the smallest possible set, call it $\{s_3\}$, is trivially $\{1, 1, 2\}$, since c_1 and c_2 are obviously minimized and c_3 follows as minimal. Using this as the base case, we see inductively that in general $\{s_n\}$ is the set of the first n Fibonacci numbers. To show this note that if $\{s_n\} = \{F_0, F_1, \dots, F_n\}$, then $\{s_{n+1}\} = \{F_0, F_1, \dots, F_n, c_{n+1}\}$. The smallest possible value for c_{n+1} is the sum of the two greatest values of $\{s_n\}$ which are F_{n-1} and F_n . But these sum to F_{n+1} so $\{s_{n+1}\} = \{F_0, F_1, \dots, F_{n+1}\}$ and our induction is complete.

Now since we know that the Fibonacci set is the smallest possible set which does not satisfy the conditions of the problem, then any set $\{c\}$ whose greatest term is less than F_{n-1} must satisfy the conditions. And since $\{c\}$ is bounded between 1 and n^2 , then the conditions of the problem are met if and only if $F_{n-1} > n^2$. The first n for which this restriction is satisfied is $n = 13$ and the exponential behavior of the Fibonacci numbers ensure that every n greater than 13 will also satisfy this restriction. So the final solution set is $\boxed{\{n \geq 13\}}$.

Solution 2

Outline:

1. Define the Fibonacci numbers to be $F_1 = F_2 = 1$ and $F_k = F_{k-1} + F_{k-2}$ for $k \geq 3$.
2. If the chosen n is such that $F_n \leq n^2$, then choose the sequence a_n such that $a_k = \sqrt{F_k}$ for $1 \leq k \leq n$. It is easy to verify that such a sequence satisfies the condition that the largest term is less than or equal to n times the smallest term. Also, because for any three terms $x = \sqrt{F_a}, y = \sqrt{F_b}, z = \sqrt{F_c}$ with $a < b < c$, $x^2 + y^2 = F_a + F_b \leq F_{b-1} + F_b = F_{b+1} \leq F_c = z^2$, x, y, z do not form an acute triangle. Thus, all n such that $F_n \leq n^2$ do not work.
3. It is easy to observe via a contradiction argument that all n such that $F_n > n^2$ produce an acute triangle. (If, without loss of generality, a_n is an increasing sequence, such that no three (in particular, consecutive) terms form an acute triangle, then $a_2^2 \geq F_1 a_1^2, a_3^2 \geq a_2^2 + a_1^2 \geq F_2 a_1^2$, and by induction $a_n^2 > F_n a_1^2$, a contradiction to the condition's inequality.)

4. Note that $F_{12} = 144 = 12^2$ and $F_{13} = 233 > 169 = 13^2$. It is easily to verify through strong induction that all n greater than 12 make $F_n > n^2$. Thus, $\boxed{n \geq 13}$ is the desired solution set.

3. (USAJMO 2012) Let a, b, c be positive real numbers. Prove that

$$\frac{a^3 + 3b^3}{5a + b} + \frac{b^3 + 3c^3}{5b + c} + \frac{c^3 + 3a^3}{5c + a} \geq \frac{2}{3}(a^2 + b^2 + c^2).$$

Solution

By the **Cauchy-Schwarz** inequality,

$$[a(5a + b) + b(5b + c) + c(5c + a)] \left(\frac{a^3}{5a + b} + \frac{b^3}{5b + c} + \frac{c^3}{5c + a} \right) \geq (a^2 + b^2 + c^2)^2,$$

so

$$\frac{a^3}{5a + b} + \frac{b^3}{5b + c} + \frac{c^3}{5c + a} \geq \frac{(a^2 + b^2 + c^2)^2}{5a^2 + 5b^2 + 5c^2 + ab + ac + bc}.$$

Since $a^2 + b^2 + c^2 \geq ab + ac + bc$,

$$\frac{(a^2 + b^2 + c^2)^2}{5a^2 + 5b^2 + 5c^2 + ab + ac + bc} \geq \frac{(a^2 + b^2 + c^2)^2}{6a^2 + 6b^2 + 6c^2} = \frac{1}{6}(a^2 + b^2 + c^2).$$

Hence,

$$\frac{a^3}{5a + b} + \frac{b^3}{5b + c} + \frac{c^3}{5c + a} \geq \frac{1}{6}(a^2 + b^2 + c^2).$$

Again by the Cauchy-Schwarz inequality,

$$[b(5a + b) + c(5b + c) + a(5c + a)] \left(\frac{b^3}{5a + b} + \frac{c^3}{5b + c} + \frac{a^3}{5c + a} \right) \geq (a^2 + b^2 + c^2)^2,$$

so

$$\frac{b^3}{5a + b} + \frac{c^3}{5b + c} + \frac{a^3}{5c + a} \geq \frac{(a^2 + b^2 + c^2)^2}{a^2 + b^2 + c^2 + 5ab + 5ac + 5bc}.$$

Since $a^2 + b^2 + c^2 \geq ab + ac + bc$,

$$\frac{(a^2 + b^2 + c^2)^2}{a^2 + b^2 + c^2 + 5ab + 5ac + 5bc} \geq \frac{(a^2 + b^2 + c^2)^2}{6a^2 + 6b^2 + 6c^2} = \frac{1}{6}(a^2 + b^2 + c^2).$$

Hence,

$$\frac{b^3}{5a + b} + \frac{c^3}{5b + c} + \frac{a^3}{5c + a} \geq \frac{1}{6}(a^2 + b^2 + c^2).$$

Therefore,

$$\frac{a^3 + 3b^3}{5a + b} + \frac{b^3 + 3c^3}{5b + c} + \frac{c^3 + 3a^3}{5c + a} \geq \frac{1+3}{6}(a^2 + b^2 + c^2) = \frac{2}{3}(a^2 + b^2 + c^2).$$

Solution 2

Titu's Lemma: The sum of multiple fractions in the form $\frac{a_i^2}{b_i}$ where a_i and b_i are sequences of real numbers is greater than or equal to the square of the sum of all a_i divided by the sum of all b_i , where i is a whole number less than $n+1$. Titu's Lemma can be proved using the Cauchy-Schwarz Inequality after multiplying out the denominator of the RHS.

Consider the LHS of the proposed inequality. Split up the numerators and multiply both sides of the fraction by either a or $3a$ to make the LHS

$$\sum_{cyc} \frac{a^4}{5a^2 + ab} + \sum_{cyc} \frac{9a^4}{15ac + 3a^2}$$

(Cyclic notation is a special notation in which all permutations of (a,b,c) is the summation command.)

Then use Titu's Lemma on all terms:

$$\sum_{cyc} \frac{a^4}{5a^2 + ab} + \sum_{cyc} \frac{9a^4}{15ac + 3a^2} \geq \frac{16(a^2 + b^2 + c^2)^2}{8(a^2 + b^2 + c^2) + 16(ab + bc + ca)} \geq \frac{16(a^2 + b^2 + c^2)^2}{24(a^2 + b^2 + c^2)} = \frac{2}{3}(a^2 + b^2 + c^2)$$

owing to the fact that $a^2 + b^2 + c^2 \geq ab + bc + ca$, which is actually equivalent to $(a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0$!

Solution 3

We proceed to prove that

$$\frac{a^3 + 3b^3}{5a + b} \geq -\frac{1}{36}a^2 + \frac{25}{36}b^2$$

(then the inequality in question is just the cyclic sum of both sides, since

$$\sum_{cyc} \left(-\frac{1}{36}a^2 + \frac{25}{36}b^2\right) = \frac{24}{36} \sum_{cyc} a^2 = \frac{2}{3}(a^2 + b^2 + c^2)$$

)

Indeed, by AP-GP, we have

$$41(a^3 + b^3 + b^3) \geq 41 \cdot 3ab^2$$

and

$$b^3 + a^2b \geq 2ab^2$$

Summing up, we have

$$41a^3 + 83b^3 + a^2b \geq 125ab^2$$

which is equivalent to:

$$36(a^3 + 3b^3) \geq (5a + b)(-a^2 + 25b^2)$$

Dividing $36(5a + b)$ from both sides, the desired inequality is proved.

–**Lightest** 15:31, 7 May 2012 (EDT)

Solution 4

By Cauchy-Schwarz,

$$\begin{aligned}
 \left(\sum_{cyc} \frac{a^3}{5a+b} + \frac{b^3}{5a+b} + \frac{b^3}{5a+b} + \frac{b^3}{5a+b} \right) &\geq \frac{\left(\sum_{cyc} a^2 + b^2 + b^2 + b^2 \right)^2}{\left(\sum_{cyc} a(5a+b) + b(5a+b) + b(5a+b) + b(5a+b) \right)} \\
 &= \frac{(4a^2 + 4b^2 + 4c^2)^2}{(8a^2 + 8b^2 + 8c^2) + (16ab + 16bc + 16ca)} \\
 &\geq \frac{16(a^2 + b^2 + c^2)^2}{(8a^2 + 8b^2 + 8c^2) + (8a^2 + 8b^2) + (8b^2 + 8c^2) + (8c^2 + 8a^2)} \\
 \text{(by AM-GM)} &= \frac{2}{3}(a^2 + b^2 + c^2)
 \end{aligned}$$

as desired.

Solution 5

We note that if we can prove that there exists a real number x such that

$$\frac{a^3 + 3b^3}{5a+b} \geq xa^2 + \left(\frac{2}{3} - x\right)b^2$$

for all positive reals a and b , then we are done since both sides are a cyclic sum of a, b , and c . Note that if we multiply both a and b by a constant r , the left and right sides will both multiply by r^2 (since the numerator on the left will multiply by r^3 and the denominator will multiply by r). This means that if the inequality is satisfied for an ordered pair (a, b) , then it is also satisfied for (ar, br) for any positive real r . Thus, without loss of generality we can let $b = 1$, and our inequality becomes

$$\frac{a^3 + 3}{5a+1} \geq a^2x + \frac{2}{3} - x.$$

Expanding this and rearranging we get $a^3(1-5x) + a^2(-x) + a(5x - \frac{10}{3}) + x + \frac{7}{3} \geq 0$. Since 1 is a root of the left side, we can factor this as

$$(a-1)(a^2(1-5x) + a(1-6x) - x - \frac{7}{3}) \geq 0.$$

Note that for this to always be nonnegative, we must have $a^2(1-5x) + a(1-6x) - x - \frac{7}{3}$ be positive when $a > 1$ and negative when $a < 1$. Thus, it must have a root at $a = 1$, so we plug in $a = 1$ is a root and find that we must have $x = -\frac{1}{36}$. When $x = -\frac{1}{36}$, the expression factors as $(a-1)^2(\frac{41}{36}a + \frac{83}{36}) \geq 0$ which obviously is nonnegative for all positive reals a , so we are done.

Solution 6

$LHS = \sum_{cyc} \frac{a^3+3b^3}{5a+b} = \sum_{cyc} \frac{a^3}{5a+b} + \sum_{cyc} \frac{3b^3}{5a+b} = \sum_{cyc} \frac{a^4}{5a^2+ab} + \sum_{cyc} \frac{9b^4}{15ab+3b^2}$ which tells that $LHS \geq \frac{(a^2+b^2+c^2)^2}{5(a^2+b^2+c^2)+ab+bc+ac} + \frac{3(a^2+b^2+c^2)}{(a^2+b^2+c^2)+5(ab+bc+ac)} \geq (\frac{1}{2} + \frac{1}{6})(a^2 + b^2 + c^2) = RHS$, our proof is done

4. (*USAJMO 2012*) Let α be an irrational number with $0 < \alpha < 1$, and draw a circle in the plane whose circumference has length 1. Given any integer $n \geq 3$, define a sequence of points $P_1, P_2, \dots, P_n$ as follows. First select any point P_1 on the circle, and for $2 \leq k \leq n$ define P_k as the point on the circle for which the length of arc

$P_{k-1}P_k$ is α , when travelling counterclockwise around the circle from P_{k-1} to P_k . Suppose that P_a and P_b are the nearest adjacent points on either side of P_n . Prove that $a + b \leq n$.

Solution

Use mathematical induction. For $n = 3$ it is true because one point can't be closest to P_3 in both ways, and that $1 + 2 \leq 3$. Suppose that for some n , the nearest adjacent points P_a and P_b on either side of P_n satisfy $a + b \leq n$. Then consider the nearest adjacent points P_c and P_d on either side of P_{n+1} . It is by the assumption of the nearness we can see that either $(c, d) = (a + 1, b + 1)$ still holds, or P_1 jumps into the interior of the arc $P_{a+1}P_nP_{b+1}$, so that c or d equals to 1. Let's consider the following two cases.

(i) Suppose $a + b = n$.

Since the length of the arc P_nP_a is $\{(a - n)\alpha\}$ (where $\{x\}$ equals to x subtracted by the greatest integer not exceeding x) and length of the arc P_bP_n is $\{(n - b)\alpha\} = \{a\alpha\}$, we now consider a point P_0 which is defined by P_1 traveling clockwise on the circle such that the length of arc P_0P_1 is α . We claim that P_0 is in the interior of the arc $P_bP_nP_a$. Algebraically, it is equivalent to either $\{0 - n\alpha\} < \{a\alpha - n\alpha\}$ or $\{n\alpha - 0\} < \{n\alpha - b\alpha\} = \{a\alpha\}$.

Suppose the latter fails, i.e. $\{n\alpha\} \geq \{a\alpha\}$. Then suppose $n\alpha = m_1 + r_1$ and $a\alpha = m_2 + r_2$, where m_1, m_2 are integers and $0 < r_2 \leq r_1 < 1$ (r_2 is not zero because $a\alpha$ is irrational). We now have

$$\{0 - n\alpha\} = \{-m_1 - 1 + (1 - r_1)\} = 1 - r_1$$

and

$$\{a\alpha - n\alpha\} = \{m_2 - m_1 - 1 + (1 + r_2 - r_1)\} = 1 + r_2 - r_1 > 1 - r_1$$

Therefore P_0 is either closer to P_n than P_a on the P_a side, or closer to P_n than P_b on the P_b side. In other words, P_1 is the closest adjacent point of P_{n+1} on the P_{a+1} side, or the closest adjacent point of P_{n+1} on the P_{b+1} side. Hence P_c or P_d is P_1 , therefore $c + d \leq n + 1$.

(ii) Suppose $a + b \leq n - 1$. Then either $c + d = (a + 1) + (b + 1) \leq n + 1$ when $c = a + 1$ and $d = b + 1$, or $c + d \leq 1 + n$ when one of P_c or P_d is P_1 .

In either case, $c + d \leq n + 1$ is true.

–Lightest 20:27, 6 May 2012 (EDT)

5. (USAJMO 2012) For distinct positive integers $a, b < 2012$, define $f(a, b)$ to be the number of integers k with $1 \leq k < 2012$ such that the remainder when ak divided by 2012 is greater than that of bk divided by 2012. Let S be the minimum value of $f(a, b)$, where a and b range over all pairs of distinct positive integers less than 2012. Determine S .

Solution 1

First we'll show that $S \geq 502$, then we'll find an example (a, b) that have $f(a, b) = 502$.

Let x_k be the remainder when ak is divided by 2012, and let y_k be defined similarly for bk . First, we know that, if $x_k > y_k > 0$, then $x_{2012-k} \equiv a(2012 - k) \equiv 2012 - ak \equiv 2012 - x_k \pmod{2012}$ and $y_{2012-k} \equiv 2012 - y_k \pmod{2012}$. This

implies that, since $2012 - x_k \neq 0$ and $2012 - y_k \neq 0$, $x_{2012-k} < y_{2012-k}$. Similarly, if $0 < x_k < y_k$ then $x_{2012-k} > y_{2012-k}$, establishing a one-to-one correspondence between the number of k such that $x_k < y_k$. Thus, if n is the number of k such that $x_k \neq y_k$ and $y_k \neq 0 \neq x_k$, then $S \geq \frac{1}{2}n$. Now I'll show that $n \geq 1004$.

If $\gcd(k, 2012) = 1$, then I'll show you that $x_k \neq y_k$. This is actually pretty clear; assume that's not true and set up a congruence relation:

$$ak \equiv bk \pmod{2012}$$

Since k is relatively prime to 2012, it is invertible mod 2012, so we must have $a \equiv b \pmod{2012}$. Since $0 < a, b < 2012$, this means $a = b$, which the problem doesn't allow, thus contradiction, and $x_k \neq y_k$. Additionally, if $\gcd(k, 2012) = 1$, then $x_k \neq 0 \neq y_k$, then based on what we know about n from the previous paragraph, n is at least as large as the number of k relatively prime to 2012. Thus, $n \geq \phi(2012) = \phi(503 * 4) = 1004$. Thus, $S \geq 502$.

To show 502 works, consider $(a, b) = (1006, 2)$. For all even k we have $x_k = 0$, so it doesn't count towards $f(1006, 2)$. Additionally, if $k = 503, 503 * 3$ then $x_k = y_k = 1006$, so the only number that count towards $f(1006, 2)$ are the odd numbers not divisible by 503. There are 1004 such numbers. However, for all such odd k not divisible by 503 (so numbers relatively prime to 2012), we have $x_k \neq 0 \neq y_k$ and $2012 - k$ is also relatively prime to 2012. Since under those conditions exactly one of $x_k > y_k$ and $x_{2012-k} > y_{2012-k}$ is true, we have at most $1/2$ of the 1004 possible k actually count to $f(1006, 2)$, so $\frac{1004}{2} = 502 \geq f(1006, 2) \geq S \geq 502$, so $S = 502$.

Solution 2

Let $ak \equiv r_a \pmod{2012}$ and $bk \equiv r_b \pmod{2012}$. Notice that this means $a(2012 - k) \equiv 2012 - r_a \pmod{2012}$ and $b(2012 - k) \equiv 2012 - r_b \pmod{2012}$. Thus, for every value of k where $r_a > r_b$, there is a value of k where $r_b > r_a$. Therefore, we merely have to calculate $\frac{1}{2}$ times the number of values of k for which $r_a \neq r_b$ and $r_a \neq 0$.

However, the answer is NOT $\frac{1}{2}(2012) = 1006$! This is because we must count the cases where the value of k makes $r_a = r_b$ or where $r_a = 0$.

So, let's start counting.

If k is even, we have either $a \equiv 0 \pmod{1006}$ or $a - b \equiv 0 \pmod{1006}$. So, $a = 1006$ or $a = b + 1006$. We have 1005 even values of k (which is all the possible even values of k , since the two above requirements don't put any bounds on k at all).

If k is odd, if $k = 503$ or $k = 503 \cdot 3$, then $a \equiv 0 \pmod{4}$ or $a \equiv b \pmod{4}$. Otherwise, $ak \equiv 0 \pmod{2012}$ or $ak \equiv bk \pmod{2012}$, which is impossible to satisfy, given the domain $a, b < 2012$. So, we have 2 values of k .

In total, we have $2 + 1005 = 1007$ values of k which makes $r_a = r_b$ or $r_a = 0$, so there are $2011 - 1007 = 1004$ values of k for which $r_a \neq r_b$ and $r_a \neq 0$. Thus, by our reasoning above, our solution is $\frac{1}{2} \cdot 1004 = \boxed{502}$.

Solution by Invoker

Solution 3

The key insight in this problem is noticing that when ak is higher than bk , $a(2012 - k)$ is lower than $b(2012 - k)$, except at 2 (mod 4) residues*. Also, they must be equal many times. $2012 = 2^2 * 503$. We should have multiples of 503. After trying all

three pairs and getting 503 as our answer, we win. But look at the 2 (mod 4) idea. What if we just took 2 and plugged it in with 1006? We get 502.

<pre>-Va2010 11:12, 28 April 2012 (EDT)va2010</pre>

Solution 4

Say that the problem is a race track with 2012 spots. To intersect the most, we should get next to each other a lot so the negation is high. As $2012 = 2^2 * 503$, we intersect at a lot of multiples of 503.

6. (*USAJMO 2012*) Let P be a point in the plane of triangle ABC , and γ a line passing through P . Let A', B', C' be the points where the reflections of lines PA, PB, PC with respect to γ intersect lines BC, AC, AB , respectively. Prove that A', B', C' are collinear.

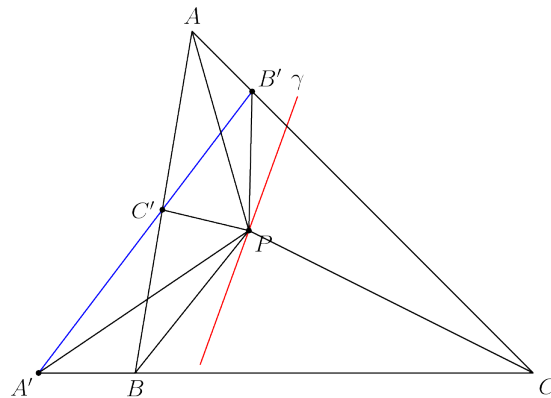
Solution

By the **sine law** on triangle $AB'P$,

$$\frac{AB'}{\sin \angle APB'} = \frac{AP}{\sin \angle AB'P},$$

so

$$AB' = AP \cdot \frac{\sin \angle APB'}{\sin \angle AB'P}.$$



Similarly,

$$\begin{aligned} B'C &= CP \cdot \frac{\sin \angle CPB'}{\sin \angle CB'P}, \\ CA' &= CP \cdot \frac{\sin \angle CPA'}{\sin \angle CA'P}, \\ A'B &= BP \cdot \frac{\sin \angle BPA'}{\sin \angle BA'P}, \\ BC' &= BP \cdot \frac{\sin \angle BPC'}{\sin \angle BC'P}, \\ C'A &= AP \cdot \frac{\sin \angle APC'}{\sin \angle AC'P}. \end{aligned}$$

Hence,

$$\frac{AB'}{B'C} \cdot \frac{CA'}{A'B} \cdot \frac{BC'}{C'A}$$

$$= \frac{\sin \angle APB'}{\sin \angle AB'P} \cdot \frac{\sin \angle CB'P}{\sin \angle CPB'} \cdot \frac{\sin \angle CPA'}{\sin \angle CA'P} \cdot \frac{\sin \angle BAP'}{\sin \angle BPA'} \cdot \frac{\sin \angle BPC'}{\sin \angle BC'P} \cdot \frac{\sin \angle AC'P}{\sin \angle APC'}.$$

Since angles $\angle AB'P$ and $\angle CB'P$ are supplementary or equal, depending on the position of B' on AC ,

$$\sin \angle AB'P = \sin \angle CB'P.$$

Similarly,

$$\begin{aligned} \sin \angle CA'P &= \sin \angle BA'P, \\ \sin \angle BC'P &= \sin \angle AC'P. \end{aligned}$$

By the reflective property, $\angle APB'$ and $\angle BPA'$ are supplementary or equal, so

$$\sin \angle APB' = \sin \angle BPA'.$$

Similarly,

$$\begin{aligned} \sin \angle CPA' &= \sin \angle APC', \\ \sin \angle BPC' &= \sin \angle CPB'. \end{aligned}$$

Therefore,

$$\frac{AB'}{B'C} \cdot \frac{CA'}{A'B} \cdot \frac{BC'}{C'A} = 1,$$

so by [Menelaus's theorem](#), A' , B' , and C' are collinear.

Solution 2, Barycentric (Modified by Evan Chen)

We will perform barycentric coordinates on the triangle PCC' , with $P = (1, 0, 0)$, $C' = (0, 1, 0)$, and $C = (0, 0, 1)$. Set $a = CC'$, $b = CP$, $c = C'P$ as usual. Since A , B , C' are collinear, we will define $A = (p : k : q)$ and $B = (p : \ell : q)$.

Claim: Line γ is the angle bisector of $\angle APA'$, $\angle BPB'$, and $\angle CPC'$. This is proved by observing that since $A'P$ is the reflection of AP across γ , etc.

Thus B' is the intersection of the isogonal of B with respect to $\angle P$ with the line CA ; that is,

$$B' = \left(\frac{p}{k} \frac{b^2}{\ell} : \frac{b^2}{\ell} : \frac{c^2}{q} \right).$$

Analogously, A' is the intersection of the isogonal of A with respect to $\angle P$ with the line CB ; that is,

$$A' = \left(\frac{p}{\ell} \frac{b^2}{k} : \frac{b^2}{k} : \frac{c^2}{q} \right).$$

The ratio of the first to third coordinate in these two points is both $b^2 pq : c^2 k \ell$, so it follows A' , B' , and C' are collinear.

Solution 3, Cartesian

We will use Cartesian coordinates. Without loss of generality, let $P = (0, 0)$ and let γ be the line $x = 0$. Let $A = (x_1, y_1)$, $B = (x_2, y_2)$, and $C = (x_3, y_3)$. Then A' is the intersection of the lines

$$\begin{aligned} y &= -\frac{y_1}{x_1}x \\ y - y_2 &= \frac{y_2 - y_3}{x_2 - x_3}(x - x_2). \end{aligned}$$

Solving the system of equations, we see it is

$$\left(\frac{x_1(x_2y_3 - y_2x_3)}{x_1y_2 + y_1x_2 - x_1y_3 - x_3y_1}, \frac{y_1(y_2x_3 - x_2y_3)}{x_1y_2 + y_1x_2 - x_1y_3 - x_3y_1} \right).$$

To check collinearity, we need the determinant of the matrix

$$\begin{bmatrix} x_1(x_2y_3 - y_2x_3) & y_1(y_2x_3 - x_2y_3) & x_1y_2 + y_1x_2 - x_1y_3 - x_3y_1 \\ x_2(x_3y_1 - y_3x_1) & y_2(y_3x_1 - x_3y_1) & x_2y_3 + y_2x_3 - x_2y_1 - x_1y_2 \\ x_3(x_1y_2 - y_1x_2) & y_3(y_1x_2 - x_1y_2) & x_3y_1 + y_3x_1 - x_3y_2 - x_2y_3 \end{bmatrix}$$

to be zero, which is true because the columns sum to zero.

§4 USAJMO 2013

1. (*USAJMO 2013*) Are there integers a and b such that $a^5b + 3$ and $ab^5 + 3$ are both perfect cubes of integers?

Solution

No, such integers do not exist. This shall be proven by contradiction, by showing that if $a^5b + 3$ is a perfect cube then $ab^5 + 3$ cannot be.

Remark that perfect cubes are always congruent to 0, 1, or -1 modulo 9. Therefore, if $a^5b + 3 \equiv 0, 1, \text{ or } -1 \pmod{9}$, then $a^5b \equiv 5, 6, \text{ or } 7 \pmod{9}$.

If $a^5b \equiv 6 \pmod{9}$, then note that $3|b$. (This is because if $3|a$ then $a^5b \equiv 0 \pmod{9}$.) Therefore $ab^5 \equiv 0 \pmod{9}$ and $ab^5 + 3 \equiv 3 \pmod{9}$, contradiction.

Otherwise, either $a^5b \equiv 5 \pmod{9}$ or $a^5b \equiv 7 \pmod{9}$. Note that since a^6b^6 is a perfect sixth power, and since neither a nor b contains a factor of 3, $a^6b^6 \equiv 1 \pmod{9}$. If $a^5b \equiv 5 \pmod{9}$, then

$$a^6b^6 \equiv (a^5b)(ab^5) \equiv 5ab^5 \equiv 1 \pmod{9} \implies ab^5 \equiv 2 \pmod{9}.$$

Similarly, if $a^5b \equiv 7 \pmod{9}$, then

$$a^6b^6 \equiv (a^5b)(ab^5) \equiv 7ab^5 \equiv 1 \pmod{9} \implies ab^5 \equiv 4 \pmod{9}.$$

Therefore $ab^5 + 3 \equiv 5, 7 \pmod{9}$, contradiction.

Therefore no such integers exist.

Amkan2022

Solution 2

We shall prove that such integers do not exist via contradiction. Suppose that $a^5b + 3 = x^3$ and $ab^5 + 3 = y^3$ for integers x and y . Rearranging terms gives $a^5b = x^3 - 3$ and $ab^5 = y^3 - 3$. Solving for a and b (by first multiplying the equations together and taking the sixth root) gives $a = (x^3 - 3)^{\frac{5}{24}}(y^3 - 3)^{\frac{-1}{24}}$ and $b = (y^3 - 3)^{\frac{5}{24}}(x^3 - 3)^{\frac{-1}{24}}$. Consider a prime p in the prime factorization of $x^3 - 3$ and $y^3 - 3$. If it has power r_1 in $x^3 - 3$ and power r_2 in $y^3 - 3$, then $5r_1 - r_2$ is a multiple of 24 and $5r_2 - r_1$ also is a multiple of 24.

Adding and subtracting the divisions gives that $r_1 - r_2$ divides 12. (actually, $r_1 - r_2$ is a multiple of 4, as you can verify if $\begin{cases} 5r_1 - r_2 = 24 \\ 5r_2 - r_1 = 48 \end{cases}$. So the rest of the proof is invalid.) Because $5r_1 - r_2$ also divides 12, $4r_1$ divides 12 and thus r_1 divides 3. Repeating this trick for all primes in $x^3 - 3$, we see that $x^3 - 3$ is a perfect cube, say q^3 . Then

$x^3 - q^3 = 3$, and $(x - q)(x^2 + xq + q^2) = 3$, so that $x - q = 1$ and $x^2 + xq + q^2 = 3$. Clearly, this system of equations has no integer solutions for x or q , a contradiction, hence completing the proof.

Therefore no such integers exist.

Solution 3

Let $a^5b + 3 = x^3$ and $ab^5 + 3 = y^3$. Then, $a^5b = x^3 - 3$, $ab^5 = y^3 - 3$, and

$$(ab)^6 = (x^3 - 3)(y^3 - 3)$$

Now take mod 9 (recall that perfect cubes $\equiv -1, 0, 1 \pmod{9}$ and perfect sixth powers $\equiv 0, 1 \pmod{9}$) on both sides. There are $3 \times 3 = 9$ cases to consider on what values mod 9 that x^3 and y^3 take. Checking these 9 cases, we see that only $x^3 \equiv y^3 \equiv 0 \pmod{9}$ or $x \equiv y \equiv 0 \pmod{3}$ yield a valid residue mod 9 (specifically, $(x^3 - 3)(y^3 - 3) \equiv 0 \pmod{9}$). But this means that $3 \mid ab$, so $729 \mid (ab)^6$ so

$$729 \mid (x^3 - 3)(y^3 - 3) \iff 729 \mid (27x'^3 - 3)(27y'^3 - 3) \iff 81 \mid (9x'^3 - 1)(9y'^3 - 1)$$

contradiction.

Solution 4

If $a^5b + 3$ is a perfect cube, then a^5b can be one of $5, 6, 7 \pmod{9}$, so $(a^5b)^5 = a^{25}b^5$ can be one of $5^5 \equiv 2, 6^5 \equiv 0$, or $7^5 \equiv 4 \pmod{9}$. If a were divisible by 3, we'd have $a^5b \equiv 0 \pmod{9}$, which we've ruled out. So $\gcd(a, 9) = 1$, which means $a^6 \equiv 1 \pmod{9}$, and therefore $a^{25}b^5 \equiv ab^5 \pmod{9}$.

We've shown that ab^5 can be one of $0, 2, 4 \pmod{9}$, so $ab^5 + 3$ can be one of $3, 5, 7 \pmod{9}$. None of these are possibilities for a perfect cube, so if $a^5b + 3$ is a perfect cube, $ab^5 + 3$ cannot be.

Solution 5

As in previous solutions, notice $ab^5, a^5b \equiv 5, 6, 7 \pmod{9}$. Now multiplying gives a^6b^6 , which is only $0, 1 \pmod{9}$, so after testing all cases we find that $ab^5 \equiv a^5b \equiv 6 \pmod{9}$. Then since $\phi(9) = 6$, $ab^5 \equiv \frac{a}{b} \pmod{9}$ and $a^5b \equiv \frac{b}{a} \pmod{9}$ (Note that a, b cannot be $0 \pmod{9}$). Thus we find that the inverse of 6 is itself under modulo 9, a contradiction.

Solution 6

I claim there are no such a or b such that both expressions are cubes.

Assume to the contrary $a^5b + 3$ and $ab^5 + 3$ are cubes.

Lemma 1: If $a^5b + 3$ and $ab^5 + 3$ are cubes, then $ab^5, a^5b \equiv 5, 7 \pmod{9}$

Proof Since cubes are congruent to any of $0, 1, -1 \pmod{9}$, $ab^5, a^5b \equiv 5, 6, 7 \pmod{9}$. But if $ab^5 \equiv 6 \pmod{9}$, $3 \mid a$, so $a^5b \equiv 0 \pmod{9}$, contradiction. A similar argument can be made for $ab^5 \not\equiv 6 \pmod{9}$.

Lemma 2: If k is a perfect 6th power, then $k \equiv 0, 1 \pmod{9}$

Proof: Since cubes are congruent to $0, 1, -1 \pmod{9}$, we can square, and get 6th powers are congruent to $0, 1 \pmod{9}$.

Since $ab^5 \cdot a^5b = a^6b^6 = (ab)^6$, which is a perfect 6th power, by lemma 2, $ab^5 \cdot a^5b \equiv 0, 1 \pmod{9}$.

But, by lemma 1, $ab^5 \cdot a^5b \equiv 5 \cdot 5, 5 \cdot 7, 7 \cdot 7 \equiv 7, 8, 4 \pmod{9}$.

So, $ab^5 \cdot a^5b$, which is an integer, can't go into any of the possible residue classes modulo 9, without breaking one of these lemmas. This is a contradiction, and the proof is complete. ■

-AlexLikeMath

Solution 7

Note that $x^3 \equiv 0, 1, -1 \pmod{9}$. So $a^5b+3, ab^5+3 \equiv 0, 1, -1 \pmod{9} \Rightarrow a^5b, ab^5 \equiv 5, 6, 7 \pmod{9}$.

$$a^5b \equiv 5 \pmod{9}; \quad ab^5 \equiv 5 \pmod{9}$$

$$a^5b \equiv$$

Multiplying a^5b and ab^5 we get $(ab)^6$. By substituting values for a^5b and ab^5 we can see that $(ab)^6 \equiv 0, 1, 3, 6, 7, 8 \pmod{9}$ but because cubes are not $3, 6, 7 \pmod{9}$, and squares are not $-1 \pmod{9}$ so $(ab)^6 \equiv 0, 1 \pmod{9}$.

- Case I: $(ab)^6 \equiv 0 \pmod{9}$. This is only true when $3|ab$. If $3|a, b$ then $a^5b+3 \equiv 3 \pmod{9}$, which is not a perfect cube. The same can be seen with if $3|a$ then $a^5b+3 \equiv 3 \pmod{9}$ and if $3|a$ then $b^5a+3 \equiv 3 \pmod{9}$. So 6 mod 9 case is eliminated from the above relations of congruences
- Case II: $(ab)^6 \equiv 1 \pmod{9}$ for this we'll take a look at the above information regarding a^5b and ab^5 . If $a^5b \equiv 5 \pmod{9}$ then,

$$(ab)^6 = a^5b \cdot ab^5 \equiv 5(ab^5) \pmod{9} \Rightarrow ab^5 \equiv 2 \pmod{9}$$

which is not possible. Similarly, if $a^5b \equiv 7 \pmod{9}$ then,

$$(ab)^6 = a^5b \cdot ab^5 \equiv 7(ab^5) \pmod{9} \Rightarrow ab^5 \equiv 4 \pmod{9}$$

which is not possible. So, no such integers.

-Lakshya Pamecha

Solution 8

We proceed by contradiction. Assuming $a^5b+3 = x^3$ for some integer x and $ab^5+3 = y^3$ for another integer y , we can multiply the two equations to get $a^6b^6+3a^5b+3ab^5+9 = x^3y^3$. We can rearrange this to $3a^5b+3ab^5 = x^3y^3 - a^6b^6 + 9$.

Looking at this $\pmod{9}$, this becomes $3a^5b+3ab^5 \equiv x^3y^3 - a^6b^6 \pmod{9}$. Looking at the right hand side, the first term is either $-1, 0$, or $1 \pmod{9}$. The second is 0 or $1 \pmod{9}$, because it is a square of a cube. Additionally, since the LHS is divisible by three, so is the RHS.

The only achievable value for the LHS that fits this is 0 . So, $3a^5b+3ab^5 \equiv 0 \pmod{9}$. Dividing by three, making sure to divide the modulus as well, $a^5b+ab^5 \equiv 0 \pmod{3}$. Noticing $a^5b \equiv ab \pmod{3}$ and $ab^5 \equiv ab \pmod{3}$, we get, $2ab \equiv 0 \pmod{3}$.

This means either $3|a$ or $3|b$, or both. WLOG, $3|a$. Looking at the original expression mod 9, we get $a^5b+3 \equiv -1, 0$, or $1 \pmod{9}$. Since a is divisible by three, a^5 is divisible by 243, so it is divisible by 9. This reduces the equation to $3 \equiv -1, 0$, or $1 \pmod{9}$, which obviously can't be true, a contradiction. Therefore, our original assumption, that both expressions were perfect cubes, is incorrect.

So, no, no such integers exist.

-sillybone

Note After you get that $a^5b \equiv 5, 6$, or $7 \pmod{9}$, you can also bash by listing all possible residues for a^5 and their corresponding values for b . At that point, you

can easily see that all the solutions for a and b do not work in the other equation, meaning that there are no solutions that satisfy both equations.

2. (*USAJMO 2013*) Each cell of an $m \times n$ board is filled with some nonnegative integer. Two numbers in the filling are said to be *adjacent* if their cells share a common side. (Note that two numbers in cells that share only a corner are not adjacent). The filling is called a *garden* if it satisfies the following two conditions:
- (i) The difference between any two adjacent numbers is either 0 or 1.
 - (ii) If a number is less than or equal to all of its adjacent numbers, then it is equal to 0.

Determine the number of distinct gardens in terms of m and n .

Solution

We claim that any configuration of 0's produces a distinct garden. To verify this claim, we show that, for any cell that is nonzero, the value of that cell is its distance away from the nearest zero, where distance means the shortest chain of adjacent cells connecting two cells. Now, since we know that any cell with a nonzero value must have a cell adjacent to it that is less than its value, there is a path that goes from this cell to the 0 that is decreasing, which means that the value of the cell must be its distance from the 0 $\rightarrow$ as the path must end. From this, we realize that, for any configuration of 0's, the value of each of the cells is simply its distance from the nearest 0, and therefore one garden is produced for every configuration of 0's.

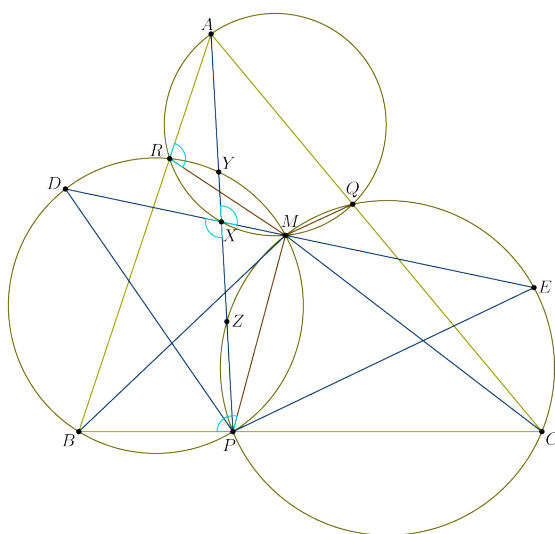
However, we also note that there must be at least one 0 in the garden, as otherwise the smallest number in the garden, which is less than or equal to all of its neighbors, is > 0 , which violates condition (ii). There are 2^{mn} possible configurations of 0 and not 0 in the garden, one of which has no 0's, so our total amount of configurations is $\boxed{2^{mn} - 1}$.

Similar Second Solution and Inspiration Note: "bordering" and "surrounding" mean that two cells have to touch on a side; one vertex is not good enough.

We note that there is no real step to begin the problem, so start by constructing the base case: put ALL the zeroes into the board. Then note that all squares bordering it (corners alone don't count) have to be 1. After doing that, what can a cell bordering a 1 have as its value? Either 1 or 2, based on (i). But if it were 1, then all cells surrounding it would have at least 1 as their value by (i). And by (ii) the value would have to be 0, which contradicts our initial construction. Therefore, all the cells bordering a 1 have to be 2. Looks too simple for JMO, right? We apply the same logic to a cell bordering value n : either it is n or $n + 1$. If it is n , however, by constraint (i) and (ii) we realize that this cell has to be 0 (because neighbors cannot be less than it), contradicting our construction again! Therefore, that cell has to be $n + 1$. And so on until the grid is filled. Basically the problem reduces to finding all 0s, surrounding them with 1s, and surrounding the 1s with 2s, etc. Beautiful! I have established a bijection between the zeros placed in the grid and the arrangement, therefore there are 2^{mn} solutions right? NO! Take the smallest cell. It has to be 0 via criterion (ii). So a case with zero zeros is apocryphal. Our answer is $2^{mn} - 1$.

3. (*USAJMO 2013*) In triangle ABC , points P, Q, R lie on sides BC, CA, AB respectively. Let $\omega_A, \omega_B, \omega_C$ denote the circumcircles of triangles AQR, BRP, CPQ , respectively. Given the fact that segment AP intersects $\omega_A, \omega_B, \omega_C$ again at X, Y, Z respectively, prove that $YX/XZ = BP/PC$.

Solution 1



In this solution, all lengths and angles are directed.

Firstly, it is easy to see by that $\omega_A, \omega_B, \omega_C$ concur at a point M (the Miquel point). Let XM meet ω_B, ω_C again at D and E , respectively. Then by Power of a Point, we have

$$XM \cdot XE = XZ \cdot XP \quad \text{and} \quad XM \cdot XD = XY \cdot XP$$

Thusly

$$\frac{XY}{XZ} = \frac{XD}{XE}$$

But we claim that $\triangle XDP \sim \triangle PBM$. Indeed,

$$\angle XDP = \angle MDP = \angle MBP = -\angle PBM$$

and

$$\angle DXP = \angle MXY = \angle MXA = \angle MRA = 180^\circ - \angle MRB = \angle MPB = -\angle BPM$$

Therefore, $\frac{XD}{XP} = \frac{PB}{PM}$. Analogously we find that $\frac{XE}{XP} = \frac{PC}{PM}$ and we are done.

Solution 2

DiagramRefer to the Diagram link.

By Miquel's Theorem, there exists a point at which $\omega_A, \omega_B, \omega_C$ intersect. We denote this point by M . Now, we angle chase:

$$\begin{aligned}\angle YMX &= 180^\circ - \angle YXM - \angle XYM \\ &= 180^\circ - \angle AXM - \angle PYM \\ &= (180^\circ - \angle ARM) - \angle PRM \\ &= \angle BRM - \angle PRM \\ &= \angle BRP = \angle BMP.\end{aligned}$$

In addition, we have

$$\begin{aligned}
 \angle ZMX &= 180^\circ - \angle MZY - \angle ZYM - \angle YMX \\
 &= 180^\circ - \angle MZP - \angle PYM - \angle BMP \\
 &= 180^\circ - \angle MCP - \angle PBM - \angle BMP \\
 &= (180^\circ - \angle PBM - \angle BMP) - \angle MCP \\
 &= \angle BPM - \angle MCP \\
 &= 180^\circ - \angle MPC - \angle MCP \\
 &= \angle CMP.
 \end{aligned}$$

Now, by the Ratio Lemma, we have

$$\begin{aligned}
 \frac{XY}{XZ} &= \frac{MY}{MZ} \cdot \frac{\sin \angle YMX}{\sin \angle ZMX} \\
 &= \frac{\sin \angle YZM}{\sin \angle ZYM} \cdot \frac{\sin \angle BMP}{\sin \angle CMP}
 \end{aligned}$$

(by the Law of Sines in $\triangle MZY$)

$$\begin{aligned}
 &= \frac{\sin \angle PZM}{\sin \angle PYM} \cdot \frac{\sin \angle BMP}{\sin \angle CMP} \\
 &= \frac{\sin \angle PCM}{\sin \angle PBM} \cdot \frac{\sin \angle BMP}{\sin \angle CMP} \\
 &= \frac{MB}{MC} \cdot \frac{\sin \angle BMP}{\sin \angle CMP}
 \end{aligned}$$

(by the Law of Sines in $\triangle MBC$)

$$= \frac{PB}{PC}$$

by the Ratio Lemma. The proof is complete.

Solution 3

Use directed angles modulo π .

Lemma. $\angle XRY \equiv \angle XQZ$.

Proof.

$$\angle XRY \equiv \angle XRA - \angle YRA \equiv \angle XQA + \angle YRB \equiv \angle XQA + \angle CPY = \angle XQA + \angle AQZ = \angle XQZ.$$

Now, it follows that (now not using directed angles)

$$\frac{XY}{YZ} = \frac{\frac{XY}{\sin \angle XRY}}{\frac{YZ}{\sin \angle XQZ}} = \frac{\frac{RY}{\sin \angle RXY}}{\frac{QZ}{\sin \angle QXZ}} = \frac{BP}{PC}$$

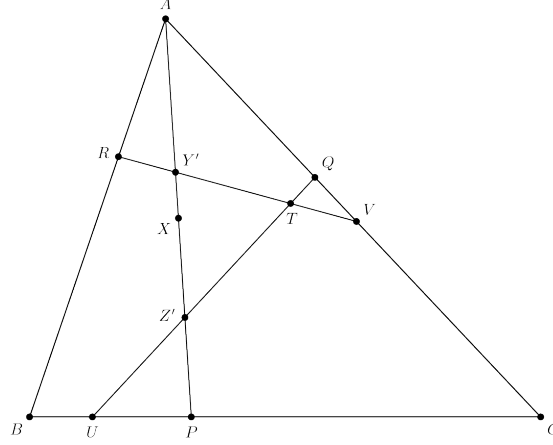
using the facts that ARY and APB , AQZ and APC are similar triangles, and that $\frac{RA}{\sin \angle RXA} = \frac{QA}{\sin \angle QXA}$ equals twice the circumradius of the circumcircle of AQR .

Solution 4

We will use some construction arguments to solve the problem. Let $\angle BAC = \alpha$, $\angle ABC = \beta$, $\angle ACB = \gamma$, and let $\angle APB = \theta$. We construct lines through the points Q , and R that intersect with $\triangle ABC$ at the points Q and R , respectively,

and that intersect each other at T . We will construct these lines such that $\angle CQV = \angle ARV = \theta$.

Now we let the intersections of AP with RV and QU be Y' and Z' , respectively. This construction is as follows.



We know that $\angle BRY' = 180^\circ - \angle ARY' = 180^\circ - \theta$. Hence, we have,

$$\begin{aligned}\angle BRY' + \angle BPY' &= 180^\circ - \theta + \theta \\ &= 180^\circ.\end{aligned}$$

Since the opposite angles of quadrilateral $RY'PB$ add up to 180° , it must be cyclic. Similarly, we can also show that quadrilaterals $CQZ'P$, and $AQTR$ are also cyclic. Since points Y' and Z' lie on AP , we know that,

$$Y' = \omega_B \cap AP$$

and that

$$Z' = \omega_C \cap AP.$$

Hence, the points Y' and Z' coincide with the given points Y and Z , respectively. Since quadrilateral $AQTR$ is also cyclic, we have,

$$\begin{aligned}\angle Y'TZ' &= 180^\circ - \angle RTQ \\ &= 180^\circ - (180^\circ - \angle RAQ) \\ &= \angle RAQ \\ &= \alpha.\end{aligned}$$

Similarly, since quadrilaterals $CQZ'P$, and $AQTR$ are also cyclic, we have,

$$\begin{aligned}\angle TY'Z' &= 180^\circ - \angle RY'P \\ &= 180^\circ - (180^\circ - \angle RBP) \\ &= \angle RBP \\ &= \beta,\end{aligned}$$

and,

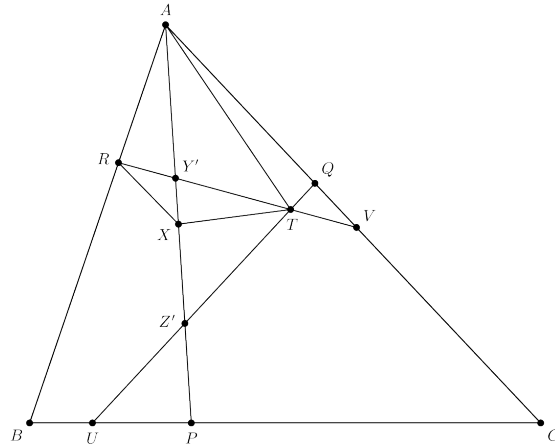
$$\angle Y'Z'T = 180^\circ - \angle PZ'Q$$

$$\begin{aligned} &= 180^\circ - (180^\circ - \angle PCQ) \\ &= \angle PCQ \\ &= \gamma. \end{aligned}$$

Since these three angles are of $\triangle TY'Z'$, and they are equal to corresponding angles of $\triangle ABC$, by AA similarity, we know that $\triangle TY'Z' \sim \triangle ABC$.

We now consider the point $X = \omega_c \cap AC$. We know that the points A, Q, T , and R are concyclic. Hence, the points A, T, X , and R must also be concyclic.

Hence, quadrilateral $AQTX$ is cyclic.



Since the angles $\angle ART$ and $\angle AXT$ are inscribed in the same arc $\widehat{6.0\text{ptAT}}$, we have,

$$\begin{aligned}\angle AXT &= \angle ART \\ &= \theta.\end{aligned}$$

Consider by this result, we can deduce that the homothety that maps ABC to $TY'Z'$ will map P to X . Hence, we have that,

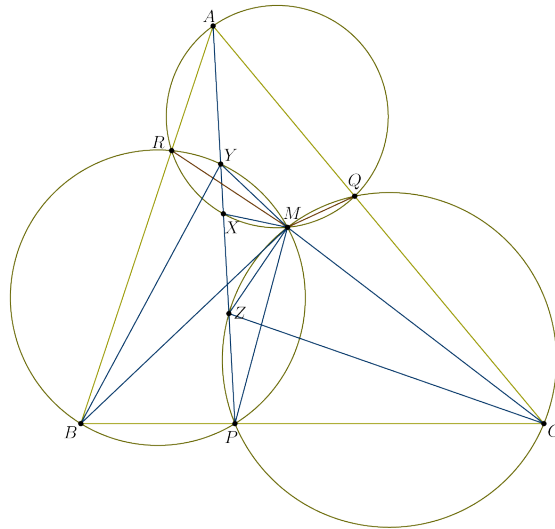
$$Y'X/XZ' = BP/PC.$$

Since $Y' = Y$ and $Z' = Z$ hence,

$$YX/XZ = BP/PC,$$

as required.

Solution 5 (Simple Rotational Homotethy)



We begin again by noting that the three circumcircles intersect at point M by Miquel's theorem. In addition, we state that the angle $\angle MQC = \alpha$, hence $\angle MPC = \angle MZC = 180 - \alpha$, as well as $\angle AQM$, from which follows that $\angle ARM = \alpha$, so $\angle BRM = 180 - \alpha$, and $\angle BPM = \alpha$. We shall prove that the triangles $\triangle MZC$ and $\triangle XMP$ and $\triangle YMB$ are similar, which will imply a rotational homothety with angle $\angle APB$ about the point M , that takes Y, X, Z to B, P, C , thus proving the problem. (In essence, just imagine we rotate $YMZX$ around M and lengthen things out and get $MCPB$ - the ratios will remain identical.)

We do this by angle chasing. Denote angle $\angle ZMP = \beta$. From the angles labeled before, we now know $\angle ZPB = \alpha - \beta = \angle ZMC$. In addition, $\angle ZPM = \angle ZCM$. So the angles in triangle $\triangle ZMC$ are $\alpha - \beta, \beta$ and thus $180 - \alpha$, with $\alpha - \beta$ at the point M . In addition, $\angle YRM = \angle YPM = \beta$, so $\angle ARY = \alpha - \beta$, so $\angle YMB = \alpha - \beta$. Since $\angle YBM = \angle YPM = \beta$, the triangle $\triangle YMB$ has angles $\alpha - \beta, \beta, 180 - \alpha$ also, with $\alpha - \beta$ at M . Finally, In triangle $\triangle XPM$, we already know the angle $\angle XPM$ to be β ; we also can find that $\angle AQM = 180 - \alpha$, so $\angle AXM = \alpha$, so $\angle MXP = 180 - \alpha$, so $\angle XMP = \alpha - \beta$. Thus, the three triangles are similar have a common point M , which proves that there is a rotational homothety around M that maps Y, X, Z to B, P, C as desired.

Solution by SimilarTriangle.

4. (USAJMO 2013) Let $f(n)$ be the number of ways to write n as a sum of powers of 2, where we keep track of the order of the summation. For example, $f(4) = 6$ because 4 can be written as 4, $2 + 2$, $2 + 1 + 1$, $1 + 2 + 1$, $1 + 1 + 2$, and $1 + 1 + 1 + 1$. Find the smallest n greater than 2013 for which $f(n)$ is odd.

Solution

First of all, note that $f(n) = \sum_{i=0}^k f(n - 2^i)$ where k is the largest integer such that $2^k \leq n$. We let $f(0) = 1$ for convenience.

From here, we proceed by induction, with our claim being that the only n such that $f(n)$ is odd are n representable of the form $2^a - 1, a \in \mathbb{Z}$

We induct on a . It is trivially true for $a = 0$ and $a = 1$. From here, we show that, if the only numbers $n \leq 2^{a-1} - 1$ where $f(n)$ is odd are of the form described above, then the only numbers $n \leq 2^a - 1$ that are odd are of that form. We first consider all numbers b , such that $2^{a-1} \leq b \leq 2^a - 2$, going from the lower bound to the upper

bound (a mini induction, you might say). We know that $f(b) = \sum_{i=0}^{a-1} f(b - 2^i)$. For a number in this summation to be odd, $b - 2^i = 2^m - 1 \rightarrow b = 2^i + 2^m - 1$. However, we know that $b > 2^{a-1}$, so m must be equal to $a - 1$, or else b cannot be in that interval. Now, from this, we know that $i < a - 1$, as $b < 2^a - 1$. Therefore, i and m are distinct, and thus $f(b - 2^i)$ and $f(b - 2^{a-1})$ are odd; since there are just two odd numbers, the ending sum for any b is even. Finally, considering $2^a - 1$, the only odd number is $f(2^a - 1 - 2^{a-1})$, so the ending sum is odd. $\square$

The smallest n greater than 2013 expressible as $2^d - 1, d \in \mathbb{N}$ is $2^{11} - 1 = \boxed{2047}$

Solution with Explanations

Of course, as with any number theory problem, use actual numbers to start, not variables! By plotting out the first few sums (do it!) and looking for patterns, we observe that $f(n) = \sum_{\text{power}=0}^{\text{pow}_{\text{larg}}} f(n - 2^{\text{power}})$, where pow_{larg} represents the largest power of 2 that is smaller than n . I will call this sum the Divine Sign, or DS.

But wait a minute... we are trying to determine odd/even of $f(n)$. Why not call all the evens 0 and odds 1, basically using mod 2? Sounds so simple. Draw a small table for the values: as n goes up from 0, you get: 1, 1, 0, 1, 0, 0, 0, 1, 0.... We have to set $f(0) = 1$ for this to work. Already it looks like $f(n)$ is only odd if $n = 2^{\text{power}} - 1$.

The only tool here is induction. The base case is clearly established. Then let's assume we successfully made our claim up to $2^n - 1$. We need to visit numbers from 2^n to $2^{n+1} - 1$. Realize that 2^n has 0 for f because there will be two numbers in DS that give a f of one: 2^{n-1} and 1.

But to look at whether a value of $f(\text{number})$ is 1 or 0, we need to revisit our first equation. We can answer this rather natural question: When will a number to be inducted upon, say $2^n + k$, ever have a 1 as $f(\text{number})$ in the DS equation? Well—because by our assumption of the claim up to $2^n - 1$, we know that the only way for that to happen is if $2^n + k - 2^{\text{power}}$ in the DS is equal to $2^{\text{Some power}} - 1$. Clearly $1 \leq k \leq 2^n - 1$.

Finally, we can simplify. Using our last equation, $2^n + k - 2^{\text{power}} = 2^{\text{Some power}} - 1$, regrouping gives $2^n + k = 2^{\text{power}} + 2^{\text{Some power}} - 1$.

Most importantly, realize that power can be from 0 to n , because of the restraints on k mentioned earlier. Same with Some power. Immediately at least one of power and Some power has to be n . If both were smaller, LHS is greater, contradiction. If both were greater, RHS is greater, contradiction.

Therefore, by setting one of power or Some power to n , we realize $k = 2^{\text{A certain power}} - 1$.

The conclusion is clear, right? Each k from 1 to $2^{n-1} - 1$ yields two distinct cases: one of power and Some power is equal to n , while the other is LESS THAN n . But for $k = 2^n - 1$, there is ONE CASE: BOTH values have to equal n . Therefore, the only k that has $f(2^n + k)$ as odd must only be $2^n - 1$, because the other ones yield a f of $1+1=0$ in our mod. That proves our induction for a new power of 2, namely $n + 1$, meaning that $f(\text{number})$ is only odd if number $= 2^{\text{Power of two}} - 1$, and we are almost done...

Thus, the answer is $2^{11} - 1 = \boxed{2047}$.

This was pretty intuitive, and realizing quite *elementary* steps about how powers of 2 work gave us the solution! Estimated time: 50 minutes.

minor edits mathboy100

5. (*USAJMO 2013*) Quadrilateral $XABY$ is inscribed in the semicircle ω with diameter XY . Segments AY and BX meet at P . Point Z is the foot of the perpendicular from P to line XY . Point C lies on ω such that line XC is perpendicular to line AZ . Let Q be the intersection of segments AY and XC . Prove that

$$\frac{BY}{XP} + \frac{CY}{XQ} = \frac{AY}{AX}.$$

Solution

Using the Law of Sines and simplifying, we have

$$\frac{BY}{XP} + \frac{CY}{XQ} = \frac{\sin \angle PXY \sin \angle XPY + \sin \angle QXY \sin \angle XQY}{\sin \angle AYX}.$$

It is easy to see that $APZX$ is cyclic. Also, we are given $XQ \perp AZ$. Then we have

$$\begin{aligned} \frac{\sin \angle PXY \sin \angle XPY + \sin \angle QXY \sin \angle XQY}{\sin \angle AYX} &= \frac{\sin \angle YAZ \sin \angle XZA + \cos \angle XZA \cos \angle YAZ}{\sin \angle AYX} \\ &= \frac{\cos(\angle XZA - \angle YAZ)}{\sin \angle AYX} \\ &= \frac{\cos \angle AYX}{\sin \angle AYX} \\ &= \cot \angle AYX \\ &= \frac{AY}{AX}, \end{aligned}$$

and we are done.

Solution 1

Let us use coordinates. Let O , the center of the circle, be $(0,0)$. WLOG the radius of the circle is 1, so set $Y(1,0)$ and $X(-1,0)$. Also, for arbitrary constants a and b set $A(\cos a, \sin a)$ and $B(\cos b, \sin b)$. Now, let's use our coordinate tools. It is easily derived that the equation of BX is $y = \frac{\sin b}{1+\cos b}(x+1) = v(x+1)$ and the equation of AY is $y = \frac{\sin a}{1-\cos a}(x-1) = u(x-1)$, where u and v are defined appropriately. Thus, by equating the y 's in the equation we find the intersection of these lines, P , is $\left(\frac{u-v}{u+v}, \frac{2uv}{u+v}\right)$. Also, $Z\left(\frac{u-v}{u+v}, 0\right)$. It shall be left to the reader to find the slope of AZ , the coordinates of Q and C , and use the distance formula to verify that $\frac{BY}{XP} + \frac{CY}{XQ} = \frac{AY}{AX}$.

Solution 2

First of all

$$\angle BXY = \angle PAZ = \angle AXQ = \angle AXC$$

since the quadrilateral $APZX$ is cyclic, and triangle AXQ is rectangle, and CX is orthogonal to AZ . Now

$$\angle BXY = \angle BAY = \angle AXC$$

because $XABY$ is cyclic and we have proved that

$$\angle AXC = \angle BXY$$

so BC is parallel to AY , and

$$AC = BY, CY = AB$$

Now by Ptolomey's theorem on $APZX$ we have

$$(AX)(PZ) + (AP)(XZ) = (AZ)(PX)$$

we see that triangles PXZ and QXA are similar since

$$\angle QAX = \angle PZX = 90$$

and

$$\angle AXC = \angle BXY$$

is already proven, so

$$(AX)(PZ) = (AQ)(XZ)$$

Substituting yields

$$(AQ)(XZ) + (AP)(XZ) = (AZ)(PX)$$

dividing by $(PX)(XZ)$ We get

$$\frac{AQ + AP}{XP} = \frac{AZ}{XZ}$$

Now triangles AYZ , and XYP are similar so

$$\frac{AY}{AZ} = \frac{XY}{XP}$$

but also triangles XPY and XZB are similar and we get

$$\frac{XY}{XP} = \frac{XB}{XZ}$$

Comparing we have,

$$\frac{AY}{XB} = \frac{AZ}{XZ}$$

Substituting,

$$\frac{AQ + AP}{XP} = \frac{AY}{XB}$$

Dividing the new relation by AX and multiplying by XB we get

$$\frac{XB(AQ + AP)}{(XP)(AX)} = \frac{AY}{AX}$$

but

$$\frac{XB}{AX} = \frac{XY}{XQ}$$

since triangles AXB and QXY are similar, because

$$\angle AYX = \angle ABX$$

and

$$\angle AXB = \angle CXY$$

since $CY = AB$ Substituting again we get

$$\frac{XY(AQ) + XY(AP)}{(XP)(XQ)} = \frac{AY}{AX}$$

Now since triangles ACQ and XYQ are similar we have

$$XY(AQ) = AC(XQ)$$

and by the similarity of APB and XPY , we get

$$AB(CP) = XY(AP)$$

so substituting, and separating terms we get

$$\frac{AC}{XP} + \frac{AB}{XQ} = \frac{AY}{AX}$$

In the beginning we prove that $AC = BY$ and $AB = CY$ so

$$\frac{BY}{XP} + \frac{CY}{XQ} = \frac{AY}{AX}$$

■

Solution 3

It is obvious that

$$\angle AXB = \angle CXY = \alpha$$

for some value α . Also, note that $\angle BYA = \alpha$. Set

$$\angle BXC = \angle BYC = \beta.$$

We have

$$\frac{XC}{CY} = \tan \angle CYZ = \tan(90 - \alpha)$$

and

$$\frac{CQ}{CY} = \tan \angle CYQ = \tan(\alpha + \beta).$$

This gives

$$\frac{CY}{XQ} = \frac{1}{\tan(90 - \alpha) - \tan(\alpha + \beta)}.$$

Similarly, we can deduce that

$$\frac{BY}{XP} = \frac{1}{\tan(90 - \alpha - \beta) - \tan \alpha}.$$

Adding gives

$$\frac{\tan \alpha}{1 - \tan(\alpha + \beta) \tan \beta} + \frac{\tan(\alpha + \beta)}{1 - \tan(\alpha + \beta) \tan \beta} = \tan(2\alpha + \beta) = \frac{AY}{AX}.$$

Solution 4

First, since XY is the diameter and A , B , and C lie on the circle,

$$\angle XAY = \angle XBY = \angle XCY = 90$$

. Next, because AZ and CY are both perpendicular to CX , we have AZ to be parallel to CY .

Now looking at quadrilateral $APZX$, we see that this is cyclic because

$$\angle PAX + \angle PZX = 90 + 90 = 180.$$

Set $\alpha = \angle BXA = \angle BYA$, and $\beta = \angle BXC = \angle BYC$. Now,

$$\angle AYC = \angle YAZ$$

since AZ and CY are parallel. Also,

$$\angle PAZ = \angle PXZ = \alpha + \beta.$$

That means

$$\angle PXZ = \angle PXQ + \angle QXZ = \beta + \angle QXZ = \alpha + \beta$$

so

$$\angle QXZ = \alpha.$$

This means $\angle QXZ = \angle YBC = \alpha$, so BC and AY are parallel. Finally, we can look at the equation. We know

$$XP \cos \alpha = AX,$$

so $XP = \frac{AX}{\cos \alpha}$. We also know

$$XQ \cos(\alpha + \beta) = AX,$$

so $XQ = \frac{AX}{\cos(\alpha + \beta)}$. Plugging this into the LHS of the equation, we get

$$\frac{BY \cos \alpha}{AX} + \frac{CY \cos(\alpha + \beta)}{AZ}.$$

Now, let H be the point on AY such that BH is perpendicular to AY . Also, since $\angle AYB = \angle CXY$, their arcs have equal length, and $AB = CY$. Now, the LHS is simplified even more to

$$\frac{AB \cos \alpha}{AX} + \frac{CY \cos(\alpha + \beta)}{AZ}$$

which is equal to

$$\frac{AH + YH}{AZ},$$

which is equal to

$$\frac{AY}{AX}.$$

This completes the proof.

6. (*USAJMO 2013*) Find all real numbers $x, y, z \geq 1$ satisfying

$$\min(\sqrt{x + xyz}, \sqrt{y + xyz}, \sqrt{z + xyz}) = \sqrt{x - 1} + \sqrt{y - 1} + \sqrt{z - 1}.$$

Solution with Thought Process

Without loss of generality, let $1 \leq x \leq y \leq z$. Then $\sqrt{x + xyz} = \sqrt{x-1} + \sqrt{y-1} + \sqrt{z-1}$.

Suppose $x = y = z$. Then $\sqrt{x + x^3} = 3\sqrt{x-1}$, so $x + x^3 = 9x - 9$. It is easily verified that $x^3 - 8x + 9 = 0$ has no solution in positive numbers greater than 1. Thus, $\sqrt{x + xyz} \geq \sqrt{x-1} + \sqrt{y-1} + \sqrt{z-1}$ for $x = y = z$. We suspect if the inequality always holds.

Let $x = 1$. Then we have $\sqrt{1 + yz} \geq \sqrt{y-1} + \sqrt{z-1}$, which simplifies to

$$1 + yz \geq y + z - 2 + 2\sqrt{(y-1)(z-1)}$$

and hence

$$yz - y - z + 3 \geq 2\sqrt{(y-1)(z-1)}$$

Let us try a few examples: if $y = z = 2$, we have $3 > 2$; if $y = z$, we have $y^2 - 2y + 3 \geq 2(y-1)$, which reduces to $y^2 - 4y + 5 \geq 0$. The discriminant $(16 - 20)$ is negative, so in fact the inequality is strict. Now notice that $yz - y - z + 3 = (y-1)(z-1) + 2$. Now we see we can let $u = \sqrt{(y-1)(z-1)}$. Thus,

$$u^2 - 2u + 2 = (u-1)^2 + 1 > 0$$

and the claim holds for $x = 1$.

If $x > 1$, we see the $\sqrt{x-1}$ will provide a huge obstacle when squaring. But, using the identity $(x + y + z)^2 = x^2 + y^2 + z^2 + xy + yz + xz$:

$$x + xyz \geq x - 1 + y - 1 + z - 1 + 2\sqrt{(x-1)(y-1)} + 2\sqrt{(y-1)(z-1)} + 2\sqrt{(x-1)(z-1)}$$

which leads to

$$xyz \geq y + z - 3 + 2\sqrt{(x-1)(y-1)} + 2\sqrt{(y-1)(z-1)} + 2\sqrt{(x-1)(z-1)}$$

Again, we experiment. If $x = 2$, $y = 3$, and $z = 3$, then $18 > 7 + 4\sqrt{6}$.

Now, we see the finish: setting $u = \sqrt{x-1}$ gives $x = u^2 + 1$. We can solve a quadratic in u ! Because this problem is a #6, the crown jewel of USAJMO problems, we do not hesitate in computing the messy computations:

$$u^2(yz) - u(2\sqrt{y-1} + 2\sqrt{z-1}) + (3 + yz - y - z - 2\sqrt{(y-1)(z-1)}) \geq 0$$

Because the coefficient of u^2 is positive, all we need to do is to verify that the discriminant is nonpositive:

$$b^2 - 4ac = 4(y-1) + 4(z-1) - 8\sqrt{(y-1)(z-1)} - yz(12 + 4yz - 4y - 4z - 8\sqrt{(y-1)(z-1)})$$

Let us try a few examples. If $y = z$, then the discriminant $D = 8(y-1) - 8(y-1) - yz(12 + 4y^2 - 8y - 8(y-1)) = -yz(4y^2 - 16y + 20) = -4yz(y^2 - 4y + 5) < 0$.

We are almost done, but we need to find the correct argument. (How frustrating!) Success! The discriminant is negative. Thus, we can replace our claim with a strict one, and there are no real solutions to the original equation in the hypothesis.

-Thinking Process by sul

§5 USAJMO 2014

1. (*USAJMO 2014*) Let a, b, c be real numbers greater than or equal to 1. Prove that

$$\min \left(\frac{10a^2 - 5a + 1}{b^2 - 5b + 10}, \frac{10b^2 - 5b + 1}{c^2 - 5c + 10}, \frac{10c^2 - 5c + 1}{a^2 - 5a + 10} \right) \leq abc.$$

Solution

Since $(a - 1)^5 \geq 0$,

$$a^5 - 5a^4 + 10a^3 - 10a^2 + 5a - 1 \geq 0$$

or

$$10a^2 - 5a + 1 \leq a^3(a^2 - 5a + 10)$$

Since $a^2 - 5a + 10 = \left(a - \frac{5}{2}\right)^2 + \frac{15}{4} > 0$,

$$\frac{10a^2 - 5a + 1}{a^2 - 5a + 10} \leq a^3$$

Also note that $10a^2 - 5a + 1 = 10\left(a - \frac{1}{4}\right)^2 + \frac{3}{8} > 0$. We conclude

$$0 \leq \frac{10a^2 - 5a + 1}{a^2 - 5a + 10} \leq a^3$$

Similarly,

$$0 \leq \frac{10b^2 - 5b + 1}{b^2 - 5b + 10} \leq b^3$$

$$0 \leq \frac{10c^2 - 5c + 1}{c^2 - 5c + 10} \leq c^3$$

So

$$\left(\frac{10a^2 - 5a + 1}{a^2 - 5a + 10} \right) \left(\frac{10b^2 - 5b + 1}{b^2 - 5b + 10} \right) \left(\frac{10c^2 - 5c + 1}{c^2 - 5c + 10} \right) \leq a^3 b^3 c^3$$

or

$$\left(\frac{10a^2 - 5a + 1}{b^2 - 5b + 10} \right) \left(\frac{10b^2 - 5b + 1}{c^2 - 5c + 10} \right) \left(\frac{10c^2 - 5c + 1}{a^2 - 5a + 10} \right) \leq (abc)^3$$

Therefore,

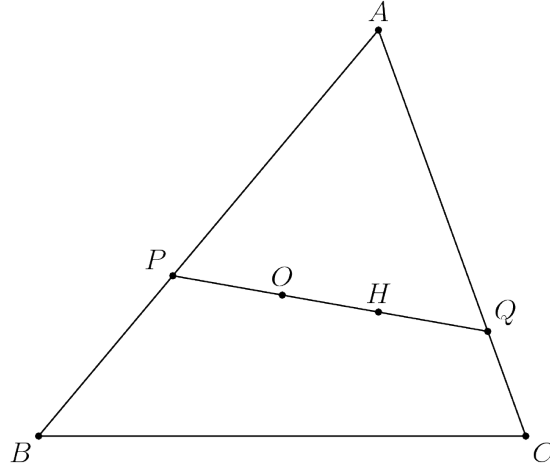
$$\min \left(\frac{10a^2 - 5a + 1}{b^2 - 5b + 10}, \frac{10b^2 - 5b + 1}{c^2 - 5c + 10}, \frac{10c^2 - 5c + 1}{a^2 - 5a + 10} \right) \leq abc.$$

2. (*USAJMO 2014*) Let $\triangle ABC$ be a non-equilateral, acute triangle with $\angle A = 60^\circ$, and let O and H denote the circumcenter and orthocenter of $\triangle ABC$, respectively.

(a) Prove that line OH intersects both segments AB and AC .

(b) Line OH intersects segments AB and AC at P and Q , respectively. Denote by s and t the respective areas of triangle APQ and quadrilateral $BPQC$. Determine the range of possible values for s/t .

Solution



Lemma: H is the reflection of O over the angle bisector of $\angle BAC$ (henceforth 'the' reflection)

Proof: Let H' be the reflection of O , and let B' be the reflection of B .

Then reflection takes $\angle ABH'$ to $\angle AB'O$.

$\triangle ABB'$ is equilateral, and O lies on the perpendicular bisector of $\overline{AB}$

It's well known that O lies strictly inside $\triangle ABC$ (since it's acute), meaning that $\angle ABH' = \angle AB'O = 30^\circ$, from which it follows that $\overline{BH'} \perp \overline{AC}$. Similarly, $\overline{CH'} \perp \overline{AB}$. Since H' lies on two altitudes, H' is the orthocenter, as desired.

End Lemma

So $\overline{OH}$ is perpendicular to the angle bisector of $\angle OAH$, which is the same line as the angle bisector of $\angle BAC$, meaning that $\triangle APQ$ is equilateral.

Let its side length be s , and let $PH = t$, where $0 < t < s, t \neq s/2$ because O lies strictly within $\angle BAC$, as must H , the reflection of O . Also, it's easy to show that if $O = H$ in a general triangle, it's equilateral, and we know $\triangle ABC$ is not equilateral. Hence H is not on the bisector of $\angle BAC \implies t \neq s/2$. Let $\overline{BH}$ intersect $\overline{AC}$ at P_B .

Since $\triangle H P_B Q$ and $\triangle B P_B A$ are 30-60-90 triangles, $AB = 2AP_B = 2(s - QP_B) = 2(s - HQ/2) = 2s - HQ = 2s - (s - t) = s + t$

Similarly, $AC = 2s - t$

The ratio $\frac{[APQ]}{[ABC] - [APQ]}$ is $\frac{AP \cdot AQ}{AB \cdot AC - AP \cdot AQ} = \frac{s^2}{(s+t)(2s-t) - s^2}$. The denominator equals $(1.5s)^2 - (.5s - t)^2 - s^2$ where $.5s - t$ can equal any value in $(-.5s, .5s)$ except 0. Therefore, the denominator can equal any value in $(s^2, 5s^2/4)$, and the ratio is any value in $\left(\frac{4}{5}, 1\right)$.

Note: It's easy to show that for any point H on $\overline{PQ}$ except the midpoint, Points B and C can be validly defined to make an acute, non-equilateral triangle.

Solution 2

Let J be the farthest point on the circumcircle of $\triangle ABC$ from line BC . Lemma:

Line $AJ \parallel$ Line OH Proof: Set $b = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$ and $c = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$, and a on the unit circle. It is well known that $o = 0$ and $h = a + b + c$, so we have $h - o = a + b + c = a - 1 = a - j$, so $\frac{a - j}{h - o}$ is real and thus the 2 lines are parallel.

WLOG let A be in the first quadrant. Clearly by the above lemma OH must intersect line BC closer to B than to C . Intersect AJ and BC at D and OH and BC at E . We clearly have $0 < \frac{6.0\text{pt}JC - 6.0\text{pt}AB}{2} = \frac{120 - 6.0\text{pt}AC}{2} = \angle ADB = \angle OEC < 30 = \angle OBC$, OH must intersect AB . We also have, letting the intersection of line OH and line AC be Q , and letting intersection of OH and AB be P , $\angle OQA = \angle JAB = \frac{6.0\text{pt}JB}{2} = 60$. Since $\angle OCA = \angle BCA - \angle BCO < 90 - 30 = \angle OQA$, and $\angle OQC = 120 > \angle OAC$, OH also intersects AC . We have $\angle OPA = \angle PAJ = 60$, so $\triangle APQ$ is equilateral. Letting $AJ = 2x$, and letting the foot of the perpendicular from O to AJ be L , we have $OL = \sqrt{1 - x^2}$, and since OL is an altitude of $\triangle APQ$, we have $[APQ] = \frac{OL^2 \sqrt{3}}{3} = \frac{\sqrt{3}(1 - x^2)}{3}$. Letting the foot of the perpendicular from A to OJ be K , we have $\triangle JKA \sim \triangle JLO$ by AA with ratio $\frac{JA}{JO} = 2x$. Therefore, $JK = 2x(JL) = 2x^2$. Letting D be the foot of the altitude from J to BC , we have $KD = JD - JK = \frac{3}{2} - 2x^2$, since $re(B) = re(C) \implies JD = re(j) - (-\frac{1}{2}) = \frac{3}{2}$. Thus, since $BC = \sqrt{3}$ we have $[ABC] = \frac{(\frac{3}{2} - 2x^2)(\sqrt{3})}{2} = \frac{(3 - 4x^2)(\sqrt{3})}{2}$, so $[PQCB] = [ABC] - [APQ] = \frac{(5 - 8x^2)(\sqrt{3})}{12}$, so $\frac{[APQ]}{[PQCB]} = \frac{4 - 4x^2}{5 - 8x^2} = \frac{1}{2} + \frac{3}{10 - 16x^2}$. We have $x = \sin(\frac{JOA}{2})$, with $0 < 120 - 2\angle ACB = \angle JOA < 60$, so x can be anything in the interval $(0, \frac{1}{2})$. Therefore, the desired range is $(\frac{4}{5}, 1)$.

Solution by Shaddoll

3. (USAJMO 2014) Let $\mathbb{Z}$ be the set of integers. Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$xf(2f(y) - x) + y^2f(2x - f(y)) = \frac{f(x)^2}{x} + f(yf(y))$$

for all $x, y \in \mathbb{Z}$ with $x \neq 0$.

Solution

Note: This solution is kind of rough. I didn't want to put my 7-page solution all over again. It would be nice if someone could edit in the details of the expansions.

Lemma 1: $f(0) = 0$. Proof: Assume the opposite for a contradiction. Plug in $x = 2f(0)$ (because we assumed that $f(0) \neq 0$), $y = 0$. What you get eventually reduces to:

$$4f(0) - 2 = \left(\frac{f(2f(0))}{f(0)} \right)^2$$

which is a contradiction since the LHS is divisible by 2 but not 4.

Then plug in $y = 0$ into the original equation and simplify by Lemma 1. We get:

$$x^2f(-x) = f(x)^2$$

Then:

$$x^6f(x) = x^4(x^2f(x))$$

$$\begin{aligned}
 &= x^4((-x)^2 f(-(-x))) \\
 &= x^4(-x)^2 f(-(-x)) \\
 &= x^4 f(-x)^2 \\
 &= f(x)^4
 \end{aligned}$$

Therefore, $f(x)$ must be 0 or x^2 .

Now either $f(x)$ is x^2 for all x or there exists $a \neq 0$ such that $f(a) = 0$. The first case gives a valid solution. In the second case, we let $y = a$ in the original equation and simplify to get:

$$xf(-x) + a^2 f(2x) = \frac{f(x)^2}{x}$$

But we know that $xf(-x) = \frac{f(x)^2}{x}$, so:

$$a^2 f(2x) = 0$$

Since a is not 0, $f(2x)$ is 0 for all x (including 0). Now either $f(x)$ is 0 for all x , or there exists some $m \neq 0$ such that $f(m) = m^2$. Then m must be odd. We can let $x = 2k$ in the original equation, and since $f(2x)$ is 0 for all x , stuff cancels and we get:

$$y^2 f(4k - f(y)) = f(yf(y))$$

for $k \neq 0$. Now, let $y = m$ and we get:

$$m^2 f(4k - m^2) = f(m^3)$$

Now, either both sides are 0 or both are equal to m^6 . If both are m^6 then:

$$m^2(4k - m^2)^2 = m^6$$

which simplifies to:

$$4k - m^2 = \pm m^2$$

Since $k \neq 0$ and m is odd, both cases are impossible, so we must have:

$$m^2 f(4k - m^2) = f(m^3) = 0$$

Then we can let k be anything except 0, and get $f(x)$ is 0 for all $x \equiv 3 \pmod{4}$ except $-m^2$. Also since $x^2 f(-x) = f(x)^2$, we have $f(x) = 0 \Rightarrow f(-x) = 0$, so $f(x)$ is 0 for all $x \equiv 1 \pmod{4}$ except m^2 . So $f(x)$ is 0 for all x except $\pm m^2$. Since $f(m) \neq 0$, $m = \pm m^2$. Squaring, $m^2 = m^4$ and dividing by m , $m = m^3$. Since $f(m^3) = 0$, $f(m) = 0$, which is a contradiction for $m \neq 1$. However, if we plug in $x = 1$ with $f(1) = 1$ and y as an arbitrary large number with $f(y) = 0$ into the original equation, we get $0 = 1$ which is a clear contradiction, so our only solutions are $f(x) = 0$ and $f(x) = x^2$.

Solution 2

Given that the range of f consists entirely of integers, it is clear that the LHS must be an integer and $f(yf(y))$ must also be an integer, therefore $\frac{f(x)^2}{x}$ is an integer. If x divides $f(x)^2$ for all integers $x \neq 0$, then x must be a factor of $f(x)$, therefore $f(0) = 0$. Now, by setting $y = 0$ in the original equation, this simplifies to $xf(-x) = \frac{f(x)^2}{x}$. Assuming $x \neq 0$, we have $x^2 f(-x) = f(x)^2$. Substituting in $-x$

for x gives us $x^2 f(x) = f(-x)^2$. Substituting in $\frac{f(x)^2}{x^2}$ in for $f(-x)$ in the second equation gives us $x^2 f(x) = \frac{f(x)^4}{x^4}$, so $x^6 f(x) = f(x)^4$. In particular, if $f(x) \neq 0$, then we have $f(x)^3 = x^6$, therefore $f(x)$ is equivalent to 0 or x^2 for every integer x . Now, we shall prove that if for some integer $t \neq 0$, if $f(t) = 0$, then $f(x) = 0$ for all integers x . If we assume $f(y) = 0$ and $y \neq 0$ in the original equation, this simplifies to $xf(-x) + y^2 f(2x) = \frac{f(x)^2}{x}$. However, since $x^2 f(-x) = f(x)^2$, we can rewrite this equation as $\frac{f(x)^2}{x} + y^2 f(2x) = \frac{f(x)^2}{x}$, $y^2 f(2x)$ must therefore be equivalent to 0. Since, by our initial assumption, $y \neq 0$, this means that $f(2x) = 0$, so, if for some integer $y \neq 0$, $f(y) = 0$, then $f(x) = 0$ for all integers x . The contrapositive must also be true, i.e. If $f(x) \neq 0$ for all integers x , then there is no integral value of $y \neq 0$ such that $f(y) = 0$, therefore $f(x)$ must be equivalent for x^2 for every integer x , including 0, since $f(0) = 0$. Thus, $f(x) = 0, x^2$ are the only possible solutions.

Solution 3

Let's assume $f(0) \neq 0$. Substitute $(x, y) = (2f(0), 0)$ to get

$$2f(0)^2 = f(2f(0))^2 / 2f(0) + f(0)$$

$$2f(0)^2(2f(0) - 1) = f(2f(0))^2$$

This means that $2(2f(0) - 1)$ is a perfect square. However, this is impossible, as it is equivalent to 2 (mod 4). Therefore, $f(0) = 0$. Now substitute $x \neq 0, y = 0$ to get

$$xf(-x) = \frac{f(x)^2}{x} \implies x^2 f(-x) = f(x)^2.$$

Similarly,

$$x^2 f(x) = f(-x)^2.$$

From these two equations, we can find either $f(x) = f(-x) = 0$, or $f(x) = f(-x) = x^2$. Both of these are valid solutions on their own, so let's see if there are any solutions combining the two.

Let's say we can find $f(x) = x^2, f(y) = 0$, and $x, y \neq 0$. Then

$$xf(-x) + y^2 f(2x) = f(x)^2 / x.$$

$$y^2 f(2x) = x - x^3.$$

(NEEDS FIXING: $f(x)^2/x = x^4/x = x^3$, so the RHS is 0 instead of $x - x^3$.)

If $f(2x) = 4x^2$, then $y^2 = \frac{x-x^3}{4x^2} = \frac{1-x^2}{4x}$, which is only possible when $y = 0$. This contradicts our assumption. Therefore, $f(2x) = 0$. This forces $x = \pm 1$ due to the right side of the equation. Let's consider the possibility $f(2) = 0, f(1) = 1$. Substituting $(x, y) = (2, 1)$ into the original equation yields

$$0 = 2f(0) + 1f(2) = 0 + f(1) = 1,$$

which is impossible. So $f(2) = f(-2) = 4$ and there are no solutions "combining" $f(x) = x^2$ and $f(x) = 0$.

Therefore our only solutions are $\boxed{f(x) = 0}$ and $\boxed{f(x) = x^2}$.

Solution 4

Let the given assertion be $P(x, y)$. We try $P(x, 0)$ and get $xf(2c-x) = f(x)^2/x + c$, where $f(0) = c$. We plug in $x = c$ and get $cf(c) = f(c)^2/c + c$. Rearranging and

solving for c^2 gives us $c^2 = \frac{f(c)^2}{f(c)-1}$. Obviously, the only c that works such that the RHS is an integer is $c = 0$, and thus $f(0) = 0$.

We use this information on assertion $P(x, 0)$ and obtain $xf(-x) = f(x^2)/x$, or $f(-x) = \frac{f(x)^2}{x^2}$. Thus, $f(x)$ is an even function. It follows that $f(x) = 0, x^2$ for each x . We now prove that $f(x) = x^2$, $f(x)=0$ are the only solutions. [in progress] SigmaPiE

4. (USAJMO 2014) Let $b \geq 2$ be an integer, and let $s_b(n)$ denote the sum of the digits of n when it is written in base b . Show that there are infinitely many positive integers that cannot be represented in the form $n + s_b(n)$, where n is a positive integer.

Solution

Define $S(n) = n + s_b(n)$, and call a number *unrepresentable* if it cannot equal $S(n)$ for a positive integer n . We claim that in the interval $(b^p, b^{p+1}]$ there exists an unrepresentable number, for every positive integer p .

If b^{p+1} is unrepresentable, we're done. Otherwise, time for our lemma:

Lemma: Define the function $f(p)$ to equal the number of integers x less than b^p such that $S(x) \geq b^p$. If $b^{p+1} = S(y)$ for some y , then $f(p+1) > f(p)$.

Proof: Let $F(p)$ be the set of integers x less than b^p such that $S(x) \geq b^p$. Then for every integer in $F(p)$, append the digit $(b-1)$ to the front of it to create a valid integer in $F(p+1)$. Also, notice that $(b-1) \cdot b^p \leq y < b^{p+1}$. Removing the digit $(b-1)$ from the front of y creates a number that is not in $F(p)$. Hence, $F(p) \rightarrow F(p+1)$, but there exists an element of $F(p+1)$ not corresponding with $F(p)$, so $f(p+1) > f(p)$.

Note that our lemma combined with the Pigeonhole Principle essentially proves the claim. Therefore, because there are infinitely many disjoint intervals containing an unrepresentable number, there are infinitely many unrepresentable numbers.

Solution 2 (Simple)

Let $f(n) = n + s_b(n)$. It is easy to see that $f(b^2) = b^2 + 1$, and $f(b^2 - b + 1) = b^2 - b + 1 + b - 1 + 1 = b^2 + 1$, because the digits of $b^2 - b + 1$ are $b-1$ and 1 . Additionally, if we take any number, let's say x , and multiply it by b^3 , and then add it to b^2 and $b^2 - b + 1$, we still obtain $f(b^3x + b^2) = f(b^3x + b^2 - b + 1)$, because we are simply adding $b^3x + s_b(x)$ to each value.

Thus, we conclude that there are infinitely many numbers that are counted twice in $f(n)$, and that these numbers come b^3 apart. Finally, it is clear that for a large N , the number of numbers that are counted at least twice is at least $\left\lfloor \frac{N}{b^3} \right\rfloor$. Because $f(n) > n$ for all n , this means that there are at least $\left\lfloor \frac{N}{b^3} \right\rfloor$ numbers uncounted, and because this is unbounded, the proof is complete.

Operation Intuitive Number Theory (INT)- Solution 2 I hope this solution is quite intuitive, because it is without complicated notation. It didn't take me very long to discover. As with Solution 1, define $S(n) = n + s_b(n)$. We will use numbers from b^k to $b^{k+1} - 1$ for induction. Call this interval Class k .

Start with $b^0 = 1$ and go up to $b - 1$. There are $b - 1$ numbers covered. Easily $S(n)$ ranges from $1 + 1 = 2$ to $b - 1 + b - 1 = 2b - 2$, or a range of $2b - 3$. Thus, there are at LEAST $b - 2$ numbers lacking coverage. Now, in order to fill up these gaps, we consult Class 1: $b^1 = b$ to $b^2 - 1$. If we are to fill up all these gaps, then we need at least $b - 2$ numbers in the next series with their S values. Unfortunately, there are only at MOST $b - 3$ numbers that can satisfy a value, from b to $2b - 3$, otherwise the S value is too big

(note that none of them are zero)! Thus, between Class 0 and 1, there is at least one value lacking.

After setting up the base case, consider the Class k numbers from b^k to $b^{k+1} - 1$ there are clearly $b^{k+1} - b^k$ integers. Yet the $S(n)$ can range from $b^k + 1 = b^k + 1$ to $(b^{k+1} - 1) + (k + 1)(b - 1) = b^{k+1} + b(k + 1) - (k + 1)$. This gives a range of $b^{k+1} - b^k + b(k + 1) - (k + 1)$. This leaves an extra $b(k + 1) - (k + 1)$ numbers. Now, we invoke the next class: $k + 1$. Again, to fill in the gap there are sadly only $b(k + 1) - (k + 2)$ numbers available: b^{k+1} to $b^{k+1} + b(k + 1) - (k + 3)$, because $s_b(n)$ is at least 1.

By induction we are done. Because each Class k for integral k at least 0 misses at least one S value, we miss an infinite number of numbers. Game over!

Note: If you are unconvinced of the range of $S(n)$ values we go from the smallest value of both the number and the $s_b(n)$, or b^k and 1, to the greatest of both, i.e. $b^{k+1} - 1$ and $k(b + 1)$, because in that series the largest possible sum is a bunch of $b - 1$ s.

5. (*USAJMO 2014*) Let k be a positive integer. Two players A and B play a game on an infinite grid of regular hexagons. Initially all the grid cells are empty. Then the players alternately take turns with A moving first. In his move, A may choose two adjacent hexagons in the grid which are empty and place a counter in both of them. In his move, B may choose any counter on the board and remove it. If at any time there are k consecutive grid cells in a line all of which contain a counter, A wins. Find the minimum value of k for which A cannot win in a finite number of moves, or prove that no such minimum value exists.

Solution

The answer is $k = 6$. We prove that A can win for $k = 5$ (which hence proves it for $k < 5$ as well) and show that B can thwart A for $k \geq 6$.

Arrange the board so that a pair of opposite sides are horizontal. Create a coordinate system on the board by setting the center of some hexagon as the origin and setting the hexagons directly above and above-and-right as $(0, 1)$ and $(1, 0)$, respectively. Then, for example, the below-and-right hexagon touching the origin is $(1, -1)$. So two hexagons touch if their coordinate difference is one of these.

Now for $k = 5$, person A places his counters only in $\{0, 1, 2, 3, 4\} \times \{0, 1\}$. Note that if, at A 's turn, there are 4 counters in either column, then A can win immediately, so let us assume that in both columns there are at least 2 missing, meaning that at most 6 counters are on the board. We would like to find when A cannot play under these circumstances. If we look at the disjoint sets

$$\{(0, 0), (0, 1), (1, 0)\}, \{(1, 1), (2, 0), (2, 1)\}, \{(3, 0), (3, 1), (4, 0)\}, \{(4, 1)\}$$

we see that either some set has at least 2 hexagons without counters, in which case A can move, or all four sets have exactly 1 missing counter. Similarly for the sets

$$\{(0, 0)\}, \{(0, 1), (1, 0), (1, 1)\}, \{(2, 0), (2, 1), (3, 0)\}, \{(3, 1), (4, 0), (4, 1)\}$$

So both $(0, 0), (4, 1)$ have no token on them. This means that $(0, 1), (1, 0), (3, 1), (4, 0)$ do. Thus $(3, 0)$ and $(1, 1)$ do not. So this is the only situation in which A can neither win immediately nor play in only these 10 hexagons.

So A plays only in these 10 hexagons until either he has a win or he can't anymore. If he wins, then we're done. Otherwise, A plays in hexagons $(5, 0)$ and $(5, 1)$. Then B either removes $(5, 1)$ so that A can win at $(2, 0)$, or B removes $(5, 0)$ and A plays at $(5, 0)$ and $(4, 1)$ and then at either $(3, 0)$ or $(1, 1)$ the next turn, or B removes one of

$\{0, 1, 2, 3, 4\} \times \{0, 1\}$ in which case A can either win immediately at $(3, 0)$ or can play in both columns, and then win the next turn.

Now if $k \geq 6$, then if A plays on anything in the lattice generated by $(2, -1)$ and $(1, 1)$, that is $(2a + b, b - a)$ for a, b integers, then B removes it. Otherwise, B removes any of A 's counters. This works because in order for A to win, there must be at least 2 counters in this lattice, but A can only put a counter on 1 at any time, so there's at most 1 on the lattice at any time.

So A wins if $k \leq 5$, and B wins if $k \geq 6$.

6. (USAJMO 2014) Let ABC be a triangle with incenter I , incircle γ and circumcircle Γ . Let M, N, P be the midpoints of sides $\overline{BC}$, $\overline{CA}$, $\overline{AB}$ and let E, F be the tangency points of γ with $\overline{CA}$ and $\overline{AB}$, respectively. Let U, V be the intersections of line EF with line MN and line MP , respectively, and let X be the midpoint of arc BAC of Γ .

(a) Prove that I lies on ray CV .

(b) Prove that line XI bisects $\overline{UV}$.

Solution 1

(a) Let $V' = \overline{CI} \cap \overline{EF}$. We claim $BIV'F$ is cyclic. This follows by Angle chasing:

$$\angle V'FB \equiv \angle V'FE \equiv 90^\circ - \frac{A}{2} \pmod{180^\circ} \text{ and } \angle V'IB \equiv \frac{B}{2} + \frac{C}{2} \equiv 90^\circ - \frac{A}{2} \pmod{180^\circ}$$

Thus, $\angle BV'C = \angle BV'I = \angle BFI = 90^\circ$. It follows that $(BV'C)$ is a circle centered at M , so $\angle V'MB = 2\angle V'CB = C$.

But, $\angle PMC = C$, so $V' \in \overline{MP}$ and thus $V' = V$, as needed.

(b) Apply a homothety at I sending $\overline{XI} \cap \overline{EF}$ to X . Notice that $\overline{EF} \parallel \overline{AX}$, so the homothety sends $\overleftrightarrow{EF}$ to $\overleftrightarrow{AX}$.

Thus, $U' = \overline{AX} \cap \overline{BI} = I_B$ and $V' = \overline{AX} \cap \overline{CI} = I_C$, where I_B, I_C are the respective excenters.

By duality, A, B, C are the feet of the altitude of $\triangle I_A I_B I_C$, so Γ is the nine-point circle of $I_A I_B I_C$.

But, $X = \overline{I_B I_C} \cap \Gamma$ other than A , so X must be the midpoint of $I_B I_C$, as needed.

Note 1: In (a), we use directed angles with half-angles, which typically doesn't work. I circumvent this issue by considering the half-angles themselves mod 180, and this means there is no need to half any of the angles.

Note 2: In (b), you can prove more generally that B, C, I_B, I_C sit on a circle centered at X . Maybe call it Fact 6.

-sillybone

Solution 2



Solution 2: First we show that the intersection V' of MP with the internal angle bisector of C is the same as the intersection V'' of EF with the internal angle bisector of C . Let D denote the intersection of AB with the internal angle bisector of C , and let a, b, c denote the side lengths of BC, AC, AB . By Menelaus on V'', F, E , with respect to $\triangle ADC$,

Similarly,

Since V' and V'' divide CD in the same ratio, they must be the same point. Now, since $\frac{b-a}{b+a} > -1$, I lies on ray CV . ■

(b)

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Note that X lies on both the circumcircle and the perpendicular bisector of segment BC . Let D be the midpoint of UV ; our goal is to prove that points X , D , and I are collinear, which equates to proving X lies on ray ID .

Because MD is also an altitude of triangle MVU , and MD and IA are both perpendicular to EF , $\overline{MD} \parallel \overline{IA}$. Furthermore, we have $\angle VMD = \angle UMD = x$ because $APMN$ is a parallelogram. (incomplete)

Solution 2: Let I_A , I_B , and I_C be the excenters of ABC . Note that the circumcircle of ABC is the nine-point circle of $I_AI_BI_C$. Since AX is the external angle bisector of $\angle BAC$, X is the midpoint of I_BI_C . Now UV and I_BI_C are parallel since both are perpendicular to the internal angle bisector of $\angle BAC$. Since IX bisects I_BI_C , it bisects UV as well.

Solution 3: Let X' be the antipode of X with respect to the circumcircle of triangle ABC . Then, by the Incenter-Excenter lemma, X' is the center of a circle containing B , I , and C . Because XX' is a diameter, XB and XC are tangent to the aforementioned circle; thus by a well-known symmedian lemma (the Tangent Construction Lemma), XI is the I -symmedian of triangle IBC . From part (a); we know that $BVUC$ is cyclic with diameter BC (we can derive a similar argument for point U); thus, from similar triangles, we know that XI is also the median of triangle VIU , and we are done.

Solution 4: Let M_b, M_c be the midpoints of arcs CA, AB respectively, and let T_a be the tangency point between the A -mixtilinear incircle of ABC and Γ . It's well-known that $T_a \in XI$, T_aX bisects M_bM_c , $AI \perp EF$, and $AI \perp M_bM_c$.

Now, it's easy to see $EF \parallel M_bM_c$, so IUV and IM_bM_c are homothetic at I . But $T_aX \equiv IX$ bisects M_bM_c , so Part B follows directly from the homothety. ike.chen

Another Solution for Part (a) Let $\angle CAI = \angle BAI = x$, $\angle ACI = \angle BCI = z$, and $\angle ABI = \angle CBI = y$. In this solution, we use degrees.

Since $AE = AF$ (they are both tangents), we know that $\triangle AEF$ is isosceles with vertex angle bisected by $\overline{AI}$. Therefore, $\overline{AI} \perp \overline{EF}$. This means that $\angle AEF = 90 - x$, so $\angle CEF = 90 + x$.

Let V be the intersection of $\overleftrightarrow{EF}$ and $\overleftrightarrow{CI}$, so

$$\angle EVC = 180 - (90 + x) - z = 90 - x - z = \frac{180 - 2x - 2z}{2} = \frac{2y}{2} = y,$$

so

$$\angle FVI = 180 - y = 180 - \angle FBI.$$

Therefore, $BFVI$ is cyclic, meaning

$$\angle BVC = \angle BVI = \angle BFI = 90.$$

This means that M is the circumcenter of $\triangle BVC$, so $MV = CM$, so $\angle CVM = z \rightarrow \angle CMV = 180 - 2z$.

Then $\angle CMV$ is supplementary to $\angle ACB$, so $\overleftrightarrow{VM} \parallel \overleftrightarrow{AC} \parallel \overleftrightarrow{MP}$, so V is on $\overleftrightarrow{MP}$, so the three lines $\overleftrightarrow{EF}, \overleftrightarrow{CI}, \overleftrightarrow{MP}$ are concurrent at V , meaning that I is on $\overleftrightarrow{CI}$, as desired.

§6 USAJMO 2015

1. (USAJMO 2015) Given a sequence of real numbers, a move consists of choosing two terms and replacing each with their arithmetic mean. Show that there exists a

sequence of 2015 distinct real numbers such that after one initial move is applied to the sequence – no matter what move – there is always a way to continue with a finite sequence of moves so as to obtain in the end a constant sequence.

Solution 1

We claim $\{-1007, -1006, \dots, 1006, 1007\}$ works. Let the original two swapped terms be i and j and let their average be x .

Average out all pairs of inverses other than $\{-j, -i, x, x\}$, so we are left with at most 4 non-zero terms.

Case 1: $i, j \neq 0$

If $i = -j$, then we are already done. Otherwise, average $-i, -j$, so the non-zero terms are $\{x, x, -x, -x\}$ and now we cancel inverses.

Case 2: $i = 0$

Then, the non-zero terms are $\{\frac{j}{2}, \frac{j}{2}, -j\}$. Average $-j$ with 0, so now the non-zero terms are $\{x, x, -x, -x\}$, and cancel inverses.

-sillybone

Solution 2

Let the set be $-1007, -1006, \dots, 0, 1, 2, \dots, 1006, 1007$, namely all the consecutive integers from -1007 to 1007 . Notice that the operation we are applying

2. (*USAJMO 2015*) Solve in integers the equation

$$x^2 + xy + y^2 = \left(\frac{x+y}{3} + 1\right)^3.$$

Solution

We first notice that both sides must be integers, so $\frac{x+y}{3}$ must be an integer.

We can therefore perform the substitution $x + y = 3t$ where t is an integer.

Then:

$$\begin{aligned} (3t)^2 - xy &= (t+1)^3 \\ 9t^2 + x(x-3t) &= t^3 + 3t^2 + 3t + 1 \\ 4x^2 - 12xt + 9t^2 &= 4t^3 - 15t^2 + 12t + 4 \\ (2x-3t)^2 &= (t-2)^2(4t+1) \end{aligned}$$

$4t+1$ is therefore the square of an odd integer and can be replaced with $(2n+1)^2 = 4n^2 + 4n + 1$.

By substituting using $t = n^2 + n$ we get:

$$\begin{aligned} (2x - 3n^2 - 3n)^2 &= [(n^2 + n - 2)(2n + 1)]^2 \\ 2x - 3n^2 - 3n &= \pm(2n^3 + 3n^2 - 3n - 2) \\ x &= n^3 + 3n^2 - 1 \text{ or } x = -n^3 + 3n + 1 \end{aligned}$$

Using substitution we get the solutions: $(n^3 + 3n^2 - 1, -n^3 + 3n + 1) \cup (-n^3 + 3n + 1, n^3 + 3n^2 - 1)$

3. (*USAJMO 2015*) Quadrilateral $APBQ$ is inscribed in circle ω with $\angle P = \angle Q = 90^\circ$ and $AP = AQ < BP$. Let X be a variable point on segment $\overline{PQ}$. Line AX meets ω again at S (other than A). Point T lies on arc AQB of ω such that $\overline{XT}$ is perpendicular to $\overline{AX}$. Let M denote the midpoint of chord $\overline{ST}$. As X varies on segment $\overline{PQ}$, show that M moves along a circle.

Solution 1

We claim the M lies on the circle centered at the midpoint of $\overline{AO}$ that passes through P .

Proceed via complex numbers. $a = 1, b = -1$ and s, t are free parameters. Then,

$$x = \frac{1}{2}(s + 1 + t - s\bar{t}) \text{ and } \Re(p) = \Re(x) \implies p + \bar{p} = x + \bar{x}$$

Putting it together,

$$\begin{aligned} \left| \frac{1}{2} - \left(\frac{s+t}{2} \right) \right| = \left| \frac{1}{2} - p \right| &\iff (s+t-1)(\bar{s}+\bar{t}-1) = 5 - 2(p+\bar{p}) = 5 - 2(x+\bar{x}) \\ &\iff 3 + s\bar{t} + \bar{s}t - s - t - \bar{s} - \bar{t} = 5 - (s+1+t-s\bar{t} + \bar{s}+1+\bar{t}-\bar{s}t) \end{aligned}$$

which is clear by expanding. Thus, the proof is complete.

-sillybone

Solution 2

We will use coordinate geometry.

Without loss of generality, let the circle be the unit circle centered at the origin,

$$A = (1, 0), P = (1 - a, b), Q = (1 - a, -b)$$

, where $(1 - a)^2 + b^2 = 1$.

Let angle $\angle XAB = A$, which is an acute angle, $\tan A = t$, then $X = (1 - a, at)$.

Angle $\angle BOS = 2A$, $S = (-\cos(2A), \sin(2A))$. Let $M = (u, v)$, then $T = (2u + \cos(2A), 2v - \sin(2A))$.

The condition $TX \perp AX$ yields: $(2v - \sin(2A) - at)/(2u + \cos(2A) + a - 1) = \cot A$. (E1)

Use identities $(\cos A)^2 = 1/(1 + t^2)$, $\cos(2A) = 2(\cos A)^2 - 1 = 2/(1 + t^2) - 1$, $\sin(2A) = 2 \sin A \cos A = 2t^2/(1 + t^2)$, we obtain $2vt - at^2 = 2u + a$. (E1')

The condition that T is on the circle yields $(2u + \cos(2A))^2 + (2v - \sin(2A))^2 = 1$, namely $v \sin(2A) - u \cos(2A) = u^2 + v^2$. (E2)

M is the mid-point on the hypotenuse of triangle STX , hence $MS = MX$, yielding $(u + \cos(2A))^2 + (v - \sin(2A))^2 = (u + a - 1)^2 + (v - at)^2$. (E3)

Expand (E3), using (E2) to replace $2(v \sin(2A) - u \cos(2A))$ with $2(u^2 + v^2)$, and using (E1') to replace $a(-2vt + at^2)$ with $-a(2u + a)$, and we obtain $u^2 - u - a + v^2 = 0$, namely $(u - \frac{1}{2})^2 + v^2 = a + \frac{1}{4}$, which is a circle centered at $(\frac{1}{2}, 0)$ with radius $r = \sqrt{a + \frac{1}{4}}$.

Solution 3

Let the midpoint of AO be K . We claim that M moves along a circle with radius KP .

We will show that $KM^2 = KP^2$, which implies that $KM = KP$, and as KP is fixed, this implies the claim.

$$KM^2 = \frac{AM^2 + OM^2}{2} - \frac{AO^2}{4} \text{ by the median formula on } \triangle AMO.$$

$$KP^2 = \frac{AP^2 + OP^2}{2} - \frac{AO^2}{4} \text{ by the median formula on } \triangle APO.$$

$$KM^2 - KP^2 = \frac{1}{2}(AM^2 + OM^2 - AP^2 - OP^2).$$

As $OP = OT$, $OP^2 - OM^2 = MT^2$ from right triangle OMT . (1)

$$\text{By (1), } KM^2 - KP^2 = \frac{1}{2}(AM^2 - MT^2 - AP^2).$$

Since M is the circumcenter of $\triangle XTS$, and MT is the circumradius, the expression $AM^2 - MT^2$ is the power of point A with respect to (XTS) . However, as $AX * AS$ is also the power of point A with respect to (XTS) , this implies that $AM^2 - MT^2 = AX * AS$. (2)

$$\text{By (2), } KM^2 - KP^2 = \frac{1}{2}(AX * AS - AP^2)$$

Finally, $\triangle APX \sim \triangle ASP$ by AA similarity ($\angle XAP = \angle SAP$ and $\angle APX = \angle AQP = \angle ASP$), so $AX * AS = AP^2$. (3)

By (3), $KM^2 - KP^2 = 0$, so $KM^2 = KP^2$, as desired. *QED*

More Solutions https://artofproblemsolving.com/wiki/index.php/2015_USAMO_Problems/Problem

4. (USAJMO 2015) Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that

$$f(x) + f(t) = f(y) + f(z)$$

for all rational numbers $x < y < z < t$ that form an arithmetic progression. ($\mathbb{Q}$ is the set of all rational numbers.)

Solution 1

According to the given, $f(x - a) + f(x + 0.5a) = f(x - 0.5a) + f(x)$, where x and a are rational. Likewise $f(x - 0.5a) + f(x + a) = f(x + 0.5a) + f(x)$. Hence $f(x + a) - f(x) = f(x) - f(x - a)$, namely $2f(x) = f(x - a) + f(x + a)$. Let $f(0) = C$, then consider $F(x) = f(x) - C$, where $F(0) = 0$, $2F(x) = F(x - a) + F(x + a)$.

$F(2x) = F(x) + [F(x) - F(0)] = 2F(x)$, $F(3x) = F(2x) + [F(2x) - F(x)] = 3F(x)$. Easily, by induction, $F(nx) = nF(x)$ for all integers n . Therefore, for nonzero integer m , $(1/m)F(mx) = F(x)$, namely $F(x/m) = (1/m)F(x)$. Hence $F(n/m) = (n/m)F(1)$. Let $F(1) = k$, we obtain $F(x) = kx$, where k is the slope of the linear functions, and $f(x) = kx + C$.

Solution 2

We have

$$f(x - 3d) + f(x + 3d) = f(x - d) + f(x + d)$$

and

$$f(x) + f(x + 3d) = f(x + d) + f(x + 2d).$$

Subtracting these two and rearranging gives

$$f(x - 3d) + f(x + 2d) = f(x) + f(x - d),$$

and since $f(x + 2d) = f(x + d) + f(x) - f(x - d)$ we get

$$f(x - 3d) + f(x + d) = 2f(x - d)$$

from which we get

$$f(x-d) + f(x+d) = 2f(x).$$

Then we have $f(x) + f(y) = f(0) + f(x+y) = 2f\left(\frac{x+y}{2}\right)$. Setting $f(0) = c$, we let $f(x) = g(x) + c$ to get $g(x) + g(y) = g(x+y)$. This is Cauchy's functional equation, so it has solutions at $g(x) = kx$, so the answer is $\boxed{f(x) = kx + c}$.

5. (USAJMO 2015) Let $ABCD$ be a cyclic quadrilateral. Prove that there exists a point X on segment $\overline{BD}$ such that $\angle BAC = \angle XAD$ and $\angle BCA = \angle XCD$ if and only if there exists a point Y on segment $\overline{AC}$ such that $\angle CBD = \angle YBA$ and $\angle CDB = \angle YDA$.

Solution 1

Note that lines AC, AX are isogonal in $\triangle ABD$, so an inversion centered at A with power $r^2 = AB \cdot AD$ composed with a reflection about the angle bisector of $\angle DAB$ swaps the pairs (D, B) and (C, X) . Thus,

$$\frac{AD}{XD} \cdot \frac{XD}{CD} = \frac{AC}{BC} \cdot \frac{AB}{CA} \implies (A, C; B, D) = -1$$

so that $ACBD$ is a harmonic quadrilateral. By symmetry, if Y exists, then $(B, D; A, C) = -1$. We have shown the two conditions are equivalent, whence both directions follow. ■

Solution 2

All angles are directed. Note that lines AC, AX are isogonal in $\triangle ABD$ and CD, CE are isogonal in $\triangle CDB$. From the law of sines it follows that

$$\frac{DX}{XB} \cdot \frac{DE}{ED} = \left(\frac{AD}{DB}\right)^2 = \left(\frac{DC}{BC}\right)^2.$$

Therefore, the ratio equals $\frac{AD \cdot DC}{DB \cdot BC}$.

Now let Y be a point of AC such that $\angle ABE = \angle CBY$. We apply the above identities for Y to get that $\frac{CY}{YA} \cdot \frac{CE}{EA} = \left(\frac{CD}{DA}\right)^2$. So $\angle CDY = \angle EDA$, the converse follows since all our steps are reversible.

Beware that directed angles, or angles mod 180, are not standard olympiad material. If you use them, provide a definition.

6. (USAJMO 2015) Steve is piling $m \geq 1$ indistinguishable stones on the squares of an $n \times n$ grid. Each square can have an arbitrarily high pile of stones. After he finished piling his stones in some manner, he can then perform stone moves, defined as follows. Consider any four grid squares, which are corners of a rectangle, i.e. in positions $(i, k), (i, l), (j, k), (j, l)$ for some $1 \leq i, j, k, l \leq n$, such that $i < j$ and $k < l$. A stone move consists of either removing one stone from each of (i, k) and (j, l) and moving them to (i, l) and (j, k) respectively, or removing one stone from each of (i, l) and (j, k) and moving them to (i, k) and (j, l) respectively.

Two ways of piling the stones are equivalent if they can be obtained from one another by a sequence of stone moves.

How many different non-equivalent ways can Steve pile the stones on the grid?

Solution

Let the number of stones in row i be r_i and let the number of stones in column i be c_i . Since there are m stones, we must have $\sum_{i=1}^n r_i = \sum_{i=1}^n c_i = m$

Lemma 1: If any 2 pilings are equivalent, then r_i and c_i are the same in both pilings $\forall i$.

Proof: We suppose the contrary. Note that r_i and c_i remain invariant after each move, therefore, if any of the r_i or c_i are different, they will remain different.

Lemma 2: Any 2 pilings with the same r_i and $c_i \forall i$ are equivalent.

Proof: Suppose piling 1 and piling 2 not the same piling. Call a stone in piling 1 wrong if the stone occupies a position such that there are more stones in that position in piling 1 than piling 2. Similarly define a wrong stone in piling 2. Let a wrong stone be at (a, b) in piling 1. Since c_b is the same for both pilings, we must have a wrong stone in piling 2 at column b , say at (c, b) , such that $c \neq a$. Similarly, we must have a wrong stone in piling 1 at row c , say at (c, d) where $d \neq b$. Clearly, making the move $(a, b); (c, d) \implies (c, b); (a, d)$ in piling 1 decreases the number of wrong stones in piling 1. Therefore, the number of wrong stones in piling 1 must eventually be 0 after a sequence of moves, so piling 1 and piling 2 are equivalent.

Lemma 3: Given the sequences g_i and h_i such that $\sum_{i=1}^n g_i = \sum_{i=1}^n h_i = m$ and $g_i, h_i \geq 0 \forall i$, there is always a piling that satisfies $r_i = g_i$ and $c_i = h_i \forall i$.

Proof: We take the lowest i, j , such that $g_i, h_j > 0$ and place a stone at (i, j) , then we subtract g_i and h_j by 1 each, until g_i and h_i become 0 $\forall i$, which will happen when m stones are placed, because $\sum_{i=1}^n g_i$ and $\sum_{i=1}^n h_i$ are both initially m and decrease by 1 after each stone is placed. Note that in this process $r_i + g_i$ and $c_i + h_i$ remains invariant, thus, the final piling satisfies the conditions above.

By the above lemmas, the number of ways to pile is simply the number of ways to choose the sequences r_i and c_i such that $\sum_{i=1}^n r_i = \sum_{i=1}^n c_i = m$ and $r_i, c_i \geq 0 \forall i$. By stars and bars, the number of ways is $\binom{n+m-1}{m}^2$.

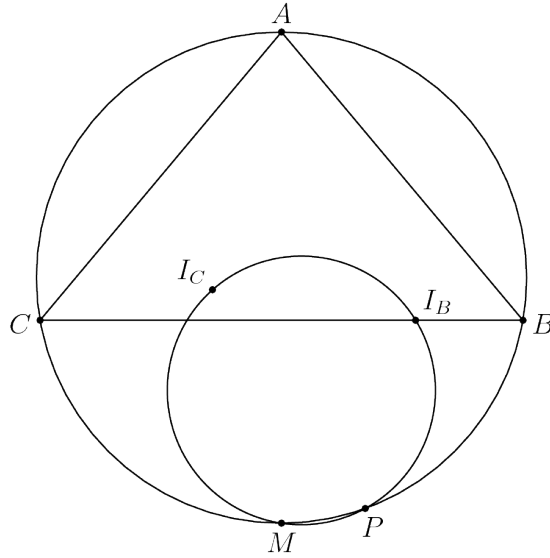
Solution by Shaddoll

§7 USAJMO 2016

1. (*USAJMO 2016*) The isosceles triangle $\triangle ABC$, with $AB = AC$, is inscribed in the circle ω . Let P be a variable point on the arc $\widehat{BC}$ that does not contain A , and let I_B and I_C denote the incenters of triangles $\triangle ABP$ and $\triangle ACP$, respectively.

Prove that as P varies, the circumcircle of triangle $\triangle PI_B I_C$ passes through a fixed point.

Solution 1



We claim that M (midpoint of arc BC) is the fixed point. We would like to show that M, P, I_B, I_C are cyclic.

We extend PI_B to intersect ω again at R . We extend PI_C to intersect ω again at S . We invert around a circle centered at P with radius 1 (for convenience). (I will denote X' as the reflection of X for all the points) The problem then becomes: Prove I'_B, I'_C , and M' are collinear.

Now we look at triangle $\triangle PR'S'$. We apply Menelaus (the version where all three points lie outside the triangle). It suffices to show that

$$\frac{PI'_B}{I'_B R'} \cdot \frac{R' M'}{M' S'} \cdot \frac{S' I'_C}{I'_C P} = 1$$

By inversion, we know $PX' = \frac{1}{PX}$ for any point X and $X'Y' = \frac{XY}{PX \cdot PY}$ for any points X and Y .

Plugging this into our Menelaus equation we obtain that it suffices to show

$$\frac{\frac{1}{PI_B}}{\frac{RI_B}{PI_B \cdot PR}} \cdot \frac{\frac{RM}{PR \cdot PM}}{\frac{SM}{PS \cdot PM}} \cdot \frac{\frac{SI_C}{PI_C \cdot PS}}{\frac{1}{PI_C}} = 1$$

We cancel out the like terms and rewrite. It suffices to show

$$\frac{RM \cdot SI_C}{SM \cdot RI_B} = 1$$

We know that AM is the diameter of ω because $\triangle ABC$ is isosceles and AM is the angle bisector. We also know $\angle RMA = \frac{\angle ACB}{2} = \frac{\angle ABC}{2} = \angle SMA$ so R and S are symmetric with respect to AM so $RM = SM$.

Thus, it suffices to show $\frac{SI_C}{RI_B} = 1$. This is obvious because $RI_B = RA = SA = SI_C$. Therefore we are done. ■

Solution 2

We will use complex numbers as mentioned [here](#). Set the circumcircle of $\triangle ABC$ to be the unit circle. Let

$$A = 1, B = w^2, C = \frac{1}{w^2}, P = p^2,$$

such that

$$I_B = w + wp - p, I_C = p - \frac{p}{w} + \frac{1}{w}.$$

We claim that the circumcircle of $\triangle PI_B I_C$ passes through $M = -1$. This is true if

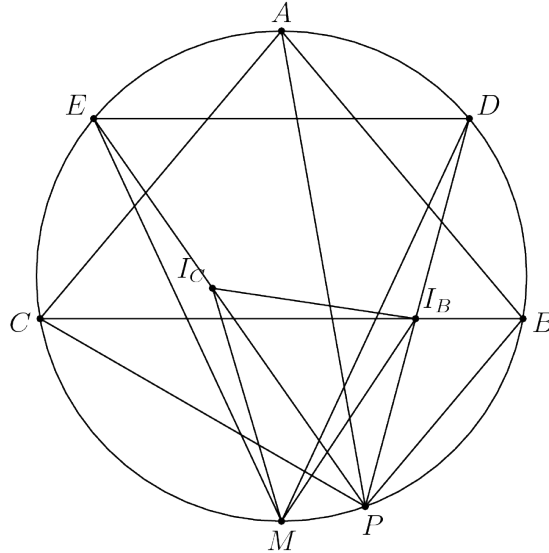
$$k = \frac{(I_B - M)(I_C - P)}{(I_B - P)(I_C - M)} = \frac{(w + wp - p + 1)(p + \frac{1}{w} - \frac{p}{w} - p^2)}{(w + wp - p - p^2)(p + \frac{1}{w} - \frac{p}{w} + 1)} = \frac{(wp + 1)(1 - p)}{(p + 1)(w - p)}$$

is real. Now observe that

$$\bar{k} = \frac{(1 + \frac{1}{wp})(1 - \frac{1}{p})}{(\frac{1}{p} + 1)(\frac{1}{w} - \frac{1}{p})} = k,$$

so k is real and we are done. ■

Solution 3



Let M be the midpoint of arc BC . Let D be the midpoint of arc AB . Let E be the midpoint of arc AC . Then, P, I_B , and D are collinear and P, I_C , and E are collinear.

We'll prove $MPI_B I_C$ is cyclic. (Intuition: we'll show that M is the Miquel's point of quadrilateral $DEI_C I_B$).

D is the center of the circle $AI_B B$ (the P -excenter of PAB is also on the same circle). Therefore $DI_B = DB$. Similarly $EI_C = EC$. Since $AB = AC$, $DB = EC$. Therefore $DI_B = EI_C$. Obviously $ME = MD$ and $\angle MEI_C = \angle MEP = \angle MDP = \angle MDI_B$. Thus by SAS, $\triangle MEI_C \cong \triangle MDI_B$.

Hence $\angle I_B M I_C = \angle DME = \angle DPE = \angle I_B P I_C$, so $MPI_B I_C$ is cyclic and we are done.

2. (*USAJMO 2016*) Prove that there exists a positive integer $n < 10^6$ such that 5^n has six consecutive zeros in its decimal representation.

Solution

Let digit 1 of a number be the units digit, digit 2 be the tens digit, and so on. Let the 6 consecutive zeroes be at digits $k - 5$ through digit k . The criterion is then obviously equivalent to

$$5^n \bmod 10^k < 10^{k-6}$$

We will prove that $n = 20 + 2^{19}, k = 20$ satisfies this, thus proving the problem statement (since $n = 20 + 2^{19} < 10^6$).

We want

$$5^{20+2^{19}} \pmod{10^{20}}$$

We can split this into its prime factors. Obviously, it is a multiple of 5^{20} , so we only need to consider it $\bmod 2^{20}$.

$$5^{20} \cdot 5^{2^{19}} \pmod{2^{20}}$$

$$5^{20} \cdot 5^{\varphi(2^{20})} \pmod{2^{20}}$$

(φ is the Euler Totient Function.) By Euler's Theorem, since $(5, 2^{20}) = 1$,

$$5^{\varphi(2^{20})} \equiv 1 \pmod{2^{20}}$$

so

$$5^{20} \cdot 5^{\varphi(2^{20})} \equiv 5^{20} \pmod{2^{20}}$$

Since $2^{20} > 10^6, 5^{20} < 10^{14} = 10^{20-6}$,

$$5^n \bmod 10^k < 10^{k-6}$$

for $n = 20 + 2^{19}$ and $k = 20$, and thus the problem statement has been proven. ■

Motivation for Solution Modifying our necessary and sufficient inequality, we get:

$$5^n \bmod 10^k < 10^{k-6}$$

$$5^k \left(5^{n-k} \bmod 2^k \right) < 10^{k-6}$$

$$5^{n-k} \bmod 2^k < \frac{2^{k-6}}{5^6}$$

Since $\gcd(5^{n-k}, 2^k) = 1$ if $k \neq 0$ (which is obviously true) and $n \neq k$ which is also true given that $5^k < 10^k$, we need the RHS to be greater than 1:

$$\frac{2^{k-6}}{5^6} > 1$$

$$2^{k-6} > 5^6$$

$$2^k > 10^6$$

The first k that satisfies this inequality is $k = 20$, so we let $k = 20$:

$$5^{n-20} \pmod{2^{20}} < \frac{2^{14}}{5^6}$$

$$5^{n-20} \pmod{2^{20}} \leq 1$$

$$5^{n-20} \pmod{2^{20}} = 1$$

From this, Euler's Theorem comes to mind and we see that if $n - 20 = \varphi(2^{20})$, the equality is satisfied. Thus, we get that $n = 20 + 2^{19}$, which is less than 10^6 , and we should be done. However, this requires slightly more formalization, and can be proven directly more easily if $n = 20 + 2^{19}$ is known or suspected.

3. (*USAJMO 2016*) Let $X_1, X_2, \dots, X_{100}$ be a sequence of mutually distinct nonempty subsets of a set S . Any two sets X_i and X_{i+1} are disjoint and their union is not the whole set S , that is, $X_i \cap X_{i+1} = \emptyset$ and $X_i \cup X_{i+1} \neq S$, for all $i \in \{1, \dots, 99\}$. Find the smallest possible number of elements in S .

Solution 1

The answer is that $|S| \geq 8$.

First, we provide a inductive construction for $S = \{1, \dots, 8\}$. Actually, for $n \geq 4$ we will provide a construction for $S = \{1, \dots, n\}$ which has $2^{n-1} + 1$ elements in a line. (This is sufficient, since we then get 129 for $n = 8$.) The idea is to start with the following construction for $|S| = 4$:

$$34 \quad 1 \quad 23 \quad 4 \quad 12 \quad 3 \quad 14 \quad 2 \quad 13 .$$

Then inductively, we do the following procedure to move from n to $n + 1$: take the chain for n elements, delete an element, and make two copies of the chain (which now has even length). Glue the two copies together, joined by $\emptyset$ in between. Then place the element $n + 1$ in alternating positions starting with the first (in particular, this hits $n + 1$). For example, the first iteration of this construction gives:

$$\begin{array}{cccccccc} 345 & 1 & 235 & 4 & 125 & 3 & 145 & 2 & 5 \\ 34 & 15 & 23 & 45 & 12 & 35 & 14 & 25 & \end{array}$$

Now let's check $|S| \geq 8$ is sufficient. Consider a chain on a set of size $|S| = 7$. (We need $|S| \geq 7$ else $2^{|S|} < 100$.) Observe that there are sets of size ≥ 4 can only be neighbored by sets of size ≤ 2 , of which there are $\binom{7}{1} + \binom{7}{2} = 28$. So there are ≤ 30

sets of size ≥ 4 . Also, there are $\binom{7}{3} = 35$ sets of size 3. So the total number of sets in a chain can be at most $30 + 28 + 35 = 93 < 100$.

Solution 2

My proof that $|S| \geq 8$ is basically the same as the one above. Here is another construction for $|S| = 8$ that I like because it works with remainders and it's pretty intuitive. The basic idea is to assign different subsets to different remainders when divided by particular numbers, and then to use the Chinese Remainder Theorem to show that all of the subsets are distinct. The motivation for this comes from the fact that we want X_i and X_{i+1} to always be disjoint, so remainders are a great way to systematically make that happen, since i and $i + 1$ do not have the same remainder modulo any positive integer greater than 1. Anyway, here is the construction:

Let $S = \{1, 2, \dots, 8\}$. For $i = 1, 2, \dots, 98$, we will choose which elements of the set $\{1, 2, 3, 4\}$ belong to X_i based on the remainder of i modulo 9, and we will choose which elements of the set $\{5, 6, 7, 8\}$ belong to X_i based on the remainder of i modulo 11. We do this as follows:

$$X_i \cap \{1, 2, 3, 4\} = \begin{array}{ll} \emptyset & i \equiv 0 \pmod{9} \\ \{1, 2\} & i \equiv 1 \pmod{9} \\ \{3\} & i \equiv 2 \pmod{9} \\ \{1, 4\} & i \equiv 3 \pmod{9} \\ \{2\} & i \equiv 4 \pmod{9} \\ \{3, 4\} & i \equiv 5 \pmod{9} \\ \{1\} & i \equiv 6 \pmod{9} \\ \{2, 3\} & i \equiv 7 \pmod{9} \\ \{4\} & i \equiv 8 \pmod{9} \end{array}$$

$$X_i \cap \{5, 6, 7, 8\} = \begin{array}{ll} \emptyset & i \equiv 0 \pmod{11} \\ \{5, 6\} & i \equiv 1 \pmod{11} \\ \{7, 8\} & i \equiv 2 \pmod{11} \\ \{5\} & i \equiv 3 \pmod{11} \\ \{6, 8\} & i \equiv 4 \pmod{11} \\ \{5, 7\} & i \equiv 5 \pmod{11} \\ \{6\} & i \equiv 6 \pmod{11} \\ \{5, 8\} & i \equiv 7 \pmod{11} \\ \{6, 7\} & i \equiv 8 \pmod{11} \\ \{8\} & i \equiv 9 \pmod{11} \\ \{7\} & i \equiv 10 \pmod{11} \end{array}$$

Finally, we specially define $X_{99} = \{1, 2, 3\}$ and $X_{100} = \{5, 6, 7\}$.

It is relatively easy to see that this configuration satisfies all of the desired conditions. We see that $X_{98} = \{4, 7\}$, so X_{98} and X_{99} are disjoint, as are X_{99} and X_{100} . The remainder configuration above takes care of the rest, so any two consecutive sets are disjoint. Then, by the Chinese Remainder Theorem, no two integers from 1 to 98 have the same combination of residues modulo 9 and modulo 11, so all of the sets X_i are distinct for $i = 1, 2, \dots, 98$. It is also easy to verify that none of these match X_{99} or X_{100} , since they all have at most two elements of $\{1, 2, 3, 4\}$ and at most two elements of $\{5, 6, 7, 8\}$, whereas X_{99} and X_{100} do not satisfy this; hence all of the sets are distinct. Finally, notice that, for any pair of consecutive sets, at least one of them has at most 3 elements, while the other has at most 4. Thus, their union always has at most 7 elements, so $X_i \cup X_{i+1} \neq S$ for all $i = 1, 2, \dots, 99$.

All of the conditions are satisfied, so this configuration works. We thus conclude that $\min(|S|) = 8$.

4. (*USAJMO 2016*) Find, with proof, the least integer N such that if any 2016 elements are removed from the set $\{1, 2, \dots, N\}$, one can still find 2016 distinct numbers among the remaining elements with sum N .

Solution

Since any 2016 elements are removed, suppose we remove the integers from 1 to 2016. Then the smallest possible sum of 2016 of the remaining elements is

$$2017 + 2018 + \dots + 4032 = 1008 \cdot 6049 = 6097392$$

so clearly $N \geq 6097392$. We will show that $N = 6097392$ works.

$\{1, 2, \dots, 6097392\}$ contain the integers from 1 to 6048, so pair these numbers as follows:

$$1, 6048$$

$$2, 6047$$

$$3, 6046$$

...

$$3024, 3025$$

When we remove any 2016 integers from the set $\{1, 2, \dots, N\}$, clearly we can remove numbers from at most 2016 of the 3024 pairs above, leaving at least 1008 complete pairs. To get a sum of N , simply take these 1008 pairs, all of which sum to 6049. The sum of these 2016 elements is $1008 \cdot 6049 = 6097392$, as desired.

We have shown that N must be at least 6097392, and that this value is attainable. Therefore our answer is $\boxed{6097392}$.

5. (*USAJMO 2016*) Let $\triangle ABC$ be an acute triangle, with O as its circumcenter. Point H is the foot of the perpendicular from A to line $\overleftrightarrow{BC}$, and points P and Q are the feet of the perpendiculars from H to the lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{AC}$, respectively.

Given that

$$AH^2 = 2 \cdot AO^2,$$

prove that the points O , P , and Q are collinear.

Solution 1

6. (*USAJMO 2016*) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all real numbers x and y ,

$$(f(x) + xy) \cdot f(x - 3y) + (f(y) + xy) \cdot f(3x - y) = (f(x + y))^2.$$

Step 1: Set $x = y = 0$ to obtain $f(0) = 0$.

Step 2: Set $x = 0$ to obtain $f(y)f(-y) = f(y)^2$.

In particular, if $f(y) \neq 0$ then $f(y) = f(-y)$.

In addition, replacing $y \rightarrow -t$, it follows that $f(t) = 0 \implies f(-t) = 0$ for all $t \in \mathbb{R}$.

Step 3: Set $x = 3y$ to obtain $[f(y) + 3y^2] f(8y) = f(4y)^2$.

In particular, replacing $y \rightarrow t/8$, it follows that $f(t) = 0 \implies f(t/2) = 0$ for all $t \in \mathbb{R}$.

Step 4: Set $y = -x$ to obtain $f(4x) [f(x) + f(-x) - 2x^2] = 0$.

In particular, if $f(x) \neq 0$, then $f(4x) \neq 0$ by the observation from Step 3, because $f(4x) = 0 \implies f(2x) = 0 \implies f(x) = 0$. Hence, the above equation implies that $2x^2 = f(x) + f(-x) = 2f(x)$, where the last step follows from the first observation from Step 2.

Therefore, either $f(x) = 0$ or $f(x) = x^2$ for each x .

Looking back on the equation from Step 3, it follows that $f(y) + 3y^2 \neq 0$ for any nonzero y . Therefore, replacing $y \rightarrow t/4$ in this equation, it follows that $f(t) = 0 \implies f(2t) = 0$.

Step 5: If $f(a) = f(b) = 0$, then $f(b - a) = 0$.

This follows by choosing x, y such that $x - 3y = a$ and $3x - y = b$. Then $x + y = \frac{b-a}{2}$, so plugging x, y into the given equation, we deduce that $f\left(\frac{b-a}{2}\right) = 0$. Therefore, by the third observation from Step 4, we obtain $f(b - a) = 0$, as desired.

Step 6: If $f \not\equiv 0$, then $f(t) = 0 \implies t = 0$.

Suppose by way of contradiction that there exists a nonzero t with $f(t) = 0$. Choose x, y such that $f(x) \neq 0$ and $x + y = t$. The following three facts are crucial:

1. $f(y) \neq 0$. This is because $(x + y) - y = x$, so by Step 5, $f(y) = 0 \implies f(x) = 0$, impossible.
2. $f(x - 3y) \neq 0$. This is because $(x + y) - (x - 3y) = 4y$, so by Step 5 and the observation from Step 3, $f(x - 3y) = 0 \implies f(4y) = 0 \implies f(2y) = 0 \implies f(y) = 0$, impossible.
3. $f(3x - y) \neq 0$. This is because by the second observation from Step 2, $f(3x - y) = 0 \implies f(y - 3x) = 0$. Then because $(x + y) - (y - 3x) = 4x$, Step 5 together with the observation from Step 3 yield $f(3x - y) = 0 \implies f(4x) = 0 \implies f(2x) = 0 \implies f(x) = 0$, impossible.

By the second observation from Step 4, these three facts imply that $f(y) = y^2$ and $f(x - 3y) = (x - 3y)^2$ and $f(3x - y) = (3x - y)^2$. By plugging into the given equation, it follows that

$$(x^2 + xy)(x - 3y)^2 + (y^2 + xy)(3x - y)^2 = 0.$$

But the above expression miraculously factors into $(x + y)^4$! This is clearly a contradiction, since $t = x + y \neq 0$ by assumption. This completes Step 6.

Step 7: By Step 6 and the second observation from Step 4, the only possible solutions are $f \equiv 0$ and $f(x) = x^2$ for all $x \in \mathbb{R}$. It's easy to check that both of these work, so we're done.

Alternative Solution 1 From steps 1 and 2 of Solution 1 we have that $f(0) = 0$, and $f(-y) \cdot f(y) = (f(y))^2$. Therefore, if $f(y) \neq 0$, then $f(y) = f(-y)$. Furthermore, setting $y = -x$ gives us $(f(x) - x^2) \cdot f(4x) + (f(-x) - x^2) \cdot f(4x) = f(0) = 0$. The LHS can be factored as $(f(x) + f(-x) - 2x^2) \cdot f(4x) = 0$. In particular, if $f(4x) \neq 0$, then we have $f(x) + f(-x) = 2x^2$. However, since we have from step 2 that $f(x) = f(-x)$, assuming $f(x) \neq 0$, the equation becomes $2f(x) = 2x^2$, so for every x , $f(x)$ is equivalent to either 0 or x^2 . From step 6 of Solution 1, we can prove that $f(x) = 0$, and $f(x) = x^2$ are the only possible solutions.

Solution 2

Step 1: $x = y = 0 \implies f(0) = 0$

Step 2: $x = 0 \implies f(y)f(-y) = f(y)^2$. Now, assume $y \neq 0$. Then, if $f(y) = 0$, we substitute in $-y$ to get $f(y)f(-y) = f(-y)^2$, or $f(y) = f(-y) = 0$. Otherwise, we divide both sides by $f(y)$ to get $f(y) = f(-y)$. If $y = 0$, we obviously have $f(0) = f(0)$. Thus, the function is even. Step 3: $y = -x \implies 2f(4x)(f(x) - x^2) = 0$. Thus, $\forall x$, we have $f(4x) = 0$ or $f(x) = x^2$.

Step 4: We now assume $f(x) \neq 0$, $x \neq 0$. We have $f(\frac{x}{4}) = \frac{x^2}{16}$. Now, setting $x = y = \frac{x}{4}$, we have $f(\frac{x}{2}) = \frac{x^2}{4}$ or $f(\frac{x}{2}) = 0$. The former implies that $f(x) = 0$ or x^2 . The latter implies that $f(x) = 0$ or $f(x) = \frac{x^2}{2}$. Assume the latter. $y = -2x \implies -\frac{3y^2}{2}f(7y) - 2y^2f(5y) = \frac{x^2}{2}$. Clearly, this implies that $f(x)$ is negative for some m . Now, we have $f(\frac{m}{4}) = \frac{m^2}{16} \implies f(\frac{m}{2}) = 0, \frac{m^2}{4} \implies f(m) \geq 0$, which is a contradiction. Thus, $\forall x f(x) = 0$ or $f(x) = x^2$.

Step 5: We now assume $f(x) = 0$, $f(y) = y^2$ for some $x, y \neq 0$. Let m be sufficiently large integer, let $z = |4^m x|$ and take the absolute value of y (since the function is even). Choose c such that $3z - c = y$. Note that we have $\frac{c}{z} \rightarrow 3$ and $\frac{y}{z} \rightarrow 0$. Note that

$f(z) = 0$. Now, $x = z, y = c \implies$ LHS is positive, as the second term is positive and the first term is nonnegative and thus the right term is equal to $(z + c)^4 - 256z^4$. Now if $f(z - 3c) = 0$, the second term of the LHS/RHS clearly $\rightarrow 0$ as $m \rightarrow \infty$. if $f(z - 3c) = 0$, then we have LHS/RHS $\rightarrow 0$, otherwise, we have LHS/RHS $\frac{8^2 \cdot 3z^4}{256z^4} = \frac{3}{4}$, a contradiction, as we're clearly not dividing by 0, and we should have LHS/RHS=1.

§8 USAJMO 2017

1. (*USAJMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Prove that there are infinitely many distinct pairs (a, b) of relatively prime positive integers $a > 1$ and $b > 1$ such that $a^b + b^a$ is divisible by $a + b$.

Solution 1

Let $a = 2n - 1$ and $b = 2n + 1$. We see that $(2n \pm 1)^2 = 4n^2 \pm 4n + 1 \equiv 1 \pmod{4n}$. Therefore, we have $(2n + 1)^{2n-1} + (2n - 1)^{2n+1} \equiv 2n + 1 + 2n - 1 = 4n \equiv 0 \pmod{4n}$, as desired.

(Credits to mathmaster2012)

Solution 2

Let x be odd where $x > 1$. We have $x^2 - 1 = (x - 1)(x + 1)$, so $x^2 - 1 \equiv 0 \pmod{2x + 2}$. This means that $x^{x+2} - x^x \equiv 0 \pmod{2x + 2}$, and since x is odd, $x^{x+2} + (-x)^x \equiv 0 \pmod{2x + 2}$, or $x^{x+2} + (x + 2)^x \equiv 0 \pmod{2x + 2}$, as desired.

Solution 3

Because problems such as this usually are related to expressions along the lines of $x \pm 1$, it's tempting to try these. After a few cases, we see that $(a, b) = (2x - 1, 2x + 1)$ is convenient due to the repeated occurrence of $4x$ when squared and added. We rewrite the given expressions as:

$$(2x - 1)^{2x+1} + (2x + 1)^{2x-1}, (2x - 1) + (2x + 1) = 4x.$$

After repeatedly factoring the initial equation, we can get:

$$(2x - 1)^2 (2x - 1)^2 \dots (2x - 1) + (2x + 1)^2 (2x + 1)^2 \dots (2x + 1).$$

Expanding each of the squares, we can compute each product independently then sum them:

$$(4x^2 - 4x + 1) (4x^2 - 4x + 1) \dots (2x - 1) \equiv (1) (1) \dots (2x - 1) \equiv 2x - 1 \pmod{4x},$$

$$(4x^2 + 4x + 1) (4x^2 + 4x + 1) \dots (2x + 1) \equiv (1) (1) \dots (2x + 1) \equiv 2x + 1 \pmod{4x}.$$

Now we place the values back into the expression:

$$(2x - 1)^{2x+1} + (2x + 1)^{2x-1} \equiv (2x - 1) + (2x + 1) \equiv 0 \pmod{4x}.$$

Plugging any positive integer value for x into $(a, b) = (2x - 1, 2x + 1)$ yields a valid solution, because there is an infinite number of positive integers, there is an infinite number of distinct pairs (a, b) . $\square$

-fatant

Solution 4

Let $a = 2x + 1$ and $b = 2x - 1$, where x leaves a remainder of 1 when divided by 4. We seek to show that $(2x + 1)^{2x-1} + (2x - 1)^{2x+1} \equiv 0 \pmod{4x}$ because that will show that there are infinitely many distinct pairs (a, b) of relatively prime integers $a > 1$ and $b > 1$ such that $a^b + b^a$ is divisible by $a + b$.

Claim 1: $(2x + 1)^{2x-1} + (2x - 1)^{2x+1} \equiv 0 \pmod{4}$. We have that the remainder when $2x + 1$ is divided by 4 is 3 and the remainder when $2x - 1$ is divided by 4 is always 1. Therefore, the remainder when $(2x + 1)^{2x-1} + (2x - 1)^{2x+1}$ is divided by 4 is always going to be $(-1)^{2x-1} + 1^{2x+1} = 0$.

Claim 2: $(2x + 1)^{2x-1} + (2x - 1)^{2x+1} \equiv 0 \pmod{x}$. We know that $(2x + 1) \pmod{x} \equiv 1$ and $(2x - 1) \pmod{x} \equiv -1$, so the remainder when $(2x + 1)^{2x-1} + (2x - 1)^{2x+1}$ is divided by x is always going to be $(-1)^{2x-1} + 1^{2x+1} = 0$.

Claim 3: $(2x + 1)^{2x-1} + (2x - 1)^{2x+1} \equiv 0 \pmod{4x}$. Trivial given claim 1, 2. $\square$

Solution 5

I claim $(a, b) = (2n - 1, 2n + 1)$, $n \in \mathbb{N}$, $n \geq 2$ always satisfies above conditions.

Note: We could have also substituted $2n$ with $2^k n$ or $4n, 8n, \dots$ any sequence of numbers such that they are all even. The proof will work the same.

Proof:

Since there are infinitely many integers larger than or equal to 2, there are infinitely many distinct pairs (a, b) .

We only need to prove:

$$a^b + b^a \equiv 0 \pmod{a + b}$$

We can expand $a^b + b^a = (2n - 1)^{2n+1} + (2n + 1)^{2n-1}$ using binomial theorem. However, since $a + b = 2n - 1 + 2n + 1 = 4n$, all the $2n$ terms (with more than 2 powers of) when evaluated modulo $4n$ equal to 0, and thus can be omitted. We are left with the terms: $(2n + 1)(2n)^1 - 1 + (2n - 1)(2n)^1 + 1 = 4n \cdot 2n$, which is divisible by $4n$.

$$(2n - 1)^{2n+1} + (2n + 1)^{2n-1} \equiv (2n + 1)(2n) - 1 + (2n - 1)(2n) + 1 = 4n \cdot 2n \equiv 0 \pmod{4n}$$

The proof is complete. $\square$

-AlexLikeMath

Solution 6 (Motivation for Solution)

Note that

$$a^b + b^a = a^b - a^a + a^a + b^a.$$

To get rid of the $a^a + b^a$ part $\pmod{a+b}$, we can use the sum of powers factorization. However, a must be odd for us to do this. If we assume that a is odd,

$$a^b - a^a + a^a + b^a \equiv a^b - a^a + (a+b)(\text{an integer}) \equiv a^b - a^a \equiv a^a (a^{b-a} - 1) \pmod{a+b}.$$

Because a and b are relatively prime, $a + b$ cannot divide a^a . Thus, we have to show that there exists an integer b such that for odd a ,

$$a^{b-a} \equiv 1 \pmod{a+b}.$$

Suppose that $a = 2n - 1$. To keep the powers small, we try $b = 2n + k$ for small values of k . We can find that $b = 2n$ does not work. $b = 2n + 1$ works though, as $a + b = 4n$ and

$$(2n - 1)^{2n+1-2n+1} \equiv (2n - 1)^2 \equiv 4n^2 - 4n + 1 \equiv 4n(n - 1) + 1 \equiv 1 \pmod{4n}.$$

Because a is odd, $b = a + 2$ is relatively prime to a . Thus,

$$(a, b) = (2n - 1, 2n + 1)$$

is a solution for positive $n \geq 2$. There are infinitely many possible values for n , so the proof is complete. ■

BS2012

2. (*USAJMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Consider the equation

$$(3x^3 + xy^2)(x^2y + 3y^3) = (x - y)^7.$$

- (a) Prove that there are infinitely many pairs (x, y) of positive integers satisfying the equation.
 (b) Describe all pairs (x, y) of positive integers satisfying the equation.

Solution 1

We have $(3x^3 + xy^2)(yx^2 + 3y^3) = (x - y)^7$, which can be expressed as $xy(3x^2 + y^2)(x^2 + 3y^2) = (x - y)^7$. At this point, we think of substitution. A substitution of form $a = x + y, b = x - y$ is slightly derailed by the leftover x and y terms, so instead, seeing the xy in front, we substitute $x = a + b, y = a - b$. This leaves us with:

$$\begin{aligned} (a^2 - b^2)(4a^2 + 4ab + 4b^2)(4a^2 - 4ab + 4b^2) &= 128b^7 \\ (a^2 - b^2)(a^2 + ab + b^2)(a^2 - ab + b^2) &= 8b^7 \\ a^6 - b^6 &= 8b^7 \end{aligned}$$

Rearranging, we have $b^6(8b + 1) = a^6$. To satisfy this equation in integers, $8b + 1$ must obviously be a 6th power, and further inspection shows that it must also be odd. Also, since it is a square and all odd squares are 1 mod 8, every odd sixth power gives a solution. Since the problem asks for positive integers, the pair $(a, b) = (0, 0)$ does not work. We go to the next highest odd 6th power, 3^6 or 729. In this case, $b = 91$, so the LHS is $91^6 \cdot 3^6 = 273^6$, so $a = 273$. Using the original substitution yields $(x, y) = (364, 182)$ as the first solution. We have shown part a by showing that there are an infinite number of positive integer solutions for (a, b) , which can then be manipulated into solutions for (x, y) .

To solve part (b), we look back at the original method of generating solutions. Define a_n and b_n to be the pair representing the n th solution.

Since the n odd number is $2n + 1$, $b_n = \frac{(2n+1)^6-1}{8}$. It follows that $a_n = (2n + 1)b_n = \frac{(2n+1)^7-(2n+1)}{8}$. From our original substitution, $(x, y) = \left(\frac{(2n+1)^7+(2n+1)^6-2n-2}{8}, \frac{(2n+1)^7-(2n+1)^6-2n}{8} \right)$.

Solution 2 (and motivation)

First, we shall prove a lemma:

LEMMA:

$$(3x^3 + xy^2)(x^2y + 3y^3) = \frac{(x+y)^6 - (x-y)^6}{4}$$

PROOF: Expanding and simplifying the right side, we find that

$$\begin{aligned} \frac{(x+y)^6 - (x-y)^6}{4} &= \frac{12x^5y + 40x^3y^3 + 12xy^5}{4} \\ &= 3x^5y + 10x^3y^3 + 3xy^5 \\ &= (3x^3 + xy^2)(x^2y + 3y^3) \end{aligned}$$

which proves our lemma.

Now, we have that

$$\frac{(x+y)^6 - (x-y)^6}{4} = (x-y)^7$$

Rearranging and getting rid of the denominator, we have that

$$(x+y)^6 = 4(x-y)^7 + (x-y)^6$$

Factoring, we have

$$(x+y)^6 = (x-y)^6(4(x-y) + 1)$$

Dividing both sides, we have

$$\left(\frac{x+y}{x-y} \right)^6 = 4x - 4y + 1$$

Now, since the LHS is the 6th power of a rational number, and the RHS is an integer, the RHS must be a perfect 6th power. Define $a = \frac{x+y}{x-y}$. By inspection, a must be a positive odd integer satisfying $a \geq 3$. We also have

$$a^6 = 4x - 4y + 1$$

Now, we can solve for x and y in terms of a : $x - y = \frac{a^6-1}{4}$ and $x + y = a(x - y) = \frac{a(a^6-1)}{4}$. Now we have:

$$(x, y) = \left(\frac{(a+1)(a^6-1)}{8}, \frac{(a-1)(a^6-1)}{8} \right)$$

and it is trivial to check that this parameterization works for all such a (to keep x and y integral), which implies part (a).

MOTIVATION FOR LEMMA: I expanded the LHS, noticed the coefficients were $(3, 10, 3)$, and immediately thought of binomial coefficients. Looking at Pascal's triangle, it was then easy to find and prove the lemma.

-sunfishho

Solution where

So named because it is a mix of solutions 1 and 2 but differs in other aspects. After fruitless searching, let $x + y = a$, $x - y = b$. Clearly $a, b > 0$. Then,

$$\begin{aligned}x &= \frac{a+b}{2} \\y &= \frac{a-b}{2} \\xy &= \frac{a^2-b^2}{4} \\x^2 &= \frac{a^2+2ab+b^2}{4} \\y^2 &= \frac{a^2-2ab+b^2}{4} \\x^2+y^2 &= \frac{a^2+b^2}{2}\end{aligned}$$

Change the LHS of the original equation to

$$\begin{aligned}&xy(3x^2+y^2)(x^2+3y^2) \\&= (a-b)(a+b)(a^2+ab+b^2)(a^2-ab+b^2) \cdot \frac{1}{4} \\&= (a^3-b^3)(a^3+b^3) \cdot \frac{1}{4} \\&= (a^6-b^6) \cdot \frac{1}{4},\end{aligned}$$

and change the RHS to b^7 . Therefore $\left(\frac{a}{b}\right)^6 = 4b + 1$. Let $n = \frac{a}{b}$, and note that n is an integer. Therefore $b = \frac{n^6-1}{4}$, $a = bn = n\left(\frac{n^6-1}{4}\right)$. Because $n^6 \equiv 1 \pmod{4}$, n is odd, and thus is greater than 1, since $b > 0$. Therefore, substituting for x and y , we get:

$$\begin{aligned}x &= \frac{(n^4+n^2+1)(n-1)(n+1)^2}{8} \\y &= \frac{(n+1)(n-1)^2(n^4+n^2+1)}{8}\end{aligned}$$

Simplified Evan Chen's Solution Let $x = da$, $y = db$, such that $\gcd(a, b) = 1$. It's obvious $x > y$, so $a > b$

By plugging it in to the original equation, and simplifying, we get

$$d = \frac{ab(a^2+3b^2)(3a^2+b^2)}{(a-b)^7}$$

Since d is an integer, we got:

$$(a-b)^7 \mid ab(a^2+3b^2)(3a^2+b^2) \quad (*)$$

1. Since $\gcd(a, b) = 1$, a and b can't both be even

2. If both a and b are odd. $a - b$ will be even. The left hand side (*) has at least 7 factors of 2. Evaluating the (*) RHS term by term, we get:

$$a \equiv 1 \pmod{2}$$

$$b \equiv 1 \pmod{2}$$

$$a^2 + 3b^2 \equiv 4 \pmod{8}$$

$$3a^2 + b^2 \equiv 4 \pmod{8}$$

the (*) RHS has only 4 factors of 2. Contradiction.

3. Thus, $a - b$ has to be odd. We want to prove $a - b = 1$

We know:

$$a \equiv b \pmod{a - b}$$

So

$$ab(a^2 + 3b^2)(3a^2 + b^2) \equiv 16a^6 \pmod{a - b}$$

We can assume if $a - b \neq 1$

Since $a - b$ is odd, $\gcd(16, a - b) = 1$

Since $\gcd(a, a - b) = 1$, $\gcd(a^6, a - b) = 1$

$$ab(a^2 + 3b^2)(3a^2 + b^2) \equiv 16a^6 \not\equiv 0 \pmod{a - b}$$

contradiction. So, $a - b = 1$.

If $a - b = 1$, obviously (*) will work, d will be an integer.

So (*) $\Leftrightarrow a - b = 1$. The proof is complete.

-Alexlikemath

Solution 4

Part a:

Let $a = 1 + \frac{1}{n}$, where n is a positive integer. We will show that there is precisely one solution (x, y) to the equation such that $x = ay$.

If $x = ay$, we have

$$\begin{aligned} (3a^3y^3 + ay^3)(a^2y^3 + 3y^3) &= ((a - 1)y)^7 \\ y^6(3a^5 + 10a^3 + a) &= (a - 1)^7y^7 \\ \frac{3a^5 + 10a^3 + a}{(a - 1)^7} &= y. \end{aligned}$$

The numerator is a multiple of $\frac{1}{n^5}$, so y is an integer multiple of n^2 . Thus, $x = \frac{n+1}{n} \cdot y$ is also an integer, and we conclude that this pair (x, y) satisfies the system of equations. Because this works for any positive integer n , we conclude that there are infinitely many solutions to the equation.

Part b:

We will now prove that these are the only possible solutions. Suppose the contrary, that there are solutions with a different ratio between x and y . As before, set $a = 1 + \frac{1}{n}$, but this time $n = \frac{p}{q}$ in simplest terms. Then,

$$\begin{aligned}
 y &= n^2(3(n+1)^5 + 10n^2(n+1)^3 + 3n^4(n+1)) \\
 &= \frac{p^2}{q^7}(16p^5 + 48p^4q + 60p^3q^2 + 40p^2q^3 + 15pq^4 + 3q^5).
 \end{aligned}$$

For this to be rational, q must divide the expression in brackets and thus must divide $16p^5$. However, p and q are relatively prime, so q must divide 16, and p is therefore odd.

Next, set $q = 2^k$, where k is a positive integer between 1 and 4, inclusive. We have

$$y = \frac{p^2}{2^{7k}}(2^4p^5 + 3 \cdot 2^{4+k}p^4 + 15 \cdot 2^{2+2k}p^3 + 5 \cdot 2^{3+3k}p^2 + 15 \cdot 2^{4k}p + 3 \cdot 2^5k).$$

The sum in the brackets must be divisible by 2^{7k} , which is a power of 2 greater than or equal to 2^7 . Let z be this sum. Then,

$$z \equiv 16 + 0 + v + 0 + w + 0 = 16 + v + w \pmod{32},$$

where v represents $15 \cdot 2^{2+2k}p^3$ and w represents $15 \cdot 2^{4k}p$. Therefore, we must have

$$15(2^{2+k}p^3 + 2^{4k}p) \equiv 16 \pmod{32}.$$

For $k > 1$, this is equal to 0 (mod 32), so we conclude that $k = 1$ (and thus $q = 2$). Then,

$$\begin{aligned}
 y &= \frac{p^2}{2^7}(16p^5 + 96p^4 + 240p^3 + 320p^2 + 240p + 96) \\
 &= \frac{p^2}{2^3}(p^5 + 6p^4 + 15p^3 + 20p^2 + 15p + 6).
 \end{aligned}$$

For this to be integral, the expression in brackets must be a multiple of 8. However, there are three terms that are odd in this expression, p^5 , $15p^3$, and $15p$. Thus, we have a contradiction, and we conclude that the only solutions (x, y) are of the form stated above. $\square$

3. (*USAJMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Let ABC be an equilateral triangle and let P be a point on its circumcircle. Let lines PA and BC intersect at D ; let lines PB and CA intersect at E ; and let lines PC and AB intersect at F . Prove that the area of triangle DEF is twice the area of triangle ABC .

Solution 1

Proceed via Barycentrics. Let P be $(x : y : z)$, so $D = (0 : y : z)$, $E = (x : 0 : z)$, $F = (x : y : 0)$. Normalizing, we want to prove

$$2 = \frac{[DEF]}{[ABC]} = \left(\frac{1}{y+z}\right) \left(\frac{1}{x+z}\right) \left(\frac{1}{x+y}\right) \begin{vmatrix} 0 & y & z \\ x & 0 & z \\ x & y & 0 \end{vmatrix}$$

The determinant is $2xyz$, but since P is on the circumcircle of $\triangle ABC$, one of the coordinates is negative. Thus, we should use the positive $-2xyz$ in our calculations. To finish, we note that $\triangle ABC$ being equilateral implies $xy + xz + yz = 0$, so

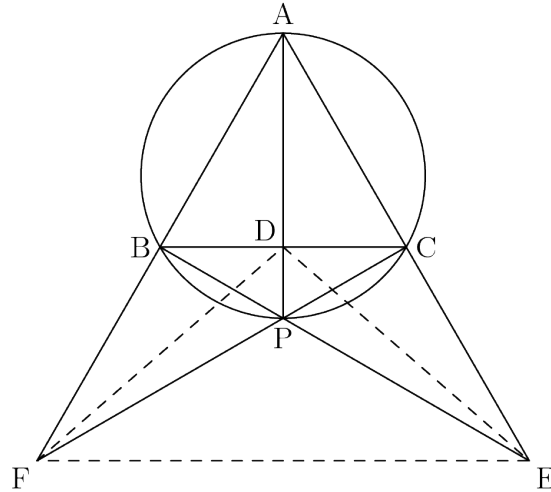
$$2(y+z)(x+z)(x+y) = -2xyz \iff x^2y + x^2z + y^2x + y^2z + z^2x + z^2y + 3xyz = 0$$

$$\iff (x+y+z)(xy+xz+yz) = 0$$

as needed $\square$

-sillybone

Solution 2(No Bash)



Extend DP to hit EF at K . Then note that $[DEF] \cdot \frac{AK}{DK} \cdot \frac{AB}{AF} \cdot \frac{AC}{AE} = [ABC]$. Letting $BF = x$ and $PF = y$, we have that $\frac{x+AB}{y} = \frac{y+PC}{x} = \frac{AC}{BP}$. Solving and simplifying using LoC on $\triangle BPC$ gives $\frac{AB}{AF} = \frac{PC}{PB+PC}$. Similarly, $\frac{AC}{AE} = \frac{PB}{PB+PC}$.

Now we find $\frac{AK}{DK}$. Note that $\frac{AD}{DP} = \frac{AD}{BD} \cdot \frac{BD}{DP} = \frac{AC}{PB} \cdot \frac{AB}{PC} = \frac{AB^2}{PB \cdot PC}$. Now let $E' = DE \cap AF$ and $F' = DF \cap AE$. Then by an area/concurrence theorem, we have that $\frac{DK}{AK} + \frac{DE'}{EE'} + \frac{DF'}{FF'} = 1$, or $\frac{DK}{AK} + (1 - \frac{DP}{AP} - \frac{DC}{BC}) + (1 - \frac{DP}{AP} - \frac{BD}{BC}) = 1$. Thus we have that $\frac{DK}{AK} = 2 \cdot \frac{DP}{AP}$.

Manipulating these gives $\frac{AK}{DK} = \frac{(PB+PC)^2}{2 \cdot PB \cdot PC}$. Thus $\frac{AK}{DK} \cdot \frac{AB}{AF} \cdot \frac{AC}{AE} = \frac{1}{2}$, and we are done.

Solution 3

WLOG, let $AB = 1$. Let $[ABD] = X$, $[ACD] = Y$, and $\angle BAD = \theta$. After some angle chasing, we find that $\angle BCF \cong \angle BEC \cong \theta$ and $\angle FBC \cong \angle BCE \cong 120^\circ$. Therefore, $\triangle FBC \sim \triangle BCE$.

Lemma 1: If $BF = k$, then $CE = \frac{1}{k}$. This lemma results directly from the fact that $\triangle FBC \sim \triangle BCE$; $\frac{BF}{BC} = \frac{BC}{CE} = \frac{1}{CE}$, or $CE = \frac{1}{BF}$.

Lemma 2: $[AEF] = (k + \frac{1}{k} + 2)(X + Y)$. We see that $[AEF] = (X + Y) \frac{[AEF]}{[ABC]} = (k + 1)(1 + \frac{1}{k})(X + Y) = (k + \frac{1}{k} + 2)(X + Y)$, as desired.

Lemma 3: $\frac{X}{Y} = k$. We see that

$$\frac{X}{Y} = \frac{\frac{1}{2}(AB)(AD)\sin(\theta)}{\frac{1}{2}(AC)(AD)\sin(60^\circ - \theta)} = \frac{\sin(\theta)}{\sin(60^\circ - \theta)}.$$

However, after some angle chasing and by the Law of Sines in $\triangle BCF$, we have $\frac{k}{\sin(\theta)} = \frac{1}{\sin(60^\circ - \theta)}$, or $k = \frac{\sin(\theta)}{\sin(60^\circ - \theta)}$, which implies the result.

By the area lemma, we have $[BDF] = kX$ and $[CDF] = \frac{Y}{k}$.

We see that $[DEF] = [AEF] - [ABC] - [BDF] - [DCE] = Xk + Yk + \frac{X}{k} + \frac{Y}{k} + 2X + 2Y - X - Y - Xk - \frac{Y}{k} = X + Y + \frac{X}{k} + Yk$. Thus, it suffices to show that $X + Y + \frac{X}{k} + Yk = 2X + 2Y$, or $\frac{X}{k} + Yk = X + Y$. Rearranging, we find this to be equivalent to $\frac{X}{Y} = k$, which is Lemma 3, so the result has been proven.

Solution 4

We will use barycentric coordinates and vectors. Let $\vec{X}$ be the position vector of a point X . The point (x, y, z) in barycentric coordinates denotes the point $x\vec{A} + y\vec{B} + z\vec{C}$. For all points in the plane of $\triangle ABC$, we have $x + y + z = 1$. It is clear that $A = (1, 0, 0)$; $B = (0, 1, 0)$; and $C = (0, 0, 1)$.

Define the point P as $P = (x_P, y_P, z_P)$. The fact that P lies on the circumcircle of $\triangle ABC$ gives us $x_P^2 + y_P^2 + z_P^2 = 1$. This, along with the condition $x_P + y_P + z_P = 1$ inherent to barycentric coordinates, gives us $x_P y_P + y_P z_P + z_P x_P = 0$.

We can write the equations of the following lines:

$$BC : x = 0$$

$$CA : y = 0$$

$$AB : z = 0$$

$$PA : \frac{y}{y_P} = \frac{z}{z_P}$$

$$PB : \frac{x}{x_P} = \frac{z}{z_P}$$

$$PC : \frac{x}{x_P} = \frac{y}{y_P}.$$

We can then solve for the points D, E, F :

$$D = \left(0, \frac{y_P}{y_P + z_P}, \frac{z_P}{y_P + z_P}\right)$$

$$E = \left(\frac{x_P}{x_P + z_P}, 0, \frac{z_P}{x_P + z_P}\right)$$

$$F = \left(\frac{x_P}{x_P + y_P}, \frac{y_P}{x_P + y_P}, 0\right).$$

The area of an arbitrary triangle XYZ is:

$$[XYZ] = \frac{1}{2} |\vec{XY} \times \vec{XZ}|$$

$$[XYZ] = \frac{1}{2} |(\vec{X} \times \vec{Y}) + (\vec{Y} \times \vec{Z}) + (\vec{Z} \times \vec{X})|.$$

To calculate $[DEF]$, we wish to compute $(\vec{D} \times \vec{E}) + (\vec{E} \times \vec{F}) + (\vec{F} \times \vec{D})$. After a lot of computation, we obtain the following:

$$(\vec{D} \times \vec{E}) + (\vec{E} \times \vec{F}) + (\vec{F} \times \vec{D}) = \frac{2x_P y_P z_P}{(x_P + y_P)(y_P + z_P)(z_P + x_P)} [(\vec{A} \times \vec{B}) + (\vec{B} \times \vec{C}) + (\vec{C} \times \vec{A})].$$

Evaluating the denominator,

$$(x_P + y_P)(y_P + z_P)(z_P + x_P) = (1 - z_P)(1 - y_P)(1 - x_P)$$

$$(x_P + y_P)(y_P + z_P)(z_P + x_P) = 1 - (x_P + y_P + z_P) + (x_P y_P + y_P z_P + z_P x_P) - x_P y_P z_P.$$

Since $x_P + y_P + z_P = 1$ and $x_P y_P + y_P z_P + z_P x_P = 0$, it follows that:

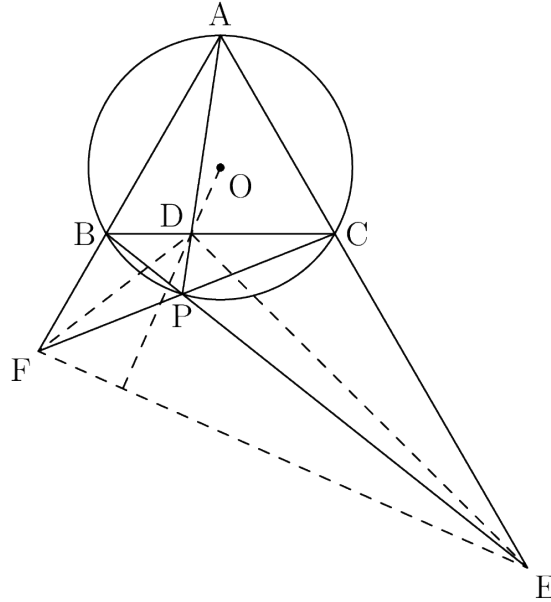
$$(x_P + y_P)(y_P + z_P)(z_P + x_P) = -x_P y_P z_P.$$

We thus conclude that:

$$(\vec{D} \times \vec{E}) + (\vec{E} \times \vec{F}) + (\vec{F} \times \vec{D}) = -2[(\vec{A} \times \vec{B}) + (\vec{B} \times \vec{C}) + (\vec{C} \times \vec{A})].$$

From this, it follows that $[DEF] = 2[ABC]$, and we are done.

Solution 5



We'll use coordinates and shoelace. Let the origin be the midpoint of BC . Let $AB = 2$, and $BF = 2x$, then $F = (-x - 1, -x\sqrt{3})$. Using the facts $\triangle CBP \sim \triangle CFB$ and $\triangle BCP \sim \triangle BEC$, we have $BF * CE = BC^2$, so $CE = \frac{1}{2x}$, and $E = (\frac{1}{x} + 1, -\frac{\sqrt{3}}{x})$.

The slope of FE is

$$k = \frac{-\frac{\sqrt{3}}{x} + x\sqrt{3}}{2 + \frac{1}{x} + x}$$

It is well-known that $\triangle DFE$ is self-polar, so FE is the polar of D , i.e., OD is perpendicular to FE . Therefore, the slope of OD is $-\frac{1}{k}$. Since $O = (0, \frac{1}{\sqrt{3}})$, we get the x-coordinate of D , $x_D = \frac{k}{\sqrt{3}}$, i.e., $D = (\frac{k}{\sqrt{3}}, 0)$. Using shoelace,

$$\begin{aligned} 2[\triangle FDE] &= \frac{k}{\sqrt{3}}(-x\sqrt{3}) + (-x - 1)(-\frac{\sqrt{3}}{x}) - (-x\sqrt{3})(\frac{1}{x} + 1) - (-\frac{\sqrt{3}}{x})\frac{k}{\sqrt{3}} \\ &= 2\sqrt{3} + \sqrt{3}(\frac{1}{x} + x) + k(\frac{1}{x} - x) \end{aligned}$$

$$\begin{aligned}
 &= 2\sqrt{3} + \sqrt{3} \left(\frac{2\left(\frac{1}{x} + x\right) + \left(\frac{1}{x} + x\right)^2 - \left(\frac{1}{x} - x\right)^2}{2 + \frac{1}{x} + x} \right) \\
 &= 2\sqrt{3} + \sqrt{3} \left(\frac{2\left(\frac{1}{x} + x\right) + 4}{2 + \frac{1}{x} + x} \right) \\
 &= 4\sqrt{3}
 \end{aligned}$$

So $[\triangle FDE] = 2\sqrt{3} = 2[\triangle ABC]$. Q.E.D

By Mathdummy.

Solution 6 Without the nasty computations

Note that $\angle APB = \angle FPB = \angle EPC = \angle APC = 60^\circ$. We will use a special version of Stewart's theorem for angle bisectors in triangle with an 120° angle to calculate various side lengths.

Let $BP = x$ and $CP = y$. Then, From Law of Cosine, $BC^2 = x^2 + y^2 + xy$.

From Ptolemy's theorem, $APBC = xAC + yAB$, so $AP = x + y$.

Lemma 1: In Triangle ABC with side lengths a, b, c and $\angle A = 120^\circ$, the length of the angle bisector of A is

$$d = \frac{bc}{b + c}$$

This can be easily proved with Stewart's and Law of Cosine.

Using Lemma 1, we have

$$\begin{aligned}
 PD &= \frac{xy}{x + y} \\
 x &= \frac{FPAP}{FP + AP} \\
 y &= \frac{EPAP}{EP + AP}
 \end{aligned}$$

Plug in $AP = x + y$, we get:

$$\begin{aligned}
 PD &= \frac{xy}{x + y} \\
 FP &= \frac{x(x + y)}{y} \\
 EP &= \frac{y(x + y)}{x}
 \end{aligned}$$

Then

$$\begin{aligned}
 [\triangle FDE] &= \frac{1}{2} \sin(120^\circ) (PDFP + FPPE + EPPD) \\
 &= \frac{\sqrt{3}}{4} (x^2 + (x + y)^2 + y^2) \\
 &= \frac{\sqrt{3}}{2} (x^2 + xy + y^2) \\
 &= \frac{\sqrt{3}}{2} BC^2 \\
 &= 2[\triangle ABC]
 \end{aligned}$$

By Mathdummy.

4. (*USAJMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Are there any triples (a, b, c) of positive integers such that $(a-2)(b-2)(c-2) + 12$ is a prime that properly divides the positive number $a^2 + b^2 + c^2 + abc - 2017$?

Solution 1

The answer is no. Let $(x, y, z) = (a-2, b-2, c-2)$. Then,

$$\begin{aligned} \mathbb{Z}^+ \ni \frac{a^2 + b^2 + c^2 + abc - 2017}{(a-2)(b-2)(c-2) + 12} &= \frac{(x-2)^2 + (y-2)^2 + (z-2)^2 + (x-2)(y-2)(z-2) - 2017}{xyz + 12} \\ &= \frac{(x+y+z-41)(x+y+z+49)}{xyz + 12} + 1 \end{aligned}$$

It follows that $x + y + z > 41$. Also, since $xyz + 12$ is prime, this implies

$$xyz + 12 \mid x + y + z - 41 \quad (1) \quad \text{or} \quad xyz + 12 \mid x + y + z + 49 \quad (2)$$

Now, we do casework based on the number -1 's in (x, y, z) .

Case 1: No -1 's.

Then, note that two of (x, y, z) must be 1, since otherwise $x + y + z > 41$ implies $xyz \geq 2(x + y + z) > x + y + z + 37$.

Let $(x, y, z) = (t, 1, 1)$. Clearly, (1) can't be true. Then, we have $t + 12 \mid 39$. But, $t + 2 > 41$, a contradiction.

Case 2: 2 -1 's

Then, let $(x, y, z) = (t, -1, -1)$. Again, (1) can't be true, so $t + 12 \mid t + 47$. But, as with before, $t + 2 > 41$, a contradiction.

-sillybone

Solution 2

The answer is no. Substitute $x = a - 2, y = b - 2, z = c - 2$. This means that $x, y, z \geq -1$. Then

$$a^2 + b^2 + c^2 + abc - 2017 = (x + y + z - 41)(x + y + z + 49) + xyz + 12.$$

It is given in the problem that this is positive. Now, suppose for the sake of contradiction that $xyz + 12$ is a prime. Clearly $x, y, z \neq 0$. Then we have

$$\frac{(x + y + z - 41)(x + y + z + 49)}{xyz + 12}$$

is an integer greater than or equal to 1. This also implies that $x + y + z > 41$. Since $xyz + 12$ is prime, we must have

$$xyz + 12 \mid x + y + z - 41 \quad \text{or} \quad xyz + 12 \mid x + y + z + 49.$$

Additionally, x, y, z must be odd, so that $xyz + 12$ is odd while $x + y + z - 41, x + y + z + 49$ are even. So, if

$$xyz + 12 \mid x + y + z - 41 \quad \text{or} \quad xyz + 12 \mid x + y + z + 49,$$

we must have

$$2(xyz + 12) \leq x + y + z - 41 \text{ or } 2(xyz + 12) \leq x + y + z + 49.$$

Now suppose WLOG that $x = -1$ and $y, z > 0$. Then we must have $yz \leq 10$, impossible since $x + y + z > 41$. Again, suppose that $x, y = -1$ and $z > 0$. Then we must have

$$2(z + 12) \leq z - 43 \text{ or } 2(z + 12) \leq z + 47,$$

and since in this case we must have $z > 43$, this is also impossible. Then the final case is when x, y, z are positive odd numbers. Note that if $xyz > x + y + z$ for positive integers x, y, z , then $abc > a + b + c$ for positive integers a, b, c where $a > x, b > y, c > z$. Then we only need to prove the case where $x + y + z = 43$, since $x + y + z$ is odd. Then one of

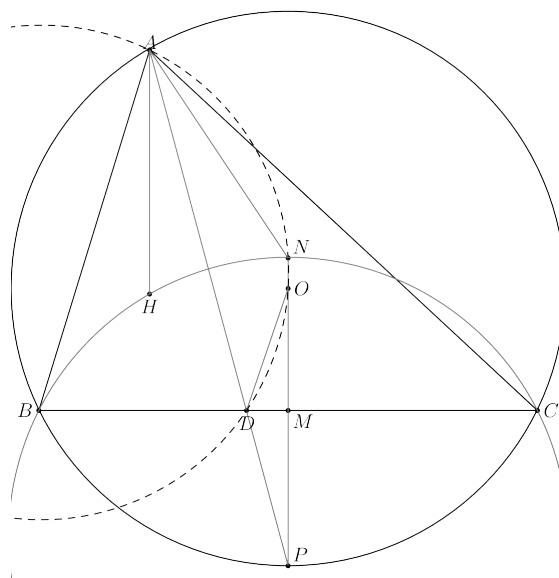
$$2(xyz + 12) \leq 2 \text{ and/or } 2(xyz + 12) \leq 92$$

is true, implying that $xyz \leq -11$ or $xyz \leq 34$. But if $x + y + z = 43$, then xyz is minimized when $x = 1, y = 1, z = 41$, so that $xyz \geq 41$. This is a contradiction, so we are done.

5. (*USAJMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Let O and H be the circumcenter and the orthocenter of an acute triangle ABC . Points M and D lie on side BC such that $BM = CM$ and $\angle BAD = \angle CAD$. Ray MO intersects the circumcircle of triangle BHC in point N . Prove that $\angle ADO = \angle HAN$.

Solution 1



It is well known that (BHC) is the reflection of (ABC) over $\overline{BC}$ (if you don't know this, consider the reflection of H over $\overline{BC}$)

Thus, if we let P be the midpoint of BC , N is the reflection of P over $\overline{BC}$. Also, A, D, P are collinear.

Claim: $ANOD$ is cyclic

Proof: We have $DN = DP$ and $OA = OP$, so

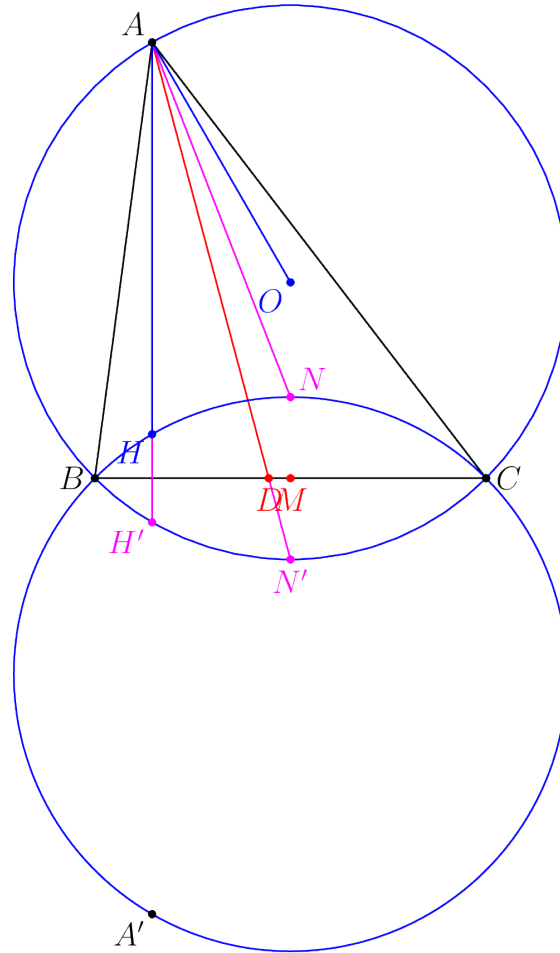
$$\angle DNO = \angle DPO = \angle DAO \quad \blacksquare$$

To finish, note that $\overline{AH} \parallel \overline{NP}$, so

$$\angle ADO = 180^\circ - \angle ANO = \angle HAN \quad \square$$

-sillybone

Solution 2



(original diagram by [integralarefun](#))

It's well known that the reflection of H across $\overline{BC}$, H' , lies on (ABC) . Then (BHC) is just the reflection of $(BH'C)$ across $\overline{BC}$, which is equivalent to the reflection of (ABC) across $\overline{BC}$. Reflect points A and N across $\overline{BC}$ to points A' and N' , respectively. Then N' is the midpoint of minor arc $\widehat{BC}$, so A, D, N' are collinear in that order. It suffices to show that $\angle AA'N' = \angle ADO$.

Claim: $\triangle AA'N' \sim \triangle ADO$. The proof easily follows.

Proof: Note that $\angle BAA' = \angle CAO = 90^\circ - \angle ABC$. Then we have $\angle A'AN' = \angle BAD - \angle BAA' = \angle CAD - \angle CAO = \angle DAO$. So, it suffices to show that

$$\frac{AA'}{AN'} = \frac{AD}{AO} \rightarrow AA' \cdot AO = AN' \cdot AD.$$

Notice that $\triangle ABA' \sim \triangle AOC$, so that

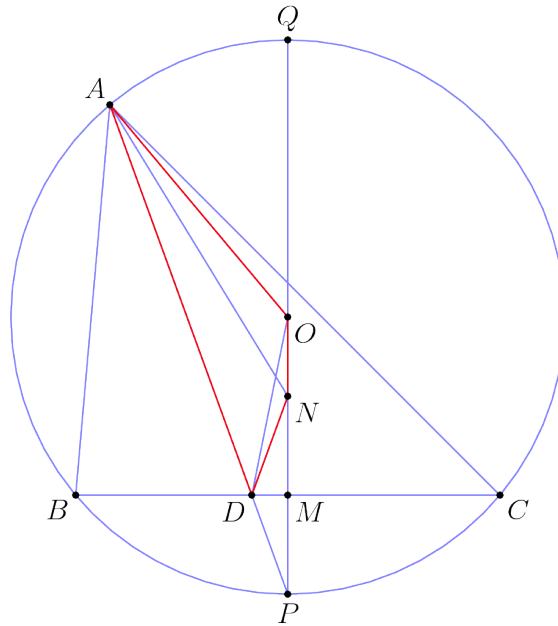
$$\frac{AB}{AA'} = \frac{AO}{AC} \rightarrow AA' \cdot AO = AB \cdot AC.$$

Therefore, it suffices to show that

$$AB \cdot AC = AN' \cdot AD \rightarrow \frac{AB}{AN'} = \frac{AD}{AC}.$$

But it is easy to show that $\triangle BAN' \sim \triangle DAC$, implying the result. ■

Solution 3



Suppose ray OM intersects the circumcircle of BHC at N' , and let the foot of the A -altitude of ABC be E . Note that $\angle BHE = 90 - \angle HBE = 90 - 90 + \angle C = \angle C$. Likewise, $\angle CHE = \angle B$. So, $\angle BHC = \angle BHE + \angle CHE = \angle B + \angle C$. $BHCN'$ is cyclic, so $\angle BN'C = 180 - \angle BHC = 180 - \angle B - \angle C = \angle A$. Also, $\angle BAC = \angle A$. These two angles are on different circles and have the same measure, but they point to the same line BC ! Hence, the two circles must be congruent. (This is also a well-known result)

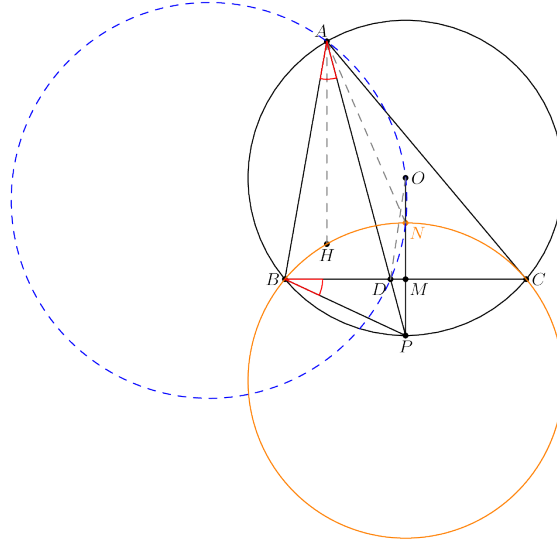
We know, since M is the midpoint of BC , that OM is perpendicular to BC . AH is also perpendicular to BC , so the two lines are parallel. AN is a transversal, so $\angle HAN = \angle ANO$. We wish to prove that $\angle ANO = \angle ADO$, which is equivalent to $AOND$ being cyclic.

Now, assume that ray OM intersects the circumcircle of ABC at a point P . Point P must be the midpoint of $\widehat{BC}$. Also, since AD is an angle bisector, it must also hit the circle at the point P . The two circles are congruent, which implies $MN = MP \implies ND = DP \implies NDP$ is isosceles. Angle ADN is an exterior angle, so $\angle ADN = \angle DNP + \angle DPO = 2\angle DPO$. Assume WLOG that $\angle B > \angle C$. So, $\angle DPO = \angle APO = \frac{\angle B + \angle C}{2} - \angle C = \frac{\angle B - \angle C}{2}$. In addition, $\angle AON = \angle AOP = \angle AOB + \angle BOP = 2\angle C + \angle A$. Combining these two equations, $\angle AON + \angle ADN = \angle B - \angle C + 2\angle C + \angle A = \angle A + \angle B + \angle C = 180$.

Opposite angles sum to 180, so quadrilateral $AOND$ is cyclic, and the condition is proved.

-william122

Solution 4



Let P be the midpoint of the arc BC not containing A . It is well-known that lines OM and AD intersect at P . Additionally, the circumcircles of $\triangle BHC$ and $\triangle ABC$ are known to be identical. (This property can be exhibited by showing that the circumradii of both circles are equivalent via the extended sine law.)

This means that the circumcircle of $\triangle BHC$ is the reflection of the circumcircle of $\triangle ABC$ about line BC . Thus, N is the reflection of P across BC , so M is the midpoint of PN ; $PN = 2PM$.

As is the intent of Solution 2, the crucial finding deserving of the pedestal is the observation that $ADNO$ is a cyclic quadrilateral. Distinctly, to prove it, we will show that $PD \cdot PA = PN \cdot PO$ so that the converse of the intersecting secants theorem validates the observation claimed.

Notice that $\angle PBC = \angle PAC = \angle PAB = \angle BAC/2$. Consequently, $\triangle PBD \sim \triangle PAB$ as both are equiangular. Evaluating their ratios suggests

$$\frac{PD}{PB} = \frac{PB}{PA} \implies PD \cdot PA = PB^2 \quad (1)$$

and assuming $R = OB = OP$ is the circumradius of $\triangle ABC$, by applications of Pythagoras' theorem,

$$\begin{aligned} PB^2 &= PM^2 + BM^2 \\ &= PM^2 + R^2 - OM^2 \\ &= R^2 + (PM - OM) \underbrace{(PM + OM)}_R \\ &= R(R + PM - OM) \\ &= R(PM + PM) \\ &= R \cdot PN \end{aligned}$$

$$= PO \cdot PN \quad (2)$$

Therefore, $PD \cdot PA = PN \cdot PO$ is indeed true, and $ADNO$ is concyclic. ✓

At last, we can complete the proof through a walk in the park. As both $AH, ON \perp BC$, $AH \parallel ON$, so $\angle HAN = \angle ANO$. Furthermore, the inscribed angle theorem in $ADNO$ implies $\angle ANO = \angle ADO$.

$$\boxed{\therefore \angle HAN = \angle ADO}$$

■

6. (*USAJMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $P_1, \dots, P_{2n}$ be $2n$ distinct points on the unit circle $x^2 + y^2 = 1$ other than $(1, 0)$. Each point is colored either red or blue, with exactly n of them red and exactly n of them blue. Let $R_1, \dots, R_n$ be any ordering of the red points. Let B_1 be the nearest blue point to R_1 traveling counterclockwise around the circle starting from R_1 . Then let B_2 be the nearest of the remaining blue points to R_2 traveling counterclockwise around the circle from R_2 , and so on, until we have labeled all the blue points $B_1, \dots, B_n$. Show that the number of counterclockwise arcs of the form $R_i \rightarrow B_i$ that contain the point $(1, 0)$ is independent of the way we chose the ordering $R_1, \dots, R_n$ of the red points.

Solution

I define a sequence to be, starting at $(1, 0)$ and tracing the circle counterclockwise, and writing the color of the points in that order - either R or B. For example, possible sequences include $RB, RBBR, BBBBBB, BRBBBBBR$, etc. Note that choosing an R_1 is equivalent to choosing an R in a sequence, and B_1 is defined as the B closest to R_1 when moving rightwards. If no B s exist to the right of R_1 , start from the far left. For example, if I have the above example $RBBR$, and I define the 2nd R to be R_1 , then the first B will be B_1 . Because no R or B can be named twice, I can simply remove R_1 and B_1 from my sequence when I choose them. I define this to be a move. Hence, a possible move sequence of $BBBBBR$ is: $BBR_1RRB_1 \implies B_2BRR_2 \implies B_3R_3$

Note that, if, in a move, B_n appears to the left of R_n , then $\widehat{R_n B_n}$ intersects $(1, 0)$

Now, I define a commencing B to be a B which appears to the left of all R s, and a terminating R to be a R which appears to the right of all B s. Let the amount of commencing B s be j , and the amount of terminating R s be k , I claim that the number of arcs which cross $(1, 0)$ is constant, and it is equal to $\max(j, k)$. I will show this with induction.

Base case is when $n = 1$. In this case, there are only two possible sequences - RB and BR . In the first case, $\widehat{R_1 B_1}$ does not cross $(1, 0)$, but both j and k are 0, so $\max(j, k) = 0$. In the second example, $j = 1, k = 1$, so $\max(j, k) = 1$. $\widehat{R_1 B_1}$ crosses $(1, 0)$ since B_1 appears to the left of R_1 , so there is one arc which intersects. Hence, the base case is proved.

For the inductive step, suppose that for a positive number n , the number of arcs which cross $(1, 0)$ is constant, and given by $\max(j, k)$ for any configuration. Now, I will show it for $n + 1$.

Suppose I first choose R_1 such that B_1 is to the right of R_1 in the sequence. This implies that $\widehat{R_1 B_1}$ does not cross $(1, 0)$. But, neither R_1 nor B_1 is a commencing B or terminating R . These numbers remain constant, and now after this move we have a sequence of length $2n$. Hence, by assumption, the total amount of arcs is $0 + \max(j, k) = \max(j, k)$.

- Here is a counter-case. $BRR_1 B_1 RR : j = 1, k = 2 \Rightarrow BRRR : j = 1, k = 3$. These numbers may not remain constant.

Thus, this solution probably doesn't work.

Now suppose that R_1 appears to the right of B_1 , but B_1 is not a commencing B . This implies that there are no commencing B s in the series, because there are no B s to the left of B_1 , so $j = 0$. Note that this arc does intersect $(1, 0)$, and R_1 must be a terminating R . R_1 must be a terminating R because there are no B s to the right of R_1 , or else that B would be B_1 . The $2n$ length sequence that remains has 0 commencing B s and $k - 1$ terminating R s. Hence, by assumption, the total amount of arcs is $1 + \max(0, k - 1) = 1 + k - 1 = k = \max(j, k)$.

Finally, suppose that R_1 appears to the right of B_1 , and B_1 is a commencing B . We know that this arc will cross $(1, 0)$. Analogous to the previous case, R_1 is a terminating R , so the $2n$ length sequence which remains has $j - 1$ commencing B s and $k - 1$ terminating R s. Hence, by assumption, the total amount of arcs is $1 + \max(j - 1, k - 1) = 1 + \max(j, k) - 1 = \max(j, k)$.

There are no more possible cases, hence the induction is complete, and the number of arcs which intersect $(1, 0)$ is indeed a constant which is given by $\max(j, k)$.

-william122

Solution 2

Lemma: If we switch the ordering of two consecutive R_k, R_{k+1} , the number of arcs crossing $(1, 0)$ stays invariant.

Proof: There are two situations. If the two arcs don't cross this is simple because the actual arcs stay the same, and only the number order of the arcs change. Otherwise, if B_k is to the left of R_{k+1} , the two arcs will have some overlap. The order of the points counterclockwise must be $RRBB$ or some rotation of it. Notice how in both the old and the new ordering, the arc between the second R and the first B is counted twice and the arcs between the same colors are only counted once. Thus no matter where $(1, 0)$ is the number of arcs containing it will stay invariant.

Lemma 2: We can swap the place of any two points R_i and R_j while keeping the number of crossings invariant.

Proof: By Lemma 1 we can use consecutive arc moves to swap R_i with R_{i+1} , then the new R_{i+1} with R_{i+2} and so on until it swaps with R_j . Now, we can do the reverse swaps to move the new R_{j-1} to R_{j-2} , and so on into the original position of R_i . Since the other points were moved once in one direction and once in the opposite, all the other points are kept in place.

Now, let I be the sequence where R_1 through R_n is numbered counterclockwise. We show that any ordering has the same number of crossings as I . We can swap the current place of R_1 with the desired position in I . Now we can ignore R_1 and swap the current place of R_2 with the place of R_2 in I , and so on until all the points are in the desired position. Thus, all orderings of R_1 to R_n have the same number of crossings as I .

-tigershark22

§9 USAJMO 2018

1. (*USAJMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

For each positive integer n , find the number of n -digit positive integers that satisfy both of the following conditions:

- no two consecutive digits are equal, and
- the last digit is a prime.

Solution 1

The answer is $\boxed{\frac{2}{5} (9^n + (-1)^{n+1})}$.

Suppose a_n denotes the number of n -digit numbers that satisfy the condition. We claim $a_n = 4 \cdot 9^{n-1} - a_{n-1}$, with $a_1 = 4$. *Proof.* It is trivial to show that $a_1 = 4$. Now, we can do casework on whether or not the tens digit of the n -digit integer is prime. If the tens digit is prime, we can choose the digits before the units digit in a_{n-1} ways and choose the units digit in 3 ways, since it must be prime and not equal to the tens digit. Therefore, there are $3a_{n-1}$ ways in this case.

If the tens digit is not prime, we can use complementary counting. First, we consider the number of $(n-1)$ -digit integers that do not have consecutive digits. There are 9 ways to choose the first digit and 9 ways to choose the remaining digits. Thus, there are 9^{n-1} integers that satisfy this. Therefore, the number of those $(n-1)$ -digit integers whose units digit is not prime is $9^{n-1} - a_{n-1}$. It is easy to see that there are 4 ways to choose the units digit, so there are $4(9^{n-1} - a_{n-1})$ numbers in this case. It follows that

$$a_n = 3a_{n-1} + 4(9^{n-1} - a_{n-1}) = 4 \cdot 9^{n-1} - a_{n-1},$$

and our claim has been proven.

Then, we can use induction to show that $a_n = \frac{2}{5} (9^n + (-1)^{n+1})$. It is easy to see that our base case is true, as $a_1 = 4$. Then,

$$a_{n+1} = 4 \cdot 9^n - a_n = 4 \cdot 9^n - \frac{2}{5} (9^n + (-1)^{n+1}) = 4 \cdot 9^n - \frac{2}{5} \cdot 9^n - \frac{2}{5} (-1)^{n+1},$$

which is equal to

$$a_{n+1} = \left(4 - \frac{2}{5}\right) \cdot 9^n - \frac{2}{5} \cdot \frac{(-1)^{n+1}}{(-1)} = \frac{18}{5} \cdot 9^n + \frac{2}{5} \cdot (-1)^{n+2} = \frac{2}{5} (9^{n+1} + (-1)^{n+2}),$$

as desired. $\square$

Solution by TheUltimate123.

Solution 2

The answer is $\boxed{\frac{2}{5} (9^n - (-1)^n)}$.

As in the first solution, let a_n denote the number of n -digit numbers that satisfy the condition. Clearly $a_1 = 4$ and $a_2 = 32$. We claim that for $n \geq 3$ we have the recurrence $a_n = 8a_{n-1} + 9a_{n-2}$.

To prove this, we split the n -digit numbers satisfying the conditions into cases depending on whether or not the second digit is 0. If the second digit is nonzero, our number is formed from one of the a_{n-1} numbers with one fewer digit satisfying the conditions, times 8 possible choices of adding a digit to the left. If the second digit is zero, our number is formed from one of the a_{n-2} numbers with two fewer digits satisfying the conditions, times 9 possible choices of adding 0 and then any nonzero digit to the left. This proves our claim.

This gives us a linear three-term recurrence. It is well-known that its solution is of the form $c_1r_1^n + c_2r_2^n$, where the c_i are constants to be determined from the initial conditions a_1 and a_2 , and r_1 and r_2 are the roots of the corresponding quadratic equation $x^2 = 8x + 9$. We solve and get $(x - 9)(x + 1) = 0$, so our roots are $r_1 = 9$ and $r_2 = -1$. Now, we use our conditions $a_1 = 4$ and $a_2 = 32$ to derive the system of linear equations

$$9c_1 - c_2 = 4$$

$$81c_1 + c_2 = 32$$

Solving this system yields $c_1 = \frac{2}{5}$ and $c_2 = -\frac{2}{5}$, and we are done.

Solution by putnam-lowell.

Solution 3

The answer is $\boxed{\frac{2}{5}(9^n - (-1)^n)}$.

As in the first solution, let a_n denote the number of n -digit numbers that satisfy the condition. Clearly $a_1 = 4$ and $a_2 = 32$.

From here, we proceed by complementary counting. We first count the total amount of numbers that satisfy both bullets but that may have zero as its first digit. The units digit can be one of four primes, and each digit to the left will have 9 choices (any digit but the one that was just used). Then the total for this group of numbers is just $4 * 9^{n-1}$.

Now we must subtract all numbers in the above group that have 0 as its first digit. This is just a_{n-1} because for each number in the first group beginning with a zero, we could take away the zero, leaving us with a number that works for the case $n - 1$ (this is true because the next number would not be zero, or the consecutive digit requirement would be violated). Then we have the recursive formula $a_n = 4 * 9^{n-1} - a_{n-1}$.

To simplify this, we take a look at the first few terms. We see that

$$a_2 = 4 * 9^1 - a_1 = 36 - 4 = 4(9 - 1) = 4(9^1 - 9^0)$$

$$a_3 = 4 * 9^2 - a_2 = 4(81) - 4(9 - 1) = 4(9^2 - 9^1 + 9^0)$$

We see a pattern where $a_n = 4(9^{n-1} - 9^{n-2} + 9^{n-3} \dots + (-1)^{n-1}9^0)$ and we can prove that it holds for all a_n because subtracting a_{n-1} from $4(9^n)$ is the same thing as reversing all previous signs of the preceding powers of 9. This constitutes an

alternating pattern, which we can calculate as a geometric series. The first term is $4(9^{n-1})$ and the common ratio is $-\frac{1}{9}$, so

$$a_n = \frac{4(9^{n-1})(1 - (-\frac{1}{9})^n)}{1 - (-\frac{1}{9})} = \frac{4(9^{n-1} - (-1)^n(\frac{1}{9}))}{\frac{10}{9}} = \frac{4(9^n - (-1)^n)}{10} = \frac{2}{5}(9^n - (-1)^n)$$

We are done.

Solution by aopsal

Solution 4

Let $f(n)$ denote the number of n -digit positive integers satisfying the conditions listed in the problem. Claim 1: $f(n) = 8f(n-1) + 9f(n-2)$. To prove this, let k be the leftmost digit of the n -digit positive integer. When k ranges from 1 to 9, the allowable second-to-leftmost digits is the set $\{0, 1, 2, \dots, 9\}$, with k excluded. Note that since 1, 2, $\dots$, 9 are all repeated 8 times and using our definition of $f(n)$ (since n -digit positive integers must begin with a positive digit), we have proved that $8f(n-1)$ part. Since 0 is repeated 9 times, and since digits that are allowed to be adjacent to the 0 (the third-to-leftmost digit of our positive integer) are integers ranging from 1 to 9, we have proved the $9f(n-2)$ part similarly. We can see that $\frac{f(1)}{4} = 1$ and $\frac{f(2)}{4} = 8$, by simple computation. These will be our base cases. Now, let $s(n) = \frac{f(n)}{4}$. Claim 2: $s(n+1) = 9s(n) + (-1)^n$. We can prove this using induction. Let $s(a) = k$ and $s(a+1) = 9k - 1$. By plugging k and $9k - 1$ into our formula $s(a+2) = 8s(a+1) + 9s(a)$, we obtain

$$s(a+2) = 8(9k - 1) + 9k \implies s(a+2) = 81k - 8 = 9s(a+1) + 1.$$

Now, by plugging $9k - 1$ and $81k - 8$ into the same formula, we obtain $s(a+3) = 8(81k - 8) + 9(9k - 1) = 729k - 73 = 9s(a+2) - 1$. Since with $a+2$ and $a+3$ we're back where we started (this is since $s(a+3) = 9s(a+2) - 1$ just like $s(a+1) = 9s(a) - 1$), and using our base cases of $s(1) = 1$ and $s(2) = 8$, our induction is complete. Claim 3: $s(n) = \frac{9^n + (-1)^{n+1}}{10}$. We can prove that

$$s(n) = \underbrace{9(9(9(\dots 9(9-1) + 1) \dots - 1) + 1) - 1}_{n-1 \text{ nested parentheses}} = \sum_{i=0}^{n-1} 9^i (-1)^{n-i-1}$$

by simply using our base case of $s(1) = 1$ and repeatedly using the distributive property to obtain further values of $s(n)$. For the case where n is even, we can manipulate the summation into $\sum_{i=0}^{\frac{n}{2}-1} 8 \cdot 9^{2i}$ which equals $\frac{8 \cdot 9^n - 1}{9^2 - 1} = \frac{9^n - 1}{10}$, by geometric series. For the case where n is odd, we can manipulate the summation into $1 + \sum_{i=0}^{\frac{n-1}{2}-1} 8 \cdot 9^{2i+1}$ which equals

$$9 \cdot \frac{8 \cdot 9^{n-1} - 1}{80} + 1 = \frac{9^n - 9}{10} + 1 = \frac{9^n + 1}{10}.$$

Thus, we have proved Claim 3 for both odd and even n . Since $f(n) = 4s(n)$, our answer is $4 \cdot \frac{9^n + (-1)^{n+1}}{10} = \boxed{\frac{2}{5}(9^n + (-1)^{n+1})}$.

-fidgetboss_4000

Solution 5

Let a_n be the number of n -digit numbers that satisfy the condition that most-left digit different than 0. Let b_n be the number of n -digit numbers that satisfy the condition that most-left digit is 0. We can write that

$$\begin{cases} a_n + b_n = 4 \cdot 9^{n-1} \\ b_n = a_{n-1} \end{cases}$$

and so we find that

$$a_n + a_{n-1} = 4 \cdot 9^{n-1} \quad (1)$$

$x + 1$ is the characteristic polynomial of the homogeneous recurrence relation $a_n + a_{n-1} = 0$. Then $x = -1$ is a root. Also, by the non-homogeneous part of $4 \cdot 9^{n-1}$, we understand that $x = 9$ is another root of the recurrence relation (1) with the complementary solution idea. Therefore, (a_n) sequence in the form

$$a_n = C_1 9^n + C_2 (-1)^n.$$

We can calculate that $C_1 = \frac{2}{5}, C_2 = -\frac{2}{5}$ with using $a_1 = 4, a_2 = 32$. Thus we get:

$$a_n = \frac{2}{5} (9^n - (-1)^n)$$

Lokman GÖKÇE

2. (*USAJMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let a, b, c be positive real numbers such that $a + b + c = 4\sqrt[3]{abc}$. Prove that

$$2(ab + bc + ca) + 4 \min(a^2, b^2, c^2) \geq a^2 + b^2 + c^2.$$

Solution 1

https://wiki-images.artofproblemsolving.com//6/69/IMG_8946.jpg

-srisainandan6

Solution 2

Similarly to Solution 2, we will prove homogeneity but we will use that to solve the problem differently. Let $f(a, b, c) = a + b + c - 4\sqrt[3]{abc}$. Note that $f(a, b, c) = f(ka, kb, kc)$, thus proving homogeneity.

WLOG, we can scale down all variables such that the lowest one is 1. WLOG, let this be $a = 1$. We now have $1 + b + c = 4\sqrt[3]{bc}$, and we want to prove $2bc + 2b + 2c + 4 \geq 1 + b^2 + c^2$. Adding $2bc$ to both sides and subtracting $2b + 2c$ gives us $4bc + 4 \geq 1 + (b + c)(b + c - 2)$, or $4bc + 3 \geq (b + c)(b + c - 2)$. Let $\sqrt[3]{bc} = x$. Now, we have

$$4x^3 + 3 \geq (4x - 1)(4x - 3)$$

$$4x^3 - 16x^2 + 16x \geq 0$$

$$4x^2 - 16 + 16 \geq 0$$

$$4(x - 2)^2 \geq 0$$

By the trivial inequality, this is always true. Since all these steps are reversible, the proof is complete. SigmaPiE

Solution 3

WLOG, let $a \leq b$ and $a \leq c$ and add $2(ab + bc + ca)$ to both sides to make the left side a square.

$$4(ab + bc + ca) + 4a^2 \geq (a + b + c)^2$$

$$4bc + 4a(a + b + c) \geq (a + b + c)^2$$

Now we perform the substitution for $a + b + c$.

$$4bc + 4a(4(abc)^{1/3}) \geq 16(abc)^{2/3}$$

Multiply both sides by a because we want all the terms to have a factor of $(abc)^{1/3}$.

$$4abc + 16a^2(abc)^{1/3} \geq 16a(abc)^{2/3}$$

Divide both sides by $4(abc)^{1/3}$.

$$(abc)^{2/3} + 4a^2 \geq 4a(abc)^{1/3}$$

$$((abc)^{1/3} - 2a)^2 \geq 0$$

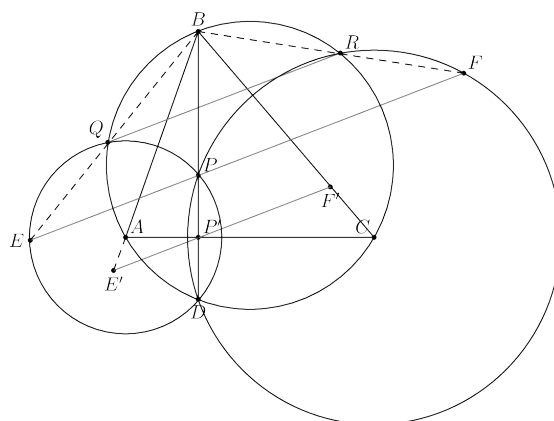
This is true by the trivial inequality. Lastly, all the steps are reversible so the given inequality has been proved.

-Themathcanadian

3. (*USAJMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Let $ABCD$ be a quadrilateral inscribed in circle ω with $\overline{AC} \perp \overline{BD}$. Let E and F be the reflections of D over lines BA and BC , respectively, and let P be the intersection of lines BD and EF . Suppose that the circumcircle of $\triangle EPD$ meets ω at D and Q , and the circumcircle of $\triangle FPD$ meets ω at D and R . Show that $EQ = FR$.

Solution 1



Claim 1: P is the orthocenter of $\triangle ABC$

Apply a homothety at D with factor $\frac{1}{2}$. This sends E, F to the feet of the perpendiculars from D to sides $\overline{AB}, \overline{AC}$, respectively.

However, this implies $P' = \overline{E'F'} \cap \overline{PD}$ is the foot of the altitude from D to $\overline{AC}$, since $\overleftrightarrow{E'F'}$ is the Simpson Line from D .

Thus, P is the reflection of D over $\overline{AC}$, implying P is the orthocenter of $\triangle ABC$.

Claim 2: Q and R are the reflections of P over $\overline{AB}$ and $\overline{AC}$.

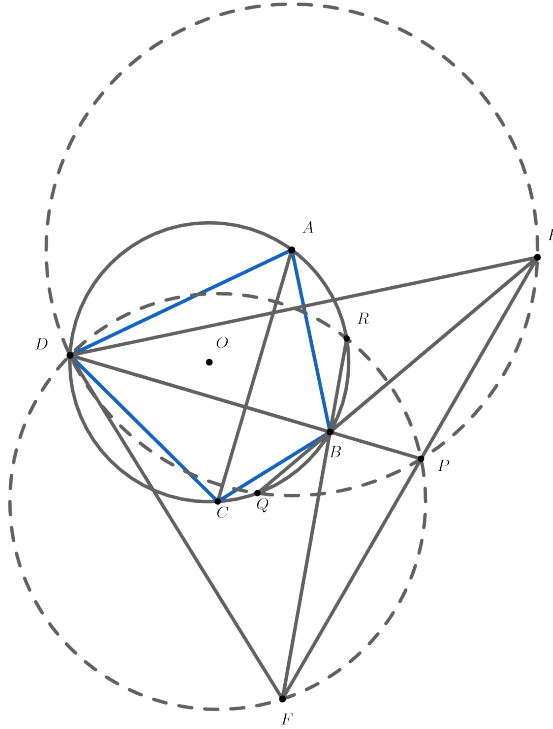
Note that the information of reflections tells us that $(DPE), (DPF)$ are centered at A, C , respectively. Let Q', R' be the reflections of P over the respective sides. Then, $Q' \in \omega$, but also $AQ' = AP = AD = AE$ implies $Q' \in (DPE)$. Thus, $Q' = \omega \cap (DPE) = Q$. Similarly, $R' = R$, as needed.

Thus, $\overline{QE}, \overline{RF}$ are reflections of $\overline{PD}$ over the sides $\overline{AB}, \overline{AC}$. To finish,

$$EQ = DP = FR \quad \square$$

-sillybone

Solution 2



First we have that $BE = BD = BF$ by the definition of a reflection. Let $\angle DEB = \alpha$ and $\angle DFB = \beta$. Since $\triangle DBE$ is isosceles we have $\angle BDE = \alpha$. Also, we see that $\angle BDE = \angle CAB = \angle CDB = \alpha$, using similar triangles and the property of cyclic quadrilaterals. Similarly,

$$\angle DFB = \angle FDB = \angle ACB = \angle ADB = \beta.$$

Now, from $BE = BD = BF$ we know that B is the circumcenter of $\triangle DEF$. Using the properties of the circumcenter and some elementary angle chasing, we find that

$$\angle DPE = 90^\circ + \beta - \alpha.$$

Now, we claim that Q is the intersection of ray $\overrightarrow{EB}$ and the circumcircle of $ABCD$. To prove this, we just need to show that $DEPQ$ is cyclic by this definition of Q . We have that

$$\angle DQE = \angle DCB = \angle DCA + \angle ACB = (90^\circ - \alpha) + \beta.$$

We also have from before that

$$\angle DPE = 90 + \beta - \alpha,$$

so $\angle DQE = \angle DPE$ and this proves the claim.

We can use a similar proof to show that F, B, R are collinear.

Now, DP is the radical axis of the circumcircles of $\triangle EDP$ and $\triangle FDP$. Since B lies on DP , and E, Q lie on the circumcircle of $\triangle EPD$ and F, R lie on the circumcircle of $\triangle FPD$, we have that

$$BF \cdot BR = BE \cdot BQ.$$

However, $BF = BE$, so $BR = BQ$. Since E, B, Q are collinear and so are F, B, R we can add these 2 equations to get

$$EQ = BE + BQ = BF + BR = FR,$$

which completes the proof.

Solution 3

We begin with the following claims.

Claim. B is the circumcenter of $\triangle DEF$. Proof. By reflection $BD = BE = BF$.

Claim. A is the circumcenter of $\triangle EPD$. Proof. First, we have

$$\angle DPE = 180^\circ - \angle PDE - \angle PED = 180^\circ - \angle BED - \frac{1}{2}\angle EAD$$

Then

$$\angle ADE = \angle ADP - \angle EDP = \angle ADB - \angle EDB = \angle ACB - \angle ADB$$

Then $2\angle ADE = 180^\circ - 2(180^\circ - \angle DPE) \implies \angle DAE = 2(180^\circ - \angle DPE)$. This is enough to imply what we desire.

Claim. C is the circumcenter of $\triangle FPD$. Proof. Similar to above.

Claim. E, B, Q are collinear. Proof. We have

$$\angle ABE = \angle ABD = \angle AQD = \angle ADQ = 180^\circ - \angle ABQ$$

Claim. F, B, R are collinear. Proof. Similar to above.

Since $DEPQ$ is cyclic, $\triangle EBP \sim \triangle DBQ$. However, $BE = BD$, so $BP = BQ$. Similarly, $BP = BR$. Finishing, we have

$$EQ = EB + BQ = BD + BP = FB + BR = FR$$

, as desired. ■ MSC

Solution 4

Let J, K, L be the feet of the altitude from D to BA, BC, CA , respectively. Note that $L = AC \cap BD$. J, K, L are collinear by the Simson line. Let h be the homothety with center D and scale factor of 2. We see $h(J) = E$ and $h(K) = F$. Since E, F, P are collinear and P lies on DL , we have that $h(L) = P$. Therefore, P is the reflection of D over AC . Let O_1 and O_2 be the centers of (EPD) and (FPD) , respectively. From earlier, the perpendicular bisector of DP is AC . The perpendicular bisectors of DE and DF are BA and BC , respectively. Then, $O_1 = AC \cap BA$ and $O_2 = AC \cap BC$, so $O_1 = A$ and $O_2 = C$. Let O be the center of $(ABCD)$. DQ is the radical axis of (EPD) and $(ABCD)$, so $\angle OAD = \angle OAQ$.

$$\angle AOD = 2\angle ABD = 2(90^\circ - \angle BAC) \Rightarrow \angle DAQ = \angle OAD + \angle OAQ = 2\angle OAD = 180^\circ - \angle AOD = 2\angle B$$

Then, $\angle BAC = \angle BAQ + \angle CAD \Rightarrow \angle BAP + \angle CAP = \angle BAQ + \angle CAD \Rightarrow \angle BAP = \angle BAQ$. Therefore, Q is the reflection of P across BA because $AP = AQ$. Reflections preserve length, so $DP = EQ$. Similarly, $DP = FR$, so $EQ = FR$. ■
IS

4. (USAJMO 2018) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Triangle ABC is inscribed in a circle of radius 2 with $\angle ABC \geq 90^\circ$, and x is a real number satisfying the equation $x^4 + ax^3 + bx^2 + cx + 1 = 0$, where $a = BC, b = CA, c = AB$. Find all possible values of x .

Solution

Notice that

$$x^4 + ax^3 + bx^2 + cx + 1 = \left(x^2 + \frac{a}{2}x\right)^2 + \left(\frac{c}{2}x + 1\right)^2 + \left(b - \frac{a^2}{4} - \frac{c^2}{4}\right)x^2.$$

Thus, if $b > \frac{a^2}{4} + \frac{c^2}{4}$, then the expression above is strictly greater than 0 for all x , meaning that x cannot satisfy the equation $x^4 + ax^3 + bx^2 + cx + 1 = 0$. It follows that $b \leq \frac{a^2}{4} + \frac{c^2}{4}$.

Since $\angle ABC \geq 90^\circ$, we have $b^2 \geq a^2 + c^2$. From this and the above we have $4b \leq a^2 + c^2 \leq b^2$, so $4b \leq b^2$. This is true for positive values of b if and only if $b \geq 4$. However, since $\triangle ABC$ is inscribed in a circle of radius 2, all of its side lengths must be at most the diameter of the circle, so $b \leq 4$. It follows that $b = 4$.

We know that $4b \leq a^2 + c^2 \leq b^2$. Since $4b = b^2 = 16$, we have $4b = a^2 + c^2 = b^2 = 16$.

The equation $x^4 + ax^3 + bx^2 + cx + 1 = 0$ can be rewritten as $(x^2 + \frac{a}{2}x)^2 + (\frac{c}{2}x + 1)^2 = 0$, since $b = \frac{a^2}{4} + \frac{c^2}{4}$. This has a real solution if and only if the two separate terms have zeroes in common. The zeroes of $(x^2 + \frac{a}{2}x)^2$ are 0 and $-\frac{a}{2}$, and the zero of $(\frac{c}{2}x + 1)^2 = 0$ is $-\frac{2}{c}$. Clearly we cannot have $0 = -\frac{2}{c}$, so the only other possibility is $-\frac{a}{2} = -\frac{2}{c}$, which means that $ac = 4$.

We have a system of equations: $ac = 4$ and $a^2 + c^2 = 16$. Solving this system gives $(a, c) = (\sqrt{6} + \sqrt{2}, \sqrt{6} - \sqrt{2}), (\sqrt{6} - \sqrt{2}, \sqrt{6} + \sqrt{2})$. Each of these gives solutions for x as $-\frac{\sqrt{6} + \sqrt{2}}{2}$ and $-\frac{\sqrt{6} - \sqrt{2}}{2}$, respectively. Now that we know that any valid

value of x must be one of these two, we will verify that both of these values of x are valid.

First, consider a right triangle ABC , inscribed in a circle of radius 2, with side lengths $a = \sqrt{6} + \sqrt{2}$, $b = 4$, $c = \sqrt{6} - \sqrt{2}$. This generates the polynomial equation

$$x^4 + (\sqrt{6} + \sqrt{2})x^3 + 4x^2 + (\sqrt{6} - \sqrt{2})x + 1 = \left(\frac{\sqrt{6} - \sqrt{2}}{2}x + 1\right)^2 \left(\left(\frac{\sqrt{6} + \sqrt{2}}{2}x\right)^2 + 1\right) = 0.$$

This is satisfied by $x = -\frac{\sqrt{6} + \sqrt{2}}{2}$.

Second, consider a right triangle ABC , inscribed in a circle of radius 2, with side lengths $a = \sqrt{6} - \sqrt{2}$, $b = 4$, $c = \sqrt{6} + \sqrt{2}$. This generates the polynomial equation

$$x^4 + (\sqrt{6} - \sqrt{2})x^3 + 4x^2 + (\sqrt{6} + \sqrt{2})x + 1 = \left(\frac{\sqrt{6} + \sqrt{2}}{2}x + 1\right)^2 \left(\left(\frac{\sqrt{6} - \sqrt{2}}{2}x\right)^2 + 1\right) = 0.$$

This is satisfied by $x = -\frac{\sqrt{6} - \sqrt{2}}{2}$.

It follows that the possible values of x are $-\frac{\sqrt{6} + \sqrt{2}}{2}$ and $-\frac{\sqrt{6} - \sqrt{2}}{2}$.

Fun fact: these solutions correspond to a 15-75-90 triangle.

(sujaykazi)

5. (*USAJMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let p be a prime, and let $a_1, \dots, a_p$ be integers. Show that there exists an integer k such that the numbers

$$a_1 + k, a_2 + 2k, \dots, a_p + pk$$

produce at least $\frac{1}{2}p$ distinct remainders upon division by p .

Solution

Lemma: For fixed $i \neq j$, where $i, j \in \{1, 2, \dots, p\}$, the statement $a_i + ik \equiv a_j + jk \pmod{p}$ holds for exactly one $k \in 1, 2, \dots, p$.

Proof: Notice that the left side minus the right side is congruent to $(a_i - a_j) + (i - j)k$ modulo p . For this difference to equal 0, there is a unique solution for k modulo p given by $k \equiv (a_j - a_i)(i - j)^{-1} \pmod{p}$, where we have used the fact that every nonzero residue modulo p has a unique multiplicative inverse. Therefore, there is exactly one $k \in 1, 2, \dots, p$ that satisfies $a_i + ik \equiv a_j + jk \pmod{p}$ for any fixed $i \neq j$. **End Lemma**

Suppose that you have p graphs $G_1, G_2, \dots, G_p$, and graph G_k consists of the vertices (i, k) for all $1 \leq i \leq p$. Within any graph G_k , vertices (i_1, k) and (i_2, k) are connected by an edge if and only if $a_{i_1} + i_1k \equiv a_{i_2} + i_2k \pmod{p}$. Notice that the number of disconnected components of any graph G_k equals the number of distinct remainders when divided by p given by the numbers $a_1 + k, a_2 + 2k, \dots, a_p + pk$.

These p graphs together have exactly one edge for every unordered pair of elements of $\{1, 2, \dots, p\}$, so they have a total of exactly $\frac{p(p-1)}{2}$ edges. Therefore, there exists

at least one graph G_k that has strictly fewer than $\frac{p}{2}$ edges, meaning that it has more than $\frac{p}{2}$ disconnected components. Therefore, the collection of numbers $\{a_i + ik : 1 \leq i \leq p\}$ for this particular value of k has at least $\frac{p}{2}$ distinct remainders modulo p . This completes the proof.

(sujaykazi)

6. (*USAJMO 2018*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Karl starts with n cards labeled $1, 2, 3, \dots, n$ lined up in a random order on his desk. He calls a pair (a, b) of these cards swapped if $a > b$ and the card labeled a is to the left of the card labeled b . For instance, in the sequence of cards $3, 1, 4, 2$, there are three swapped pairs of cards, $(3, 1)$, $(3, 2)$, and $(4, 2)$.

He picks up the card labeled 1 and inserts it back into the sequence in the opposite position: if the card labeled 1 had i card to its left, then it now has i cards to its right. He then picks up the card labeled 2 and reinserts it in the same manner, and so on until he has picked up and put back each of the cards $1, 2, \dots, n$ exactly once in that order. (For example, the process starting at $3, 1, 4, 2$ would be $3, 1, 4, 2 \rightarrow 3, 4, 1, 2 \rightarrow 2, 3, 4, 1 \rightarrow 2, 4, 3, 1 \rightarrow 2, 3, 4, 1$.)

Show that, no matter what lineup of cards Karl started with, his final lineup has the same number of swapped pairs as the starting lineup.

Solution

Note that in this solution, the term "inversions" is used synonymously with "swapped pairs."

We define a new process P' where, when re-inserting card i , we additionally change its label from i to $n + i$. For example, an example of P' also starting with 3142 is:

$$3142 \longrightarrow 3452 \longrightarrow 6345 \longrightarrow 6475 \longrightarrow 6785.$$

Note that now, each step of P' preserves the number of inversions. Moreover, the final configuration of P' is the same as the final configuration of P with all cards incremented by n , and thus, of course, has the same number of inversions.

§10 USAJMO 2019

1. (*USAJMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

There are $a + b$ bowls arranged in a row, numbered 1 through $a + b$, where a and b are given positive integers. Initially, each of the first a bowls contains an apple, and each of the last b bowls contains a pear.

A legal move consists of moving an apple from bowl i to bowl $i + 1$ and a pear from bowl j to bowl $j - 1$, provided that the difference $i - j$ is even. We permit multiple fruits in the same bowl at the same time. The goal is to end up with the first b bowls each containing a pear and the last a bowls each containing an apple. Show that this is possible if and only if the product ab is even.

Solution

Claim: If a and b are both odd, then the end goal of achieving b pears followed by a apples is impossible.

Proof: Let A_1 and P_1 denote the number of apples and the number of pears in odd-numbered positions, respectively. We show that $A_1 - P_1$ is invariant. Notice that if i and j are odd, then A_1 and P_1 both decrease by 1, as one apple and one pear are both moved from odd-numbered positions to even-numbered positions. If i and j are even, then A_1 and P_1 both increase by 1.

Because the starting configuration $aa \dots ab \dots b$ has $\lfloor \frac{a}{2} \rfloor + 1$ odd-numbered apples and $\lfloor \frac{b}{2} \rfloor$ odd-numbered pears, the initial value of $A_1 - P_1$ is $\lfloor \frac{a}{2} \rfloor - \lfloor \frac{b}{2} \rfloor + 1$. But the desired ending configuration has $\lfloor \frac{b}{2} \rfloor + 1$ odd-numbered pears and $\lfloor \frac{a}{2} \rfloor$ odd-numbered apples, so $A_1 - P_1 = \lfloor \frac{a}{2} \rfloor - \lfloor \frac{b}{2} \rfloor - 1$. As $A_1 - P_1$ is invariant, it is impossible to attain the desired ending configuration. ■

Claim: If at least one of a and b is even, then the end goal of achieving b pears followed by a apples is possible.

Proof: Without loss of generality, assume a is even. If only b is even, then we can number the bowls in reverse, and swap apples with pears. We use two inductive arguments to show that $(a, b) = (2, b)$ is achievable for all $b \geq 1$, then to show that (a, b) is achievable for all even a and all $b \geq 1$.

Base case: $(a, b) = (2, 1)$. Then we can easily achieve the ending configuration in two moves, by moving the apple in bowl #1 and the pear in bowl #3 so that everything is in bowl #2, then finishing the next move.

Inductive step: suppose that for $a \geq 2$ (even) apples and $b \geq 1$ pears, that with a apples and b pears, the ending configuration is achievable. We will show two things: i) achievable with a apples and $b + 1$ pears, and ii) achievable with $a + 2$ apples and b pears.

i) Apply the process on the a apples and first b pairs to get a configuration $bb \dots ba \dots ab$. Now we will "swap" the leftmost apple with the last pear by repeatedly applying the move on just these two fruits (as a is even, the difference $i - j$ is even). This gives a solution for a apples and $b + 1$ pears. In particular, this shows that $(a, b) = (2, b)$ for all $b \geq 1$ is achievable.

ii) To show $(a + 2, b)$ is achievable, given that (a, b) is achievable, apply the process on the last a apples and b pears to get the configuration $aabbb \dots baaa \dots a$. Then, because we have shown that 2 apples and b pears is achievable in i), we can now reverse the first two apples and b pears.

Thus for ab even, the desired ending configuration is achievable. ■

Combining the above two claims, we see that this is possible if and only if ab is even. -scrabbler94

Solution 2

"If" part: Note that two opposite fruits can be switched if they have even distance. If one of a, b is odd and the other is even, then switch 1 with $a + b$, 2 with $a + b - 1$, 3 with $a + b - 2$... until all of one fruit is switched. If both are even, then if $a > b$, then switch 1 with $a + 1$, 2 with $a + 2$, 3 with $a + 3$... until all of one fruit is switched; if $a < b$ then switch a with $a + b$, $a - 1$ with $a + b - 1$, $a - 2$ with $a + b - 2$... until all of one fruit is switched. Each of these processes achieve the end goal.

"Only if" part: Assign each apple a value of 1 and each pear a value of -1 . At any given point in time, let E be the sum of the values of the fruit in even-numbered bowls. Because a and b both are odd, at the beginning there are $\frac{a-1}{2}$ apples in even bowls and $\frac{b+1}{2}$ pears in even bowls, so at the beginning $E = \frac{a-b}{2} - 1$. After the end goal is achieved, there are $\frac{a+1}{2}$ apples in even bowls and $\frac{b-1}{2}$ pears in even bowls, so after the end goal is achieved $E = \frac{a-b}{2} + 1$. However, because two opposite fruits must have the same parity to move and will be the same parity after they move, we see that E is invariant, which is a contradiction; hence, it is impossible for the end goal to be reached if ab is odd.

Stormersyle

Solution 3

We define a cycle to be when 2 fruits in bowls i and j are swapped with each other, provided that bowl i contains an apple, bowl j contains a pear, and the difference $i - j$ is even.

We claim that a cycle consists of only the valid moves described in the problem statement. Proof: We can move the apple and pear one step at a time to eventually reach the other's bowl (i.e. the apple moves to $i + 1$ and the pear moves to bowl $j - 1$ at the beginning) using only valid moves. By a simple parity argument (if k moves have already been done, $i + k - (j - k) = i + j + 2k = 0 + 2k = 2k \equiv 0 \pmod{2}$), these moves are always valid, so we are done. ■

We now split the "if" part of this proof into cases based on the parities of a and b .

Case 1: a even, b odd In this case we can simply cycle the fruit in bowl $k + 1$ with the fruit in bowl $a + b - k$ for all integer k in the range $[0, b - 1]$ to achieve our desired goal. The difference $a + b - k - k - 1 = a + b - 1$ is even, so the move is valid.

Case 2: b even, a odd Using the same cycle as the previous case works here.

Case 3: a even, b even Here, we can cycle the fruits in bowls k and $a + b - k$ for all integer k in the range $[1, b]$. Then, we finish by cycling the fruits in bowls a and $a + b$. These are both valid moves by similar parity arguments as done in Case 1.

Since we have exhausted all possible cases for the "if" part of this proof, we have completed the if part. We will proceed with the "only if" part.

We let the number of apples in odd positions be A_1 , and the number of pears in odd positions be A_2 . Obviously $A_1 - A_2$ is invariant when using valid moves only. We can now use proof by contradiction (i.e. showing that this invariant does not occur when $a, b \equiv 1 \pmod{2}$ and our desired goal is achieved) to find that we have that $0 = 2$, which is obviously false, so we have completed the proof for the "only if" part.

We are now done, since we have completed our proofs for the "if" and "only if" parts. ■

Solution 4

Begin by defining a *swap* as a set of $i - j$ legal moves taking an apple in $i \rightarrow j$ and the pear in $j \rightarrow i$. Now we begin by proving if ab is even, then the final configuration is possible.

WLOG let b be even. Then there are two cases to consider. $a < b, a \geq b$ to provide valid constructions for. In the first case, consider the apple/pear pairs

$(1, b+1), (2, b+2), \dots, (i, b+i), \dots, (a, b+a)$ and swap them. This results in the transformation,

$$\begin{array}{c} \underbrace{AA \cdots A}_{a \text{ apples}} \underbrace{PP \cdots P}_{b \text{ pears}} \\ \underbrace{PP \cdots P}_{a \text{ pears}} \underbrace{PP \cdots P}_{b-a \text{ pears}} \underbrace{AA \cdots A}_{a \text{ apples}} \\ \underbrace{PP \cdots P}_{b \text{ pears}} \underbrace{AA \cdots A}_{a \text{ apples}} \end{array}$$

as desired.

Now consider the case where $a \geq b$. Also consider the pairing, $(a-b+1, a+1), (a-b+2, a+2), \dots, (a-b+i, a+i), \dots, (a, a+b)$ and swap them. This results in the transformation,

$$\begin{array}{c} \underbrace{AA \cdots A}_{a \text{ apples}} \underbrace{PP \cdots P}_{b \text{ pears}} \\ \underbrace{AA \cdots A}_{a-b \text{ apples}} \underbrace{PP \cdots P}_{b \text{ pears}} \underbrace{AA \cdots A}_{b \text{ apples}}. \end{array}$$

Now just consider the leftmost a bowls. There are $a-b$ apples, and then b pears. We have effectively reduced $(a, b) \rightarrow (a-b, b)$. Reducing in this fashion recursively will lead to $(a \bmod b, b)$ which can be constructed with the $a < b$ case. This finishes the construction for $a \geq b$.

Now assume ab, a, b are odd. FTSOC, let this case satisfy the claim. Then, for each apple, define its *position* as the sum of the distance from it to all other pairs. So for apple in bowl $i \leq a$ originally, the position of i is $\sum_{j=a+1}^b j - i$.

Every time a legal move is made, the position of the apple decreases by $b-1+2 = b+1$ (from the $b-1$ pears not part of the move and the decrease in 2 by the corresponding pear). Also note that there needs to be exactly ab moves in order to attain the final configurations. So the difference in sum of positions of the apples at the beginning and end is $ab(b+1)$. Mathematically,

$$\begin{aligned} \sum_{i=1}^a \sum_{j=a+1}^{a+b} j - i - \sum_{i=b+1}^{b+a} \sum_{j=1}^b j - i &= ab(b+1) \\ \left(\frac{ab(2a+b+1)}{2} - \frac{ab(a+1)}{2} \right) - \left(\frac{ab(b+1)}{2} - \frac{ab(a+2b+1)}{2} \right) &= \\ ab(a+b) &= ab(b+1) \\ a &= 1. \end{aligned}$$

A symmetric argument for the position of pears gives $b = 1$. Clearly the $a = b = 1$ does not have a solution, therefore we have contradiction. $\square$

2. (*USAJMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $\mathbb{Z}$ be the set of all integers. Find all pairs of integers (a, b) for which there exist functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ and $g : \mathbb{Z} \rightarrow \mathbb{Z}$ satisfying

$$f(g(x)) = x + a \quad \text{and} \quad g(f(x)) = x + b$$

for all integers x .

Solution 1

We claim that the answer is $|a| = |b|$.

Proof: f and g are surjective because $x + a$ and $x + b$ can take on any integral value, and by evaluating the parentheses in different order, we find $f(g(f(x))) = f(x + b) = f(x) + a$ and $g(f(g(x))) = g(x + a) = g(x) + b$. We see that if $a = 0$ then $g(x) = g(x) + b$ to $b = 0$ as well, so similarly if $b = 0$ then $a = 0$, so now assume $a, b \neq 0$.

We see that if $x = |b|n$ then $f(x) \equiv f(0) \pmod{|a|}$, if $x = |b|n + 1$ then $f(x) \equiv f(1) \pmod{|a|}$, if $x = |b|n + 2$ then $f(x) \equiv f(2) \pmod{|a|}$... if $x = |b|(n + 1) - 1$ then $f(x) \equiv f(|b| - 1) \pmod{|a|}$. This means that the b -element collection $\{f(0), f(1), f(2), \dots, f(|b| - 1)\}$ contains all $|a|$ residues mod $|a|$ since f is surjective, so $|b| \geq |a|$. Doing the same to g yields that $|a| \geq |b|$, so this means that only $|a| = |b|$ can work.

For $a = b$ let $f(x) = x$ and $g(x) = x + a$, and for $a = -b$ let $f(x) = -x$ and $g(x) = -x - a$, so $|a| = |b|$ does work and are the only solutions, as desired.

-Stormersyle

Solution 2

We claim that f and g exist if and only if $|a| = |b|$.

Only If:

For some fixed j , let $f(j) = k$.

If $b = 0$, then $g(k) = j$. Suppose $a \neq 0$. Then $f(j) = f(g(k)) = k + a \neq k$, a contradiction. Thus, $a = 0$. Similarly, if $a = 0$, then $b = 0$, satisfying $|a| = |b|$.

Otherwise, $a, b \neq 0$. We know that $g(k) = g(f(j)) = j + b$, $f(j + b) = f(g(k)) = k + a$, $g(k + a) = j + 2b$, and so on: $f(j + nb) = k + na$ and $g(k + na) = j + (n + 1)b$ for $n \geq 0$.

Consider the value of $g(k - a)$. Suppose $g(k - a) = j' \neq j$. Then $f(j') = k$ and $g(f(j')) = j + b \neq j' + b$, a contradiction. Thus, $g(k - a) = j$. We repeat with $f(j - b)$. Suppose $f(j - b) = k' - b \neq k - b$. Then $g(k' - b) = j$ and $f(g(k' - b)) = k \neq k'$, a contradiction. Thus, $f(j - b) = k - b$. Continuing, $g(k - 2a) = j - a$, and so on: $f(j + nb) = k + na$ and $g(k + na) = j + (n + 1)b$ now for all n .

This defines $f(x)$ for all $x \equiv j \pmod{|b|}$ and $g(x)$ for all $x \equiv f(j) \pmod{|a|}$.

This means that $x \equiv j \pmod{|b|} \implies f(x) \equiv f(j) \pmod{|a|}$, and $y \equiv f(j) \pmod{|a|} \implies g(y) \equiv j \pmod{|b|}$ which implies $f(x) \equiv f(j) \pmod{|a|} \implies x \equiv j \pmod{|b|}$.

As a result, $f(x)$ maps each residue mod $|b|$ to a unique residue mod $|a|$, so $|a| \geq |b|$. Similarly, $g(x)$ maps each residue mod $|a|$ to a unique residue mod $|b|$, so $|b| \geq |a|$. Therefore, $|a| = |b|$.

If:

$|a| = |b|$ means that either $a = b$ or $a = -b$. $f(x) = x, g(x) = x + a$ works for the former and $f(x) = -x, g(x) = -x - a$ works for the latter, and we are done.

3. (*USAJMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Let $ABCD$ be a cyclic quadrilateral satisfying $AD^2 + BC^2 = AB^2$. The diagonals of $ABCD$ intersect at E . Let P be a point on side $\overline{AB}$ satisfying $\angle APD = \angle BPC$. Show that line PE bisects $\overline{CD}$.

Solution 1

Let $PE \cap DC = M$. Also, let N be the midpoint of AB .

Note that only one point P satisfies the given angle condition. With this in mind, construct P' with the following properties:

$$AP' \cdot AB = AD^2 \quad \text{and} \quad BP' \cdot AB = CD^2$$

Claim: $P = P'$

Proof:

The conditions imply the similarities $ADP \sim ABD$ and $BCP \sim BAC$ whence $\angle APD = \angle BDA = \angle BCA = \angle CPB$ as desired. $\square$

Claim: PE is a symmedian in AEB

Proof:

We have

$$\begin{aligned} AP \cdot AB = AD^2 &\iff AB^2 \cdot AP = AD^2 \cdot AB \\ &\iff \left(\frac{AB}{AD}\right)^2 = \frac{AB}{AP} \\ &\iff \left(\frac{AB}{AD}\right)^2 - 1 = \frac{AB}{AP} - 1 \\ &\iff \frac{AB^2 - AD^2}{AD^2} = \frac{BP}{AP} \\ &\iff \left(\frac{BC}{AD}\right)^2 = \left(\frac{BE}{AE}\right)^2 = \frac{BP}{AP} \end{aligned}$$

as desired. $\blacksquare$

Since P is the isogonal conjugate of N , $\angle PEA = \angle MEC = \angle BEN$. However $\angle MEC = \angle BEN$ implies that M is the midpoint of CD from similar triangles, so we are done. $\blacksquare$

Solution 2

By monotonicity, we can see that the point P is unique. Therefore, if we find another point P' with all the same properties as P , then $P = P'$

Part 1) Let N be a point on $\overline{AB}$ such that $AN \cdot AB = AD^2$, and $BN \cdot AB = BC^2$. Obviously N exists because adding the two equations gives $AN \cdot AB + BN \cdot AB = AD^2 + BC^2 = AB^2$, which is the problem statement. Notice that converse PoP gives

$$AN \cdot AB = AD^2 \implies \triangle ADN \sim \triangle ABD$$

$$BN \cdot AB = AD^2 \implies \triangle CBN \sim \triangle ABC$$

Therefore, $\angle AND = \angle ADB = \angle ACB = \angle CNB$, so N does indeed satisfy all the conditions P does, so $N = P$. Hence, $\triangle ADP \sim \triangle ABD$ and $\triangle CBP \sim \triangle ABC$.

Part 2) Define G as the midpoint of CD . Furthermore, create a point X such that $DX \parallel EC$ and $CX \parallel ED$. Obviously $XCED$ must be a parallelogram. Now we set up for Jacobi's. The problem already gives us that $\angle APD = \angle CPB$, which is good for starters. Furthermore, $\triangle ADP \sim \triangle ABD$ tells us that

$$\angle ADP = \angle ABD = \angle ACD = \angle XDC$$

This gives us our second needed angle equivalence. Lastly, $\triangle CBP \sim \triangle ABC$ will give

$$\angle BCP = \angle BAC = \angle BDC = \angle XCD$$

which is our last necessary angle equivalence to apply Jacobi's. Finally, applying Jacobi's tells us that AC , BD , and XP are concurrent $\implies X, E, P$ collinear. Additionally, since parallelogram diagonals bisect each other, X, G , and E are collinear, so finally we obtain that P, E , and G are collinear, as desired.

-jj_ca888

4. (*USAJMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Let ABC be a triangle with $\angle ABC$ obtuse. The *A-excircle* is a circle in the exterior of $\triangle ABC$ that is tangent to side BC of the triangle and tangent to the extensions of the other two sides. Let E, F be the feet of the altitudes from B and C to lines AC and AB , respectively. Can line EF be tangent to the *A-excircle*?

Solution 1

Instead of trying to find a synthetic way to describe EF being tangent to the *A-excircle* (very hard), we instead consider the foot of the perpendicular from the *A-excircle* to EF , hoping to force something via the length of the perpendicular. It would be nice if there were an easier way to describe EF , something more closely related to the *A-excircle*; as we are considering perpendicularity, if we could generate a line parallel to EF , that would be good.

So we recall that it is well known that triangle AEF is similar to ABC . This motivates reflecting BC over the angle bisector at A to obtain $B'C'$, which is parallel to EF for obvious reasons.

Furthermore, as reflection preserves intersection, $B'C'$ is tangent to the reflection of the *A-excircle* over the *A-angle* bisector. But it is well-known that the *A-excenter* lies on the *A-angle* bisector, so the *A-excircle* must be preserved under reflection over the *A-excircle*. Thus $B'C'$ is tangent to the *A-excircle*. Yet for all lines parallel to EF , there are only two lines tangent to the *A-excircle*, and only one possibility for EF , so $EF = B'C'$.

Thus as ABB' is isocetes,

$$[ABC] = \frac{1}{2} \cdot AC \cdot BE = \frac{AC}{2} \cdot \sqrt{AB^2 - AE^2} = \frac{AC}{2} \cdot \sqrt{AB^2 - AB'^2} = \frac{AC}{2} \cdot \sqrt{AB^2 - AB^2} = 0,$$

contradiction. -alifenix-

Solution 2

The answer is no.

Suppose otherwise. Consider the reflection over the bisector of $\angle BAC$. This swaps rays AB and AC ; suppose E and F are sent to E' and F' . Note that the A -excircle is fixed, so line $E'F'$ must also be tangent to the A -excircle.

Since $BEFC$ is cyclic, we obtain $\angle ECB = \angle EFB = \angle EF'E'$, so $\overline{E'F'} \parallel \overline{BC}$. However, as $\overline{EF}$ is a chord in the circle with diameter $\overline{BC}$, $EF \leq BC$.

If $EF < BC$ then $E'F' < BC$ too, so then $\overline{E'F'}$ lies inside $\triangle ABC$ and cannot be tangent to the excircle.

The remaining case is when $EF = BC$. In this case, $\overline{EF}$ is also a diameter, so $BEFC$ is a rectangle. In particular $\overline{BE} \parallel \overline{CF}$. However, by the existence of the orthocenter, the lines BE and CF must intersect, contradiction.

Solution 3

The answer is no.

Suppose for the sake of contradiction that it is possible for EF to be tangent to the A -excircle. Call the tangency point T , and let S_1, S_2 denote the contact points of AB, AC with the A -excircle, respectively. Let s denote the semiperimeter of ABC . By equal tangents, we have

$$ET = ES_2, FT = FS_1 \implies EF = ES_2 + FS_1$$

It is also well known that $AS_1 = AS_2 = \frac{s}{2}$, so

$$EF = ES_2 + FS_1 = (AS_2 - AE) + (AS_1 - AF) = s - AE - AF \implies s = AE + AF + EF$$

It is well known (by an easy angle chase) that $\triangle AEF \sim \triangle ABC$, so we must have the ratio of similitude is 2. In particular,

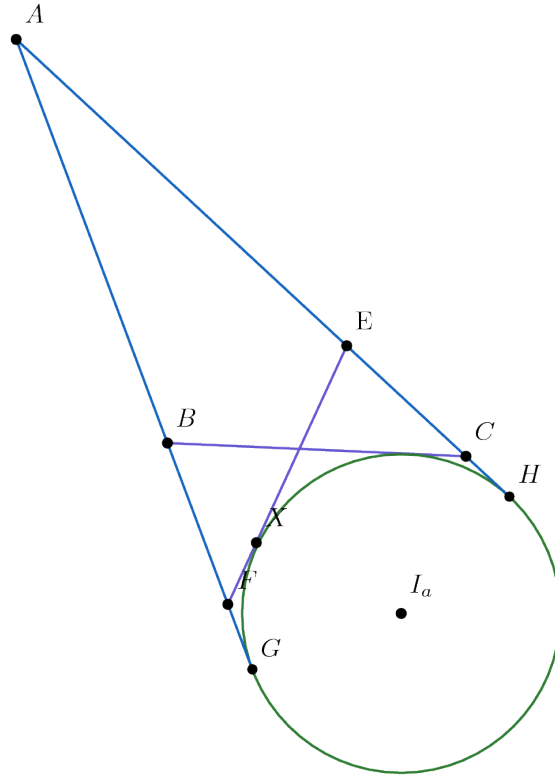
$$AB = 2 \cdot AE, AC = 2 \cdot AF$$

This results in

$$\angle ABE = 30^\circ, \angle CBF = 30^\circ \implies \angle EBC = 120^\circ$$

which is absurd since $\triangle BEC$ is a right triangle. We reached a contradiction, so we are done. ■ Mathscienceclass

Solution 4



We claim that the answer is no. We proceed with contradiction. Suppose that EF is indeed tangent to the a -excenter. Define the point of tangency to be X . Let G to be the intersection of the a -excircle with the extension of AB and H to be the intersection of the a -excircle with the extension of AC . Define I_a to be the a -excenter. It is a well-known fact that I_a lies on the angle bisector of $\angle BAC$. For convenience, let $AB = c, AC = b, BC = a, \angle CAB = A, \angle CBA = B, \angle ACB = C$. Notice that by Power of a Point:

$$FX = FG.$$

$$XE = EH.$$

Therefore, adding, we see that:

$$FE = FX + XE = FG + EH.$$

It would be rather nice if we could re-write FG . Indeed, we can:

$$FG = AG - AF = \frac{r_a}{\tan\left(\frac{A}{2}\right)} - b \cos(A)$$

and similarly for FH :

$$FH = AH - AF = \frac{r_a}{\tan\left(\frac{A}{2}\right)} - c \cos(A)$$

We now seek to re-write FE . Using the law of cosines:

$$FE^2 = AF^2 + AE^2 - 2AF \cdot AE \cos(A) = (c \cos(A))^2 + (b \cos(A))^2 - 2(bc \cos^2(A)) \cos(A) = \cos^2(A)(c^2 + b^2 - 2bc \cos(A))$$

Therefore,

$$FE = a \cos(A)$$

Putting this all together, we see that:

$$\begin{aligned} a \cos(A) &= \frac{r_a}{\tan\left(\frac{A}{2}\right)} - b \cos(A) + \frac{r_a}{\tan\left(\frac{A}{2}\right)} - c \cos(A) \iff \\ (a + b + c) \cos(A) &= \frac{2r_a}{\tan\left(\frac{A}{2}\right)} \end{aligned}$$

Now, we seek to write $\tan\left(\frac{A}{2}\right)$ in terms of the side lengths of the triangle. Notice that:

$$\begin{aligned} 2 \cos^2\left(\frac{A}{2}\right) - 1 &= \cos(A) \iff \\ 2 \cos^2\left(\frac{A}{2}\right) &= 1 + \frac{b^2 + c^2 - a^2}{2bc} = \frac{b^2 + 2bc + c^2 - a^2}{2bc} = \frac{(b + c)^2 - a^2}{2bc} = \frac{(a + b + c)(b + c - a)}{2bc} = \frac{(2s)(2s - a)}{2bc} \end{aligned}$$

where s is the semi-perimeter. We get that:

$$\cos\left(\frac{A}{2}\right) = \sqrt{\frac{s(s - a)}{bc}}$$

Using the fact that $\cos^2\left(\frac{A}{2}\right) + \sin^2\left(\frac{A}{2}\right) = 1$, we have that:

$$\sin^2\left(\frac{A}{2}\right) = 1 - \frac{b^2 + 2bc + c^2 - a^2}{4bc} = \frac{(b - c)^2 - a^2}{4bc} = \frac{(2s - 2c)(2s - 2b)}{4bc}$$

Therefore,

$$\tan\left(\frac{A}{2}\right) = \sqrt{\frac{(s - b)(s - c)}{s(s - a)}}$$

Returning to our original problem:

$$\frac{2r_a}{\sqrt{\frac{(s - b)(s - c)}{s(s - a)}}} = (a + b + c) \cos(A)$$

It is a well-known fact that $r_a = \sqrt{\frac{s(s - b)(s - c)}{s - a}}$, so:

$$\frac{2\sqrt{\frac{s(s - b)(s - c)}{s - a}}}{\sqrt{\frac{(s - b)(s - c)}{s(s - a)}}} = 2s \cos(A)$$

which implies that:

$$\cos(A) = 1 \iff A = 0$$

which is a contradiction. Hence, our original assumption that EF is tangent to the a -excircle is incorrect. ■ AopsUser101

5. (*USAJMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let n be a nonnegative integer. Determine the number of ways that one can choose $(n + 1)^2$ sets $S_{i,j} \subseteq \{1, 2, \dots, 2n\}$, for integers i, j with $0 \leq i, j \leq n$ such that:

- for all $0 \leq i, j \leq n$, the set $S_{i,j}$ has $i + j$ elements; and

- $S_{i,j} \subseteq S_{k,l}$ whenever $0 \leq i \leq k \leq n$ and $0 \leq j \leq l \leq n$

Solution 1

Note that there are $(2n)!$ ways to choose $S_{1,0}, S_{2,0} \dots S_{n,0}, S_{n,1}, S_{n,2} \dots S_{n,n}$, because there are $2n$ ways to choose which number $S_{1,0}$ is, $2n - 1$ ways to choose which number to append to make $S_{2,0}$, $2n - 2$ ways to choose which number to append to make $S_{3,0} \dots$. After that, note that $S_{n-1,1}$ contains the $n - 1$ in $S_{n-1,0}$ and 1 other element chosen from the 2 elements in $S_{n,1}$ not in $S_{n-1,0}$ so there are 2 ways for $S_{n-1,1}$. By the same logic there are 2 ways for $S_{n-1,2}$ as well so 2^n total ways for all $S_{n-1,j}$, so doing the same thing $n - 1$ more times yields a final answer of $(2n)! \cdot 2^{n^2}$.

-Stormersyle

Solution 2

There are $\frac{(2n)!}{2^n}$ ways to choose $S_{0,0}, S_{1,1} \dots S_{n,n}$. Since, there are $\binom{2n}{2}$ ways to choose $S_{1,1}$, and after that, to generate $S_{2,2}$, you take $S_{1,1}$ and add 2 new elements, getting you $\binom{2n-2}{2}$ ways to generate $S_{2,2}$. And you can keep going down the line, and you get that there are $\frac{(2n)!}{2^n}$ ways to pick $S_{0,0}, S_{1,1} \dots S_{n,n}$. Then we can fill out the rest of the grid. First, let's prove a lemma.

Lemma Claim: If we know what $S_{a,b}$ is and what $S_{a+1,b+1}$ is, then there are 2 choices for both $S_{a,b+1}$ and $S_{a+1,b}$.

Proof: Note $a \leq a + 1$ and $b \leq b + 1$, so $S_{a,b} \subseteq S_{a+1,b+1}$. Let A be a set that contains all the elements in $S_{a+1,b+1}$ that are not in $S_{a,b}$. $A = S_{a+1,b+1} - S_{a,b}$. We know $S_{a+1,b+1}$ contains total $a + b + 2$ elements. And $S_{a,b}$ contains total $a + b$ elements. That means A contains only 2 elements since $(a + b + 2) - (a + b) = 2$. Let's call these 2 elements m, n . $A = \{m, n\}$. $S_{a,b+1}$ contains 1 elements more than $S_{a,b}$ and 1 elements less than $S_{a+1,b+1}$. That 1 elements has to select from A . It's easy to see $S_{a,b+1} = S_{a,b} \cup \{m\}$ or $S_{a,b+1} = S_{a,b} \cup \{n\}$, so there are 2 choice for $S_{a,b+1}$. Same thing applies to $S_{a+1,b}$.

Filling in the rest of the grid We used our proved lemma, and we can fill in $S_{0,1}, S_{1,2} \dots S_{n-1,n}$ then we can fill in the next diagonal, until all $S_{i,j}$ are filled, where $i \leq j$. But, we haven't finished everything! Fortunately, filling out the rest of the diagonals in a similar fashion is pretty simple. And, it's easy to see that we have made $n(n + 1)$ decisions, each with 2 choices, when filling out the rest of the grid, so there are $2^{n(n+1)}$ ways to finish off.

Finishing off To finish off, we have $\frac{(2n)!}{2^n} \cdot 2^{n(n+1)}$ ways to fill in the grid, which gets us $\boxed{(2n)! \cdot 2^{n^2}}$

- AlexLikeMath

6. (USAJMO 2019) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Two rational numbers $\frac{m}{n}$ and $\frac{n}{m}$ are written on a blackboard, where m and n are relatively prime positive integers. At any point, Evan may pick two of the numbers x and y written on the board and write either their arithmetic mean $\frac{x+y}{2}$ or their harmonic mean $\frac{2xy}{x+y}$ on the board as well. Find all pairs (m, n) such that Evan can write 1 on the board in finitely many steps.

Solution

We claim that all odd m, n work if $m + n$ is a positive power of 2.

Proof: We first prove that $m + n = 2^k$ works. By weighted averages we have that $\frac{n(\frac{m}{n}) + (2^k - n)\frac{n}{m}}{2^k} = \frac{m+n}{2^k} = 1$ can be written, so the solution set does indeed work. We will now prove these are the only solutions.

Assume that $m + n \neq 2^k$, so then $m + n \equiv 0 \pmod{p}$ for some odd prime p . Then $m \equiv -n \pmod{p}$, so $\frac{m}{n} \equiv \frac{n}{m} \equiv -1 \pmod{p}$. We see that the arithmetic mean is $\frac{-1+(-1)}{2} \equiv -1 \pmod{p}$ and the harmonic mean is $\frac{2(-1)(-1)}{-1+(-1)} \equiv -1 \pmod{p}$, so if 1 can be written then $1 \equiv -1 \pmod{p}$ and $2 \equiv 0 \pmod{p}$ which is obviously impossible, and we are done.

-Stormersyle

§11 USAJMO 2020

1. (*USAJMO 2020*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $n \geq 2$ be an integer. Carl has n books arranged on a bookshelf. Each book has a height and a width. No two books have the same height, and no two books have the same width. Initially, the books are arranged in increasing order of height from left to right. In a move, Carl picks any two adjacent books where the left book is wider and shorter than the right book, and swaps their locations. Carl does this repeatedly until no further moves are possible. Prove that regardless of how Carl makes his moves, he must stop after a finite number of moves, and when he does stop, the books are sorted in increasing order of width from left to right.

Solution

Without loss of generality, let the books have widths and heights of 1, 2, 3, ..., and n . First we will show that Carl can only make finitely many moves.

Note that two books can never be swapped more than once with each other. This is because after a first swap, the wider book will be to the left of the other, so they can never be swapped back. Thus there can be at most $\binom{n}{2}$ moves.

Now we will show that the books are in increasing order of width when Carl is done making moves. Assume otherwise for the sake of contradiction. Now let $f(k)$ be the width of the book in the k th spot from the left. Since the books are not in increasing order of width, there must be i such that $f(i) > f(i+1)$, for $1 \leq i \leq n-1$.

Claim: The i th book must be shorter than the $(i+1)$ th book.

Proof: Assume otherwise for contradiction. Then the i th book must have started to the right of the $(i+1)$ th book, since the i th book is taller. However, the i th book is now to the left of the $(i+1)$ th book, so they must have been swapped at some point. But since $f(i) > f(i+1)$, the i th book is also wider. But the i th book started to the right, so since it was wider it cannot have been swapped with the $(i+1)$ th book, so we arrive at a contradiction. ■

From the claim, the i th book must be shorter than the $(i+1)$ th book. But since the i th book is also wider by definition (as $f(i) > f(i+1)$), Carl can swap those

two books, so he still has legal moves and we arrive at a contradiction, so we are done. ■

2. (*USAJMO 2020*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let ω be the incircle of a fixed equilateral triangle ABC . Let ℓ be a variable line that is tangent to ω and meets the interior of segments BC and CA at points P and Q , respectively. A point R is chosen such that $PR = PA$ and $QR = QB$. Find all possible locations of the point R , over all choices of ℓ .

Solution 1

Call a point *good* if it is a possible location for R . Let the incircle of $\triangle ABC$ touch BC at D , AC at E , and ℓ at T . Also, let the center of the incircle be I . Clearly, a point is good iff it lies on the circle containing A with center P as well as the circle containing B with center Q . Call these circles ω_1 and ω_2 , respectively.

Note that point T can only lie on minor arc $\widehat{DE}$ (excluding the endpoints).

Claim: A point X is good iff $XT \perp PQ$ and $XT = AD = BE$.

Proof: WLOG, let X be on the same side of PQ as I . Then we have that T, I, X are collinear. In particular, we have $\angle XTP = 90^\circ$. Then $TX = AD$, $PT = PD$, $\angle XTP = \angle ADP$, so that $\triangle ADP \cong \triangle XTP \rightarrow PA = PX$. Similarly, $QB = QX$, so X is good. Then X' , the reflection of X across T , is also good. But ω_1 and ω_2 have at most two intersections, so X and X' must be these intersections, and since a point is good iff it lies on both circles, we are done.

Now, we know that R, I, T are collinear. Then we have two cases:

Case 1: R and I lie on the same side of PQ . Then we have $RI + IT = AI + ID = BI + IE$, so that $RI = AI = BI$. Then we have that R, A, B lie on a circle with center I . Note that because T lies on $\widehat{DE}$, R must lie on $\widehat{AB}$. So, one of the solutions is R in arc AB (excluding the endpoints).

Case 2: R and I lie on opposite sides of PQ . Then extend ID out from D to X such that $DX = TR$, and extend IE out from E to Y such that $EY = TR$. Then we see that $IR = IX = IY$, so that R lies on arc XY (excluding the endpoints).

Solution 2

We claim that R can lie on minor arc AB of the circumcircle of triangle ABC , and it can also lie on the dilation of this arc about the center of triangle ABC with a factor of -2 .

Let D, E , and F be the feet of the angle bisectors from points A, B , and C respectively. Trivially, DEF is also the medial triangle, orthic triangle, and contact triangle (ABC is equilateral).

Let I be the incenter of ABC . Trivially, I is the centroid, orthocenter, and circumcenter of ABC (ABC is equilateral). Also, $AD = BE = CF = 3r$ where r is the radius of circle ω (This is trivial). T is the point of tangency of ω and segment $\overline{PQ}$.

R has to lie on the intersection of circles ω_1 (center P , radius PA) and ω_2 (center Q , radius QB), and for each choice of P , there exist two locations for R . The location that we claim to lie on the minor arc AB of the circumcircle of ABC shall be denoted M , and the other location shall be denoted N .

Define triangle XYZ to be the homothety of triangle ABC about I with a factor of -2.

Critical claim: M, T, I, and N are collinear.

Proof: First we shall prove that T lies on MN using phantom points.

Let the intersection of MN and PQ be denoted as K. We shall prove that K and T are the same point. Let $PT = p$ and $QT = q$. Because of the equal tangent theorem, $PD = PT = p$ and $QE = QT = q$. Hence, by the pythagorean theorem (recall $AD = BE = 3r$), $PA^2 = 9r^2 + p^2$ and $QB^2 = 9r^2 + q^2$. Since $PN = PA$ and $QN = QB$, then $PN^2 = 9r^2 + p^2$ and $QN^2 = 9r^2 + q^2$.

PQ is the perpendicular bisector of MN because MN is the radical axis of ω_1 and ω_2 . Hence, M is the reflection of N across K. Also, NK is the altitude of triangle PNQ, so $PK^2 - QK^2 = PN^2 - QN^2 = p^2 - q^2$ by using the pythagorean theorem and earlier expressions for PN^2 and QN^2 . However, $PK + QK = PQ = p + q$. Now, we have a system of equations to solve for PK and QK in terms of p and q.

Dividing the first equation by the second (we can do this because $p+q$ is always nonzero), we get $PK - QK = p - q$. Combining this with our $PK+QK$ result, we get $PK = p$ and $QK = q$. However, $PT = p$ and $QT = q$, and only one point can exist on PQ for which this result holds true. As a result, K and T are the same point, otherwise it is a contradiction. Hence M, T, and N are collinear.

$IT \parallel MN$. This is because both MN and IT are perpendicular to PQ (IT is perpendicular to PQ because PQ is a tangent with point of tangency T). However, both lines share point T, as discussed earlier. Hence, IT and MN are the same line, and M, T, I, and N are collinear.

In fact, from our earlier results from the lengths of PN and PT, we can use the pythagorean theorem to get that $NT = 3r$, a result that is always true and independent of P and Q! Also, because M is the reflection of N over K (which is the same as T), $MT = 3r$ also. However, T varies based on P and Q. On the other hand, $IT = r$ and M, T, I and N are collinear. Remembering our earlier definitions of M and N, we get that $MI = 2r$ and $IN = 4r$, with M on the opposite side of N and T from I. Hence, M can be taken to N with a homothety about I with a factor of -2, and T can be taken to M with a homothety about I with a factor of -2. Since, trivially, the circumradius of ABC is $2r$ (ABC is equilateral), it seems like M can lie anywhere on the circumcircle of ABC.

However, we must take into account the restrictions on P and Q. This limits T to only minor arc DE on the incircle of ABC, hence, because of our earlier homothety statement, M is restricted to minor arc AB on the circumcircle of ABC. Because of our homothety statement about N, N has to lie on minor arc XY on the circumcircle of triangle XYZ.

Because we defined both M and N to be possible locations for R, R can only lie on minor arc AB of the

-QED

-Solution by thanosaops

3. (*USAJMO 2020*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

An empty $2020 \times 2020 \times 2020$ cube is given, and a 2020×2020 grid of square unit cells is drawn on each of its six faces. A beam is a $1 \times 1 \times 2020$ rectangular prism. Several beams are placed inside the cube subject to the following conditions:

- The two 1×1 faces of each beam coincide with unit cells lying on opposite faces of the cube. (Hence, there are $3 \cdot 2020^2$ possible positions for a beam.)
- No two beams have intersecting interiors.
- The interiors of each of the four 1×2020 faces of each beam touch either a face of the cube or the interior of the face of another beam.

What is the smallest positive number of beams that can be placed to satisfy these conditions?

Solution

Place the cube in the xyz-coordinate, with the positive x-axis pointing forward, the positive y-axis pointing right, and the positive z-axis pointing up. Let the position of a unit cube be (x, y, z) if it is centered at (x, y, z) . Place the $2020 \times 2020 \times 2020$ cube so that the edges are parallel to the axes, and two of its corners are at $(1, 1, 1)$ and $(2020, 2020, 2020)$. Now call a beam z-oriented if its endpoints differ only in z-coordinates, and similarly call it x-oriented or y-oriented if the endpoints differ in x- or y-coordinates, respectively.

We claim that the answer is $\boxed{3030}$. First we will prove 3030 suffices. Place the first beam so its endpoints lie at $(1, 1, 1)$, and $(2020, 1, 1)$. Place the 2nd beam so its endpoints lie at $(1, 1, 2)$ and $(1, 2020, 2)$. Place the 3rd beam so its endpoints lie at $(2, 2, 1)$ and $(2, 2, 2020)$. Now it is clear that the only faces among these three beams that are not touching the faces of the $2020 \times 2020 \times 2020$ cube or the face of another beam are the top face of beam 2, and the front and right faces of beam 3.

Now place the 4th beam from $(1, 3, 3)$ to $(2020, 3, 3)$, 2 units up and right of beam 1. Then place beam 5 from $(3, 1, 4)$ to $(3, 2020, 4)$, 2 units forward and up of beam 2. Place beam 6 from $(4, 4, 1)$ to $(4, 4, 2020)$, 2 units forward and right of beam 3. Note that these 3 beams will touch the top face of beam 2, and the front and right faces of beam 3. Now the only faces that are not touching another face are the top of beam 5, and the front and right of beam 6.

We can repeat this process 1010 times, placing the beams in groups of 3 so that beam $3n + 1$ is 2 units up and right of beam $3n - 2$, beam $3n + 2$ is 2 units forward and up of beam $3n - 1$, and beam $3n + 3$ is 2 units forward and right of beam $3n$, for all $1 \leq n \leq 1009$. After placing each set of 3 as such, the faces of the previous beams will all satisfy condition 3. And for the last set of 3 (that is, beams 3028, 3029, and 3030), their faces will satisfy condition 3 as well, since the top face of beam 3029 and the front/right faces of beam 3030 will touch the cube (again, note that all other faces will be touching another beam).

Thus, we have shown that 3030 beams suffices, so we will now show that 3030 beams is the minimum. Let there be A x-oriented beams, B y-oriented beams, and C z-oriented beams. Now look each $2020 \times 1 \times 2020$ slice of the cube, parallel to the xz-plane. Note there are 2020 such slices. There are two cases. Either $B = 2020^2$, or at least one of A or C is nonzero. In the first case, we clearly have at least $2020^2 > 3030$ beams as desired. In the second case, we must have at least one beam in each $2020 \times 1 \times 2020$ slice, in order to satisfy condition 3. If there is some slice without a beam, then there will be some x- or z-oriented beam with no beams touching its top or bottom face. Additionally, notice that only x- or z-oriented

beams can fit in a $2020 \times 1 \times 2020$ slice, so $A + C \geq 2020$. Similarly, we must have $A = 2020^2$, $C = 2020^2$, or both $A + B \geq 2020$ and $B + C \geq 2020$. In every case, the number of beams, $A + B + C$, will be at least 3030 as desired. ■

4. (*USAJMO 2020*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $ABCD$ be a convex quadrilateral inscribed in a circle and satisfying $DA < AB = BC < CD$. Points E and F are chosen on sides CD and AB such that $BE \perp AC$ and $EF \parallel BC$. Prove that $FB = FD$.

Solution

Let G be the intersection of AE and $(ABCD)$ and H be the intersection of DF and $(ABCD)$.

Claim: $GH \parallel FE \parallel BC$

By Pascal's on $GDCBAH$, we see that the intersection of GH and BC , E , and F are collinear. Since $FE \parallel BC$, we know that $HG \parallel BC$ as well. ■

Note that since all cyclic trapezoids are isosceles, $HB = GC$. Since $AB = BC$ and $EB \perp AC$, we know that $EA = EC$, from which we have that $DGCA$ is an isosceles trapezoid and $DA = GC$. It follows that $DA = GC = HB$, so $BHAD$ is an isosceles trapezoid, from which $FB = FD$, as desired. ■

Solution 2

Let $G = \overline{FE} \cap \overline{BC}$, and let $H = \overline{AC} \cap \overline{BE}$. Now let $x = \angle ACE$ and $y = \angle BCA$. From $BA = BC$ and $\overline{BE} \perp \overline{AC}$, we have $AE = EC$ so $\angle EAC = \angle ECA = x$. From cyclic quadrilateral $ABCD$, $\angle ABD = \angle ACD = x$. Since $BA = BC$, $\angle BCA = \angle BAC = y$.

Now from cyclic quadrilateral $ABCD$ and $\overline{FE} \parallel \overline{BC}$ we have $\angle FAC = \angle BAC = \pi - \angle BCD = \pi - \angle FED$. Thus F, A, D , and E are concyclic, and $\angle DFG = \angle DAE = \angle DAC - \angle EAC = \angle DBC - x$. Let this be statement 1.

Now since $\overline{AH} \perp \overline{BH}$, triangle ABC gives us $\angle BAH + \angle ABG = \frac{\pi}{2}$. Thus $y + x + \angle GBE = \frac{\pi}{2}$, or $\angle GBE = \frac{\pi}{2} - x - y$.

Right triangle BHC gives $\angle HBC = \frac{\pi}{2} - y$, and $\overline{BC} \parallel \overline{FE}$ implies $\angle BEG = \angle HBC = \frac{\pi}{2} - y$.

Now triangle BGE gives $\angle BGE = \pi - \angle BEG - \angle GBE = \pi - (\frac{\pi}{2} - y) - (\frac{\pi}{2} - x - y) = x + 2y$. But $\angle FGB = \angle BGE$, so $\angle FGB = x + 2y$. Using triangle FGD and statement 1 gives

$$\begin{aligned} \angle FDG &= \pi - \angle DFG - \angle FGB \\ &= \pi - (\angle DBC - x) - (x + 2y) \\ &= \pi - (\angle GBE + \angle EBC - x) - (x + 2y) \\ &= \pi - ([\frac{\pi}{2} - x - y] + [\frac{\pi}{2} - y] - x) - (x + 2y) \\ &= x \\ &= \angle FBD \end{aligned}$$

Thus, $\angle FDB = \angle FBD$, so $\boxed{FB = FD}$ as desired. ■

Solution 3 (Angle-Chasing)

Proving that $FB = FD$ is equivalent to proving that $\angle FBD = \angle FDB$. Note that $\angle FBD = \angle ACD$ because quadrilateral $ABCD$ is cyclic. Also note that $\angle BAC = \angle ACB$ because $AB = BC$. $AE = EC$, which follows from the facts that $BE \perp AC$ and $AB = AC$, implies that $\angle CAE = \angle ACE = \angle ACD = \angle FBD$. Thus, we would like to prove that triangle FBD is similar to triangle AEC . In order for this to be true, then $\angle BFD$ must equal $\angle AEC$ which implies that $\angle AFD$ must equal $\angle AED$. In order for this to be true, then quadrilateral $AFED$ must be cyclic. Using the fact that $EF \parallel BC$, we get that $\angle AFE = \angle ABC$, and that $\angle FED = \angle BCE$, and thus we have proved that quadrilateral $AFED$ is cyclic. Therefore, triangle FDB is similar to isosceles triangle AEC from AA and thus $FB = FD$.

-xXINs1c1veXx

Solution 4

BE is perpendicular bisector of AC , so $\angle ACE = \angle EAC$. FE is parallel to BC and $ABCD$ is cyclic, so $AFED$ is also cyclic. $\angle AFD = \angle AED = 2\angle ACE = 2\angle FBD$. Hence, $\angle FBD = \angle BDF$, $FD = FB$.

Mathdummy

Solution 5

Let G be on AB such that $GD \perp DB$, and $H = GD \cap EB$. Then $\angle ADB = \angle EDB = 90^\circ - \angle ABE \implies \triangle DAE$ is the orthic triangle of $\triangle HGB$. Thus, F is the midpoint of GB and lies on the $\perp$ bisector of DB .

Solution 6

Let FE meet AC at J , BE meet AC at S , connect AE, SD . Denote that $\angle BCA = \alpha$; $AB = BC$, $\angle BAC = \angle BCA = \alpha$, since EF is parallel to BC , $\angle AJF = \angle ACB = \alpha$. $\angle AJF$ and $\angle EJS$ are vertical angle, so they are equal to each other. $BE \perp AC$, $\angle JES = 90^\circ - \alpha$, since $\angle EFB = \angle AJF + \angle FAJ = 2\alpha$, we can express $\angle FBE = 180^\circ - 2\alpha - (90^\circ - \alpha) = 90^\circ - \alpha = \angle FEB$, leads to $FE = FB$.

Notice that quadrilateral $AFED$ is a cyclic quadrilateral since $\angle ADE + \angle AFE = \angle ADE + \angle ABC = 180^\circ$.

Assume $\angle ECA = \beta$, $\triangle AES$ is congruent to $\triangle CES$ since $AS = AS$, $\angle ASE = \angle BSE$, $SE = SE$ (SAS), so we can get $\angle EAS = \beta$. Let the circumcircle of $AFED$ meet AC at Q . Now notice that $\widehat{QE} = \widehat{QE}$, $\angle QAE = \angle QDE = \beta$; similarly, $\widehat{FQ} = \widehat{FQ}$; $\angle FDQ = \angle FAQ = \alpha$.

$\angle FDE = \alpha + \beta$; $\angle FED = \angle BCD = \alpha + \beta$, it leads to $FD = FE$.

since $FE = FB$; $FD = FE$, $DF = BF$ as desired bluesoul

5. (USAJMO 2020) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Suppose that $(a_1, b_1), (a_2, b_2), \dots, (a_{100}, b_{100})$ are distinct ordered pairs of non-negative integers. Let N denote the number of pairs of integers (i, j) satisfying $1 \leq i < j \leq 100$ and $|a_i b_j - a_j b_i| = 1$. Determine the largest possible value of N over all possible choices of the 100 ordered pairs.

Solution

Call the pair (i, j) good if $1 \leq i < j \leq 100$ and $|a_i b_j - a_j b_i| = 1$. Note that we can reorder the pairs $(a_1, b_1), (a_2, b_2), \dots, (a_{100}, b_{100})$ without changing the number of good pairs. Thus, we can reorder them so that $a_1 \leq a_2 \leq \dots \leq a_{100}$. Furthermore, reorder them so that if $a_i = a_j$ for some $i < j$, then $b_i < b_j$.

Now I claim the maximum value of N is $\boxed{197}$. First, we will show $N \leq 197$.

The key claim is that for any fixed integer m , $1 \leq m \leq 100$, then there are at most 2 values of i with $1 \leq i < m$ with the pair (i, m) good. We will prove this with contradiction.

Assume there are 3 integers i, j , and k that are less than m , such that (i, m) , (j, m) , and (k, m) are all good. Note that $\gcd(a_m, b_m)$ divides $a_i b_m - a_m b_i = \pm 1$, so $\gcd(a_m, b_m) = 1$. From $a_i b_m - a_m b_i = \pm 1$, we have

$$a_i b_m \equiv \pm 1 \pmod{a_m}$$

Similarly,

$$a_j b_m \equiv \pm 1 \pmod{a_m}$$

$$a_k b_m \equiv \pm 1 \pmod{a_m}$$

From $\gcd(a_m, b_m) = 1$ we have

$$a_i \equiv \pm a_j \equiv \pm a_k \pmod{a_m}$$

But since i, j , and k are less than m , from our reordering we have $0 \leq a_i < a_m$, $0 \leq a_j < a_m$, and $0 \leq a_k < a_m$. Thus our congruence gives us that $a_j = a_i$ or $a_j = a_m - a_i$, and similar for a_k . Thus at least two of a_i, a_j, a_k are equal. WLOG let them be a_i and a_j . Now we have 4 cases.

Case 1: $a_m > 2$

We have $a_i b_m - a_m b_i = \pm 1$ and $a_j b_m - a_m b_j = \pm 1$, so since $a_i = a_j$ we can subtract the two equations to get $a_m b_i - a_m b_j = \pm 2, 0$. But $a_m > 2$, so $|a_m(b_i - b_j)| \leq 2$ implies that $b_i = b_j$. But all pairs are distinct and $a_i = a_j$, so this is a contradiction.

Case 2: $a_m = 2$

We have 2 subcases.

Subcase 2a: $a_i \equiv a_j \equiv a_k \equiv 1 \pmod{a_m}$

Clearly we must have $a_i = a_j = a_k = 1$ in this case, as all 3 variables are less than $a_m = 2$. Then $1 \cdot b_m - a_m b_i = \pm 1$, $1 \cdot b_m - a_m b_j = \pm 1$, and $1 \cdot b_m - a_m b_k = \pm 1$. So at least 2 of b_i, b_j , and b_k must be equal, but this contradicts the fact that all pairs (a_x, b_x) are distinct.

Subcase 2b: $a_i \equiv a_j \equiv a_k \equiv 0 \pmod{a_m}$

a_i must be even in this case, but since a_m is also even then $a_i b_m - a_m b_i$ must be even, which contradicts $|a_i b_m - a_m b_i| = 1$.

Case 3: $a_m = 1$

We must have $a_i = 0, 1$, and similarly for a_j and a_k . Thus at least 2 of the 3 must be equal. Now we have 2 subcases again.

Subcase 3a: 2 of the 3 are equal to 0

WLOG let $a_i = a_j = 0$. Then $a_i b_m - a_m b_i = \pm 1$ and $a_m = 1$ so $b_i = \pm 1$. Since b_i is nonnegative, $b_i = 1$. Similarly $b_j = 1$, but all pairs are distinct so we have a contradiction.

Subcase 3b: 2 of the 3 are equal to 1

WLOG let $a_i = a_j = 0$. Then $a_i b_m - a_m b_i = \pm 1$, so $b_m - b_i = \pm 1$. But $i < m$ and $a_i = a_m = 1$, so by our ordering of the pairs we have $b_i < b_m$ so $b_m - b_i = 1$. But similarly $b_m - b_j = 1$, so $b_i = b_j$, which is a contradiction to all pairs being distinct.

Case 4: $a_m = 0$

We must have $a_i = 0$, since $0 \leq a_i \leq a_m = 0$. Thus $a_i b_m - a_m b_i = 0$, but $|a_i b_m - a_m b_i| = 1$, contradiction. ■

Thus our claim is proved. Now for $m = 1$, there is clearly no i with $1 \leq i < m \leq 100$ and (i, m) good. For $m = 2$, there is at most one value of i with $1 \leq i < m \leq 100$ and (i, m) good (namely, this is when $i = 1$). By our claim, for all $2 < m \leq 100$ there are at most 2 values of i with $1 \leq i < m \leq 100$ and (i, m) good, so adding these up gives at most $2 \cdot 98 + 1 = 197$ good pairs. Now we will show that 197 good pairs is achievable.

The sequence $(1, 1), (1, 2), (2, 3), \dots, (99, 100)$ has 197 good pairs - namely, $(1, n)$ is good for all $1 < n \leq 100$, and $(i, i + 1)$ is good for all $1 < i < 100$. This adds up for a total of 197 good pairs, so we are done. ■

Solution 2

We claim the answer is $2n - 3$ (so 197), where 100 is to be replaced with n .

The problem essentially counts unordered pairs (P, Q) of lattice points so that POQ has area $\frac{1}{2}$ due to Shoelace (O is the origin).

For a construction, take the points $(0, 1), (1, 1), (1, 2), \dots, (1, n - 1)$.

We now show by induction that $2n - 3$ is the maximum, where $n = 2$ obviously holds as a base case.

To reduce to the inductive hypothesis of $n - 1$ points from n points, we are allowed to consider the point with the greatest sum of coordinates R .

The key claim is that R is part of at most 2 pairs (triangles) with area $\frac{1}{2}$.

Before we proceed, note that R must have relatively prime coordinates (m, n) , or else any triangles found can be reduced down by a greatest common factor along OR , impossible since $\frac{1}{2}$ is the minimum positive area.

Call the hypothetical point(s) X . With triangle XOR with area $\frac{1}{2}$, we know the length of OR , so indeed the locus of X is two lines evenly spaced next to OR each parallel to OR (before we consider the lattice condition).

When we add in the lattice condition, we see that each of these two lines contain at most one valid X . This is because if two lattice points with non-negative integers lied on one of these lines, one would have to be a multiple of m more than the other on the x -coordinates and a multiple of n more on the y -coordinates. This contradicts the coordinate sum maximality of R .

Thus, by deleting R , we lose at most two pairs (X, R) , completing the induction and the problem.

6. (*USAJMO 2020*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $n \geq 2$ be an integer. Let $P(x_1, x_2, \dots, x_n)$ be a nonconstant n -variable polynomial with real coefficients. Assume that whenever $r_1, r_2, \dots, r_n$ are real numbers, at least two of which are equal, we have $P(r_1, r_2, \dots, r_n) = 0$. Prove that $P(x_1, x_2, \dots, x_n)$ cannot be written as the sum of fewer than $n!$ monomials. (A monomial is a polynomial of the form $cx_1^{d_1}x_2^{d_2}\dots x_n^{d_n}$, where c is a nonzero real number and $d_1, d_2, \dots, d_n$ are nonnegative integers.)

§12 USAJMO 2021

1. (*USAJMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $\mathbb{N}$ denote the set of positive integers. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for positive integers a and b ,

$$f(a^2 + b^2) = f(a)f(b) \text{ and } f(a^2) = f(a)^2.$$

Solution

I claim that the only function f that satisfies the constraints outlined within the problem is the function $f(n) = 1$ for all positive integers n .

We will proceed with strong induction. The base case is simple, as plugging $a = 1$ into the second equation given within the problem gives $f(1) = f(1)^2$. Since $f(n)$ can only return a positive integer value, we have that $f(1) = 1$.

Now we proceed with the inductive step. If the next number n is either a perfect square or can be represented as a sum of two perfect squares, then obviously $f(n) = 1$, as it is either the product of two f -values that are both equal to 1 from the inductive assumption or is the square of an f -value that is equal to 1, again due to the inductive assumption. Otherwise, we can use the Sum of Two Squares Theorem, which tells us that n has at least one prime in its prime factorization that is $3 \pmod{4}$ and is raised to an odd power.

Lemma 1: Given that $a^2 + b^2 = c^2$ and $f(b) = f(c) = 1$, we then have $f(a) = 1$.

Proof: Note that the first condition in the problem tells us that $f(a^2 + b^2) = f(a)f(b)$, or $f(c^2) = f(a)f(b)$. Using the second condition gives us $f(c)^2 = f(a)f(b)$. Plugging in the values of $f(b)$ and $f(c)$ gives us that $f(a) = 1$.

Now we will attempt to repeatedly remove prime factors that are $3 \pmod{4}$ and taken to an odd power, and we will move from the largest prime down to the smallest prime that satisfies above conditions. The prime factors will be removed by constructing a Pythagorean Triple with the prime being the smallest leg in the form of $p, \frac{p^2-1}{2}, \frac{p^2+1}{2}$ (this will always work as $p \neq 2 \pmod{4}$, and this method works via Lemma 1). We will then prove the ending numbers that we achieve via removing all the $(\pmod{4})$ primes (which I will refer to as "tips") are equal to 1 b/c they can be expressed as the sum of two squares or is a perfect square (Sum of Two Squares Theorem). For example, if we took the number 21, we would first aim to remove the 7 by splitting it into 24 and 25, so 21 would become 72 and 75. 75 is divisible by 3 to an odd power, so we transform it into 100 and 125. These two don't have any divisors that are $3 \pmod{4}$ raised to an odd power so we leave it alone, as sum of two squares will work on them or they are perfect squares. 72

does have a prime factor that is $3 \pmod{4}$, but it is an even power so we leave it alone. In this case, the tips are 72, 100, and 125. However, now we need to prove two key facts: using this Pythagorean Triple Method will never generate another $3 \pmod{4}$ prime that is bigger than the current one we are working on or create more primes or keep the same number of primes in the tips' prime factorizations., as otherwise it could cause an infinite cycle, and also we must prove when we use Sum of Two Squares/Perfect Square given in the second condition on the tips the square root of the square(s) used will never be greater than or equal to n .

The first claim can be proved rather simply. Note that $\frac{p^2-1}{2}$ can have no prime factors greater than or equal to p , as it can be factored as $(p-1)(\frac{p+1}{2})$, which are both less than p and are integers($p+1$ must be an integer due to p must being odd(not equal to 2)). For $\frac{p^2+1}{2}$, we can prove something a bit more general.

Lemma 2: For any positive integer n , all prime factors of $n^2 + 1$ must be $1 \pmod{4}$.

Proof: Note that if $n^2 + 1 \equiv 0 \pmod{p}$, then we also must have $(n^2 + 1)(n^2 - 1) = (n^4 - 1) \equiv 0 \pmod{p}$, or $n^4 \equiv 1 \pmod{p}$. Now we can apply Fermat's Little Theorem to obtain $n^{p-1} \equiv 1 \pmod{p}$. Note that since n and n^2 are obviously not $1 \pmod{p}$, as $n^2 \equiv -1 \pmod{p}$, we have that $p-1$ is a multiple of 4, or $p \equiv 1 \pmod{4}$.

We can simply apply Lemma 2 to $p^2 + 1$ as 2 is obviously not $3 \pmod{4}$, and this means that a prime factor $3 \pmod{4}$ will never be generated from this term. This completes the first of our two claims.

Now we proceed to the second of our two claims. Note that every time we use the method on n based on prime p , we will multiply by around $\frac{p}{2}$, clearly less than p (if we multiply by p we would get p^2 which is clearly greater than $\frac{p^2+1}{2}$). We will never have to use the same prime twice in our method, so at max in the end we will multiply n by the product of a little less than all $3 \pmod{4}$ primes that divide it, which is less than n itself for n greater than or equal to 3(the smallest $3 \pmod{4}$ prime), meaning that the largest number that we must use two squares on that is generated by our method is less than n^2 . We need to prove that the two squares that sum to this are both less than n , which is quite trivial, as they are less than $\sqrt{n^2}$, which obviously means it must be less than n . This proves the second of our claims.

This completes the second case of the inductive step, and therefore completes both the induction and the problem. Solution by hyxue

Solution 2 (Taken from Twitch Solves ISL)

The Answer is $\boxed{f \equiv 1}$ which works but we want to prove that it's the only one. Claim: If $f(a) = f(b) = 1$ and $a > b$, then $f(a^2 - b^2) = f(2ab) = 1$. Proof: We can write $1 = f(a) = f(b) = f(a^2 + b^2) = \sqrt{f((a^2 + b^2)^2)} = \sqrt{f((a^2 - b^2)^2 + (2ab)^2)} = \sqrt{f(a^2 - b^2)f(2ab)}$. We set it to $a = b = 1$ and we get that $f(1) = f(2) = 1$. Easily by Induction it shows $f(n) = 1$. We take if $n = 2k$ take $(u, v) = (k, 1)$ which $2uv = n$. If $n = 2k + 1$, just take $(u, v) = (k + 1, k)$ which $u^2 - v^2 = n$. Thus the only answer is $\boxed{f \equiv 1}$ and we are done.

Solution 3 (clear Solution 2)

The answer is $\boxed{f \equiv 1}$, which works. To show it is necessary, we first get $f(1) = f(1)^2$, so $f(1) = 1$. Then, we get $f(2) = f(1^2 + 1^2) = f(1)^2 = 1$.

The following claim finishes the problem via induction:

Claim: If $f(n) = f(n-1) = 1$ for $n \geq 2$, we have $f(2n) = f(2n-1) = 1$.

Proof: Note that $f(n^2 + 1) = f(n)f(1) = 1$, so

$$1 = f(n^2 + 1)^2 = f((n^2 + 1)^2) = f(2n)f(n^2 - 1)$$

implies $f(2n) = 1$.

Note that $f(2n^2 - 2n + 1) = f(n)f(n-1) = 1$, so

$$1 = f(2n^2 - 2n + 1)^2 = f((2n^2 - 2n + 1)^2) = f(2n-1)f(2n^2 - 2n)$$

implies $f(2n-1) = 1$. $\square$

Thus, the only solution is $f \equiv 1$.

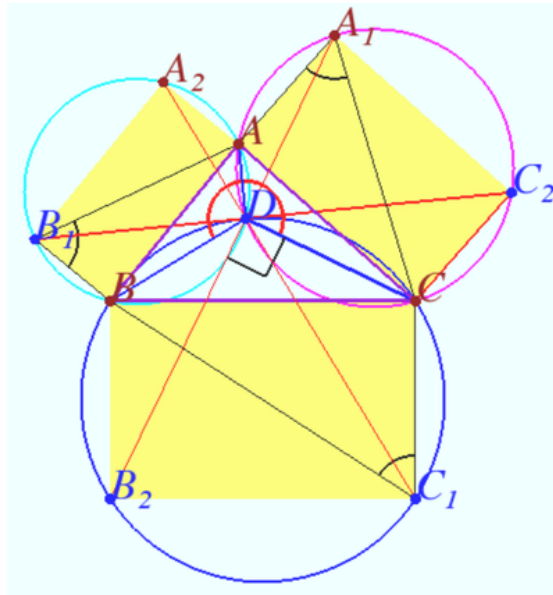
2. (*USAJMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Rectangles BCC_1B_2 , CAA_1C_2 , and ABB_1A_2 are erected outside an acute triangle ABC . Suppose that

$$\angle BC_1C + \angle CA_1A + \angle AB_1B = 180^\circ.$$

Prove that lines B_1C_2 , C_1A_2 , and A_1B_2 are concurrent.

Solution



Let D be the second point of intersection of the circles (AB_1B) and (AA_1C) . Then:

$$\angle ADB = 180^\circ \sim \angle AB_1B, \quad \angle ADC = 180^\circ \sim \angle AA_1C$$

$$\angle BDC = 360^\circ \sim \angle ADB \sim \angle ADC$$

$$= 360^\circ \sim (180^\circ \sim \angle AB_1B) \sim (180^\circ \sim \angle AA_1C)$$

$$= \angle AB_1B + \angle AA_1C$$

$$\angle BDC + \angle BC_1C = 180^\circ$$

Therefore, $BDCC_1B_2$ is cyclic with diameters BC_1 and CB_2 , and thus $\angle CDB_2 = 90^\circ$. Similarly, $\angle CDA_1 = 90^\circ$, meaning points A_1 , D , and B_2 are collinear.

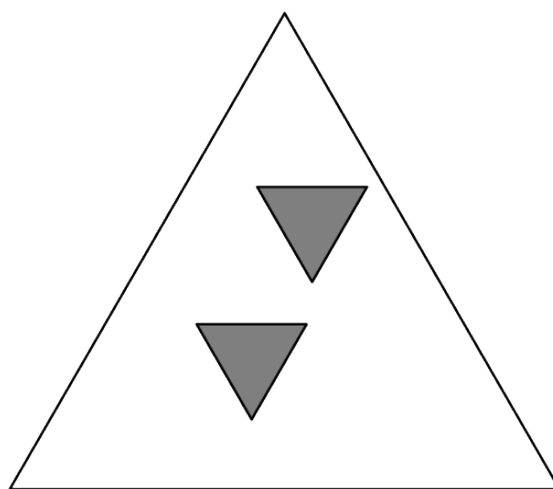
Similarly, the points A_2, D, C_1 and C_2, D, B_1 are collinear.

(After USAMO 2021 Solution Notes – Evan Chen)

vladimir.shelomovskii@gmail.com, vvsss

3. (*USAJMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

An equilateral triangle Δ of side length $L > 0$ is given. Suppose that n equilateral triangles with side length 1 and with non-overlapping interiors are drawn inside Δ , such that each unit equilateral triangle has sides parallel to Δ , but with opposite orientation. (An example with $n = 2$ is drawn below.)



Prove that

$$n \leq \frac{2}{3}L^2.$$

Solution

I will use the word "center" to refer to the centroid of any equilateral triangle.

Consider the center of each triangle we place and the surrounding area that is "denied" placement for any centers. With a little bit of testing and sliding the equilateral triangle along each side of the center triangle, it is easy to see that this region forms a hexagon of side length 1 where any other center being in this hexagon would mean that the two equilateral triangles at those centers would overlap.

Let us define a region for each center as its "personal space", where its personal space intersecting with any other center's personal space results in the two equilateral triangles intersecting. Given the above fact, it is easy to see that this personal space region encapsulates a hexagon of sidelength $\frac{1}{2}$ centered at each center by simply scaling the denied region by a factor of $\frac{1}{2}$. The area of each personal space would simply be $6\left(\frac{\sqrt{3}}{4}\left(\frac{1}{2}\right)^2\right) = \frac{3\sqrt{3}}{8}$. Since each personal space cannot intersect, the maximum amount of equilateral triangles would be bounded by

$\frac{\text{Area of Big Triangle}}{\text{Area of individual personal space}} = \frac{L^2 \frac{\sqrt{3}}{4}}{\frac{3\sqrt{3}}{8}} = \frac{2}{3}L^2$, which completes the problem. (Unrigorous solution that I hope someone else can improve on hyxue)

4. (*USAJMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Carina has three pins, labeled A, B , and C , respectively, located at the origin of the coordinate plane. In a move, Carina may move a pin to an adjacent lattice point at distance 1 away. What is the least number of moves that Carina can make in order for triangle ABC to have area 2021?

(A lattice point is a point (x, y) in the coordinate plane where x and y are both integers, not necessarily positive.)

Solution 1

The answer is 128, achievable by $A = (10, 0), B = (0, -63), C = (-54, 1)$. We now show the bound.

We first do the following optimizations:

- if you have a point goes both left and right, we may obviously delete both of these moves and decrease the number of moves by 2.
- if all of A, B, C lie on one side of the plane, for example $y > 0$, we shift them all down, decreasing the number of moves by 3, until one of the points is on $y = 0$ for the first time.

Now we may assume that $A = (a, d), B = (b, -e), C = (-c, f)$ where $a, b, c, d, e, f \geq 0$. Note we may still shift all A, B, C down by 1 if $d, f > 0$, decreasing the number of moves by 1, until one of d, f is on $y = 0$ for the first time. So we may assume one of (a, b) and (d, f) is 0, by symmetry. In particular, by shoelace the answer to 2021 JMO Problem 4 is the minimum of the answers to the following problems:

Case 1 (where $a = d = 0$) if $wx - yz = 4042$, find the minimum possible value of $w + x + y + z$.

Case 2 (else) $wy + xy + xz = (w + x)(y + z) - wz = 4042$, find the minimum possible value of $w + x + y + z$.

Note that $(m + n)^2 = 4mn + (m - n)^2$ so if $m + n$ is fixed then mn is maximized exactly when $|m - n|$ is minimized. In particular, if $m + n \leq 127$ then $mn - op \leq mn \leq 63 * 64 = 4032 < 4042$ as desired.

5. (*USAJMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

A finite set S of positive integers has the property that, for each $s \in S$, and each positive integer divisor d of s , there exists a unique element $t \in S$ satisfying $\gcd(s, t) = d$. (The elements s and t could be equal.) Given this information, find all possible values for the number of elements of S .

Solution

The answer is 0 (left out by problem author) or 2^n for any non-negative integer n .

To construct $|S| = 2^n$, take $2n$ distinct primes $p_1, q_1, p_2, q_2, \dots, p_n, q_n$ and construct 2^n elements for S by taking the product over all n indices i of either p_i or q_i for every i .

Note that (from the problem statement) each element of S has $|S|$ divisors.

Now it suffices to show that there is no element $x \in S$ such that x is divisible by more than one power of any prime.

Suppose there does exist an element x and a prime p such that $p^2 \mid x$.

This implies that there exists an element $c \in S$ that is divisible by p but not p^2 by $\gcd(x, c) = p$.

Exactly half of c 's divisors are not divisible by p , so it follows that exactly half of the elements in S are not divisible by p .

However, if $p^0, p^1, \dots, p^k$ are the powers of p dividing x for some $k \geq 2$, only $\frac{1}{k+1}$ of the divisors of x are not divisible by p .

But this means that $\frac{1}{k+1}$ of the elements of S are not divisible by p , contradiction.

Therefore, $|S|$ must be some power of 2.

6. (*USAJMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $n \geq 4$ be an integer. Find all positive real solutions to the following system of $2n$ equations:

$$\begin{aligned} a_1 &= \frac{1}{a_{2n}} + \frac{1}{a_2}, & a_2 &= a_1 + a_3, \\ a_3 &= \frac{1}{a_2} + \frac{1}{a_4}, & a_4 &= a_3 + a_5, \\ a_5 &= \frac{1}{a_4} + \frac{1}{a_6}, & a_6 &= a_5 + a_7, \\ &\vdots & & \\ a_{2n-1} &= \frac{1}{a_{2n-2}} + \frac{1}{a_{2n}}, & a_{2n} &= a_{2n-1} + a_1 \end{aligned}$$

Solution

For any odd n , we know $a_n = \frac{1}{a_{n-1}} + \frac{1}{a_{n+1}}$. Additionally, for any even m , $a_m = a_{m-1} + a_{m+1}$. Substituting gives $a_m = \frac{2}{a_m} + \frac{1}{a_{m-2}} + \frac{1}{a_{m+2}}$, other than $m = 2$ and $m = 2n$.

Consider the sequence b_k , such that $a_{2k} = b_k$. Then, say out of the set $[b_1, b_2, \dots, b_n]$ let the maximum value of an element in the set be X and the minimum value be x . So, $x \leq X$.

Then, we also know $b_k = \frac{2}{b_k} + \frac{1}{b_{k-1}} + \frac{1}{b_{k+1}}$, plug in x into b_k to get $x = \frac{2}{x} + \frac{1}{b_{k-1}} + \frac{1}{b_{k+1}}$.

We know $\frac{1}{b_k} \geq \frac{1}{X}$, so $x \geq \frac{2}{x} + \frac{2}{X}$. Similarly, plugging in X and using an inequality gives $\frac{2}{x} + \frac{2}{X} \geq X$.

This means that $x \geq \frac{2}{x} + \frac{2}{X} \geq X$, so $x \geq X$. We also showed earlier that $x \leq X$. Thus, $x = X$. So, all a_{2k} are constants and equal to each other.

Now, let $a_{2k-1} = w$ and $a_{2k} = z$, then $w = \frac{2}{z}$ and $z = 2w$. Solving gives $w = 1$ and $z = 2$.

§13 USAJMO 2022

1. (*USAJMO 2022*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

For which positive integers m does there exist an infinite arithmetic sequence of integers $a_1, a_2, \dots$ and an infinite geometric sequence of integers $g_1, g_2, \dots$ satisfying the following properties?

- $a_n - g_n$ is divisible by m for all integers $n > 1$;
- $a_2 - a_1$ is not divisible by m .

Solution

Let the arithmetic sequence be $\{a, a + d, a + 2d, \dots\}$ and the geometric sequence to be $\{g, gr, gr^2, \dots\}$. Rewriting the problem based on our new terminology, we want to find all positive integers m such that there exist integers a, d, r with $m \nmid d$ and $m \mid a + (n - 1)d - gr^{n-1}$ for all integers $n > 1$.

Note that

$$\begin{aligned} m \mid a + nd - gr^n & \quad (1), \\ m \mid a + (n + 1)d - gr^{n+1} & \quad (2), \\ m \mid a + (n + 2)d - gr^{n+2} & \quad (3), \end{aligned}$$

for all integers $n \geq 1$. From (1) and (2), we have $m \mid d - gr^{n+1} + gr^n$ and from (2) and (3), we have $m \mid d - gr^{n+2} + gr^{n+1}$. Reinterpreting both equations,

$$\begin{aligned} m \mid gr^{n+1} - gr^n - d & \quad (4), \\ m \mid gr^{n+2} - gr^{n+1} - d & \quad (5), \end{aligned}$$

for all integers $n \geq 1$. Thus, $m \mid gr^k - 2gr^{k+1} + gr^{k+2} = gr^k(r - 1)^2$ (6). Note that if $m \mid g, r$, then $m \mid gr^{n+1} - gr^n$, which, plugged into (4), yields $m \mid d$, which is invalid. Also, note that (4)+(5) gives

$$m \mid gr(r - 1)(r + 1) - 2d \quad (7),$$

so if $r \equiv \pm 1 \pmod{m}$ or $gr \equiv 0 \pmod{m}$, then $m \mid d$, which is also invalid. Thus, according to (6), $m \mid g(r - 1)^2$, with $m \nmid g, r$. Also from (7) is that $m \nmid g(r - 1)$.

Finally, we can conclude that the only m that will work are numbers in the form of xy^2 , other than 1, for integers x, y (x and y can be equal), ie. 4, 8, 9, 12, 16, 18, 20, 24, 25, $\dots$

Note: This solution has some serious logical errors and never proves sufficiency. If this is ever solved, feel free to remove this note.

Solution 1

m satisfies the conditions precisely when m is not squarefree.

Consider three consecutive terms of the arithmetic sequence modulo m : $x - d$, x , and $x + d$. To satisfy the given conditions, the arithmetic and geometric sequences must match modulo m . Thus,

$$(x - d)(x + d) \equiv x^2 \pmod{m}$$

which simplifies to

$$x^2 - d^2 \equiv x^2 \pmod{m} \implies d^2 \equiv 0 \pmod{m}.$$

If m is squarefree, the congruence $d^2 \equiv 0 \pmod{m}$ implies $m \mid d$, so the conditions are not satisfied. Thus, no solution exists for squarefree m , and it remains for us to construct a solution for all other values of m that are not squarefree.

Suppose m is divisible by some prime square p^2 . Consider the arithmetic sequence

$$1, \quad 1 + \frac{m}{p}, \quad 1 + 2\frac{m}{p}, \quad \dots$$

and the geometric sequence

$$1, \quad 1 + \frac{m}{p}, \quad \left(1 + \frac{m}{p}\right)^2, \quad \dots$$

For the geometric sequence, the first two terms in the binomial expansion of each entry match the corresponding terms in the arithmetic sequence and subsequent terms are 0 mod m because $\frac{m}{p}$ raised to a power greater than 1 contains enough p 's in its factorization to ensure that it is a multiple of m . Therefore the construction works and as originally stated, solutions exist precisely when m is not squarefree.

2. (*USAJMO 2022*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let a and b be positive integers. The cells of an $(a + b + 1) \times (a + b + 1)$ grid are colored amber and bronze such that there are at least $a^2 + ab - b$ amber cells and at least $b^2 + ab - a$ bronze cells. Prove that it is possible to choose a amber cells and b bronze cells such that no two of the $a + b$ chosen cells lie in the same row or column.

Solution 1: Pigeonhole

Let $n = a + b + 1$. There are $n!$ ways to select n cells such that no two are in the same row or column. Each such selection can be specified by $p_i = (p_{i1}, p_{i2}, \dots, p_{in})$, a permutation of $1, 2, 3, \dots, n$, such that in the k^{th} row, the cell in column p_{ik} is selected. Let $A(p_i)$ be the number of amber cells in the selection p_i . We just need to prove there exists a p_i such that

$$A(p_i) = a, \text{ or } A(p_i) = a + 1.$$

Using contradiction, assuming no such p_i exists.

In calculating $\sum A(p_i)$, each amber cell is counted $(n - 1)!$ times. We have

$$\begin{aligned} \sum A(p_i) &\geq (n - 1)!(a^2 + ab - b). \\ \Rightarrow \max(A(p_i)) &\geq \left\lceil \frac{(n - 1)!(a^2 + ab - b)}{n!} \right\rceil = \left\lceil \frac{n(a - 1) + 1}{n} \right\rceil = a. \end{aligned}$$

WLOG, let $A(p_1) = \max(A(p_i))$. With the contradiction assumption

$$A(p_1) \geq a + 2.$$

Similarly, let $A(p_2) = \min(A(p_i))$.

$$\sum A(p_i) \leq (n-1)![n^2 - (b^2 + ab - a)] \Rightarrow \min(A(p_i)) \leq a+1 \Rightarrow A(p_2) \leq a-1.$$

One can always move from p_1 to p_2 via a path in p space

$$p_1 \rightarrow q_1 \rightarrow q_2 \rightarrow \cdots \rightarrow q_k \rightarrow p_2$$

such that in each step two numbers in the starting permutation are swapped to get the next permutation. For example if

$$p_1 = (5, 4, 3, 1, 2), p_2 = (1, 2, 3, 4, 5)$$

one such path is

$$(5, 4, 3, 1, 2) \rightarrow (1, 4, 3, 5, 2) \rightarrow (1, 2, 3, 5, 4) \rightarrow (1, 2, 3, 4, 5).$$

Then the value of A along the path changes by at most 2 in each step. On the other hand since $A(p_i) \neq a, a+1$, there is a gap of at least width 3 in the range of A . A path starting above the gap and ending below the gap with max step size two should not exist. Contradiction.

Therefore there must be an i such that either $A(p_i) = a$ or $A(p_i) = a+1$. (Y)

Solution 2

We claim that it is possible to choose a amber cells such that no two lie in the same row or column.

Number the rows and columns from 1 to $a+b+1$. In cell (x, y) , write the number $x+y \pmod{a+b+1}$. Then two same-numbered cells will be on different rows and columns.

There are $a+b+1$ different numbers and $\geq a^2 + ab - b = (a-1)(a+b+1) + 1$ amber cells. Using Pigeonhole, we have $a+b+1$ holes, and cells in the same hole are in different rows and columns, and $\geq (a-1)(a+b+1) + 1$ pigeons, so there exists a amber cells such that no two lie in the same row or column. Similarly, there exists b bronze cells such that no two lie in the same row or column.

Choose those a amber cells and b bronze cells. If no two chosen cells lie in the same row or column then we are done. Otherwise, since there are $a+b+1$ rows and columns, there exists a row and a column with no chosen cell. If their intersection is an amber cell, choose that cell and unchoose an amber cell that is in the same row or column as a bronze cell. If the intersection is a bronze cell, choose that cell and unchoose a chosen bronze cell that is in the same row or column as an amber cell. Repeating this process, we can obtain a amber cells and b bronze cells such that no two lie in the same row or column.

Solution 3: Graph Theoretic

We can prove this statement by using the Hall's marriage theorem. Let us define a bipartite graph $G = (A \cup B, E)$, where A is the set of rows of the grid, B is the set of columns of the grid, and E is the set of edges between A and B . We say that a row $a \in A$ is connected to a column $b \in B$ if and only if the cell in row a and column b is colored amber.

Now, we need to show that the conditions of Hall's marriage theorem are satisfied. That is, for any subset $S \subseteq A$, the number of columns connected to S is at least $|S|$.

To prove this, let S be any subset of A , and let T be the set of columns connected to S . Then, the number of amber cells in the rows of S is at least $a|S|$, and the number of amber cells in the rows not in S is at least $b(a+b+1) - a|S|$. Therefore, the total number of amber cells is at least $a|S| + b(a+b+1) - a|S| = ab + b(a+b)$, which is at least $a^2 + ab - b$ by assumption. Hence, the number of columns connected to S is at least b .

Similarly, we can define a bipartite graph $G' = (A \cup B', E')$, where B' is the set of rows of the grid, and an edge between a row a and a column $b' \in B'$ exists if and only if the cell in row a and column b' is colored bronze. We can show that the conditions of Hall's marriage theorem are also satisfied for G' .

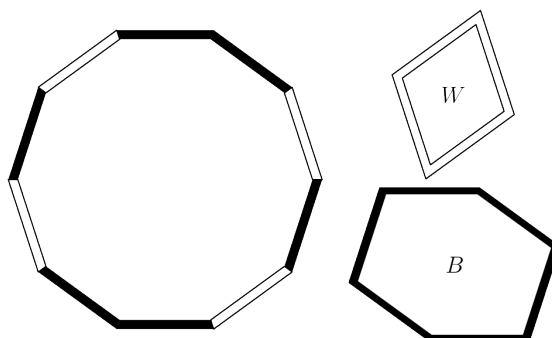
Therefore, by Hall's marriage theorem, there exist matchings M and M' in G and G' , respectively, such that each row and column in the grid is incident to at most one edge in M or M' . Let X be the set of cells in the grid that are covered by M , and Y be the set of cells in the grid that are covered by M' . Then, $|X| = |Y| = a+b$, since each matching covers a total of $a+b$ cells. Moreover, no two cells in X or Y lie in the same row or column, since each row and column is incident to at most one edge in M or M' .

Finally, we can choose a amber cells and b bronze cells from X and Y , respectively, such that no two of the chosen cells lie in the same row or column. This is because $|X| = |Y| = a+b$, and there are at least $a^2 + ab - b$ amber cells in the grid and at least $b^2 + ab - a$ bronze cells in the grid. Therefore, there are at least $a^2 + ab - b$ cells colored amber in X , and at least $b^2 + ab - a$ cells colored bronze in Y . This implies that there are at least a amber cells in X and at least b bronze cells in Y , so we can choose a subset of a amber cells from X and a subset of b bronze cells from Y , such that no two of the chosen cells lie in the same row or column.

3. (*USAJMO 2022*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $b \geq 2$ and $w \geq 2$ be fixed integers, and $n = b + w$. Given are $2b$ identical black rods and $2w$ identical white rods, each of side length 1.

We assemble a regular $2n$ -gon using these rods so that parallel sides are the same color. Then, a convex $2b$ -gon B is formed by translating the black rods, and a convex $2w$ -gon W is formed by translating the white rods. An example of one way of doing the assembly when $b = 3$ and $w = 2$ is shown below, as well as the resulting polygons B and W .



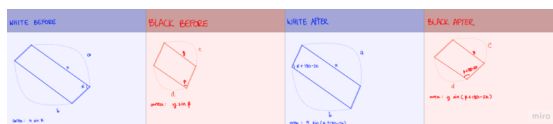
Prove that the difference of the areas of B and W depends only on the numbers b

and w , and not on how the $2n$ -gon was assembled.

Solution 1

First notice that the black rods and the white rods form polygons iff in the original $2n$ -gon, if a side is a color x , then the side that is parallel to that side in the original $2n$ -gon is also the color x .

We can prove that the difference in areas is only affected by the values of b and w by showing that for any valid arrangement of $2b$ rods and $2w$ rods, we may switch any two adjacent black and white rods (and their "parallel pairs"), and end up with the same area difference.



In the figure above (click to expand), after the switch, we can see that after removing the mutually congruent parts, we are left with two parallelograms from each color. Let x , α , y , and β be defined as shown. Notice that if we angle chase, the sides of the other parallelogram are the same, but if the angles of the original $2n$ -gon all have measure $2k$, the angles of the new parallelograms are $\alpha + 180 - 2k$ and $\beta + 180 - 2k$, as shown. We must prove that the differences between the areas are the same.

Using area formulas, the change in the difference of areas is $x \sin \alpha - y \sin \beta + y \sin (2k - \beta) - x \sin (2k - \alpha)$, which is equal to $x(\sin k(2 \cos^2 k) - 2 \sin k \cos k \cos \alpha) - y(\sin k(2 \cos^2 k)) + y(\sin k \cos k \cos \beta)$, or $2 \cos k(x \sin (\alpha - k) + y \sin (k - \beta))$. Since $\cos k$ is not 0 because $n \geq 3$, we are left with proving that $x \sin (\alpha - k) + y \sin (k - \beta) = 0$.

Now we rotate the polygon so that the vertex between the two sides that we switched is at the point $(0, 0)$, the angle bisector of that vertex is $y = 0$, and the black side is in the positive y -direction. Now think of all the sides as vectors, all pointing in the clockwise direction of the $2n$ -gon.

Notice the part labeled a in the black polygons. We have that the vector labeled x is really just the sum of all of the vectors in the part labeled a - or all the vectors in the $2n$ -gon that are in the positive y -direction excluding the one that was interchanged. Also notice that the angle of this vector x has a signed angle of $\alpha - k$ with $y = 0$ and has length x - meaning that the vertical displacement of the vector x from $y = 0$ is equal to $x \sin (\alpha - k)$! Similarly, we get that the vertical displacement of the vector y is equivalent to $y \sin (k - \beta)$.

Adding these two together, we get that $x \sin (\alpha - k) + y \sin (k - \beta)$ is simply the vertical displacement of the sum of the vectors x and y . Since the sum of the vectors x and y is equivalent to the sum of the vectors in the positive half of the polygon minus the sum of the black vector that would be switched with the white vector (the leftmost vector in the positive half of the polygon) and the rightmost vector in the positive half (which is the parallel pair of the white vector that would be interchanged later), and we know that this sum happens to have a vertical displacement of 0, along with the fact that the positive half of the polygon summed together also has a vertical displacement of 0, we get that the total vertical displacement is 0, meaning that $x \sin (\alpha - k) + y \sin (k - \beta) = 0$, and we are done.

Solution 2

Pick a pair of parallel sides of the regular $2n$ -gon and flip their color. Let those two lines be the horizontal. The area of the polygon formed by rods of the original color decreases by a parallelogram with height equal to the sum of the vertical heights of the rods of that color divided by 2 and base length 1. Similarly, the area of the polygon formed by rods of the new color increases by the sum of the vertical heights of the rods of the new color divided by 2. The sum of the heights of all the rod is equal to twice the height of the $2n$ -gon, so the difference between the areas of B and W changes by the height of the $2n$ -gon, which is fixed when n is fixed.

If we pair two pairs different-colored pairs of parallel sides of the $2n$ -gon and flip the colors of both pairs, then b and w and the difference in the areas of B and W will remain unchanged. Thus, the difference in the areas of B and W depends only on b and w .

4. (*USAJMO 2022*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Let $ABCD$ be a rhombus, and let K and L be points such that K lies inside the rhombus, L lies outside the rhombus, and $KA = KB = LC = LD$. Prove that there exist points X and Y on lines AC and BD such that $KXLY$ is also a rhombus.

Solution

(Image of the solution is here [\[1\]](#))

Let's draw (ℓ) perpendicular bisector of $\overline{KL}$. Let X, Y be intersections of ℓ with AC and BD , respectively. $KXLY$ is a kite. Let O mid-point of $\overline{KL}$. Let M mid-point of $\overline{BD}$ (and also M is mid-point of $\overline{AC}$). X, O, Y are on the line ℓ .

$BK = DL$, $BX = XD$, $XK = XL$ and so $\triangle B XK \cong \triangle DXL$ (side-side-side). By spiral similarity, $\triangle BXD \sim \triangle KXL$. Hence, we get

$$\angle XBD = \angle XDB = \angle XKL = \angle XLK = b.$$

Similarly, $AK = CL$, $YK = YL$, $YA = YC$ and so $\triangle AKY \cong \triangle CYL$ (side-side-side). From spiral similarity, $\triangle YKL \sim \triangle YAC$. Thus,

$$\angle YAC = \angle YCA = \angle YKL = \angle YLK = a.$$

If we can show that $a = b$, then the kite $KXLY$ will be a rhombus.

By spiral similarities, $\frac{BX}{BD} = \frac{XK}{KL}$ and $\frac{YA}{AC} = \frac{YK}{KL}$. Then, $KL = \frac{BD \cdot XK}{BX} = \frac{AC \cdot YK}{YA}$.

$\frac{XK}{YK} = \frac{AC \cdot BX}{AY \cdot BD}$. Then, $\frac{YK}{XK} = \frac{(BD/2) \cdot AY}{(AC/2) \cdot BX} = \frac{\sin b}{\sin a}$. Also, in the right triangles

$\triangle KXO$ and $\triangle KYO$, $\frac{YK}{XK} = \frac{OK/\cos a}{OK/\cos b} = \frac{\cos b}{\cos a}$. Therefore,

$$\frac{\sin b}{\sin a} = \frac{\cos b}{\cos a}.$$

$\sin a \cos b = \sin b \cos a \implies \sin a \cos b - \sin b \cos a = 0 \implies \sin(a - b) = 0$ and we get $a = b$.

(Lokman GÖKÇE)

5. (*USAJMO 2022*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Find all pairs of primes (p, q) for which $p - q$ and $pq - q$ are both perfect squares.

Solution 1

We first consider the case where one of p, q is even. If $p = 2$, $p - q = 0$ and $pq - q = 2$ which doesn't satisfy the problem restraints. If $q = 2$, we can set $p - 2 = x^2$ and $2p - 2 = y^2$ giving us $p = y^2 - x^2 = (y + x)(y - x)$. This forces $y - x = 1$ so $p = 2x + 1 \rightarrow 2x + 1 = x^2 + 2 \rightarrow x = 1$ giving us the solution $(p, q) = (3, 2)$.

Now assume that p, q are both odd primes. Set $p - q = x^2$ and $pq - q = y^2$ so $(pq - q) - (p - q) = y^2 - x^2 \rightarrow p(q - 1) = (y + x)(y - x)$. Since $y + x > y - x$, $p \mid (x + y)$. Note that $q - 1$ is an even integer and since $y + x$ and $y - x$ have the same parity, they both must be even. Therefore, $x + y = pk$ for some positive even integer k . On the other hand, $p > p - q = x^2 \rightarrow p > x$ and $p^2 - p > pq - q = y^2 \rightarrow p > y$. Therefore, $2p > x + y$ so $x + y = p$, giving us a contradiction.

Therefore, the only solution to this problem is $(p, q) = (3, 2)$.

6. (*USAJMO 2022*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let a_0, b_0, c_0 be complex numbers, and define

$$a_{n+1} = a_n^2 + 2b_nc_n$$

$$b_{n+1} = b_n^2 + 2c_na_n$$

$$c_{n+1} = c_n^2 + 2a_nb_n$$

for all nonnegative integers n .

Solution

Let $S_n = a_n + b_n + c_n$. Summing the three given recurrences yields:

$$S_{n+1} = a_n^2 + b_n^2 + c_n^2 + 2a_nb_n + 2b_nc_n + 2c_na_n = (a_n + b_n + c_n)^2 = S_n^2$$

By induction, we have $S_n = S_0^{2^n}$. We are given that $\max(|a_n|, |b_n|, |c_n|) \leq 2022$ for all n . By the triangle inequality,

$$|S_n| \leq |a_n| + |b_n| + |c_n| \leq 3 \times 2022 = 6066.$$

Since the sequence $|S_n|$ is bounded, we must have $|S_0| \leq 1$. (If $|S_0| > 1$, then $|S_n| \rightarrow \infty$ as $n \rightarrow \infty$, which contradicts the boundedness condition).

Next, let $\omega = e^{2\pi i/3}$ be a primitive cube root of unity, satisfying $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$. We define two new sequences:

$$\begin{aligned} T_n &= a_n + \omega b_n + \omega^2 c_n \\ U_n &= a_n + \omega^2 b_n + \omega c_n \end{aligned}$$

Squaring U_n , we obtain:

$$\begin{aligned} U_n^2 &= (a_n + \omega^2 b_n + \omega c_n)^2 \\ &= a_n^2 + \omega^4 b_n^2 + \omega^2 c_n^2 + 2\omega^2 a_n b_n + 2\omega a_n c_n + 2\omega^3 b_n c_n \end{aligned}$$

Using $\omega^3 = 1$ and $\omega^4 = \omega$, this simplifies to:

$$U_n^2 = a_n^2 + \omega b_n^2 + \omega^2 c_n^2 + 2\omega^2 a_n b_n + 2\omega a_n c_n + 2b_n c_n$$

Now, let's evaluate T_{n+1} using the given recurrences:

$$\begin{aligned} T_{n+1} &= a_{n+1} + \omega b_{n+1} + \omega^2 c_{n+1} \\ &= (a_n^2 + 2b_n c_n) + \omega(b_n^2 + 2c_n a_n) + \omega^2(c_n^2 + 2a_n b_n) \end{aligned}$$

Notice that T_{n+1} matches the expansion of U_n^2 perfectly. Thus, $T_{n+1} = U_n^2$. By a symmetric calculation, $U_{n+1} = T_n^2$.

Combining these two relations, we find that for the even terms:

$$\begin{aligned} T_{n+2} &= U_{n+1}^2 = (T_n^2)^2 = T_n^4 \\ U_{n+2} &= T_{n+1}^2 = (U_n^2)^2 = U_n^4 \end{aligned}$$

This implies $T_{2k} = T_0^{4^k}$ and $U_{2k} = U_0^{4^k}$. Using the same boundedness argument as before, since the terms a_n, b_n, c_n are bounded and $|\omega| = 1$, the sequences $|T_n|$ and $|U_n|$ are also strictly bounded. Therefore, their initial moduli must be at most 1:

$$|T_0| \leq 1 \quad \text{and} \quad |U_0| \leq 1$$

Finally, we utilize the absolute value squared identity in the complex plane:

$$|S_0|^2 + |T_0|^2 + |U_0|^2 = S_0 \overline{S_0} + T_0 \overline{T_0} + U_0 \overline{U_0}$$

Expanding this sum, the coefficient of any squared term (such as $a_0 \overline{a_0} = |a_0|^2$) is $1 + 1 + 1 = 3$. The coefficient of any cross term (such as $a_0 \overline{b_0}$) is $1 + \omega + \omega^2 = 0$, meaning all cross terms cancel out completely. Thus, we obtain:

$$|S_0|^2 + |T_0|^2 + |U_0|^2 = 3(|a_0|^2 + |b_0|^2 + |c_0|^2)$$

Since $|S_0| \leq 1$, $|T_0| \leq 1$, and $|U_0| \leq 1$, the maximum possible value of the left-hand side is $1^2 + 1^2 + 1^2 = 3$. Therefore:

$$\begin{aligned} 3(|a_0|^2 + |b_0|^2 + |c_0|^2) &\leq 3 \\ |a_0|^2 + |b_0|^2 + |c_0|^2 &\leq 1 \end{aligned}$$

This completes the proof.

§14 USAJMO 2023

1. (*USAJMO 2023*) Find all triples of positive integers (x, y, z) that satisfy the equation

$$2(x + y + z + 2xyz)^2 = (2xy + 2yz + 2zx + 1)^2 + 2023$$

Solution 1

We claim that the only solutions are $(2, 3, 3)$ and its permutations.

Factoring the above squares and canceling the terms gives you:

$$8(xyz)^2 + 2(x^2 + y^2 + z^2) = 4((xy)^2 + (yz)^2 + (zx)^2) + 2024$$

Jumping on the coefficients in front of the x^2 , y^2 , z^2 terms, we factor into:

$$(2x^2 - 1)(2y^2 - 1)(2z^2 - 1) = 2023$$

Realizing that the only factors of 2023 that could be expressed as $(2x^2 - 1)$ are 1, 7, and 17, we simply find that the only solutions are $(2, 3, 3)$ by inspection.

-Max

Alternatively, a more obvious factorization is:

$$2(x + y + z + 2xyz)^2 = (2xy + 2yz + 2zx + 1)^2 + 2023$$

$$(\sqrt{2}x + \sqrt{2}y + \sqrt{2}z + 2\sqrt{2}xyz)^2 - (2xy + 2yz + 2zx + 1)^2 = 2023$$

$$(2\sqrt{2}xyz + 2xy + 2yz + 2zx + \sqrt{2}x + \sqrt{2}y + \sqrt{2}z + 1)(2\sqrt{2}xyz - 2xy - 2yz - 2zx + \sqrt{2}x + \sqrt{2}y + \sqrt{2}z - 1) = 2023$$

$$(\sqrt{2}x + 1)(\sqrt{2}y + 1)(\sqrt{2}z + 1)(\sqrt{2}x - 1)(\sqrt{2}y - 1)(\sqrt{2}z - 1) = 2023$$

$$(2x^2 - 1)(2y^2 - 1)(2z^2 - 1) = 2023$$

Proceed as above. eevee9406

note: idk gang i think the first factorization is more readily apparent but the second factorization is very orz indeed Aarush12

2. (*USAJMO 2023*) In an acute triangle ABC , let M be the midpoint of $\overline{BC}$. Let P be the foot of the perpendicular from C to AM . Suppose that the circumcircle of triangle ABP intersects line BC at two distinct points B and Q . Let N be the midpoint of $\overline{AQ}$. Prove that $NB = NC$.

Solution 1

Let D be the foot of the altitude from A onto BC . We want to show that $DM = MQ$ for obvious reasons.

Notice that $ADPC$ is cyclic and that M lies on the radical axis of $(ABPQ)$ and $(ADPC)$. By Power of a Point, $(CM)(DM) = (BM)(MQ)$. As $BM = CM$, we have $DM = MQ$, as desired.

- Leo.Euler

Solution 2

Proceed by Barycentrics. $A = (1, 0, 0)$, $B = (0, 1, 0)$, $C = (0, 0, 1)$. Let D be the foot of the altitude from A to BC , so $D = (0 : S_C : S_B)$. Then, let $P = (t, 1, 1)$. The equation of (ACP) is

$$-a^2yz - b^2xz - c^2xy + (vy)(x + y + z) = 0$$

where $(t+2)v = t(b^2 + c^2) + a^2$. Then, the equation of (ABP) is

$$-a^2yz - b^2xz - c^2xy + (wz)(x + y + z) = 0$$

where $(t+2)w = t(b^2 + c^2) + a^2$. Thus, $v = w$. But D is also on (ACP) , so we find $w = v = S_B$. Then, if $Q = (0 : y_1 : z_1)$, we find $y_1 = \frac{w}{a^2} = \frac{S_B}{a^2}$. Therefore, $Q = (0 : S_B : a^2 - S_B)$. Note that $D + Q = (0 : a^2 : a^2) = M$, so M is the midpoint of DQ . Thus, a homothety at Q of factor 2 maps $M \mapsto D$ and $N \mapsto A$, so $MN \perp BC$, as needed.

Alternatively, one could continue with Barycentrics, finding $N = \left(1 : \frac{S_B}{a^2} : 1 - \frac{S_B}{a^2}\right)$. Then,

$$a^2(y - z) + x(b^2 - c^2) = a^2\left(\frac{2S_B}{a^2} - 1\right) + (b^2 - c^2) = 2S_B - (a^2 + c^2 - b^2) = 0$$

implying N is on the perpendicular bisector, as desired.

-sillybone

Solution 3

The condition is solved only if $\triangle NBC$ is isosceles, which in turn only happens if $\overline{MN}$ is perpendicular to $\overline{BC}$. Now, draw the altitude from A to $\overline{BC}$, and call that point X . Because of the Midline Theorem, the only way that this condition is met is if $\triangle AXQ \sim \triangle NMQ$, or if $\overline{XM} = \overline{MQ}$.

By AA similarity, $\triangle AXM \sim \triangle CPM$. Using similarity ratios, we get that $\frac{\overline{AM}}{\overline{XM}} = \frac{\overline{CM}}{\overline{PM}}$. Rearranging, we get that $\overline{AM} \cdot \overline{MP} = \overline{XM} \cdot \overline{MC}$. This implies that $AXPC$ is cyclic.

Now we start using Power of a Point. We get that $\overline{BX} \cdot \overline{XQ} = \overline{AM} \cdot \overline{MP}$, and $\overline{AM} \cdot \overline{MP} = \overline{XM} \cdot \overline{MC}$ from before. This leads us to get that $\overline{BX} \cdot \overline{XQ} = \overline{XM} \cdot \overline{MC}$.

Now we assign variables to the values of the segments. Let $\overline{BX} = a$, $\overline{XM} = b$, $\overline{MQ} = c$, and $\overline{QC} = d$. The equation from above gets us that $(a+b)c = b(c+d)$. As $a+b = c+d$ from the problem statements, this gets us that $b = c$ and $\overline{XC} = \overline{CQ}$, and we are done.

-dragoon and rhydon516 (:

note: i think a lot of places you used the wrong point names Aarush12

Solution 4 (Less technical bary)

We are going to use barycentric coordinates on $\triangle ABC$. Let $A = (1, 0, 0)$, $B = (0, 1, 0)$, $C = (0, 0, 1)$, and $a = BC$, $b = CA$, $c = AB$. We have $M = \left(0, \frac{1}{2}, \frac{1}{2}\right)$ and $P = (x : 1 : 1)$ so $\overrightarrow{CP} = \left(\frac{x}{x+2}, \frac{1}{x+2}, \frac{1}{x+2} - 1\right)$ and $\overrightarrow{AM} = \left(-1, \frac{1}{2}, \frac{1}{2}\right)$. Since $\overrightarrow{CP} \perp \overrightarrow{AM}$, it follows that

$$a^2 \left(\frac{1}{2} \cdot \frac{1}{x+2} + \frac{1}{2} \left(\frac{1}{x+2} - 1 \right) \right) + b^2 \left(\frac{1}{2} \cdot \frac{x}{x+2} - \left(\frac{1}{x+2} - 1 \right) \right) + c^2 \left(\frac{1}{2} \cdot \frac{x}{x+2} - \frac{1}{x+2} \right) = 0.$$

Solving this gives

$$x = \frac{2b^2 - 2c^2}{a^2 - 3b^2 - c^2}$$

so

$$P = \left(\frac{b^2 - c^2}{a^2 - 2b^2 - 2c^2}, \frac{a^2 - 3b^2 - c^2}{2a^2 - 4b^2 - 4c^2}, \frac{a^2 - 3b^2 - c^2}{2a^2 - 4b^2 - 4c^2} \right).$$

The equation for (ABP) is

$$-a^2yz - b^2zx - c^2xy + ux + vy + wz = 0.$$

Plugging in A and B gives $u = v = 0$. Plugging in P gives

$$\begin{aligned} & -a^2 \left(\frac{a^2 - 3b^2 - c^2}{2a^2 - 4b^2 - 4c^2} \right)^2 - b^2 \cdot \frac{a^2 - 3b^2 - c^2}{2a^2 - 4b^2 - 4c^2} \cdot \frac{b^2 - c^2}{a^2 - 2b^2 - 2c^2} \\ & - c^2 \cdot \frac{b^2 - c^2}{a^2 - 2b^2 - 2c^2} \cdot \frac{a^2 - 3b^2 - c^2}{2a^2 - 4b^2 - 4c^2} + w \cdot \frac{a^2 - 3b^2 - c^2}{2a^2 - 4b^2 - 4c^2} = 0 \end{aligned}$$

so

$$w = \frac{2b^4 - 2c^4 + a^4 - 3a^2b^2 - a^2c^2}{2a^2 - 4b^2 - 4c^2} = \frac{a^2}{2} - \frac{b^2}{2} + \frac{c^2}{2}.$$

Now let $Q = (0, t, 1 - t)$ where

$$\begin{aligned} -a^2t(1 - t) + w(1 - t) &= 0 \\ \implies t &= \frac{w}{a^2} \end{aligned}$$

so $Q = \left(0, \frac{w}{a^2}, 1 - \frac{w}{a^2}\right)$. It follows that $N = \left(\frac{1}{2}, \frac{w}{2a^2}, 1 - \frac{w}{2a^2}\right)$. It suffices to prove that $\overrightarrow{ON} \perp \overrightarrow{BC}$. Setting $\vec{O} = 0$, we get $\vec{N} = \left(\frac{1}{2}, \frac{w}{2a^2}, 1 - \frac{w}{2a^2}\right)$. Furthermore we have $\overrightarrow{CB} = (0, 1, -1)$ so it suffices to prove that

$$\begin{aligned} a^2 \left(-\frac{w}{2a^2} + \frac{1}{2} - \frac{u}{2a^2} \right) + b^2 \left(-\frac{1}{2} \right) + c^2 \left(\frac{1}{2} \right) &= 0 \\ \implies w &= \frac{a^2}{2} - \frac{b^2}{2} + \frac{c^2}{2} \end{aligned}$$

which is valid. $\square$

Solution 5 (Less bashy bary)

We employ barycentric coordinates. Set AMC as the reference triangle with $A = (1, 0, 0)$, $M = (0, 1, 0)$, and $C = (0, 0, 1)$. We immediately have,

$$P = (S_B : S_A : 0); B = (0, 2, -1)$$

Since it passes through A , for some v, w , the equation of circle (ABP) is,

$$(ABP) : -a^2yz - b^2zx - c^2xy + (vy + wz)(x + y + z) = 0$$

Plugging in P ,

$$\begin{aligned} -c^2S_{AB} + (vS_A)(c^2) &= 0 \\ \iff v &= S_B \end{aligned}$$

Plugging in B ,

$$\begin{aligned} 2a^2 + (2v - w) &= 0 \\ \iff 2a^2 + 2S_B &= 3a^2 - b^2 + c^2 = w \end{aligned}$$

In conclusion the circle has formula,

$$(ABP) : -a^2yz - b^2zx - c^2xy + ((S_B)y + (3a^2 - b^2 + c^2)z)(x + y + z) = 0$$

Q is the second intersection of circle (ABP) with $\overline{CM}$. We let $Q = (0, 1 - t, t)$ for some $t \neq -1$. Plugging this in,

$$-a^2(1 - t)t + ((S_B)(1 - t) + (3a^2 - b^2 + c^2)t) = 0$$

We claim that $t = -\frac{S_B}{a^2}$ is the other solution.

$$\begin{aligned} & \left(1 + \frac{S_B}{a^2}\right)S_B + \left((S_B)\left(1 + \frac{S_B}{a^2}\right) - (3a^2 - b^2 + c^2)\frac{S_B}{a^2}\right) = 0 \\ \iff & \left(\frac{3a^2 - b^2 + c^2}{2a^2}\right) + \left(\left(\frac{3a^2 - b^2 + c^2}{2a^2}\right) - (3a^2 - b^2 + c^2)\frac{1}{a^2}\right) = 0 \end{aligned}$$

Factoring out the $\frac{3a^2 - b^2 + c^2}{2a^2}$, this is clearly true.

We also check that, these are not the same value.

$$\begin{aligned} & -\frac{S_B}{a^2} = -1 \\ \iff & a^2 - b^2 + c^2 = 2a^2 \\ \iff & c^2 = a^2 + b^2 \end{aligned}$$

The triangle is acute, so this is impossible.

Since we had a quadratic in t with at most two solutions, the second intersection Q is indeed,

$$Q = \left(0, 1 + \frac{S_B}{a^2}, -\frac{S_B}{a^2}\right)$$

Therefore,

$$N = \frac{A + Q}{2} = \left(\frac{1}{2}, \frac{1}{2} - \frac{S_B}{2a^2}, \frac{S_B}{2a^2}\right)$$

Solution 6 (complex)

Set $\triangle AMC$ to be unit circle, where $\overline{CM}$ is horizontal.

$$A = a, C = c, M = -\frac{1}{c}, P = \frac{1}{2}(a + c + \frac{a}{c^2} - \frac{1}{c}), B = \frac{-2}{c} - c$$

Now we want to prove that $\overline{NB} = \overline{NC}$, which really means that N lies on the perpendicular bisector of $\overline{BC}$. Now, shift the coordinate system by $+\frac{1}{c}$

$$A = a + \frac{1}{c}, C = c + \frac{1}{c}, M = 0, P = \frac{1}{2}(a + c + \frac{a}{c^2} + \frac{1}{c}), B = \frac{-1}{c} - c$$

$$N = ki \text{ for } k \in R, Q = 2ki - a - \frac{1}{c}$$

The imaginary part of $Q = 0$, so $2ki = \frac{(a - \frac{1}{a} - c + \frac{1}{c})}{2}$

$$Q = \frac{-a - \frac{1}{a} - c - \frac{1}{c}}{2}$$

Now all we need to do is to check that A, B, P, Q are conc-cyclic.

$$\frac{(A-P)(B-Q)}{(A-Q)(B-P)} \in R$$

Throwing out the half and some negative terms for simplicity.

$$\frac{(a + \frac{1}{c} - c - \frac{a}{c^2})(a + \frac{1}{a} - c - \frac{1}{c})}{(3a + \frac{3}{c} + \frac{1}{a} + c)(\frac{3}{c} + 3c + a + \frac{a}{c^2})} \in R$$

So we need to prove that:

$$\frac{(a+\frac{1}{c}-c-\frac{a}{c^2})(a+\frac{1}{a}-c-\frac{1}{c})}{(3a+\frac{3}{c}+\frac{1}{a}+c)(\frac{3}{c}+3c+a+\frac{a}{c^2})} = \frac{(\frac{1}{a}+c-\frac{1}{c}-\frac{c^2}{a})(a+\frac{1}{a}-c-\frac{1}{c})}{(\frac{3}{a}+3c+a+\frac{1}{c})(3c+\frac{3}{c}+\frac{1}{a}+\frac{c^2}{a})}$$

On the RHS, we can multiply the numerator and denominator by $\frac{a}{c}$ to get:

$$(3a + \frac{3}{c} + \frac{1}{a} + c)(\frac{3}{c} + 3c + a + \frac{a}{c^2}) = (\frac{3}{a} + 3c + a + \frac{1}{c})(3a + \frac{3a}{c^2} + \frac{1}{c} + c)$$

$$(3 + \frac{c}{a})(a + \frac{1}{c})(3 + \frac{a}{c})(c + \frac{1}{c}) = (a + \frac{1}{c})(1 + \frac{3c}{a})(c + \frac{1}{c})(\frac{3a}{c} + 1)$$

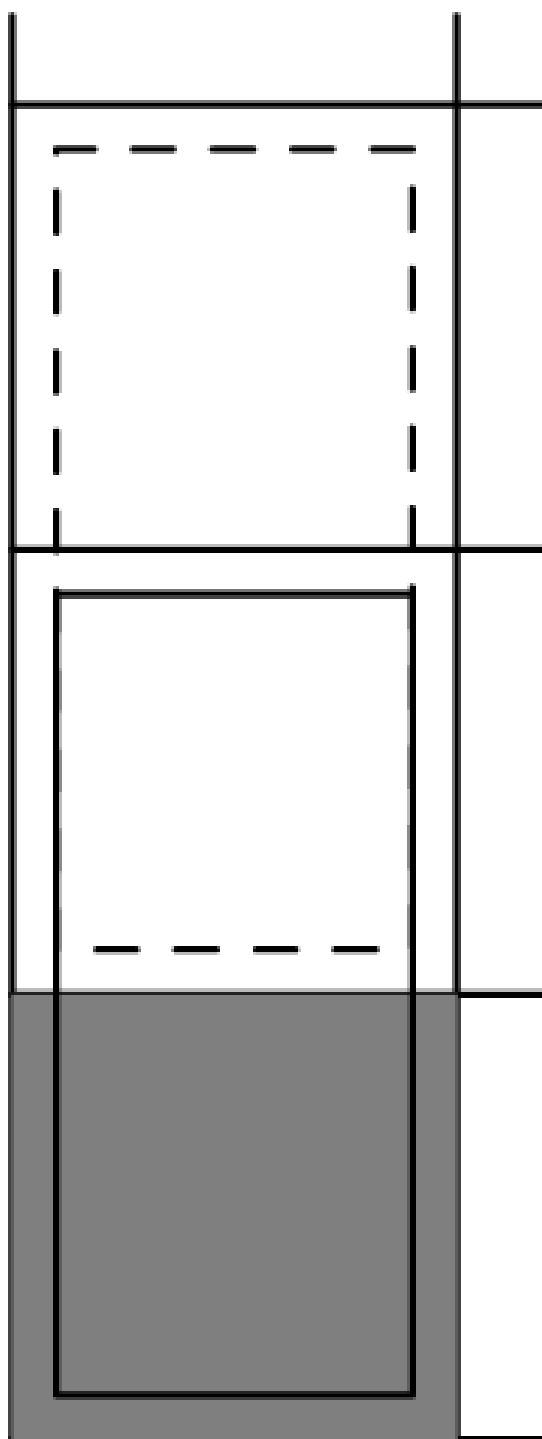
$$(3 + \frac{c}{a})(3 + \frac{a}{c}) = (1 + \frac{3c}{a})(\frac{3a}{c} + 1)$$

Which is true if you multiply ac on both sides. Thus, we are done (note we could have factored earlier which would have saved tremendous time).

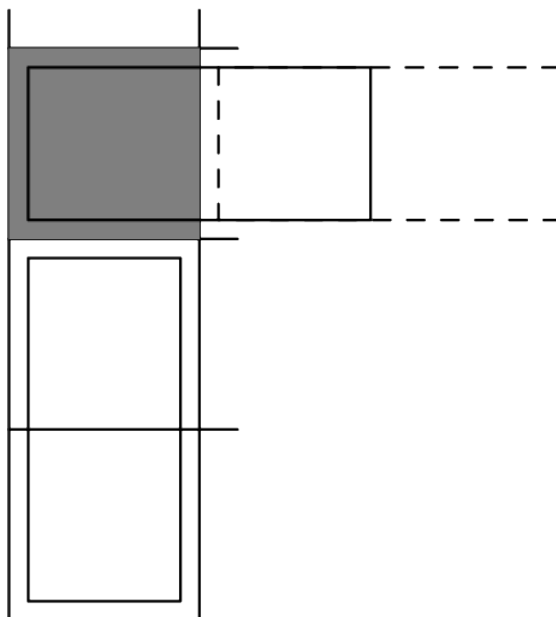
3. (*USAJMO 2023*) Consider an n -by- n board of unit squares for some odd positive integer n . We say that a collection C of identical dominoes is a maximal grid-aligned configuration on the board if C consists of $(n^2 - 1)/2$ dominoes where each domino covers exactly two neighboring squares and the dominoes don't overlap: C then covers all but one square on the board. We are allowed to slide (but not rotate) a domino on the board to cover the uncovered square, resulting in a new maximal grid-aligned configuration with another square uncovered. Let $k(C)$ be the number of distinct maximal grid-aligned configurations obtainable from C by repeatedly sliding dominoes. Find the maximum value of $k(C)$ as a function of n .

Solution

To start off, we put the initial non-covered square in a corner (marked by the shaded square). Let's consider what happens when our first domino slides over the empty square. We will call such a move where we slide a domino over the uncovered square a "step":

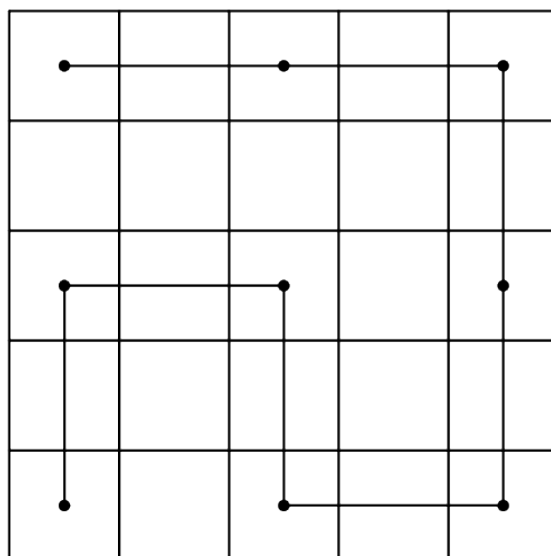


When the vertically-oriented domino above the shaded square moved down to cover the shaded square, it had uncovered a new square two squares above the original one. Next, we consider the second another direction the uncovered square can move to:



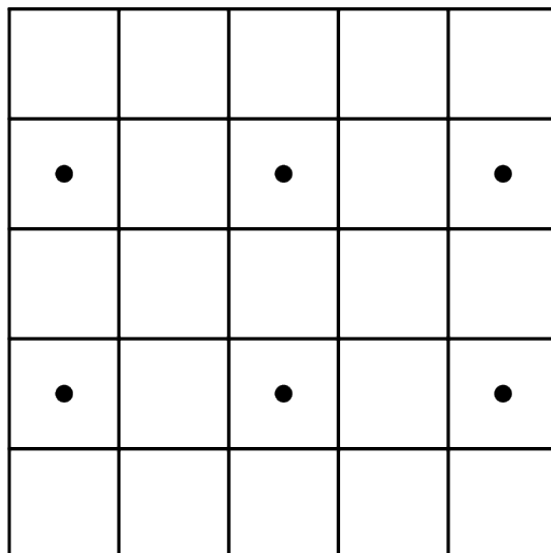
In this case, the horizontally-oriented domino besides the shaded square moved to cover the previous shaded square. The new uncovered square, as a result, would be either two squares to the left, or to the right of the previous shaded square.

So, to summarize, during each "move", the uncovered square can move either two steps up, down, left, or right. We can make a path for which the shaded square takes:

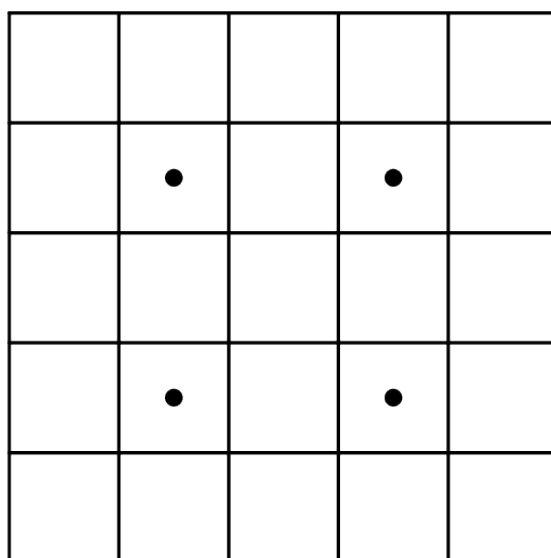


We put the original shaded square was in the corner, so, the possible number of squares that could be uncovered is $\left(\frac{n+1}{2}\right)^2 = \frac{n^2+2n+1}{4}$, all of which can be connected by a single path. Let those $\frac{n^2+2n+1}{4}$ belong to a set S of squares.

If the initial uncovered square was not an element of S , then it could either be 1 or $\sqrt{2}$ away from a square in S . If the initial square was a distance of 1 from a element of S , then there would be $\left(\frac{n+1}{2}\right)\left(\frac{n-1}{2}\right)$ possible uncovered squares:



If the initial square was a distance of $\sqrt{2}$ away from a square in S , then there would be $\left(\frac{n-1}{2}\right)^2$ possible squares that could be uncovered:



So, that means the maximal number of squares that can be uncovered is $\frac{n^2+2n+1}{4}$, thus making a maxmial of $k(C) = \left\lfloor \frac{n^2+2n+1}{4} \right\rfloor$ achievable configurations from C (C 's uncovered square must belong in S).

Remark: I believe that this solution does not address the problem when two configurations lead to the same placement of the uncovered square. To address this problem, we can prove that the paths don't make a cycle, bebebe

4. (*USAJMO 2023*) Two players, B and R , play the following game on an infinite grid of unit squares, all initially colored white. The players take turns starting with B . On B 's turn, B selects one white unit square and colors it blue. On R 's turn, R selects two white unit squares and colors them red. The players alternate until B decides to end the game. At this point, B gets a score, given by the number of unit

squares in the largest (in terms of area) simple polygon containing only blue unit squares. What is the largest score B can guarantee?

(A simple polygon is a polygon (not necessarily convex) that does not intersect itself and has no holes.)

Solution

It is clear that B can guarantee a score of 4 squares. We will show that R has a strategy to limit blue to 4 squares, thus solving the problem.

Partition the grid into 2×2 squares. Red's strategy is as follows:

- If B plays in a 2×2 square, play the two adjacent squares to B 's square that are not in the 2×2 square.- If one (or both) of these moves are blocked, instead play a square a megaparsec away from the rest of the moves. This move can only benefit you and will not change the outcome of the game.

By induction, it is clear that no two blue squares that are adjacent are not in the same 2×2 square. Thus, we conclude that R has limited blue to a maximum score of $2^2 = 4$, and the proof is complete. $\square$

5. (*USAJMO 2023*) A positive integer a is selected, and some positive integers are written on a board. Alice and Bob play the following game. On Alice's turn, she must replace some integer n on the board with $n + a$, and on Bob's turn he must replace some even integer n on the board with $n/2$. Alice goes first and they alternate turns. If on his turn Bob has no valid moves, the game ends.

After analyzing the integers on the board, Bob realizes that, regardless of what moves Alice makes, he will be able to force the game to end eventually. Show that, in fact, for this value of a and these integers on the board, the game is guaranteed to end regardless of Alice's or Bob's moves.

Solution

We claim that the game will always end if and only if $\nu_2(n) < \nu_2(a)$ for all n that are in the list of positive integers.

First, we will prove the if direction. Notice that if Alice adds a to n , since we have

$$\nu_2(ka + n) = \nu_2(n) < \nu_2(a)$$

for all integers $k > 0$, eventually Bob will decrease $\nu_2(n)$ by 1, and Alice will not be able to change $\nu_2(n)$, and so $\nu_2(n)$ will eventually become 0.

Similarly, all of the other numbers on the list will have the same fate as n , and so, no matter what Bob or Alice do, the game will end.

Now, we complete the only if direction, i.e. if $\nu_2(n) \geq \nu_2(a)$, where n is one of the numbers in the list, we will prove that Alice can keep the game going forever.

Notice that Bob can only decrease $\nu_2(n)$ by 1 at a time, and so if $\nu_2(n) \leq \nu_2(a)$, we need $\nu_2(n) = \nu_2(a)$ at some point. But, then, if this is true, take $n = 2^k(2m + 1)$, and $a = 2^k(2\ell + 1)$, and notice that

$$a + n = 2^k(2m + 1) + 2^k(2\ell + 1) = 2^k(2m + 2\ell + 2) = 2^{k+1}(m + \ell + 1),$$

so $\nu_2(a + n) > \nu_2(n)$. Thus, if Bob gets $\nu_2(n)$ equal to $\nu_2(a)$, Alice can simply add a to n , and then $\nu_2(n) > \nu_2(a)$ again. Thus, Alice can keep the game going forever, and hence we are done.

6. (*USAJMO 2023*) Isosceles triangle ABC , with $AB = AC$, is inscribed in circle ω . Let D be an arbitrary point inside BC such that $BD \neq DC$. Ray AD intersects ω again at E (other than A). Point F (other than E) is chosen on ω such that $\angle DFE = 90^\circ$. Line FE intersects rays AB and AC at points X and Y , respectively. Prove that $\angle XDE = \angle EDY$.

Solution

All angle and side length names are defined as in the figures below. Figure 1 is the diagram of the problem while Figure 2 is the diagram of the Ratio Lemma. Do note that the point names defined in the Ratio Lemma are not necessarily the same defined points on Figure 1.

First, we claim the Ratio Lemma:

$$\frac{\sin \theta_2}{\sin \theta'_2} = \frac{PB}{PC} \cdot \frac{b}{c}.$$

We prove this as follows:

By the Law of Sines of $\triangle ABP$, we get:

$$\frac{\sin \theta_2}{BP} = \frac{\sin \theta_1}{c} = \frac{\sin B}{AP}.$$

By the Law of Sines of $\triangle ACP$, we get:

$$\frac{\sin \theta'_2}{PC} = \frac{\sin \theta'_1}{b} = \frac{\sin C}{AP}.$$

Dividing each of the corresponding equations, we get:

$$\frac{PC}{PB} \cdot \frac{\sin \theta_2}{\sin \theta_1} = \frac{b}{c} \cdot \frac{\sin \theta_1}{\sin \theta'_1} = \frac{\sin B}{\sin C}.$$

Noting that $\sin \theta_1 = \sin(180^\circ - \theta'_1) = \sin \theta'_1$, we have:

$$\frac{\sin \theta_2}{\sin \theta_1} = \frac{PB}{PC} \cdot \frac{b}{c}.$$

We transition into our problem.

By Power of a Point on circle ω , we have:

$$XK \cdot XD = XF \cdot XE$$

$$YL \cdot YD = YF \cdot YE.$$

By power of a point on circle ω_1 , we have:

$$XB \cdot XA = XF \cdot XE$$

$$YC \cdot YA = YF \cdot YE.$$

Using the transitive property, we have:

$$XK \cdot XD = XB \cdot XA$$

$$YL \cdot YD = YC \cdot YA.$$

Thus, by reverse Power of a Point, we have that quadrilaterals $ADKB$ and $ACLD$ are cyclic.

Now, we move to angle chasing.

- $\angle ABD = \angle AKD = \alpha$ (from cyclic $ADKB$)
- $\angle ABD = \alpha = \angle ACD$ (from the isosceles triangle)
- $\angle ACD = \alpha = \angle ALD$ (from cyclic $ACLD$)

It is from here in which we focus on quadrilateral $AKEL$.

Let $\angle KDE = \beta$ and $\angle EDL = \gamma$. Now realize that all we want to show, as given by the problem statement, is $\beta = \gamma$.

Thus, $\angle KED = 90^\circ - \beta$, $\angle DEL = 90^\circ - \gamma$.

We also have:

$$\begin{aligned}\angle LAD &= 180^\circ - (\angle ADL + \alpha) \\ &= 180^\circ - ((180^\circ - \gamma) + \alpha) \\ &= \gamma - \alpha.\end{aligned}$$

Similarly, $\angle KAD = \beta - \alpha$.

Calling back to what we want to show - if we want to show that $\beta = \gamma$, it suffices to show that $\angle KAD = \angle LAD$ and $\angle KED = \angle LED$. But this is equivalent to showing $\triangle AKE \sim \triangle ALE$.

However, applying our Ratio Lemma to $\triangle AKE$, we have:

$$\frac{\sin \alpha}{\sin 90^\circ} = \frac{AD}{DE} \cdot \frac{KE}{KA}.$$

Applying our Ratio Lemma to $\triangle ALE$, we have:

$$\frac{\sin \alpha}{\sin 90^\circ} = \frac{AD}{DE} \cdot \frac{LE}{LA}.$$

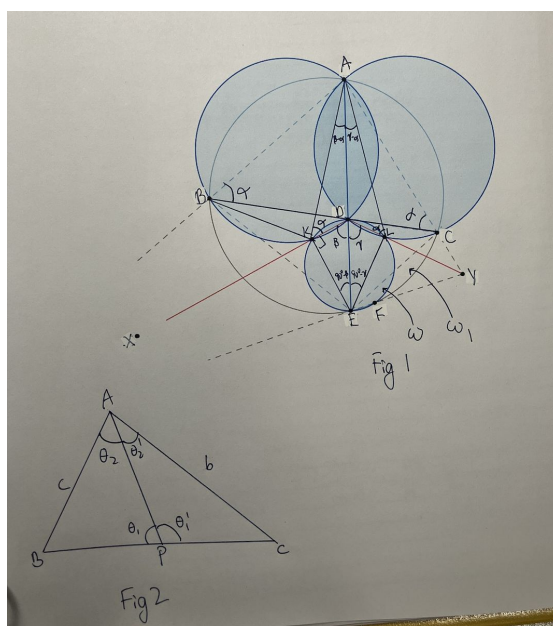
From the transitive property, we have:

$$\frac{KE}{KA} = \frac{LE}{LA}.$$

Now, realize that we already have a shared side AE and equivalent angles $\angle ALE = 90^\circ + \alpha = \angle AKE$. Thus, we establish a stronger bound between these two triangles - a congruence.

Thus, these two triangles are similar, and our proof is complete.

mathboy282



(cred. to v_Enhance diagram)

Solution 2 (Inversion)

We realize that $\angle DFE = 90$ is quite annoying in calculations. Therefore, taking an inversion at D will just make $AE \perp EF$, which is easier to incorporate. Also, many lines pass through D . Therefore, we have intuition to invert at D with some radius.

Take an inversion at D . We are not going to denote the inverted points with primes for simplicity. Note that $AE \perp EF$. Also, X is the intersection of (ABD) and (DFE) , which means it lies on the radical axis of (ABD) and (DFE) . Likewise, Y lies on the radical axis of (ADC) and (DFE) . However, note that $\overline{EF}$ is the radical axis of (ABC) and (DEF) , and $\overline{AC}$ is the radical axis of (ABC) and (ACD) , so by concurrency of radical axis, $\overline{AC} \cup \overline{EF}$ lies on $\overline{AX}$ (wow)! Meaning, we don't need to consider circle intersections, just line intersections. Now, we need to analyze what happens to the isosceles triangle.

Consider $\triangle ACD$ before inversion. Once you invert it, angle C goes to angle A . Same with $\triangle ADB$. However, angle C is equal to angle B by isosceles! This means AD is now the angle bisector of $\angle BAC$. A miracle!

Now we can phrase the question as follows:

For $\triangle ABC$, $\overline{AD}$ is the angle bisector of $\angle A$. Let $\overline{AD}$ intersect the circumcircle at E , and draw a line perpendicular to $\overline{AE}$ at E and call it l . Let $\overline{AC} \cup l = X$, and let $\overline{AB} \cup l = Y$. Prove $\angle YDE = \angle XDE$

One glance and one realizes the question is extremely easy. $\overline{AE}$ is a angle bisector and altitude of $\triangle AXY$, meaning that $\overline{YE} = \overline{EX}$

Therefore, $\overline{DE}$ is an altitude of $\triangle DYX$, and $\overline{YE} = \overline{EX}$, so of course $\overline{DE}$ is the angle bisector of $\angle YDX$! The problem is proven!

§15 USAJMO 2024

1. (*USAJMO 2024*) Let $ABCD$ be a cyclic quadrilateral with $AB = 7$ and $CD = 8$. Points P and Q are selected on segment AB such that $AP = BQ = 3$. Points R and S are selected on segment CD such that $CR = DS = 2$. Prove that $PQRS$ is a cyclic quadrilateral.

Solution 1

If $\overline{AB} \parallel \overline{CD}$, then $PQRS$ is an isosceles trapezoid, and thus is cyclic. Otherwise, let $X = \overleftrightarrow{AB} \cap \overleftrightarrow{CD}$, WLOG with A, C on segments $\overline{XB}, \overline{XD}$. By Power of a Point,

$$XA \cdot XB = XC \cdot XD \implies (XA)(XA+7) = (XC)(XC+8) \implies XA^2 + 7XA = XC^2 + 8XC$$

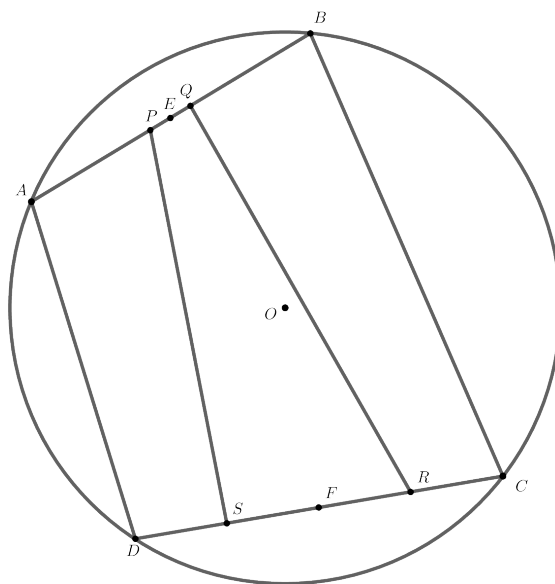
$$\implies XA^2 + 7XA + 12 = XC^2 + 8XC + 12 \implies (XA+3)(XA+4) = (XC+2)(XC+6) \implies XP \cdot XQ =$$

as desired $\square$

-sillybone

Solution 2

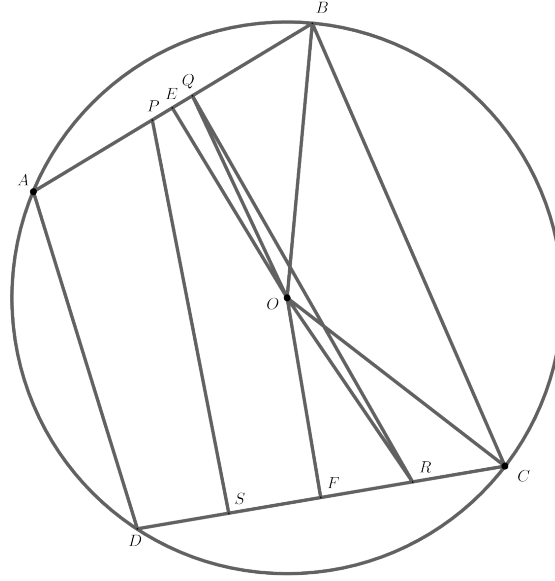
First, let E and F be the midpoints of AB and CD , respectively. It is clear that $AE = BE = 3.5$, $PE = QE = 0.5$, $DF = CF = 4$, and $SF = RF = 2$. Also, let O be the circumcenter of $ABCD$.



By properties of cyclic quadrilaterals, we know that the circumcenter of a cyclic quadrilateral is the intersection of its sides' perpendicular bisectors. This implies that $OE \perp AB$ and $OF \perp CD$. Since E and F are also bisectors of PQ and RS , respectively, if $PQRS$ is indeed a cyclic quadrilateral, then its circumcenter is also at O . Thus, it suffices to show that $OP = OQ = OR = OS$.

Notice that $PE = QE$, $EO = EO$, and $\angle QEO = \angle PEO = 90^\circ$. By SAS congruency, $\triangle QOE \cong \triangle POE \implies QO = PO$. Similarly, we find that $\triangle SOF \cong \triangle ROF$ and $OS = OR$. We now need only to show that these two pairs are equal to each other.

Draw the segments connecting O to B , Q , C , and R .



Also, let r be the circumradius of $ABCD$. This means that $AO = BO = CO = DO = r$. Recall that $\angle BEO = 90^\circ$ and $\angle CFO = 90^\circ$. Notice the several right triangles in our figure.

Let us apply Pythagorean Theorem on $\triangle BEO$. We can see that $EO^2 + EB^2 = BO^2 \implies EO^2 + 3.5^2 = r^2 \implies EO = \sqrt{r^2 - 12.25}$.

Let us again apply Pythagorean Theorem on $\triangle QEO$. We can see that $QE^2 + EO^2 = QO^2 \implies 0.5^2 + r^2 - 12.25 = QO^2 \implies QO = \sqrt{r^2 - 12}$.

Let us apply Pythagorean Theorem on $\triangle CFO$. We get $CF^2 + OF^2 = OC^2 \implies 4^2 + OF^2 = r^2 \implies OF = \sqrt{r^2 - 16}$.

We finally apply Pythagorean Theorem on $\triangle RFO$. This becomes $OF^2 + FR^2 = OR^2 \implies r^2 - 16 + 2^2 = OR^2 \implies OR = \sqrt{r^2 - 12}$.

This is the same expression as we got for QO . Thus, $OQ = OR$, and recalling that $OQ = OP$ and $OR = OS$, we have shown that $OP = OQ = OR = OS$, and we are done.

Solution 3

We can consider two cases: $AB \parallel CD$ or $AB \not\parallel CD$. The first case is trivial, as $PQ \parallel RS$ and we are done due to symmetry. For the second case, WLOG, assume that A and C are located on XB and XD respectively. Extend AB and CD to a point X , and by Power of a Point, we have

$$XA \cdot XB = XC \cdot XD,$$

which may be written as

$$XA \cdot (XA + 7) = XC \cdot (XC + 8),$$

or

$$XA^2 + 7XA = XC^2 + 8XC.$$

We can translate this to

$$XA^2 + 7XA + 12 = XC^2 + 8XC + 12,$$

so

$$XP \cdot XQ = (XA + 3)(XA + 4) = (XC + 2)(XC + 6) = XR \cdot XS,$$

and therefore by the Converse of Power of a Point $PQRS$ is cyclic, and we are done.

- spectraldragon8

Solution 4

All 4 corners of $PQRS$ have equal power of a point (12) with respect to the circle $(ABCD)$, with center O .

Draw diameters (of length AQ) of circle $(ABCD)$ through Q and S , with length A . Let q be the distance from Q to the circle along a diameter, and likewise s be distance from S to the circle.

Then $q(AQ - q) = s(AQ - s) = 12$ and $q, s < AQ/2$ (radius). Therefore, $q = s$ and $AQ/2 - q = AQ/2 - s$. But $AQ - q = OQ$, $AQ - s = OS$, $OQ = OP$ and $OS = OR$ by symmetry around the perpendicular bisectors of PQ and RS , so P, Q, R, S are all equidistant from O , forming a circumcircle around $PQRS$.

-BraveCobra22aops, oinava

Solution 4 (Coord Bash)

Let $A(2a_1, 2a_2)$, $B(2b_1, 2b_2)$, $C(2c_1, 2c_2)$, $D(2d_1, 2d_2)$, and the circumcenter of quadrilateral $ABCD$ be $O(0, 0)$.

Let's list what we know from the givens: Since the radii of a circle are equal in length, we can let

$$k = a_1^2 + a_2^2 = b_1^2 + b_2^2 = c_1^2 + c_2^2 = d_1^2 + d_2^2.$$

From the distance formula on AB and CD , we can simplify and get

$$a_1^2 + a_2^2 + b_1^2 + b_2^2 - 2(a_1b_1 + a_2b_2) = \frac{49}{4} \Rightarrow 2(a_1b_1 + a_2b_2) = 2k - \frac{49}{4}$$

and

$$c_1^2 + c_2^2 + d_1^2 + d_2^2 - 2(c_1d_1 + c_2d_2) = 16 \Rightarrow 2(c_1d_1 + c_2d_2) = 2k - 16$$

using the above substitution with k .

Since P , Q , R , and S are weighted points on AB and CD , we can get

$$P\left(\frac{8a_1 + 6b_1}{7}, \frac{8a_2 + 6b_2}{7}\right) \text{ and } Q\left(\frac{6a_1 + 8b_1}{7}, \frac{6a_2 + 8b_2}{7}\right)$$

along with

$$R\left(\frac{3c_1 + d_1}{2}, \frac{3c_2 + d_2}{2}\right) \text{ and } S\left(\frac{c_1 + 3d_1}{2}, \frac{c_1 + 3d_1}{2}\right).$$

Now, we want to show that the perpendicular bisectors of the sides of quadrilateral $PQRS$ are concurrent to prove that it is cyclic. Moreover, we suspect that the circumcenter of quadrilateral $PQRS$ is the circumcenter of quadrilateral $ABCD$, both of them being $(0, 0)$.

We only list the linear equations of the perpendicular bisectors of lines PS and QR , as the perpendicular bisectors of lines PQ and RS are the same as the perpendicular bisectors of lines AB and CD , respectively.

First, we analyze the perpendicular bisector of line PS . Notice that it has slope

$$-\frac{16a_1 + 12b_1 - 7c_1 - 21d_1}{16a_2 + 12b_2 - 7c_2 - 21d_2}$$

and point

$$\left(\frac{16a_1 + 12b_1 + 7c_1 + 21d_1}{28}, \frac{16a_2 + 12b_2 + 7c_2 + 21d_2}{28} \right).$$

The point-slope form equation would thus be

$$y - \frac{16a_2 + 12b_2 + 7c_2 + 21d_2}{28} = -\frac{16a_1 + 12b_1 - 7c_1 - 21d_1}{16a_2 + 12b_2 - 7c_2 - 21d_2} \left(x - \frac{16a_1 + 12b_1 + 7c_1 + 21d_1}{28} \right).$$

Since we claim that $(0, 0)$ is on the line, we substitute to get

$$(16a_1 + 12b_1 + 7c_1 + 21d_1)(16a_1 + 12b_1 - 7c_1 - 21d_1) = -(16a_2 + 12b_2 + 7c_2 + 21d_2)(16a_2 + 12b_2 - 7c_2 - 21d_2),$$

and this simplifies to

$$16 \left(16(a_1^2 + a_2^2) + 24(a_1b_1 + a_2b_2) + 9(b_1^2 + b_2^2) \right) = 49 \left((c_1^2 + c_2^2) + 6(c_1d_1 + c_2d_2) + 9(d_1^2 + d_2^2) \right).$$

Using substitutions from the first three equations, this becomes

$$\begin{aligned} 16 \left(16k + 12 \left(2k - \frac{49}{4} \right) + 9k \right) &= 49(k + 3(2k - 16) + 9k) \\ \Rightarrow 16(49k - 3 \cdot 49) &= 49(16k - 3 \cdot 16) \\ \Rightarrow k - 3 &= k - 3, \end{aligned}$$

which is true, implying that $(0, 0)$ does indeed satisfy the equation.

Analogously, we can show that the perpendicular bisector of line QR also passes through the origin.

Since the perpendicular bisectors of the sides of quadrilateral $PQRS$ all intersect at the same point, namely $(0, 0)$, which is also the circumcenter of quadrilateral $ABCD$, we can conclude that $PQRS$ is a cyclic quadrilateral.

Solution 5 (Euclidean)

Let O be the center of the circumcircle of $ABCD$, and let r be the circumradius of that quadrilateral. Also, let E be the midpoint of $\overline{AB}$ and let F be the midpoint of $\overline{CD}$. If we look at $\triangle AEO$, we see that it is a right triangle with hypotenuse r and one leg $\frac{7}{2}$. Therefore, the other leg, $\overline{EO}$, has length $\sqrt{r^2 - \frac{49}{4}}$. We have that $EP = EQ = \frac{1}{2}$, and therefore, by the Pythagorean Theorem we have that $OP = OQ = \sqrt{r^2 - 12}$.

Applying the same logic to $\overline{CD}$, we find that $OS = OR = \sqrt{r^2 - 12}$. Because $PO = QO = SO = RO$, the points P, Q, S , and R all lie on a common circle with center O and radius $\sqrt{r^2 - 12}$. Therefore, $PQRS$ is a cyclic quadrilateral.

2. (*USAJMO 2024*) Let m and n be positive integers. Let S be the set of integer points (x, y) with $1 \leq x \leq 2m$ and $1 \leq y \leq 2n$. A configuration of mn rectangles is called *happy* if each point in S is a vertex of exactly one rectangle, and all rectangles have sides parallel to the coordinate axes. Prove that the number of *happy* configurations is odd.

Solution 1

We will proceed by induction on n . For the purpose of notation, we will group the points in S into m rows and n columns and refer to them as such accordingly.

Base Case

Consider the case where $n = 1$. It is clear that any rectangle in the happy configuration is determined only by which two rows its vertices lie on, and that these two rows are automatically filled by the vertices of the rectangle. By repeatedly picking two rows to form a new rectangle from the remaining rows, the number of rectangles in this case is

$$\frac{1}{m!} \prod_{k=1}^m \binom{2k}{2} = \frac{1}{m!} \binom{2m}{2} \binom{2m-2}{2} \cdots \binom{2}{2} = \frac{(2m)!}{m! \cdot 2^m}$$

We can evaluate the parity of this expression by expanding the factorials:

$$\frac{(2m)(2m-1)(2m-2) \cdots (2)(1)}{(2m)(2m-2) \cdots (2)} = (2m-1)(2m-3) \cdots (1)$$

Since none of the factors are even, this expression must be odd. As a result, there are always an odd number of happy configurations if $n = 1$.

Inductive Step

Assume that there are an odd number of happy configurations for some m and n . We will now prove that there must also be an odd number of happy configurations for m and $n + 1$.

Consider S for m and $n + 1$. Denote the rightmost two columns of S as the "strip".

Case 1: All rectangles in the happy configuration lie entirely within the strip or entirely outside of the strip.

By the inductive hypothesis, there are an odd number of happy configurations in the region of S outside of the strip. Additionally, by the base case, there are an odd number of happy configurations in the strip. Since no rectangles extend between the two regions, the number of happy configurations in this case is just the product of the two values, which is clearly odd.

Case 2: There exists a rectangle R which contains some vertices inside of the strip and some vertices outside of the strip.

Let a happy configuration with this property be denoted as H .

First of all, because each vertex in a rectangle shares a column with another vertex and because the strip is only two columns wide, R must have exactly two vertices in the strip. Furthermore, the two vertices in the strip must lie in the same column, and the same applies for the two vertices outside of the strip.

Notice that both vertices of R must lie on rows which each contain only one other point inside of the strip. This forces the existence of another rectangle R' which

has exactly two vertices in the strip and has vertices lying on the same two rows as R . Then, it must be possible to create a new happy configuration H' by swapping the vertices of R and R_1 which in the strip between the two rectangles.

Notice that repeating this procedure on the new happy configuration results in H again. As a result, there exists a one-to-one pairing between happy configurations for which such a rectangle R exists.

However, the existence of such a pairing implies that the number of happy configurations in this case is even.

Conclusion:

The number of happy configurations for m and $n + 1$ is the sum of the number of happy configurations from each of the two cases. Since there are an odd number of happy configurations in case 1 and an even number of happy configurations in case 2, there must be an odd total number of happy configurations.

By induction, there must be an odd number of happy configurations of S for any positive integer m and n .

Note: It is possible to reduce the base case to $m = n = 1$ because the inductive step can be applied to incrementing m as well.

- pomme_de_terre__ (aka BlackDaddyUWU)

Solution 2

We can group configurations into pairs where:

- (i) 1. If it is *not* horizontally symmetric, it makes a pair with its horizontal reflection.
- (i) 2. Else, if it is *not* vertically symmetric, it makes a pair with its vertical reflection.
- (i) 3. Else, if it is *not* half-symmetric, we can similarly make a pair with its horizontal or vertical half-reflection.

The **half-reflection** is defined by reflecting only the edge of each rectangle closest to the horizontal/vertical center, while keeping the other edge fixed. For example, a vertical half-reflection would send the rectangle

$$\{(0, 0), (1, 0), (1, n - 1), (0, n - 1)\}$$

to

$$\{(0, 0), (1, 0), (1, n + 1), (0, n + 1)\}.$$

Since every configuration that is not symmetric or half-symmetric comes in pairs, we only need to look at the parity of the configurations that are both symmetric and half-symmetric. But, being symmetric and half-symmetric uniquely determines a rectangle if we know a single vertex, so there is only one such configuration.

3. (*USAJMO 2024*) Let $a(n)$ be the sequence defined by $a(1) = 2$ and $a(n + 1) = (a(n))^{n+1} - 1$ for each integer $n \geq 1$. Suppose that $p > 2$ is prime and k is a positive integer. Prove that some term of the sequence $a(n)$ is divisible by p^k .

Solution 1

Lemma 1:

Given a prime p , a positive integer k , and an even m such that $p^k | a(m)$, we must have that $p^{k+1} | a(m+2)$.

Proof of Lemma 1:

$$a(m+1) \equiv (a(m))^{m+1} - 1 \equiv -1 \pmod{p^{k+1}}$$

$$\text{Then, } a(m+2) \equiv (a(m+1))^{m+2} - 1 \equiv (-1)^{m+2} - 1 \equiv 0 \pmod{p^{k+1}}$$

Therefore, by induction, if there exists an even integer m such that $p | a(m)$, then for all integers k , $p^k | a(m+2k-2)$, so we are done if there exists an even m such that $p | a(m)$.

Now, consider the case where there is some prime $p > 2$ such that there are no even integers m such that $p | a(m)$.

Lemma 2:

In this case, we must have that $p | a(m)$ if $m \equiv -1 \pmod{p-1}$ for all integers m .

Proof of Lemma 2:

Suppose for the sake of contradiction that there exists some m such that $m \equiv -1 \pmod{p-1}$ and p does not divide $a(m)$. Then, we have $a(m+1) \equiv (a(m))^{m+1} - 1 \equiv 1 - 1 \equiv 0 \pmod{p}$, by Fermat's Little Theorem. Since for all $p > 2$, $p-1$ is even, then $m+1$ would be even. However this results in a contradiction.

Then, we get that if $m \equiv -1 \pmod{p-1}$, then $0 \equiv a(m) \equiv (a(m-1))^m - 1 \pmod{p} \implies (a(m-1))^{m+1} \equiv 1 \equiv a(m-1) \pmod{p}$.

Then, by LTE, $v_p(a(m)) = v_p((a(m-1))^m - 1) = v_p(a(m-1) - 1) + v_p(m) > v_p(m)$. Since $\gcd(p-1, p) = 1$, then $\gcd(p-1, p^k) = 1$ for all positive integers k , so then by **Chinese Remainder Theorem** there exists integers m such that $m \equiv -1 \pmod{p-1}$ and $m \equiv 0 \pmod{p^k}$, so we are done $\square$

Remark: I think this is a very cool NT problem.

-bronzetruck2016

4. (*USAJMO 2024*) Let $n \geq 3$ be an integer. Rowan and Colin play a game on an $n \times n$ grid of squares, where each square is colored either red or blue. Rowan is allowed to permute the rows of the grid, and Colin is allowed to permute the columns of the grid. A grid coloring is *orderly* if:

- no matter how Rowan permutes the rows of the coloring, Colin can then permute the columns to restore the original grid coloring; and
- no matter how Colin permutes the columns of the coloring, Rowan can then permute the rows to restore the original grid coloring;

In terms of n , how many orderly colorings are there?

Solution 1

We focus on the leftmost column for simplicity. Let m be the number of red squares in this column. We then have five cases:

1. $m = 1$

When Rowan permutes the rows of the coloring, we consider only the first column, which by the above contains $m = 1$ red colors, so there are $\binom{n}{1} = n$ ways to permute the first column's rows. Thus every other column will have to contain one different permutation of the first column; otherwise, there will be at least one permutation of which there is no corresponding column.

Furthermore, each permutation will be different, so each row will contain one and only one red square, which also fulfills the case of if Colin permutes the coloring first. Thus there are $n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1 = n!$ different colorings for this case (the same as choosing squares such as no square is in the same row or column as any other square).

2. $m = n - 1$

This is essentially the same as case 1 except for the coloring; now there is one blue square and the rest are red squares. Thus there are also $n!$ different colorings for this case.

3. $m = 0$

Since we have an entirely blue column, we are unable to have a column with 1 red square only as doing so would leave one permutation that is not covered by at least one column (that space is being taken for the blank column). We are also unable to have a completely blue column as doing so would allow for Colin to shift the columns and in doing so fail for Rowan to shift back the columns. We also cannot have a column with any other number of red squares other than 0 as will be shown below, so there is 1 case here in which the entire coloring is red.

4. $m = n$

This is the same as an entire blue column, and, similar to above, we have 1 coloring.

5. $1 < m < n - 1$

This is the final case and is equivalent to permuting for $\binom{n}{m}$ different ways. We must prove that this is greater than n to show that the columns are not able to contain every possible permutation of this column for all values of n such that $n > 3$ (when $n = 3$, there is no such positive integer m that satisfies the conditions). Note that if we have any column with a different number of red squares, it is an unattainable column and is thus not optimal.

Lemma: Given that m and n are positive integers such that $1 < m < n - 1$ and $n > 3$, it is true for all m and n that $\binom{n}{m} > n$.

Proof: Assume that $m < \frac{n-1}{2}$.

$$\Leftrightarrow m + 1 < n - m$$

$$\Leftrightarrow (m+1)!(n-m-1)! < m!(n-m)!$$

$$\Leftrightarrow \frac{n!}{m!(n-m)!} < \frac{n!}{(m+1)!(n-m-1)!}$$

$$\Leftrightarrow \binom{n}{m} < \binom{n}{m+1}$$

Similarly, we can prove that $\binom{n}{m} > \binom{n}{m+1}$ for $m > \frac{n-1}{2}$.

Now we split our proof into two cases.

Case 1: n is even.

The largest integer less than $\frac{n-1}{2}$ is $\frac{n}{2} - 1$, so we know that:

$$\binom{n}{\frac{n}{2}} > \binom{n}{\frac{n}{2}-1} > \cdots > \binom{n}{2}$$

by induction. On the other hand, the smallest integer greater than $\frac{n-1}{2}$ is $\frac{n}{2}$, so we know that:

$$\binom{n}{\frac{n}{2}} > \binom{n}{\frac{n}{2}+1} > \cdots > \binom{n}{n-2}$$

also by induction. Thus out of the given range for m we know that $\binom{n}{2}$ and $\binom{n}{n-2}$ are the minimum values, and all that is left is to prove that they are both greater than n . Furthermore, since $\binom{n}{2} = \binom{n}{n-2}$, we only have to prove that $\binom{n}{2} > n$.

We start with the given: $n > 3$

$$\Leftrightarrow \frac{n-1}{2} > 1$$

$$\Leftrightarrow \frac{n(n-1)}{2} > n$$

$$\Leftrightarrow \frac{n!}{2!(n-2)!} > n$$

$$\Leftrightarrow \binom{n}{2} > n$$

Thus we have proven the inequality for all even n .

Case 2: n is odd.

The greatest integer less than $\frac{n-1}{2}$ is $\frac{n-3}{2}$, so we know that:

$$\binom{\frac{n-1}{2}}{\frac{n-1}{2}} > \binom{\frac{n-3}{2}}{\frac{n-3}{2}} > \cdots > \binom{n}{2}$$

by induction. On the other hand, the smallest integer greater than $\frac{n-1}{2}$ is $\frac{n+1}{2}$, so we know that:

$$\binom{\frac{n+1}{2}}{\frac{n+1}{2}} > \binom{\frac{n+3}{2}}{\frac{n+3}{2}} > \cdots > \binom{n}{n-2}$$

also by induction. Since $\binom{\frac{n+1}{2}}{\frac{n+1}{2}} = \binom{\frac{n-1}{2}}{\frac{n-1}{2}}$, we know that once again, $\binom{n}{n-2} = \binom{n}{2}$ is the minimum of the given range for m , and the same proof applies. Thus, the inequality holds true for odd and in turn all positive integers $n > 3$.

As a result, due to our lemma, there are always more permutations of the columns than the number of columns itself, so there will always exist a permutation of the column such that there are no corresponding original columns of which to match with. Thus there are no solutions for this case.

In conclusion, there are a total of $2 \cdot n! + 2$ different colorings for which the above apply.

Solution 2

Lemma 1: Each row and column must have the same number of red squares.

Proof: Suppose two rows do not have the same number of red squares. Suppose Rowan permutes the two rows onto each other. Then, because the two rows have a different number of red squares, there is no way for Colin to permute the columns, which permutes the squares in the row, to match the other row because the two rows have a different number of red squares. By a similar proof, the columns must have the same number of red squares as each other. Assume that each row has m red squares. Then, there are nm red squares in total. Because each column must have the same number of red squares as the other columns, the columns also must have m red squares each.

Now, we want to find which values of m can produce a valid orderly coloring.

Lemma: $C(n, k) > n$ for $k \neq 0, 1, n-1, n$. This is needed to prove the next lemma and is the most mathematical part of the solution.

Proof: Start with $C(n, 1) = n$. Each time we increase k by 1 until $k = \lfloor 0.5n \rfloor$, we are multiplying the previous value of $C(n, k)$ by a fraction more than 1.* Thus, $C(n, k) > n$ for $2 \leq n \leq \lfloor 0.5n \rfloor$. Because $C(n, k) = C(n, n-k)$, the lemma is proven for the rest of the conjectured values.

Lemma 2: If $m \neq 0, 1, n-1, n$, the coloring is not orderly.

Proof: Let's take a look at a particular column, where assume there are $2 \leq m \leq n - 2$ red squares. When Rowan permutes the rows, the squares in this column have $C(n, m)$ possible permutations. In order for Colin to be able permute the columns to return the grid to its original state, all of these $C(n, m)$ permutations must be columns of the original grid. However, because there are only n columns. By the previous lemma, there are more permutations than columns, so the we are done.

Lemma 3: If a permutation satisfies $m = 0, 1, n - 1, n$ and satisfies lemma 1, it is orderly.

Proof: The $m = 0, n$ case is trivial: the grid never changes! For $m = 1, n - 1$, we only have to prove the $n = 1$ case because of symmetry: we can change red to blue and blue to red and it won't change the orderliness of the coloring as the overall arrangement of the different colors won't be changed. For the $m = 1$ case, when Rowan permutes the rows, all of the columns will still be different columns of the original grid (as each column must have one red square, the rows of the red squares will be different) and can be permuted back to the original arrangement by Colin. Thus, this lemma is true.

Finally, there is 1 coloring for the $m = 0$ case, and $n!$ ways to permute the red squares to see which column values are assigned to the rows in the $m = 1$ case. There are similarly $n!$ colorings for the $m = n - 1$ case, and 1 coloring for the $m = 0$ case. Thus, the answer is $\boxed{2n! + 2}$.

* Specifically, $\frac{C(n, k)}{C(n, k-1)} = \frac{n-k+1}{k-1}$ which is greater than 1 for $2 \leq n \leq \lfloor 0.5n \rfloor$.

Another way to prove that $\binom{n}{m} > n$ Suppose we have $2 \leq m \leq \lfloor \frac{n}{2} \rfloor$. Our goal is to show that

$$\binom{n}{m} > n$$

$$\frac{n}{m} \cdot \frac{n-1}{m-1} \cdot \frac{n-2}{m-2} \cdots \frac{n-m+1}{1} > n$$

$$\frac{1}{m} \cdot \frac{n-1}{m-1} \cdot \frac{n-2}{m-2} \cdots \frac{n-m+1}{1} > 1$$

$$\frac{n-1}{m-1} \cdot \frac{n-2}{m-2} \cdots \frac{n-m+1}{m} > 1$$

Each of the above fractions is greater than one because $2 \leq m \leq \lfloor \frac{n}{2} \rfloor$, so we are done.

5. (USAJMO 2024) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(x^2 - y) + 2yf(x) = f(f(x)) + f(y)$$

for all $x, y \in \mathbb{R}$.

Solution 1

We claim the solutions are $\boxed{f \equiv 0, x^2, -x^2}$. These solutions clearly work. Let the given assertion be $P(x, y)$. $P(0, 0)$ yields $f(f(0)) = 0$. Then, $P(f(0), y)$ produces

$$f(0) + f(y) = f(f(0)^2 - y)$$

Letting $y = 0, f(0)^2$ gives $2f(0) = f(f(0)^2)$ and $f(f(0)^2) = 0$, so $f(0) = 0$. Now, $P(x, x^2)$ and $f(x, 0)$ implies

$$x^2 f(x) = f(f(x)) = f(x^2) \quad (1)$$

Notice that this implies f is even, i.e. $f(x) = f(-x)$. Furthermore,

$$\begin{aligned} f(x_1) = f(x_2) &\implies f(f(x_1)) = f(f(x_2)) \implies x_1^2 f(x_1) = x_2^2 f(x_2) \\ &\implies x_1^2 = x_2^2 \text{ or } f(x_1) = f(x_2) = 0 \end{aligned} \quad (2)$$

To rule out the second case, suppose $f(c) = 0, c \neq 0$. Assuming f is not identically 0, let $f(y) \neq 0$, so $P(c, y)$ and $P(c, -y)$ gives

$$f(c^2 - y) = f(y) = f(-y) = f(c^2 + y) \implies (c^2 - y)^2 = (c^2 + y)^2 \implies y = 0$$

a contradiction. Thus, by (1) and (2), we conclude $|f(x)| = x^2$.

Lastly, to eliminate pointwise trap, assume $f(a) = a^2, f(b) = -b^2, a, b \neq 0$. Since f is even, assume $b < 0$. Then, $P(a, b)$ yields

$$f(a^2 - b) + 2a^2 b = f(a^2) - b^2$$

Casework on the signs of $f(a^2 - b)$ and $f(a^2)$ yields that either a or b is 0, a contradiction. Thus, the only solutions are $\boxed{f \equiv 0, x^2, -x^2}$.

-sillybone

Solution 2

Plugging in y as 0 :

$$f(x^2) = f(f(x)) + f(0) \quad (1)$$

Plugging in x, y as 0 :

$$f(0) = f(f(0)) + f(0)$$

or

$$f(f(0)) = 0$$

Plugging in x as 0 :

$$f(-y) + 2yf(0) = f(f(0)) + f(y),$$

but since $f(f(0)) = 0$,

$$f(-y) + 2yf(0) = f(y) \quad (2)$$

Plugging in y^2 instead of y in the given equation:

$$f(x^2 - y^2) + 2y^2 f(x) = f(f(x)) + f(y^2)$$

Replacing y and x :

$$f(y^2 - x^2) + 2x^2 f(y) = f(f(y)) + f(x^2)$$

The difference would be:

$$f(x^2 - y^2) - f(y^2 - x^2) + 2y^2 f(x) - 2x^2 f(y) = f(f(x)) - f(x^2) - (f(f(y)) - f(y^2)) \quad (3)$$

The right-hand side would be $f(0) - f(0) = 0$ by (1). Also,

$$f(x^2 - y^2) - f(y^2 - x^2) = 2(x^2 - y^2)f(0)$$

by (2) So, (3) is reduced to:

$$2(x^2 - y^2)f(0) + 2y^2 f(x) - 2x^2 f(y) = 0$$

Regrouping and dividing by 2:

$$y^2(f(x) - f(0)) = x^2(f(y) - f(0))$$

$$\frac{f(x) - f(0)}{x^2} = \frac{f(y) - f(0)}{y^2}$$

Because this holds for all x and y , $\frac{f(x) - f(0)}{x^2}$ is a constant. So, $f(x) = cx^2 + f(0)$. This function must be even, so $f(y) - f(-y) = 0$. So, along with (2), $2yf(0) = 0$ for all y , so $f(0) = 0$, and $f(x) = cx^2$. Plugging in cx^2 for $f(x)$ in the original equation, we get:

$$c(x^4 - 2x^2y + y^2) + 2cx^2y = c^3x^4 + cy^2$$

$$c(x^4 + y^2) = c(c^2x^4 + y^2)$$

So, $c = 0$ or $c^2 = 1$. All of these solutions work, so the solutions are $f(x) = -x^2, 0, x^2$.

-codemaster11

Solution 3

Let our equation be $P(x, y)$. We start by plugging in some initial values:

$$y = x^2 : f(0) + 2x^2 f(x) = f(f(x)) + f(x^2) \quad (1)$$

$$y = 0 : f(x^2) = f(f(x)) + f(0) \quad (2)$$

$$x = 0 : f(-y) + 2yf(0) = f(f(0)) + f(y) \quad (3)$$

Plugging in $x = 1$ into (3) gives

$$f(1) = f(f(1)) + f(0) \quad (4).$$

From (1), we get

$$f(0) + 2x^2 f(x) = 2f(x^2) - f(0) \implies x^2 f(x) + f(0) = f(x^2)$$

Substituting in what we have in (3) gives

$$x^2 f(x) + f(0) = f(0) = f(f(x)) \implies x^2 f(x) = f(f(x)).$$

Plugging in $x = 1$ gives

$$f(1) = f(f(1)) \quad (5).$$

As a result, (4) becomes $f(0) = 0$.

Now, (3) becomes

$$f(x^2) = f(f(x)) \quad (6)$$

and (2) becomes

$$f(y) = f(-y) \quad (7).$$

Note that $f \equiv 0$ is a solution. Now, assume $f(x) \neq 0$.

Claim: f is injective over $\mathbb{R}^+$.

Let $f(a) = f(b) \neq 0$ with $a, b > 0$. Plugging in $x = a, y = b^2$ and $x = b, y = a^2$ into P gives us

$$\begin{aligned} f(a^2 - b^2) + 2b^2 f(a) &= f(a^2) + f(b^2) \\ f(b^2 - a^2) + 2a^2 f(b) &= f(b^2) + f(a^2) \end{aligned}$$

Subtracting, and using (7) gives us $2(a^2 - b^2)f(a) = 0$, which implies that either $f(a) = 0$ or $a = \pm b$. Either way leads to contradiction. Thus, f is injective. $\square$

As a result, (6) becomes $f(x) = \pm x^2$. Piecing everything yields $f(x) = 0, \pm x^2$.

It just remains to verify these solutions work, and doing so is quite trivial;

$$\begin{aligned} f(x) = 0 : 0 + 0 &= 0 + 0, \\ f(x) = x^2 : (x^2 - y)^2 + 2yx^2 &= x^4 + y^2, \\ f(x) = -x^2 : -(x^2 - y)^2 - 2yx^2 &= -x^4 - y^2, \end{aligned}$$

all of which are obviously true.

Solution 4

Start with $y = 0$ to get: $f(x^2) = f(f(x)) + f(0)$. Then substitute $x = 0$ to get: $f(-y) + 2yf(0) = f(f(0)) + f(y)$. We will show that $f(f(0)) = 0$. Substitute $x = y = 0$ to get $f(0) = f(f(0)) + f(0)$ which yields the desired $f(f(0)) = 0$. Hence, $f(-x) + 2xf(0) = f(x)$ since y is just a dummy variable. Now we solve for $f(0)$ in our two equations we found. We have that $f(0) = f(x^2) - f(f(x))$ from the first equation and that $f(0) = \frac{f(x) - f(-x)}{2x}$ from the second equation.

Hence, we set these equal to get:

$$2xf(x^2) - 2xf(f(x)) = f(x) - f(-x)$$

Replace x with $-x$ to get:

$$-2xf(x^2) + 2xf(f(-x)) = f(-x) + f(x)$$

Now add to get:

$$2x(f(f(-x)) - f(f(x))) = 0$$

Now since x can't obviously be equal to 0, we need $f(f(-x)) = f(f(x))$.

Case 1: f is an injective function

Hence, $f(-x) = f(x)$ and thus $f(x)$ is an even function.

Claim: f cannot have degree at least 4

Proof:

Assume f is a polynomial. Assume there exists a solution of which f has degree at least 4. Hence, $f(x) = a_n x^{2n} + a_{n-1} x^{2n-2} + \dots + a_1 x^2 + a_0$ where $n \geq 2$. If we go back to the original equation in the problem statement and plug in $y = x^2$, then we have:

$$f(0) + 2x^2 f(x) = f(f(x)) + f(x^2)$$

So we have:

$$a_0 + 2x^2 f(x) = a_n f(x)^{2n} + \dots + a_1 f(x)^2 + a_0$$

$$\text{Hence, } 2x^2 f(x) = a_n f(x)^{2n} + \dots + a_1 f(x)^2$$

But we know f has degree $2n$ so the LHS will have degree $2 + 2n$ but the RHS will have degree $(2n)(2n) = 4n^2$. Solving $4n^2 = 2n + 2$ will have no integer solutions. Recall that $n \geq 2$. Clearly, no functions will satisfy this equation and the claim is proven. $\square$

We have shown that $f(x)$ can have degree at most 2 if it is an injective function. Hence, let $f(x) = ax^2 + bx + c$. Recall that we had shown earlier that $f(f(0)) = 0$. Hence, $af(0)^2 + bf(0) + c = 0$ but we know $f(0) = c$. Hence, $ac^2 + bc + c = 0$ which yields $c(ac + b + 1) = 0$. Clearly, $c = 0$ has to occur. Hence, $f(x) = ax^2 + bx$. Now, recall we showed that $f(x^2) = f(f(x)) + f(0)$. Hence, $ax^4 + bx^2 = af(x)^2 + bf(x) + f(0)$. Plugging and simplifying, we eventually get:

$$x^4(a - a^3) - 2a^2bx^3 - abx^2(b + 1) = f(0) = \text{constant}$$

Hence, all the x terms must cancel out for that to be a constant. This can be achieved by either $a = 0$ and b can be anything OR $b = 0$ and $a - a^3 = 0$. Now if $a = 0$, then $f(x) = bx$ but recall that $f(x)$ is an even function and hence $b = 0$ is the only suitable option for a linear function to turn into an even function. Hence, $f(x) = 0$ is a solution. If $b = 0$ and $a - a^3 = 0$, then $a = 1, -1, 0$ and hence $f(x) = x^2, -x^2$ are two more new solutions.

Now we have to consider if $f(x)$ is not a polynomial.

Claim: f has to be a polynomial to satisfy the given problem statement

Proof:

We will first show that $f(f(x)) = x^2 f(x)$. Substitute $y = \frac{x^2}{2}$:

$$f\left(\frac{x^2}{2}\right) + x^2 f(x) = f(f(x)) + f\left(\frac{x^2}{2}\right). \text{ Hence, the lemma. } \square$$

We have already shown that if f is injective, then f is even and this proof lies in the case of when f is injective. Hence, $f(x)$ is an even function. Recall that $f(x^2) = f(f(x)) + f(0)$. Hence, $f(x^2) = x^2 f(x) + f(0)$. Now we plug in $x = 1$ to get $f(1) = f(1) + f(0)$ and hence $f(0) = 0$. In other words, $f(x^2) = x^2 f(x)$. (Haven't finished proof yet)

Case 2: f is not an injective function

We can prove that f has to be injective. This is shown in sml1809's solution above in which we claim $f(a) = f(b) \neq 0$ and substitute $x = a, y = b^2$ and $x = b, y = a^2$ to prove the claim that f has to be injective.

Solution 5(Another solution)

We will show $x^2 f(x) = f(f(x))$. First, the original equation is

$$f(x^2 - y) + 2yf(x) = f(f(x)) + f(y)$$

We will make the substitution $y = x^2 - 1$

$$f(1) + 2(x^2 - 1)f(x) = f(f(x)) + f(x^2 - 1)$$

We will now take the original equation again and substitute $y = 1$

$$f(x^2 - 1) + 2f(x) = f(f(x)) + f(1) \implies f(x^2 - 1) = f(f(x)) + f(1) - 2f(x)$$

Now we will substitute this new representation of $f(x^2 - 1)$ into our equation to get

$$f(1) + (2x^2 - 2)f(x) = f(f(x)) + f(f(x)) + f(1) - 2f(x) \implies (2x^2)f(x) = 2f(f(x)) \implies x^2 f(x) = f(f(x)) \square.$$

We will now plug this in to our given equation

$$f(x^2 - y) + 2yf(x) = f(f(x)) + f(y) \implies f(x^2 - y) + 2yf(x) = x^2f(x) + f(y) \implies f(x^2 - y) - f(y) = f(x)(x^2 - 2y).$$

Now we will make the substitution $x = y$

$$f(x^2 - x) - f(x) = f(x)(x^2 - 2x) \implies f(x^2 - x) = f(x)(x^2 - 2x + 1) \implies f(x(x - 1)) = f(x)(x - 1)^2. \text{ This immediately raises a conjecture. If we have two real numbers } a, b \text{ where } f \text{ is defined at these two real numbers, then}$$

$$f(ab) = f(a)b^2$$

Let's prove this. From our equation $f(x^2 - y) + 2yf(x) = x^2f(x) + f(y)$, we can substitute $y = -x$ to get

$$f(x^2 + x) - 2xf(x) = x^2f(x) + f(-x) \implies f(x^2 + x) = f(x)(x^2 + 2x) + f(-x). \text{ If we can show } f(-x) = f(x), \text{ then } f(x^2 + x) = f(x(x + 1)) = f(x)(x^2 + 2x + 1) = f(x)(x + 1)^2 \implies f(x(x + 1)) = f(x)(x + 1)^2 \text{ and we should be done as we can now prove } f(ab) = f(a)b^2 \text{ by induction. So let's prove } f(-x) = f(x).$$

The proof is simple. From our given equation $f(x^2 - y) + 2yf(x) = f(f(x)) + f(y)$, let's substitute $x = 0$ to get

$$f(-y) = f(f(0)) + f(y).$$

We will prove $f(0) = 0$. From our given equation $f(x^2 - y) + 2yf(x) = f(f(x)) + f(y)$, we will plug in $x = y = 1$ to get

$$f(0) + 2f(1) = f(f(1)) + f(1) \implies f(0) + f(1) = f(f(1)) \implies f(0) + f(1) = 1^2f(1) \implies f(0) = 0 \text{ following } f(f(x)) = x^2f(x). \text{ Henceforth, we go back to } f(-y) = f(f(0)) + f(y) \implies f(-y) = f(0) + f(y) \implies f(-y) = f(y) \implies f(-x) = f(x) \square.$$

Hence, as $f(-x) = f(x)$, we had concluded before that if this was the case,

$$f(x(x + 1)) = f(x)(x + 1)^2 \text{ and } f(x(x - 1)) = f(x)(x - 1)^2 \text{ where the second equation we had derived before proving } f(-x) = f(x). \text{ From these two equations, we can generate any two real numbers that differ by 1 where}$$

$$f(ab) = f(a)b^2 \text{ where } a - b = 1.$$

We will show this happens for all real a, b as we have proved only part of our conjecture. Note that $f(a) = f(-a)$ has been proved.

We will assume $f(ab) = f(a)b^2$ only holds true if and only if $a - b = 1$. Now let $b = -1$. We have

$f(-a) = f(a)$ which holds true for all real numbers. However, we assumed the above was true if and only if $a - b = 1$. This would imply the above was true if and only if $a = 0$ which is not the case. Hence, this is a contradiction and we have proved our conjecture. $\square$.

Now, we have shown $f(ab) = f(a)b^2$ is true for all real a, b . Let $a = 1$ to get

$$f(b) = b^2 \text{ which means } f(x) = x^2 \text{ is a solution.}$$

Now consider $ab = x \implies b = \frac{x}{a}$. Now $f(x) = f(a)\frac{x^2}{a^2} \implies \frac{f(x)}{x^2} = \frac{f(a)}{a^2}$. From here we can generate many quadratic functions. Knowing $f(x) = x^2$ is a solution gives us motivation for $f(x) = -x^2$ which we can confirm will satisfy the given equation. We can prove this also

Given the original equation $f(x^2 - y) + 2yf(x) = f(f(x)) + f(y)$, we have found $f(x) = x^2$ is a solution. We will show that if $f(x)$ is a solution, so is $-f(x)$. First,

we'll assume something different, then prove by contradiction. We claim that if $f(x)$ is a solution, $-f(x)$ is not a solution. Let's plug in $f(x) = x^2$ since we already know that's a solution. We have

$$(x^2 - y)^2 + 2x^2y = x^4 + y^2 \implies x^4 - 2x^2y + y^2 + 2x^2y = x^4 + y^2 \text{ which holds.}$$

So now since we know $f(x) = x^2$ is a solution, we had claimed that $-f(x) = -x^2$ was not a solution. Hence we can plug this in to get

$$-(x^2 - y)^2 - 2x^2y = f(-x^2) - y^2 \implies -(x^2 - y)^2 - 2x^2y = -x^4 - y^2 \text{ of which this is exactly the negation of the equation we got for } f(x) = x^2. \text{ Therefore } -f(x) \text{ is indeed a solution and we have a contradiction with } -f(x) \text{ not being a solution. Thus } -f(x) \text{ has to be a solution. } \square.$$

Now assuming f is constant, we have

$$f(x) = c \implies c + 2yc = x^2c + c \implies 2yc = x^2c \implies c = 0 \text{ is a solution. Hence } f(x) = 0 \text{ is also a solution and so } f(x) = x^2, -x^2, 0$$

Solution 6

Denote $P(x, y)$ as the problem statement's assertion. Denoting $a = f(0)$, We have the following:

$$P(x, 0) \implies f(x^2) = f(f(x)) + a \quad (1) \quad (4)$$

$$P(0, x) \implies f(-x) + 2ax = f(a) + f(x) \quad (2) \quad (5)$$

$$P\left(x, \frac{1}{2}x^2\right) \implies x^2f(x) = f(f(x)) \quad (3) \quad (6)$$

$$P(x, y^2) \implies f(x^2 - y^2) + 2y^2f(x) = f(f(x)) + f(y^2) \quad (4) \quad (7)$$

$$P(y, x^2) \implies f(y^2 - x^2) + 2x^2f(y) = f(f(y)) + f(x^2) \quad (5) \quad (8)$$

Combining (1) and (3), we have $f(x^2) = x^2f(x) + a$. Plugging in $x = 1$, we have $f(1) = f(1) + a \implies f(0) = 0$. Rewriting (1) and combining with (3) using this fact, we have

$$f(x^2) = f(f(x)) = x^2f(x).$$

Rewriting (2) using this fact, we have $f(-x) = f(x)$ (i.e., f is even). We may henceforth limit x, y to the positive reals without loss of generality. Rewriting (4) and (5) and employing the fact that f is even, we have

$$\begin{aligned} f(x^2 - y^2) + 2y^2f(x) &= x^2f(x) + y^2f(y) \\ f(y^2 - x^2) + 2x^2f(y) &= y^2f(y) + x^2f(x) \\ \implies f(x^2 - y^2) + 2y^2f(x) &= f(y^2 - x^2) + 2x^2f(y) \\ \implies 2y^2f(x) &= 2x^2f(y) \\ \implies \frac{f(x)}{x^2} &= \frac{f(y)}{y^2} \end{aligned}$$

Since the choice of x, y was arbitrary, we find that $f(x)/x^2$ is constant for all x , so $f(x) = cx^2$ for some constant c . Plugging into the original functional equation, we have

$$\begin{aligned} c(x^2 - y)^2 + 2y(cx^2) &= c(cx^2)^2 + cy^2 \\ \iff cx^4 - 2cx^2y + cy^2 + 2cx^2y &= c^3x^4 + cy^2 \\ \iff (c - c^3)x^4 &= 0 \end{aligned}$$

for all x, y . Thus, it must be that $c \in \{-1, 0, 1\}$, so the only valid solutions are

$$f(x) = 0, \pm x^2. \blacksquare$$

6. (USAJMO 2024) Point D is selected inside acute triangle ABC so that $\angle DAC = \angle ACB$ and $\angle BDC = 90^\circ + \angle BAC$. Point E is chosen on ray BD so that $AE = EC$. Let M be the midpoint of BC . Show that line AB is tangent to the circumcircle of triangle BEM .

Solution 1

Let $\angle DBT = \alpha$ and $\angle BEM = \beta$. Extend AD intersects BC at point T , then $TC = TA$, TE is perpendicular to AC

Thus, AB is the tangent of the circle BEM

Then the question is equivalent as the $\angle ABT$ is the auxiliary angle of $\angle BEM$.

continued

Solution 2

We notice that $\angle DAC = \angle ACB$, which implies symmetry about the perpendicular bisector of $\overline{AC}$. Therefore, we can extend $\overline{AD}$ to intersect the circumcircle at Q , and by symmetry about the perpendicular bisector, $ABQC$ is an isosceles trapezoid. Now, we realize that the midpoint condition is annoying, which really suggests that to simplify things, we take a homothety at B sending M to C , and we just need to prove that the circumcircle of $\triangle BE'C$ is tangent to line $\overline{AB}$, where E' is the resultant point from a homothety at E . However, funnily enough, E lies on the perpendicular bisector of $\overline{AC}$ because $\overline{AE} = \overline{EC}$, and the perpendicular bisector gets sent to an altitude from Q under homothety, which means E' is simply the intersection of $\overline{BD}$ and the altitude from Q onto $\overline{BC}$! Call that point P . We can now ignore points E and M . Note that we have not used the $\angle BDC$ condition yet, which means we should probably try angle chasing now. Call $\angle BAC = a, \angle ABC = b$. By angle condition, $\angle BDC = 90 + a, \angle CDP = 90 - a$. Now, $\angle QCA = \angle BAC = a$, so $\angle PQC = 90 - \angle ACQ = 90 - a$. A miracle occurs! $\angle CDP = \angle ACQ$, so quadrilateral $QDPC$ is cyclic! $\angle ABC = \angle AQC = b$ from isosceles trapezoid, and from cyclicity $\angle DPC = 180 - b = \angle BPC$. Now we have $\angle ABC = b$ and $\angle BPC = 180 - b$, which looks a lot like tangency condition. We realize that it simply is the tangency condition if we extend line $\overline{AB}$ to get the supplementary angle, so we are done.

§16 USAJMO 2025

1. (USAJMO 2025) Let $\mathbb{Z}$ be the set of integers, and let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ be a function. Prove that there are infinitely many integers c such that the function $g: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $g(x) = f(x) + cx$ is not bijective. Note: A function $g: \mathbb{Z} \rightarrow \mathbb{Z}$ is bijective if for every integer b , there exists exactly one integer a such that $g(a) = b$.

Solution

Note that $c = f(a) - f(a+1)$ makes $g(a) = g(a+1)$, so for all $c = f(a) - f(a+1)$ we have that $g(x)$ is not bijective. If there are infinitely many possible values of c , we are done. Otherwise, there are only finitely many possible values of $f(a) - f(a+1)$ so there are only finitely many possible values of $-(f(a) - f(a+1)) = f(a+1) - f(a)$.

This means there is a minimum d of $f(a+1) - f(a)$. It is clear for $x > 0$ that

$$f(x) = f(0) + \sum_{a=0}^{x-1} (f(a+1) - f(a)) \geq dx + f(0)$$

and for $x < 0$ that

$$f(x) = f(0) - \sum_{a=x}^{-1} (f(a+1) - f(a)) \leq dx + f(0).$$

Then, let $c > -d + 3$ be an integer. We have for integers $x > 0$, $x \geq 1$, so

$$g(x) = f(x) + cx \geq (d+c)x + f(0) > 3x + f(0) > 3 + f(0),$$

if $x = 0$ we have $g(x) = f(0)$, and if $x < 0$ we have

$$g(x) = f(x) + cx \leq (d+c)x + f(0) < 3x + f(0) < f(0).$$

This means that there is no a such that $g(a) = f(0) + 1$, so $g(x)$ is not bijective. ■

Proof 2 Take $c = f(a) - f(a+1)$ and $g(a) = g(a+1)$, which is not injective, which implies g is not bijective. We provide another approach.

Let there exist infinitely many c . This is possible if one considers all pairs $(a, a+1)$, so we are done.

If there are only finitely many c , then we prove g cannot be surjective, and therefore not bijective. That is, all sets of $c \in \{d_{\min}, \dots, d_{\max}\}$. Choose a $c = M + 1$, where $M = d_{\max}$. Then,

$$\begin{aligned} g(a+1) - g(a) &= f(a+1) - f(a) + c \\ &= -d_{\max} + c \\ &\geq 1 \end{aligned}$$

which is viable. However, if $c > M + 1$,

$$\begin{aligned} g(a+1) - g(a) &= f(a+1) - f(a) + c \\ &= -d_{\max} + c \\ &\geq 2 \end{aligned}$$

Which is indeed not surjective and thus g is not bijective.

The proof is complete. □

2. (USAJMO 2025) Let k and d be positive integers. Prove that there exists a positive integer N such that for every odd integer $n > N$, the digits in the base- $2n$ representation of n^k are all greater than d .

Solution 1

We define a remainder operation $a \bmod b$ to be the remainder when a is divided by b . Also, let $\lfloor x \rfloor$ be the usual floor function.

Base- $(2n)$ Representation:

$$n^k = a_{k-1} (2n)^{k-1} + a_{k-2} (2n)^{k-2} + \dots + a_1 (2n) + a_0,$$

where each a_i satisfies $0 \leq a_i < 2n$. Hence, the base- $(2n)$ representation of n^k is $a_{k-1} a_{k-2} \dots a_1 a_0$.

Leading Digit:

$$a_{k-1} = \left\lfloor \frac{n^k}{(2n)^{k-1}} \right\rfloor = \left\lfloor \frac{n}{2^{k-1}} \right\rfloor.$$

General Digit Formula: For $0 \leq i < k$,

$$a_i = \left\lfloor \frac{n^k \bmod (2n)^{i+1}}{(2n)^i} \right\rfloor.$$

Because n is odd, one can show

$$n^k \bmod (2n)^{i+1} \geq n^{i+1},$$

which implies

$$a_i \geq \left\lfloor \frac{n^{i+1}}{(2n)^i} \right\rfloor = \left\lfloor \frac{n}{2^i} \right\rfloor \geq 2^{k-1-i} \left\lfloor \frac{n}{2^{k-1}} \right\rfloor.$$

As n grows large, $\lfloor n/2^{k-1} \rfloor$ becomes arbitrarily big, so each digit a_i eventually exceeds any fixed d . Hence there exists an integer N such that for all odd $n > N$, the digits a_i in the base- $(2n)$ representation of n^k are all $> d$.

3. (*USAJMO 2025*) Let m and n be positive integers, and let $\mathcal{R}$ be a $2m \times 2n$ grid of unit squares.

A domino is a 1×2 or 2×1 rectangle. A subset S of grid squares in $\mathcal{R}$ is domino-tileable if dominoes can be placed to cover every square of S exactly once with no domino extending outside of S . Note: The empty set is domino tileable.

An up-right path is a path from the lower-left corner of $\mathcal{R}$ to the upper-right corner of $\mathcal{R}$ formed by exactly $2m + 2n$ edges of the grid squares.

Determine, with proof, in terms of m and n , the number of up-right paths that divide $\mathcal{R}$ into two domino-tileable subsets.

Solution 1

We claim the answer is $\binom{m+n}{m}^2$. Color $\mathcal{R}$ in a black-and-white checkerboard pattern such that the lower left square is black. Suppose an up-right path P splits $\mathcal{R}$ into two subsets S and S' such that S is below P . Call a subset balanced if there are an equal number of black and white squares.

Claim 1: S and S' are tileable iff S is balanced.

Proof: Since $\mathcal{R}$ itself is balanced and rotating it by 180° swaps S and S' , it suffices to only consider S . Note that dominoes each cover exactly one black and one white square, so S being tileable implies S is balanced.

We will proceed with the converse by attempting to remove a domino from S such that S can still be traced by a new up-right path. Indeed, so long as P contains a sequence of $\uparrow \rightarrow \rightarrow$ or $\uparrow \uparrow \rightarrow$, we can replace these with $\rightarrow \rightarrow \uparrow$ and $\rightarrow \uparrow \uparrow$, respectively, and effectively remove a domino (this also preserves balanced-ness). Repeating this until we reach the empty set suffices, as we can reconstruct using the sequence of domino removal. The only cases where this fails is when P forms a staircase and

remains on the upper or left edges of $\mathcal{R}$ for at most one edge each. However, we can trivially observe that S is then unbalanced, since if the largest diagonal in S has k squares (which must be of the same color), $k + (k - 2) + \cdots > (k - 1) + (k - 3) + \cdots$. $\square$

Assign coordinates to the gridline intersections such that the lower left corner is $(0, 0)$ and the upper right is $(2m, 2n)$. Assign each $\rightarrow$ move with starting point (x, y) a two-letter designation, where the first letter is o or e depending on the parity of $x + y$ and the second letter depends on the parity of x . (I blame this naming system on leo; he was making some really weird sounds :P)

Claim 2: S is balanced iff there are an equal number of $\rightarrow$'s with starting letter o and e .

Proof: Consider S as a union of multiple columns of squares, each topped with a $\rightarrow$. Upon inspection, oe 's give one more black square than white square, while eo 's give one more white square. There are an even number of squares, and therefore equal number of black and white squares, under an oo or ee . Thus, for S to be balanced, there must be an equal number of oe 's and eo 's; but since there are also an equal number of odd-numbered and even-numbered columns, we must have $oe + ee = eo + oo$, so $ee = oo$ and $oe + oo = eo + ee$, as desired. Reversing the logic gives the converse. $\square$

We finish by splitting moves into two sets based on the parity of $x + y$ and then ordering m $\rightarrow$'s amongst the $m + n$ moves in each set, giving the desired $\binom{m+n}{m}^2$ paths. $\square$

Solution 2

$$\boxed{\binom{m+n}{m}^2}$$

Proof. View the grid as lattice points (x, y) with $0 \leq x \leq 2n$ and $0 \leq y \leq 2m$. Partition the $2m \times 2n$ rectangle into mn disjoint 2×2 blocks

$$B_{i,j} = \{2i, 2i+1\} \times \{2j, 2j+1\}, \quad 0 \leq i \leq n-1, \quad 0 \leq j \leq m-1,$$

where (x, y) denotes the unit square with lower-left corner (x, y) .

Let P be an up-right path from $(0, 0)$ to $(2n, 2m)$. Inside each 2×2 block $B_{i,j}$, the restriction of P is a monotone lattice path from $(2i, 2j)$ to $(2i+2, 2j+2)$. There are six such possibilities; exactly two of them turn at the block center $(2i+1, 2j+1)$.

Claim 1. If P turns at $(2i+1, 2j+1)$ for some i, j , then one of the two regions cut by P is not domino-tileable.

Proof. A turn at the center of a 2×2 block forces exactly 1 or 3 of the 4 unit squares of that block to lie on one side of P . Any domino tiling covers an even number of squares, so that side cannot be tiled. $\square$

Claim 2. If P never turns at any $(2i+1, 2j+1)$, then both regions cut by P are domino-tileable.

Proof. If no block-center turn occurs, then in each 2×2 block the path leaves 0, 2, or 4 squares on each side, and when 2 squares occur they share an edge. Hence each side can be tiled independently within each block (by 0, 1, or 2 dominoes), giving a tiling of each entire region. $\square$

By Claims 1 and 2, the desired paths are exactly those with no turns at block centers.

Now group the rows into m strips of height 2: strip j consists of rows $y = 2j$ and $y = 2j + 1$. For such a path P , let $r_{j,0}$ be the number of right-steps taken along $y = 2j$, and let $r_{j,1}$ be the number of right-steps taken along $y = 2j + 1$. Avoiding block-center turns forces both $r_{j,0}$ and $r_{j,1}$ to be even. Write

$$r_{j,0} = 2a_j, \quad r_{j,1} = 2b_j$$

with $a_j, b_j \geq 0$. Since P has exactly $2n$ right-steps,

$$\sum_{j=0}^{m-1} (2a_j + 2b_j) = 2n,$$

so

$$\sum_{j=0}^{m-1} a_j + \sum_{j=0}^{m-1} b_j = n.$$

The sequence $(a_0, \dots, a_{m-1})$ uniquely encodes an up-right path from $(0, 0)$ to (n, m) by taking a_0 right-steps, then an up-step, then a_1 right-steps, and so on. Similarly, $(b_0, \dots, b_{m-1})$ encodes another up-right path from $(0, 0)$ to (n, m) .

Conversely, given any ordered pair of up-right paths in an $m \times n$ grid, doubling every right-step and placing one path on even rows and the other on odd rows produces a unique up-right path in the $2m \times 2n$ grid with no block-center turns. Thus there is a bijection between the desired paths and ordered pairs of up-right paths in an $m \times n$ grid.

The number of up-right paths from $(0, 0)$ to (n, m) is $\binom{m+n}{m}$. Therefore the number of desired paths is $\binom{m+n}{m}^2$.

4. (*USAJMO 2025*) Let n be a positive integer, and let $a_0, a_1, \dots, a_n$ be nonnegative integers such that $a_0 \geq a_1 \geq \dots \geq a_n$. Prove that

$$\sum_{i=0}^n i \binom{a_i}{2} \leq \frac{1}{2} \binom{a_0 + a_1 + \dots + a_n}{2}.$$

Note: $\binom{k}{2} = \frac{k(k-1)}{2}$ for all nonnegative integers k .

Solution 1

By Vandermonde's,

$$\frac{1}{2} \binom{\sum a_i}{2} = \frac{1}{2} \left(\sum \binom{a_i}{2} + \sum_{\text{sym}} a_i a_j \right) \geq \frac{1}{2} \sum i a_i^2 \geq \sum i \binom{a_i}{2},$$

with equality at $a_0 = 0, 1$ and $a_i = 0$ for $i > 0$. $\square$

Solution 2

We proceed by induction.

Base case: $n = 1$. Note that $\frac{1}{2} \binom{a_0 + a_1}{2} \geq \frac{1}{2} \binom{a_1 + a_1}{2} = 2a_1(2a_1 - 1)/4 \geq a_1(a_1 - 1)/2 = \binom{a_1}{2}$.

Inductive Hypothesis: Assume $\sum_{i=0}^n i \binom{a_i}{2} \leq \frac{1}{2} \binom{a_0 + a_1 + \dots + a_n}{2}$.

Inductive Step: We try to show it works for $n + 1$: $\sum_{i=0}^{n+1} i \binom{a_i}{2} \leq \frac{1}{2} \binom{a_0 + a_1 + \dots + a_{n+1}}{2}$.

By the Inductive Hypothesis all we need to show is $(n+1) \binom{a_{n+1}}{2} \leq \frac{1}{2} \binom{a_0 + a_1 + \dots + a_{n+1}}{2} - \frac{1}{2} \binom{a_0 + a_1 + \dots + a_n}{2}$.

Let $s = a_0 + a_1 + \dots + a_n$ and $c = a_{n+1}$. Now we need $(n+1)\binom{c}{2} \leq \frac{1}{2}\binom{s+c}{2} - \frac{1}{2}\binom{s}{2}$. The LHS is $\frac{1}{2}(n+1)(c)(c-1)$ the RHS is $\frac{1}{4}(2sc + c^2 - c)$. If $c = 0$ we're done, otherwise we divide by c and multiply by 4 to get $2(n+1)(c-1) = 2nc + 2c - 2n - 2$ which we are trying to show is $\leq 2s + c - 1$. Note that $s \geq (n+1)c$ by the problem statement condition. So it simplifies to $2nc + 2c - 2n - 2 \leq 2nc + 3c - 1$ which is obvious.

5. (USAJMO 2025) Let H be the orthocenter of acute triangle ABC , let F be the foot of the altitude from C to AB , and let P be the reflection of H across BC . Suppose that the circumcircle of triangle AFP intersects line BC at two distinct points X and Y . Prove that C is the midpoint of XY .

Solution 1

Let AP intersect BC at D . Extend FC to the point E on the circumcircle ω of FAP . Since H is the orthocenter of $\triangle ABC$, we know that $HD = DP$ or $HP = 2HD$, and $AH \cdot HD = CH \cdot HF$. Next we use the power of H in ω : $AH \cdot HP = CH \cdot HE$. These relations imply that $HE = 2HF$.

Hence C, D are midpoints of HE, HP respectively. By midline theorem, $CD \parallel EP$. Since $AD \perp CD$, we have $AD \perp EP$. This implies that $\angle APE = 90^\circ$. Consequently, AE is the diameter of ω . Let G be the midpoint of AE which is also the center of ω . G, C are midpoints of AE, EH respectively. By the midline theorem again, we have $GC \parallel AH$, consequently, $GC \perp BC$. This implies that GC is the perpendicular bisector of the chord XY hence C is the midpoint of XY . Dr. Shidavincimath.com

Solution 2

Denote O_1 as the center of (ABC) , O_2 as the center of FAP , K as the midpoint of AF , M as the midpoint of AC , and N as the midpoint of AP . It suffices to show that $\angle O_2CB = 90^\circ$.

Claim: O_1MO_2C is cyclic.

Proof: Since $AK = FK$ and $AM = MC$, KM is a midline of $\triangle AFC$ and $KM \parallel FC$. $KO_2 \parallel FC$ as well since $\angle AKO_2 = \angle AFC = 90^\circ$, so M lies on KO_2 . Next, note that P lies on (ABC) , so the perpendicular bisector of AP through N passes through O_1 . In other words, N, O_1 , and O_2 are collinear. Since NO_2 and BC are both perpendicular to AP , it follows that they are parallel. Since $KO_2 \parallel FC$ and $NO_2 \parallel BC$, then $\angle KO_2N = \angle FCB$. Finally, we have that

$$\angle MO_2O_1 = \angle KO_2N = \angle FCB = 90^\circ - B = \angle MCO_1,$$

and thus O_1MO_2C is cyclic. It follows that $\angle O_1O_2C = \angle O_1MC = 90^\circ$, so $\angle O_2CB = 180^\circ - \angle O_1MC = 90^\circ$, as desired.

-mop

Solution 3

Connect HP and have HP intersect XY at W . Also extend FC past point C and have it intersect with the circle at point D .

Since P is the reflection of H over BC , we know that $HP \perp XY$. Since H is the orthocenter, we can draw the altitude and tell that A, H , and P are collinear. We know $m\angle AFH = m\angle CWH = 90^\circ$ and $m\angle FHA = m\angle WHC$, so $\triangle AHF \sim \triangle CHW$ by AA, so $m\angle FAH = m\angle WCH$.

$m\angle FAP = \frac{1}{2}m\widehat{FP} = \frac{1}{2}(m\widehat{FX} + m\widehat{XP})$ and $m\angle FCX = \frac{1}{2}(m\widehat{FX} + m\widehat{DY})$. From this, we can tell that $m\widehat{XP} = m\widehat{DY}$. Therefore, $m\widehat{XD} = m\widehat{PY}$ and $XD = PY$.

If we connect HY , we can tell that that $PY = HY$ due to P being the reflection of H and WY being perpendicular to HP , so $XD = HY$. In addition, $m\angle HYW = m\angle PYW = \frac{1}{2}m\widehat{XP} = \frac{1}{2}m\widehat{DY} = m\angle YXD$. Also, $m\angle HCY = m\angle XCD$ because they are vertical angles.

So, $\triangle HCY \cong \triangle XCD$ because of SAA. From this we can conclude that $XC = CY$, so C is the midpoint of XY .

Solution 4

Let D be the foot of the altitude from A to BC . By Power of a Point, we have

$$BF \cdot BA = BX \cdot BY = (BC - CX)(BC + CY) = BC^2 + BC(CY - CX) - CX \cdot CY$$

and

$$DB \cdot DC = DX \cdot DY = (CX - CD)(CD + CY) = CX \cdot CY - CD(CY - CX) - CD^2.$$

Adding, we get

$$BF \cdot BA + DB \cdot DC = BC^2 - DC^2 + (BC - CD)(CY - CX).$$

It is well known that the reflection of H over AB , which we denote by P , lies on (ABC) . Then, let $\angle BAP = \angle BCP = \angle BCH = \theta$. We have

$$BF \cdot BA + DB \cdot DC = (BC \sin \theta) \cdot BA + DB \cdot DC = BC \cdot (BA \sin \theta) + DB \cdot DC = DB \cdot (BC + DC) = (BC -$$

Thus, $(BC - CD)(CY - CX) = 0$, and since $BC \neq CD$ we have $CY = CX$. Hence, C is the midpoint of XY . TThB0501

Solution 5

Let Q be the antipode of B . Claim — AHQ is a parallelogram, and AP CQ is an isosceles trapezoid.

Proof. As $AH \perp BC \perp CQ$ and $CF \perp AB \perp AQ$. Let M be the midpoint of QC .

Claim — Point M is the circumcenter of triangle AFP .

Proof. It's clear that $MA = MP$ from the isosceles trapezoid.

As for $MA = MF$, let N denote the midpoint of AF ; then MN is a midline of the parallelogram, so $MN \perp AF$. Since $CM \perp BC$ and M is the center of $(AF P)$, it follows $CX = CY$.

6. (USAJMO 2025) Let S be a set of integers with the following properties:

- $\{1, 2, \dots, 2025\} \subseteq S$.
- If $a, b \in S$ and $\gcd(a, b) = 1$, then $ab \in S$.
- If for some $s \in S$, $s + 1$ is composite, then all positive divisors of $s + 1$ are in S .

Prove that S contains all positive integers.

Solution 1

We will show that if $\{1, 2, 3, \dots, n\} \in S$, then $n + 1 \in S$. Note that if $n + 1$ is composite, we are done, so we assume that $n + 1$ is an odd prime.

Case 1: $n + 1 \neq 2^k \pm 1$ for some integer $k > 2$. Since $n + 2 \neq 2^k$, we have that $2^{v_2(n+2)}, \frac{n+2}{2^{v_2(n+2)}} \leq n \implies 2^{v_2(n+2)}, \frac{n+2}{2^{v_2(n+2)}} \in S \implies n + 2 \in S$. Either $v_2(n) = 1$ or $v_2(n + 2) = 1$, so $2^{v_2(n)+v_2(n+2)} = 2^{1+\max(v_2(n), v_2(n+2))} = 2 \cdot 2^{\max(v_2(n), v_2(n+2))} \leq 2 \cdot \frac{n+2}{3} \leq n \implies 2^{v_2(n)+v_2(n+2)} \in S$. Since $\gcd(n, n + 2) = 2$, we have that $\gcd\left(\frac{n}{2^{v_2(n)}}, \frac{n+2}{2^{v_2(n+2)}}\right) = 1$, so

$$2^{v_2(n)+v_2(n+2)} \cdot \frac{n}{2^{v_2(n)}} \cdot \frac{n+2}{2^{v_2(n+2)}} = n(n+2) \in S \implies n(n+2)+1 = (n+1)^2 \in S \implies n+1 \in S.$$

Case 2: $n + 1 = 2^k + 1$ for some integer $k > 2$. Note that k must be even, otherwise $2^k + 1 \equiv 0 \pmod{3}$. Then we have the following:

$$\begin{aligned} \left(2^{\frac{k}{2}} + 1\right)^2 - 1 &= 2^{\frac{k}{2}} \left(2^{\frac{k}{2}} + 2\right) = 2^{\frac{k}{2}+1} \left(2^{\frac{k}{2}-1} + 1\right) \in S \implies \left(2^{\frac{k}{2}} + 1\right)^2 \in S \\ \implies 2^{\frac{k}{2}-1} \left(2^{\frac{k}{2}} + 1\right)^2 &= \left(2^{\frac{k}{2}-1} + 1\right) \left(2^k + 1\right) - 1 \in S \implies \left(2^{\frac{k}{2}-1} + 1\right) \left(2^k + 1\right) \in S \implies 2^k + 1 \in S, \end{aligned}$$

as desired.

Case 3: $n + 1 = 2^k - 1$ for some integer $k > 2$. Let $x < k$ be a positive integer. Then

$$(2^x - 1)(2^k - 1) - 1 = 2^x (2^k - 2^{k-x} - 1) \in S \implies 2^k - 1 \in S,$$

as desired.

Hence, since $\{1, 2, 3, \dots, n\} \in S \implies n + 1 \in S$, the set S spans all positive integers.

-mop

§17 USAJMO 2026

1. (*USAJMO 2026*) Let a, b, c be distinct positive integers such that $ab + c = c^2$. Prove that $(a - b)^2 \geq 4c$.

Solution 1

Write $ab = c^2 - c = c(c - 1)$. WLOG $a > b$, so $a > \sqrt{c(c - 1)} > c - 1$. Since a and c are positive integers, this implies $a \geq c$. However, $a \neq c$, so let $a = c + k$, where k is a positive integer. We know that $c + k \mid c(c - 1)$. Let $c + k = c_1 c_2$, such that $c_1 \mid c - 1$ and $c_2 \mid c$.

We have

$$c_1 \leq \gcd(c - 1, c + k) \leq k + 1$$

$$c_2 \leq \gcd(c, c + k) \leq k$$

Thus, $c + k \leq k(k + 1) \implies k \geq \sqrt{c}$. Finishing up, $a = c + k \geq c + \sqrt{c}$ and $b = \frac{c^2 - c}{a} \leq c - \sqrt{c}$, so

$$a - b \geq 2\sqrt{c} \implies (a - b)^2 \geq 4c$$

and the proof is complete. $\square$

Solution 2

Rewrite the condition as $ab = c^2 - c$. Then examine the LHS:

$$(a - b)^2 = (a + b)^2 - 4ab = (a + b)^2 + 4c - 4c^2$$

Given that $ab = c(c - 1)$ and a, b, c are distinct meaning one of a, b is larger than c and the other is smaller than c . Let $a = c + n, b = c + m$ where $nm < 0$. So

$$ab = c^2 + (n + m)c + nm = c^2 - c \implies c = \frac{-nm}{n + m + 1}$$

Since $c > 0$ and $-nm > 0$ thus $n + m + 1 > 0$. n, m are both integers which implies $n + m \geq 0$ so $a + b \geq 2c$. Plugging this in to the inequality gives

$$(a - b)^2 = (a + b)^2 - 4ab = (a + b)^2 + 4c - 4c^2 \geq (2c)^2 + 4c - 4c^2 = 4c$$

Solution 3 (easier)

By AM-GM,

$$a + b \geq 2\sqrt{c^2 - c}$$

If $a + b \leq 2c - 2$, then $2c - 2 \geq 2\sqrt{c^2 - c} \implies 4c^2 - 8c + 4 \geq 4c^2 - 4c \implies c \leq 1$. Hence $c = 1$, so $ab = c^2 - c = 0$, contradiction.

Next, if $a + b = 2c - 1$, then a, b are the roots of the quadratic

$$t^2 - (2c - 1)t + (c^2 - c) = 0$$

which has roots at $t = c, c - 1$. But then one of a, b is equal to c , contradicting a, b, c distinct, so we conclude that $a + b \geq 2c$. Then,

$$a + b \geq 2c \implies (a + b)^2 \geq 4c^2 \implies (a - b)^2 \geq 4c^2 - 4ab = 4c$$

as desired.

Solution 4 (Easier AM-GM)

Lemma

For all real numbers a and b that satisfy $ab \leq 0$, the following inequality suffices.

$$\frac{a - b}{2} \geq \sqrt{ab}$$

.

Proof of the Lemma

Substitute $-b$ in the AM-GM inequality.

$$\frac{a - b}{2} \geq \sqrt{-ab} \quad (1.1)$$

Then the inequality $\sqrt{-ab}$ holds only if $ab \leq 0$. Then the result becomes

$$\frac{a - b}{2} \geq \sqrt{ab} \quad (1.2)$$

hence proven.

USAJMO Q1

AM-GM and rearrangement of the inequality states that

$$ab \leq \frac{(a + b)^2}{4}$$

and thus

$$4ab \leq (a+b)^2$$

.

The application of 1.1 gives

$$\begin{aligned}(a-b)^2 &\geq -4ab \\ -(a-b)^2 &\leq 4ab\end{aligned}$$

And thus the inequality

$$-(a-b)^2 \leq 4ab \leq (a+b)^2 \quad (1.3)$$

Rearranging 1.3 gives

$$\begin{aligned}-(a^2+b^2) &\leq 2ab \leq a^2+b^2 \\ -(a-b)^2 &\leq 0 \leq (a-b)^2\end{aligned}$$

It then follows that $(a-b)^2 \geq 0$ (1.4).

Applying the lemma, particularly result 1.2 and rearranging gives

$$(a-b)^2 \geq 4ab$$

Note that then

$$4ab \leq 0$$

as a direct result from 1.1. It is given that $ab = c(c-1)$. Thus,

$$c(c-1) \leq 0$$

in which equality holds if one of c or $c-1$ is 0, in which the result is trivial. Otherwise if $c-1 \neq 0$, division of $c-1$ into 0 gives $c \leq 0$. This result then shows that $4c \leq 0$.

Given that $(a-b)^2 \geq 0$ and $4c \leq 0$, it is proven that

$$(a-b)^2 \geq 4c \quad \square$$

Question to author: How is $4c \leq 0$ if it is given that a, b, c are distinct positive integers?

Because we are under the condition of the Lemma, remember! We take an integer $b \mapsto -b$. The only reason the final inequality stands is because at the end, even after obtaining $4c \leq 0$ we can still formulate a map $-c \mapsto c$ such that we shift $4c$ by some x so the inequality $4c + x \leq x$ holds. Similarly, we can shift the RHS $(a-b)^2 \geq 0$ by the same x . This unique x is the equilibrium factor in respect to $4c$, which is why the Lemma still proves the claim correct.

It is hard to see at first, maybe I should have clarified... if required, I will rewrite the solution. But as I've heard from others it is fine as is, so if I get another

question then I'll try to rewrite it. Too lazy rn though! :(. Nice question, I'll try to be more specific later on...

Same person who wrote the question: Ty, this really helped!

Remark

Another way one could have seen this inequality hold is the fact $(a - b)^2$ expands quadratically whereas $4c$ expands linearly. Both are centred correspondingly to the centre/origin hence the inequality must hold true. Indeed, a graphical approach can be used along with the function $ab = c(c - 1)$.

Solution 5 (0 tech)

Note that $4c = 4c^2 - 4ab$. Thus, it suffices to prove that $a + b \geq 2c$.

Note that by common sense, since a, b, c are positive integers, $ab = c(c - 1)$ implies that $a + b \geq 2c - 1$. However, since $a \neq b \neq c$, $a + b \neq 2c - 1$, so $a + b \geq 2c$.

Equality occurs a lot; for example, $(a, b, c) = (2, 6, 4)$ Q.E.D.

2. (USAJMO 2026) There are m foxes and n rabbits sitting in $m + n$ seats around a circular table, where all animals are distinguishable. If a fox and a rabbit are sitting next to each other, they may swap positions with each other. For a fixed starting configuration, determine the number of configurations that can be reached by a sequence of swaps.

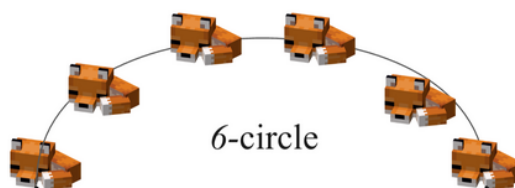
Note: Rotations and reflections of a configuration are considered distinct.

Solution 2 (Fixed, Cyclic Orientation and Number Theory)

Preliminary :

Note that there are $\binom{m+n}{m}$ total possible configurations. Without loss of generality, let $m \geq n$.

Define an z -circle to be a partial circle consisting of z members. The two foxes at the ends of the z -circle have the potential to swap. Such an example of a z -circle: $z = 6$ is shown below.



Lemma 1 :

The foxes arranged in an m -circle will have a minimum of 2 swaps.

Proof of Lemma 1 :

Trivial through definition of an m -circle, hence desired.

Letting k denote the invariant number of possible swaps, it is given that k satisfies the inequality $k \geq 2$.

Lemma 2 :

The upper bound of k is n , and there are a total of $\text{lcm}(m, n)$ swaps possible given m, n .

Proof of Lemma 2 :

We label the foxes as $1, 2, \dots, m$ and the rabbits as $1, 2, \dots, n$. Consider pairings (or swaps) between foxes and rabbits.

If every fox can be paired with every rabbit independently, then the total number of possible pairings is simply

$$mn.$$

Now suppose the system imposes a **cyclical non-totient structure**, meaning foxes are arranged in a cycle of length m and rabbits are arranged in a cycle of length n , and pairings are considered equivalent under a certain $\gcd \neq 1$ (In other words, m and n are not relatively prime)

In this case, we are no longer counting raw pairs but rotations under

$$(i, j) \rightarrow (i + 1 \pmod{m}, j + 1 \pmod{n}).$$

The idea is that through basic modular arithmetic, every single i will map to every single j (where $0 < i, j < m, n$) uniquely if $\gcd(i, j) = 1$. To prove that statement, the concept of the **Chinese Remainder Theorem** immediately results in mn . However, if this is not the case, each cycle maps onto one another by some primitive root, in which every cycle is shortened by one factor of $\gcd mn$ from its parent. To prove this statement, imagine applying **Chinese Remainder Theorem** to two elements i, j , whose elements satisfy $\gcd(i, j) \neq 1$. Obviously, to obtain some $k = \frac{i}{x}$ where $x \in \mathbb{Z}^+$ so that $\gcd(k, j) = 1$ to apply **Chinese Remainder Theorem**, we need to find the minimal value of x in primitive modulo that satisfies this. The smallest plausible that is apart of both primitive $\pmod{i}$ and $\pmod{j}$ is simply $\gcd(m, n)$, and thus the cases for this is $\frac{mn}{\gcd(m, n)} = \text{lcm}(m, n)$.

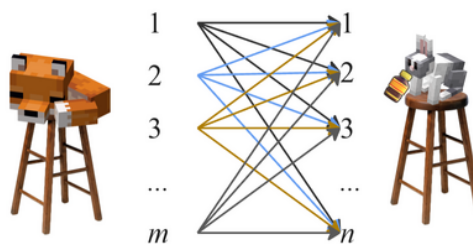
As $\text{lcm}(m, n) = mn$ holds for all $\gcd(m, n) = 1$, it follows for conveniency that every single case falls into

$$\text{lcm}(m, n)$$

and the upper bound of k is indeed n .

(Note: While my previous interpretation failed to identify the idea of cyclicity and when the greatest common divisors of m and n are not 1, it does correctly hold the fact that n is the upper bound as n is not dependent on the cyclicity. The mistake I made when constructing my and incorrectly fixing the previous solution 2 was simply not accounting for when m and n were not relatively prime, which explains why I made the mistake. For that, I would like to apologize).

The diagram below is a visual interpretation of the upper bound of k .



Claim :

The problem statement, that is, "rotations are distinct", is negligible given the above criteria.

Proof of Claim :

As we cyclically pair each fox and rabbit together, the rotations have already been accounted for, hence proved.

USAJMO Q2 :

Thus the total number of such configurations is just the number of possible configurations, each of which multiplied by the number of distinct swaps per that given section. There are a total of $\text{lcm}(m, n)$ distinct swaps through Lemma 2 and a total of $\binom{m+n}{m}$ given distinct cases. Rearrangement using commutative property derives the count

$$\binom{m+n}{m} \cdot \text{lcm}(m, n) \quad \square$$

Solution 3

We start by considering which seats can have foxes while not changing $I(F), I(R)$. The claim is that all $\binom{m+n}{m}$ different ways to arrange foxes are possible. Assume the smallest indexed fox not in the seat with its own index is i who is currently in seat j . This implies that all seats $i, i+1 \dots j-1$ are occupied by rabbits $1, 2, 3 \dots j-i$. So we swap fox i with rabbit $j-1$ then with rabbit $j-2$ so on until swapped with rabbit 1, returning it to seat i . None of these swaps involves seat $m+n$ swapping seat 1 so the order is preserved. So by induction we can place all the foxes in their respective seats (call this state T). This means from any initial configuration we can reach T . So if we consider a random configuration P we can also reach T . So if we reach T from the initial configuration then reverse all the moves needed to reach T from P we actually end up in P from the initial configuration. So all configurations are possible. Next we consider all possible orders of foxes and rabbits. we start from $I(F) = 1, 2, 3 \dots m$ and $I(R) = 1, 2, 3 \dots n$. The only way to change the order is to swap the first fox with the last rabbit resulting in $I(F) = 2, 3 \dots m, 1$ and $I(R) = n, 1, 2, 3 \dots n-1$. So each single change rotates $I(F), I(R)$ by 1 slot. Thus $I(F)$ rotates with period m and $I(R)$ rotates with period n . So the combined period of foxes and rabbits is $\text{lcm}(m, n)$. Combining the ways to seat the foxes and order the foxes rabbits, we get

$$\binom{m+n}{m} \cdot \text{lcm}(m, n)$$

3. (USAJMO 2026) Let ABC be an acute scalene triangle with no angle equal to 60° . Let ω be the circumcircle of ABC . Let $\triangle_B$ be the equilateral triangle with three vertices on ω , one of which is B . Let ℓ_B be the line through the two vertices of $\triangle_B$ other than B . Let $\triangle_C$ and ℓ_C be defined analogously. Let Y to be the intersection of AC and ℓ_B , and let Z be the intersection of AB and ℓ_C .

Suppose that the circumcircle of AYZ intersects ω at $P \neq A$, BC intersects YZ at D , and PA intersects YZ at E . Prove that $PE = PD$.

Solution 1 (Complex Numbers)

Put the problem on the complex plane, so ω is the unit circle, and let $O = 0$ be the center of ω . Define $\zeta^3 = 1, \zeta \neq 1$.

Then, B_1, B_2, C_1, C_2 are $b\zeta, b\zeta^2, c\zeta, c\zeta^2$, respectively. Letting $S_1 = a + b + c$ and $S_2 = bc + ac + ab$, we find

$$y = \frac{bS_2}{b^2 - ac} \text{ and } z = \frac{cS_2}{c^2 - ab}, \text{ while } \bar{y} = -\frac{S_1}{b^2 - ac} \text{ and } \bar{z} = -\frac{S_1}{c^2 - ab}$$

P is the center of the spiral similarity sending BC to YZ , so

$$p = \frac{by - cz}{b + y - c - z} = \frac{\frac{b^2S_2}{b^2 - ac} - \frac{c^2S_2}{c^2 - ab}}{\frac{b^2S_1}{b^2 - ac} - \frac{c^2S_1}{c^2 - ab}} = \frac{S_2}{S_1}$$

Claim 1: $\overleftrightarrow{YZ}$ is the perpendicular bisector of $\overline{OP}$

Proof: To see this, write

$$|y - p| = |y| \iff \bar{p}y + p\bar{y} = 1$$

But, $\bar{p} = \frac{1}{p} = \frac{S_1}{S_2}$, so

$$\bar{p}y + p\bar{y} = \frac{bS_1 - S_2}{b^2 - ac} = \frac{b^2 - ac}{b^2 - ac} = 1$$

as needed. Similarly, we can show $|z - p| = |z|$, proving our claim.

Now, we move onto D . Let $K = \frac{1}{(b^2 - ac)(c^2 - ab)}$. With lots of help from factoring, we derive

$$\begin{aligned} D &= \frac{(\bar{y}z - y\bar{z})(b - c) + (y - z)(\frac{c}{b} - \frac{b}{c})}{(\bar{y} - \bar{z})(b - c) + (y - z)(\frac{1}{b} - \frac{1}{c})} = \frac{(\bar{y}z - y\bar{z})bc + (y - z)(b + c)}{(\bar{y} - \bar{z})bc + (y - z)} \\ &= \frac{bcS_1S_2(b - c)K + (b + c)S_2S_2(c - b)K}{bcS_1S_1(b - c)K + S_2S_2(c - b)K} = \frac{(b + c)S_2^2 - bcS_1S_2}{S_2^2 - bcS_1^2} \end{aligned}$$

Similarly,

$$E = \frac{(a + p)S_2^2 - apS_1S_2}{S_2^2 - apS_1^2} = \frac{pS_2}{S_2 - aS_1}$$

where we have used $p = \frac{S_2}{S_1}$. To finish, we write

$$\begin{aligned} |D| = |E| &\iff \left| \frac{S_2(S_2(b + c) - S_1)}{S_2^2 - bcS_1^2} \right| = \left| \frac{S_2}{bc - a^2} \right| \\ &\iff \left| \frac{(bc - a^2)(b^2 + bc + c^2)}{S_2^2 - bcS_1^2} \right| = 1 \quad (*) \end{aligned}$$

where we have used $S_2(b + c) - S_1bc = a(b^2 + bc + c^2)$. But,

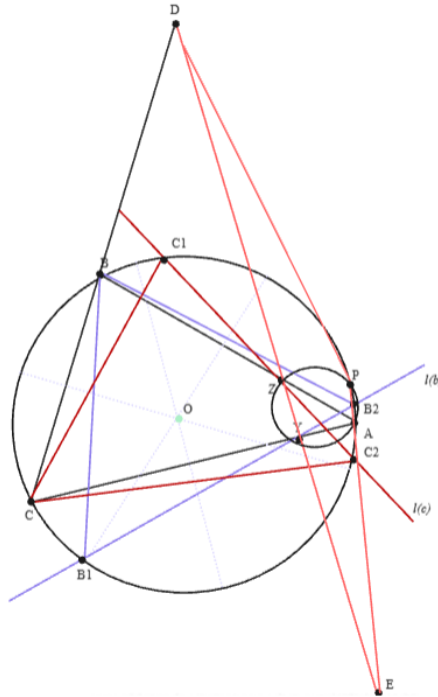
$$S_2^2 - bcS_1^2 = a^2b^2 + a^2bc + ac^2 - b^3c - b^2c^2 - bc^3 = (a^2 - bc)(b^2 + bc + c^2)$$

Thus, the fraction in $(*)$ is -1, and so we conclude $|D| = |E|$. Thus,

$$PD = OD = OE = PE \quad \square$$

-sillybone

Solution 2 (Angle Chasing, Much Shorter, Come and Go Solution)



The picture above is not to scale and is hand drawn.

We wish to show that the median down from P intersects DE at 90° .

Both triangles BB_1B_2 and CC_1C_2 lie on the same circle O . Thus, each of their vertices are separated by 120 degree arcs, and thus their sides must be congruent. Through SSS, the two triangles are congruent.

Then, quadrilateral $BB_1C_2C_1$ is cyclic. Let angle $C_1C_2B_1 = (60 + x)^\circ$ and angle $B_1BC_1 = (60 + y)^\circ$. It follows that $x + y = 60^\circ$.

Lay the quadrilateral upon BB_1 as the x -axis in the coordinate plane. Without loss of generality, one can let $BC_1 = 1$. The equation for the diagonal BB_2 is $\tan(60^\circ)x = \sqrt{3}x$.

The two equilateral triangles are congruent. All partitions of segments are congruent to their opposing segments. Thus, their corresponding segments are symmetrical about the centre of circle O , and a translation of the line BB_1 up one unit correctly translates the line as the angle bisector of CC_1C_2 . Then, the equation of the line C_1C is $\tan(-30^\circ)x + 1 = -\frac{1}{\sqrt{3}}x + 1$. The two lines are negative reciprocals of one another, and thus $BB_2 \perp C_1C$.

Let the intersection of BB_2 and C_1C be V . It follows that triangle BVC_1 is isosceles with the base angles measure y . Then, $(2y)^\circ + 90^\circ = 180^\circ$ through the Triangle Angle Sum formula, and thus $y = 45^\circ$. It follows through Opposite Angles of a Cyclic Quadrilateral that $x = 15^\circ$.

Angle $C_1ZB = 30^\circ$, and $AZC_2 = 30^\circ$ through Vertical Angles. Let the intersection of CC_2 and B_1B_2 be W . It is then true that angle $B_2WC_2 = (2x)^\circ = 30^\circ$. Thus, angle $WYC = 30^\circ$, and angle $B_2YA = 30^\circ$.

Let the intersection of B_2B and DE be N . Let the midpoint of DE be M . Extend PM through until it touches the circumcircle of AYZ at point J . Extend B_2

through Y until it touches the circumcircle of AYZ at point K . Triangle B_2YN is equilateral. The altitude down from N bisects B_2Y at L . Let the intersection of PJ and NL be L' . Let angle $ML'L$ be a , angle $L'MY$ be b , and $L'PB_2$ be c . Finding b and showing $b = 90^\circ$ will prove the statement.

Let the intersection of DE and CC_2 be K' . Extend CY until intersection with line CC_2 at point J' . We know $B_2NL = 30^\circ$ and therefore $LN Y = 30^\circ$. We also know triangle YWK' is 30-60-90 with angle $YWK' = 30^\circ$ and angle $WK'Y = 90^\circ$. Thus, $YK'J' = 90^\circ$ and through AA similarity triangles $YK'J'$ and $WK'Y$ are similar, and therefore angle $ZYJ = 30^\circ$. Then, through the converse of the Alternate Exterior Angles theorem lines LN and YJ are parallel.

Angle $PL'L = (180 - a)^\circ$. Through Corresponding Angles Theorem, $\angle PJY \cong \angle PL'L = (180 - a)^\circ$. Through Quadrilateral Angle Sum Formula, $b + a + 150 = 360$ and thus $b + a = 210$. Also, through triangle sum formula, $(180 - a) + b + 30 = 180$, and thus $b - a = -30$. It then follows that $2b = 180$, and $b = 90^\circ$.

We needed to find the measure of angle b to solve the problem, specifically showing $b = 90^\circ$, as because it is the angle inscribed within the median of P to side DE , angle b proves that the median PM is also the altitude, which can only happen if triangle DPE is isosceles, with $PE = PD$. $\square$

Solution 3 (In thought process)

(Someone plz check if this is fakesolve :sob) We place the figure on the unit circle and compute the relevant ratios in coordinates.

The strategy is as follows.

First, we put ω on the unit circle and write A, B, C in trigonometric form. Then we find the equations of the two special lines ℓ_B and ℓ_C , and from these compute the ratios $\frac{AY}{YC}$ and $\frac{AZ}{ZB}$. Using Menelaus in $\triangle ABC$, we then obtain $\frac{BD}{DC}$.

Next, since $E = AP \cap YZ$ and A, Y, P, Z are concyclic, we use power of a point together with another ratio computation to obtain $\frac{EY}{EZ}$. Comparing this with $\frac{BD}{DC}$ gives $BD \cdot DC = EY \cdot EZ$. This shows that D and E have the same power with respect to ω , so $OD = OE$.

Finally, D and E lie on a circle centered at O . Comparing $\triangle ODP$ and $\triangle OEP$ then gives $PD = PE$.

Now we carry this out.

Let ω be the unit circle centered at $O = (0, 0)$, and write $A = (1, 0)$, $B = (\cos 2C, \sin 2C)$, and $C = (\cos 2B, -\sin 2B)$. Then $\angle AOB = 2C$ and $\angle AOC = 2B$, so indeed $\angle BAC = A$.

Let $B_1 = (\cos(2C + 120^\circ), \sin(2C + 120^\circ))$ and $B_2 = (\cos(2C - 120^\circ), \sin(2C - 120^\circ))$, and let $\ell_B = B_1B_2$. Similarly, let $C_1 = (\cos(-2B + 120^\circ), \sin(-2B + 120^\circ))$ and $C_2 = (\cos(-2B - 120^\circ), \sin(-2B - 120^\circ))$, and let $\ell_C = C_1C_2$.

Define $Y = AC \cap \ell_B$, $Z = AB \cap \ell_C$, and $D = YZ \cap BC$. Let $\Gamma = (AYZ)$, and let $P \neq A$ be the second point of intersection of ω and Γ . Finally let $E = AP \cap YZ$. We must prove that $PD = PE$.

We first determine the equations of ℓ_B and ℓ_C . Since $\cos 2C \cos(2C \pm 120^\circ) + \sin 2C \sin(2C \pm 120^\circ) = \cos(\mp 120^\circ) = -\frac{1}{2}$, both B_1 and B_2 lie on the line $\cos 2C x + \sin 2C y = -\frac{1}{2}$. Hence $\ell_B : \cos 2C x + \sin 2C y = -\frac{1}{2}$. Similarly, $\ell_C : \cos 2B x - \sin 2B y = -\frac{1}{2}$.

Now parametrize the sidelines. A point on AC has the form $(x, y) = (1 + t(\cos 2B - 1), -t \sin 2B)$. Since $Y \in \ell_B$, substituting into the equation of ℓ_B gives $\cos 2C(1 +$

$t(\cos 2B - 1)) - t \sin 2C \sin 2B = -\frac{1}{2}$. Using $\cos 2C \cos 2B - \sin 2C \sin 2B = \cos 2(B + C) = \cos 2A$, we obtain $\cos 2C + t(\cos 2A - \cos 2C) = -\frac{1}{2}$, so $t = \frac{-\frac{1}{2} - \cos 2C}{\cos 2A - \cos 2C}$. Since $\frac{AY}{YC} = \frac{t}{1-t}$, a routine trigonometric simplification yields

$$(1) \quad \frac{AY}{YC} = -\frac{\sin A \sin 3C}{\sin C \sin 3A}.$$

Similarly, a point on AB has the form $(x, y) = (1 + s(\cos 2C - 1), s \sin 2C)$. Since $Z \in \ell_C$, substituting gives $\cos 2B(1 + s(\cos 2C - 1)) - s \sin 2B \sin 2C = -\frac{1}{2}$. Again using $\cos 2B \cos 2C - \sin 2B \sin 2C = \cos 2A$, we get $\cos 2B + s(\cos 2A - \cos 2B) = -\frac{1}{2}$, hence

$$(2) \quad \frac{AZ}{ZB} = \frac{s}{1-s} = -\frac{\sin A \sin 3B}{\sin B \sin 3A}.$$

Now apply Menelaus to triangle ABC with transversal YDZ : $\frac{AZ}{ZB} \cdot \frac{BD}{DC} \cdot \frac{CY}{YA} = 1$. Substituting (1) and (2), we obtain

$$(3) \quad \frac{BD}{DC} = \frac{\sin C \sin 3B}{\sin B \sin 3C}.$$

Next we turn to the point E . Since A, Y, P, Z are concyclic and $E = AP \cap YZ$, the intersecting chords theorem gives

$$(4) \quad EA \cdot EP = EY \cdot EZ.$$

To compute $\frac{EY}{EZ}$, we use the cyclic quadrilateral $ABPC$ together with $Y \in AC$ and $Z \in AB$. A sine-law chase gives

$$(5) \quad \frac{PY}{PZ} = \frac{CY}{BZ}.$$

Also, from the complete quadrilateral through A, Y, P, Z , one gets

$$(6) \quad \frac{EY}{EZ} = \frac{AY \cdot PY}{AZ \cdot PZ}.$$

Combining (5) and (6), we find

$$(7) \quad \frac{EY}{EZ} = \frac{AY \cdot CY}{AZ \cdot BZ}.$$

Now using (1) and (2), $\frac{AY}{CY} = -\frac{\sin A \sin 3C}{\sin C \sin 3A}$ and $\frac{AZ}{BZ} = -\frac{\sin A \sin 3B}{\sin B \sin 3A}$, so (7) simplifies to

$$(8) \quad \frac{EY}{EZ} = \frac{\sin B \sin 3C}{\sin C \sin 3B}.$$

Comparing (3) and (8), we obtain $\frac{BD}{DC} \cdot \frac{EY}{EZ} = 1$, hence

$$(9) \quad BD \cdot DC = EY \cdot EZ.$$

Now $D \in BC$, so the power of D with respect to ω is

$$(10) \quad \text{Pow}_\omega(D) = DB \cdot DC.$$

Also, $E \in AP$ with $A, P \in \omega$, so

$$(11) \quad \text{Pow}_\omega(E) = EA \cdot EP.$$

By (4) and (9), $EA \cdot EP = EY \cdot EZ = DB \cdot DC$. Therefore

$$(12) \quad \text{Pow}_\omega(E) = \text{Pow}_\omega(D).$$

Since ω is centered at O , equal powers imply equal distances from the center. Thus

$$(13) \quad OD = OE.$$

So D and E lie on a circle centered at O .

Now compare triangles ODP and OEP . They have $OD = OE$ by (13), and OP is common. Thus it remains to compare the included angles at O . By the symmetry of the construction along the common line AP , the angles $\angle DOP$ and $\angle EOP$ are equal. Therefore, by the Law of Cosines, $PD^2 = OP^2 + OD^2 - 2(OP)(OD) \cos \angle DOP$ and $PE^2 = OP^2 + OE^2 - 2(OP)(OE) \cos \angle EOP$ are equal. Hence $PD^2 = PE^2$, so $PD = PE$.

Therefore $PD = PE$, as desired. **Proof.** We use a direct coordinate bash.

The strategy is as follows.

First, we place B and C symmetrically on the x -axis and write the circumcircle ω in coordinates. Then we compute the equations of the two special lines ℓ_B and ℓ_C , and from them determine the coordinates of Y and Z . From the line YZ we obtain the point $D = YZ \cap BC$.

Next, we write the equation of the circle $\Gamma = (AYZ)$. Since A and P are the two common points of ω and Γ , the line AP is their radical axis, so we compute the equation of AP directly. Intersecting AP with YZ gives E , and intersecting AP with ω again gives P .

Finally, we prove two facts by explicit computation: first, $OD = OE$, so O lies on the perpendicular bisector of DE ; second, the midpoint M of DE lies on the line OP . Therefore OP is the perpendicular bisector of DE , so $PD = PE$.

Now we carry this out.

Set $B = (-1, 0)$, $C = (1, 0)$, and $A = (u, v)$ with $v > 0$. Write the circumcircle ω of $\triangle ABC$ in the form $x^2 + y^2 + by - 1 = 0$. Since $A \in \omega$, we have

$$(1) \quad u^2 + v^2 + bv = 1.$$

The center of ω is $O = \left(0, -\frac{b}{2}\right)$.

Let ℓ_B be the line through the two points of ω which, together with B , form an equilateral triangle on ω . Since ℓ_B is the chord subtending a central angle 120° , it is perpendicular to OB and its distance from O is half the radius of ω . A routine computation gives

$$(2) \quad \ell_B : x - \frac{b}{2}y = \frac{1}{2} + \frac{3b^2}{8}.$$

Similarly,

$$(3) \quad \ell_C : x + \frac{b}{2}y = -\frac{1}{2} - \frac{3b^2}{8}.$$

The sidelines are

$$(4) \quad AC : vx - (u - 1)y - v = 0,$$

$$(5) \quad AB : vx - (u + 1)y + v = 0.$$

Since $Y = AC \cap \ell_B$, solving (2) and (4) yields

$$(6) \quad Y = \left(\frac{-3b^2u + 3b^2 + 4bv - 4u + 4}{4(bv - 2u + 2)}, \frac{v(4 - 3b^2)}{4(bv - 2u + 2)} \right).$$

Similarly, since $Z = AB \cap \ell_C$, solving (3) and (5) gives

$$(7) \quad Z = \left(\frac{-3b^2u - 3b^2 - 4bv - 4u - 4}{4(bv + 2u + 2)}, \frac{v(4 - 3b^2)}{4(bv + 2u + 2)} \right).$$

Now compute the equation of the line YZ . After simplification, it is

$$(8) \quad 2u(3b^2 - 4)x + (9b^3 + 10b^2v + 20b + 8v)y + \frac{(3b^2 - 4)(3b^2 + 4bv + 4)}{4} = 0.$$

Since $D = YZ \cap BC$ and BC is the line $y = 0$, substituting $y = 0$ into (8) gives

$$(9) \quad D = \left(-\frac{3b^2 + 4bv + 4}{8u}, 0 \right).$$

Next let $\Gamma = (AYZ)$. Write its equation as $x^2 + y^2 + mx + ny + r = 0$. Substituting the three points A, Y, Z and solving for m, n, r , then subtracting the equation of ω , we obtain the radical axis of ω and Γ . Since $\omega \cap \Gamma = \{A, P\}$, this radical axis is the line AP , and its equation is

$$(10) \quad 4bux + (3b^2 + 4bv + 4)y + b^2v - 4b - 4v = 0.$$

Now $E = AP \cap YZ$, so solving (8) and (10) gives

$$(11) \quad E = \left(\frac{27b^4 + 36b^3v + 8b^2v^2 + 72b^2 + 80bv + 32v^2 - 16}{8u(9b^2 + 8bv + 4)}, -\frac{(3b + 2v)(3b^2 - 4)}{2(9b^2 + 8bv + 4)} \right).$$

Also, P is the second intersection point of AP with ω . Solving (10) together with $x^2 + y^2 + by - 1 = 0$, one root is $A = (u, v)$ and the other is

$$(12) \quad P = \left(\frac{u(3b^2 - 4)}{9b^2 + 8bv + 4}, \frac{b^2v + 8b + 4v}{9b^2 + 8bv + 4} \right).$$

We now prove that O lies on the perpendicular bisector of DE . From (9),

$$OD^2 = \left(\frac{3b^2 + 4bv + 4}{8u} \right)^2 + \left(\frac{b}{2} \right)^2.$$

Using (1), this simplifies to

$$(13) \quad OD^2 = \frac{(b^2 + 4)(9b^2 + 8bv + 4)}{64u^2}.$$

Now compute OE^2 from (11): $OE^2 = E_x^2 + \left(E_y + \frac{b}{2} \right)^2$. Again simplifying by means of (1), we obtain

$$(14) \quad OE^2 = \frac{(b^2 + 4)(9b^2 + 8bv + 4)}{64u^2}.$$

Hence

$$(15) \quad OD = OE.$$

So O lies on the perpendicular bisector of DE .

Let M be the midpoint of DE . From (9) and (11),

$$M = \left(\frac{D_x + E_x}{2}, \frac{E_y}{2} \right).$$

A direct simplification using (1) yields

$$(16) \quad M = \left(\frac{u(3b^2 - 4)}{2(9b^2 + 8bv + 4)}, -\frac{(3b + 2v)(3b^2 - 4)}{4(9b^2 + 8bv + 4)} \right).$$

Now compare the slopes of OM and OP . From (16),

$$(17) \quad \frac{M_y + \frac{b}{2}}{M_x} = \frac{9b^3 + 10b^2v + 20b + 8v}{2u(3b^2 - 4)}.$$

From (12),

$$(18) \quad \frac{P_y + \frac{b}{2}}{P_x} = \frac{9b^3 + 10b^2v + 20b + 8v}{2u(3b^2 - 4)}.$$

Thus the slopes are equal, so O, M , and P are collinear. Hence P lies on the line through O and the midpoint of DE .

But by (15), O lies on the perpendicular bisector of DE . Therefore the line OP is exactly the perpendicular bisector of DE . It follows that

$$PD = PE.$$

Therefore $PD = PE$, as desired.

Solution 4 (Complex Bash)

Place (ABC) on the unit circle of the complex plane. Let $\zeta = e^{2\pi i/3}$. Then line ℓ_B passes through $b\zeta$ and $b\zeta^2$, and line ℓ_C passes through $c\zeta$ and $c\zeta^2$.

We compute Y and Z as follows. We have $Y = AC \cap \ell_B$, so we have the system

$$\begin{aligned} y + ac\bar{y} &= a + c \\ y + b^2\bar{y} &= y + (b\zeta)(b\zeta^2)\bar{y} = b\zeta + b\zeta^2 = -b. \end{aligned}$$

Solving gives

$$y = \frac{b}{b^2 - ac}(ab + bc + ca),$$

and similarly

$$z = \frac{c}{c^2 - ab}(ab + bc + ca).$$

We also find that

$$\bar{y} = -\frac{a + b + c}{b^2 - ac}, \quad \bar{z} = -\frac{a + b + c}{c^2 - ab}.$$

Notice that $\angle PYZ = \angle PAZ = \angle PAB = \angle PCB$ and $\angle PZY = \angle PAY = \angle PAC = \angle PBC$, where $\angle$ denotes a directed angle. Thus $\triangle PYZ \sim \triangle PCB$, and P is the center of the spiral similarity $YZ \mapsto CB$. We compute p as follows:

$$\begin{aligned} p &= \frac{by - cz}{b + y - c - z} = (ab + bc + ca) \cdot \frac{\frac{b^2}{b^2 - ac} - \frac{c^2}{c^2 - ab}}{b - c + \left(\frac{b}{b^2 - ac} - \frac{c}{c^2 - ab}\right)(ab + bc + ca)} \\ &= (ab + bc + ca) \cdot \frac{b^2(c^2 - ab) - c^2(b^2 - ac)}{(b - c)(b^2 - ac)(c^2 - ab) + (b(c^2 - ab) - c(b^2 - ac))(ab + bc + ca)} \\ &= (ab + bc + ca) \cdot \frac{a(c^3 - b^3)}{(c - b)((ab + bc + ca)^2 - (b^2 - ac)(c^2 - ab))} \\ &= \frac{(ab + bc + ca)(b^2 + bc + c^2)}{ab^2 + ac^2 + c^3 + b^3 + abc + 2b^2c + 2bc^2} = \frac{ab + bc + ca}{a + b + c}. \end{aligned}$$

Of course, p is on the unit circle, so this checks out. Now we compute d and e . We have $D = BC \cap YZ$ and $E = PA \cap YZ$, so we find that

$$d = \frac{(\bar{y}z - y\bar{z})bc + (y - z)(b + c)}{(\bar{y} - \bar{z})bc + (y - z)}, \quad e = \frac{(\bar{y}z - y\bar{z})pa + (y - z)(p + a)}{(\bar{y} - \bar{z})pa + (y - z)}.$$

We first compute:

$$\begin{aligned} \bar{y}z - y\bar{z} &= \left(-\frac{a + b + c}{b^2 - ac}\right) \left(\frac{c(ab + bc + ca)}{c^2 - ab}\right) - \left(\frac{b(ab + bc + ca)}{b^2 - ac}\right) \left(-\frac{a + b + c}{c^2 - ab}\right) \\ &= -\frac{(a + b + c)(ab + bc + ca)}{(b^2 - ac)(c^2 - ab)} \cdot (c - b) \\ y - z &= \frac{b}{b^2 - ac}(ab + bc + ca) - \frac{c}{c^2 - ab}(ab + bc + ca) \\ &= \frac{(ab + bc + ca)^2}{(b^2 - ac)(c^2 - ab)} \cdot (c - b) \\ \bar{y} - \bar{z} &= \frac{a + b + c}{c^2 - ab} - \frac{a + b + c}{b^2 - ac} \end{aligned}$$

$$= -\frac{(a+b+c)^2}{(b^2-ac)(c^2-ab)} \cdot (c-b).$$

Then

$$\begin{aligned} d &= \frac{(ab+bc+ca)^2(b+c) - (a+b+c)(ab+bc+ca)bc}{(ab+bc+ca)^2 - (a+b+c)^2} \\ &= (ab+bc+ca) \cdot \frac{ab^2 + b^2c + abc + abc + bc^2 + c^2a - abc - b^2c - bc^2}{a^2b^2 + b^2c^2 + c^2a^2 + 2a^2bc + 2ab^2c + 2abc^2 - a^2bc - b^3c - bc^3 - 2ab^2c - 2abc^2 - 2} \\ &= \frac{a(ab+bc+ca)(b^2+bc+c^2)}{a^2b^2 + a^2c^2 - b^2c^2 + a^2bc - b^3c - bc^3} = \frac{a(ab+bc+ca)}{a^2 - bc}, \\ \bar{d} &= -\frac{a+b+c}{a^2 - bc}, \end{aligned}$$

and

$$\begin{aligned} e &= \frac{(ab+bc+ca)^2(p+a) - (a+b+c)(ab+bc+ca)pa}{(ab+bc+ca)^2 - (a+b+c)^2pa} \\ &= (ab+bc+ca) \cdot \frac{(ab+bc+ca)(p+a) - (ab+bc+ca)a}{(ab+bc+ca)^2 - (a+b+c)(ab+bc+ca)a} \\ &= \frac{(ab+bc+ca)^2/(a+b+c)}{(ab+bc+ca) - (a+b+c)a} \\ &= -\frac{(ab+bc+ca)^2}{(a+b+c)(a^2-bc)}, \\ \bar{e} &= \frac{a(a+b+c)^2}{(a^2-bc)(ab+bc+ca)}. \end{aligned}$$

At this point let $X = a+b+c$ and $Y = ab+bc+ca$. We wish to show that $PD = PE$. This is equivalent to the statement that

$$(p-d)(\bar{p}-\bar{d}) = (p-e)(\bar{p}-\bar{e}) \iff p(\bar{e}-\bar{d}) + \bar{p}(e-d) = e\bar{e} - d\bar{d}.$$

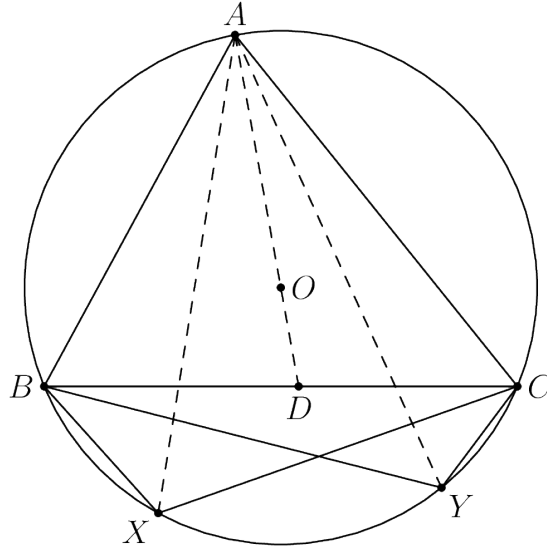
But indeed,

$$\begin{aligned} p(\bar{e}-\bar{d}) + \bar{p}(e-d) &= \frac{Y}{X} \left(\frac{aX^2}{(a^2-bc)Y} + \frac{X}{a^2-bc} \right) + \frac{X}{Y} \left(-\frac{Y^2}{(a^2-bc)X} - \frac{aY}{a^2-bc} \right) = 0 \\ e\bar{e} - d\bar{d} &= -\frac{Y^2}{(a^2-bc)X} \cdot \frac{aX^2}{(a^2-bc)Y} + \frac{aY}{a^2-bc} \cdot \frac{X}{a^2-bc} = 0, \end{aligned}$$

so both sides are equal. The result follows.

4. (USAJMO 2026) Triangle $\triangle ABC$ has circumcircle ω and circumcenter O . Lines AO and BC meet at point D . Let X be the A -excenter of $\triangle ABD$ and let Y be the A -excenter of $\triangle ACD$. Prove that if X lies on ω , then Y also lies on ω .

Solution 1



The diagram is not drawn to scale on purpose.

We proceed with angle chasing. Note that $\angle CBX = \frac{180^\circ - B}{2} = 90^\circ - \frac{\angle B}{2}$, since X is the A -excenter of $\triangle ABD$. Also,

$$\angle CAX = \angle OAX + \angle OAC = \frac{(90^\circ - \angle C)}{2} + (90^\circ - \angle B)$$

Since X is on (ABC) , we have $\angle CBX = \angle CAX$, so

$$90^\circ - \frac{\angle B}{2} = 135^\circ - \frac{\angle C}{2} - \angle B \implies \angle B + \angle C = 90^\circ$$

Now we reverse the above logic, finding that $\angle BCY = 90^\circ - \frac{\angle C}{2}$ and $\angle BAY = 135^\circ - \frac{\angle B}{2} - \angle C$. Because $\angle B + \angle C = 90^\circ$, these two are equal. Thus, quadrilateral $ABYC$ is cyclic, and Y is on (ABC) .

Solution 2

Connect segments OB, OC, AX, AY . Let N, M be points on AB, AC respectively on the opposite side of A . Let $\angle OBC = a, \angle ABO = b$. So $\angle OCB = a, \angle BAO = b, \angle OAC = 90^\circ - a - b = \angle OCA$. Since AX, BY are the bisectors of $\angle BAO, \angle CAO$ respectively,

$$\angle BAX = \frac{b}{2}, \angle YAC = \frac{90^\circ - a - b}{2}$$

Since BX, CY are the bisectors of $\angle CBN, \angle BCM$ respectively,

$$\angle XBA = a + b + \frac{180^\circ - a - b}{2} = \frac{180^\circ + a + b}{2}, \angle YCA = 135^\circ - \frac{b}{2}$$

So

$$\angle AXB = 90^\circ - \frac{a}{2} - b, \angle AYC = \frac{a}{2} + b$$

Since X is on ω , $\angle AXB = 90^\circ - \frac{a}{2} - b = \angle ACB = 90^\circ - b \implies a = 0$. This means that $\angle A = 90^\circ$. Furthermore

$$\angle AYC = \frac{a}{2} + b = b = a + b = \angle ABC$$

So $ABYC$ is a cyclic quadrilateral, thus Y is on ω .

Solution 3

Draw the angle bisectors of $\angle BAD$ and $\angle DAC$. We are given that the A -excenter of $\triangle ABD$, X , is on ω , and by definition, X is on the angle bisector of $\angle BAD$. Label the intersection of the angle bisector of $\angle DAC$ and ω as Y' . Draw rays AB and AC with endpoint A , and place points E and F on AB and AC , respectively, such that B is between A and E , and C is between A and F . Label angles $\angle CAY'$ and $\angle DAY'$ as α , and angles $\angle BAX$ and $\angle DAX$ as β . Connect CY' , XY' , and BX .

Using the inscribed angle theorem on circle ω , we can derive that

$$\angle CBX = \angle CAX = 2\alpha + \beta$$

Since X is the A -excenter of $\triangle ABD$, we know that BX bisects $\angle CBE$, so

$$\angle EBX = 2\alpha + \beta$$

Adding these two angles gives

$$\angle CBE = 180^\circ - \angle ABC = \angle BAC + \angle ACB = 4\alpha + 2\beta$$

By our definition, $\angle BAC = 2\alpha + 2\beta$, so we can subtract this from the previous equation, and we obtain

$$\angle ACB = 2\alpha$$

We already have $\angle DAC = 2\alpha$, so in $\triangle ACD$, $AD = CD$. By the definition of a circumcenter, $AO = CO$, and subtracting these two equations and rearranging gives us

$$CO + OD = CD$$

This is the segment addition postulate, which implies that O lies on CD , and by extension, it is also on BC . Thus, BC is a diameter of ω , and it follows that

$$\angle BAC = 2\alpha + 2\beta = 90^\circ$$

By the inscribed angle theorem,

$$\angle BCY' = \angle BAY' = \alpha + 2\beta$$

We add this to our result for $\angle ACB$ to get

$$\angle ACY' = 3\alpha + 2\beta$$

As F is on ray AC , $\angle ACF = 180^\circ$, which we can substitute for $4\alpha + 4\beta$. Subtracting the expression for $\angle ACY'$ from this, we can obtain

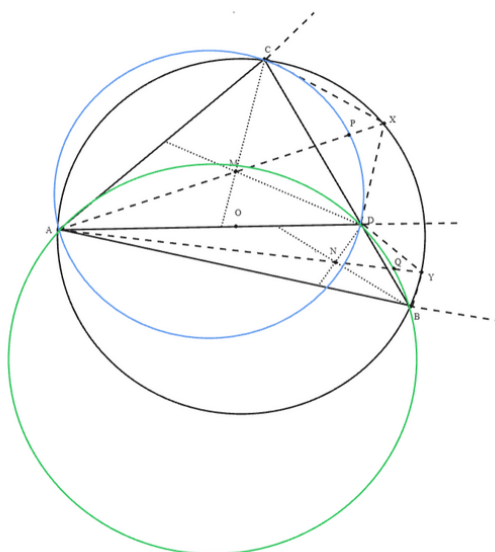
$$\angle ECY' = \alpha + 2\beta$$

and therefore,

$$\angle ECY' = \angle BCY'$$

This means CY' bisects $\angle BCD$, which, since AY' bisects $\angle CAD$, implies that Y' is the A -excenter of $\triangle ACD$. The problem gives Y as the A -excenter of that triangle, so Y' and Y are the same point. From our definition, Y must lie on ω , so we have proven the problem statement.

Solution 4 (Incentre/Excentre Lemma)



The diagram above is not to scale and is hand drawn.

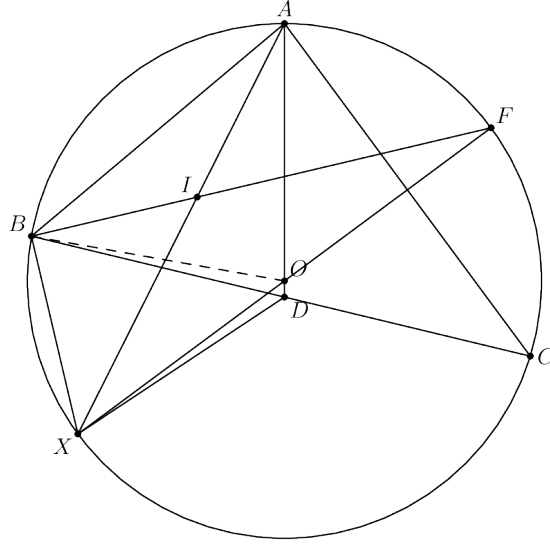
M is the incentre of triangle ACD , and N is the incentre of triangle ABD . The blue and green circles are the circumcircles of triangles ACD and ABD respectively. The incentre/excentre lemma on M and X inscribe quadrilateral $MCXD$. The incentre/excentre lemma on N and Y inscribe quadrilateral $NBYD$. Points P and Q are the centres of said circumcircles of quadrilateral $MCXD$ and $NBYD$ respectively.

NY is the diameter of the A -excircle of quadrilateral $NBYD$. The point A does not lie on the same circle but is collinear with the three points that lie on NY . A also lies on the circumcircle of ADB , and also the circumcircle of ABC . Also, both quadrilaterals $MCXD$ and $NBYD$ share point D , implying their circumcircles are tangential at point D . Because that circumcircle is tangential at point D common, both circles share D as an internal point of similitude with respect to circles P and Q . Thus, each circle is a projection of one another over the line through D parallel to the radii of circles P and Q , that being, PX and some radius QZ for some Z .

Now, realize point A is very special, as it is the quadrilateral $MCXD$ and $NBYD$ point of homothety. We wish to show the tangent line through D will pass through A . Both quadrilaterals $MBXD$ and $NBYD$ are similar, so by relating each quadrilateral to the point of homothety, we see it directly corresponds to the external point of similitude, that being A . Therefore, A is the external point of similitude. By definition, that implies A and D lie on the same line. Extend such line to intersect ω at V . Also by definition, that means if $ABXV$ is cyclic, then $AVYC$ is also cyclic. Because all four points A, B, X, V are on ω , three of the four points A, V, C are on ω , and quadrilateral $AVYC$ is cyclic, Y must also be on ω .

So, if quadrilateral $ABXD$ intersects circle ω at X , quadrilateral $ADYC$ must also intersect circle ω at Y . $\square$

Solution 5



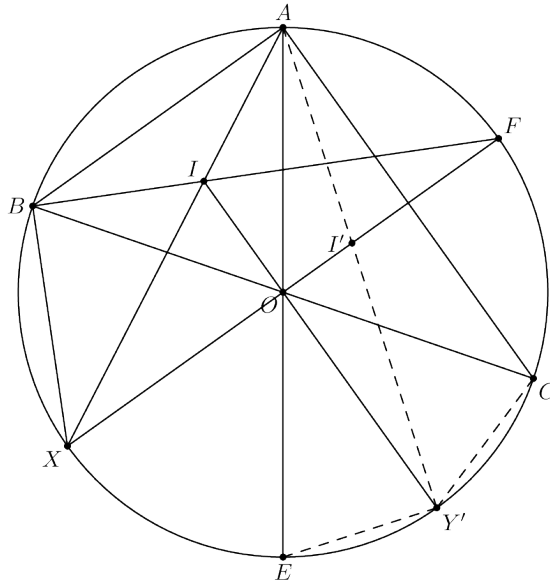
We draw the diameter $\overline{XF}$. Since $\overline{BF} \perp \overline{BX}$, we know that $\overline{BF}$ bisects $\angle ABD$. Let I be the incenter of $\triangle ABD$. Since A, I, X are collinear, we have $I = \overline{BF} \cap \overline{AX}$. Since $BO = FO = AO$, we see that $\triangle BOF$ and $\triangle AOB$ are isosceles. Using the property of incenter and the inscribed angle theorem, we get

$$\angle IAO = \angle BAI = \angle BFO = \angle FBO$$

Also, $\angle BAO = \angle ABO \iff \angle ABI + \angle FBO = \angle IAO + \angle BAI$, or

$$\angle ABI = \angle BAI \iff \angle ABI = \angle IBO$$

This means that I is also the incenter of $\triangle ABO$. We obtain $\angle IBO = \angle IBD$, and it is given that A, O, D are collinear. Thus, $D = O$.



We extend $\overline{AO}$ meeting ω at E other than A . Let Y' be the intersection of the ray IO and ω .

$\overline{OI}$ bisects $\angle AOB$. It follows that

$$\angle AOI = \angle EOY' = \angle BOI = \angle COY',$$

or $\overline{OY'}$ bisects $\angle COE$, and $\angle EOY' = \angle COY' \implies EY' = CY' \implies \angle EAY' = \angle CAY'$. We obtain that AY' bisects $\angle CAO$. By the definition of the A -excenter of $\triangle ABO$, $\overline{OX}$ bisects $\angle BOE$. We have

$$\angle BOX = \angle COF = \angle EOX = \angle AOF,$$

or $\angle AOF = \angle COF$. Since $\overline{OF}$ is the angle bisector of $\angle AOC$, we have $I' = \overline{AY'} \cap \overline{OF}$ is the incenter of $\triangle AOC$.

We know that A, I', Y are collinear, and Y lies on $\overline{OY'}$, the external bisector of $\angle AOC$. Therefore, $Y' = Y$, as desired.

5. (*USAJMO 2026*) A positive integer n is called solitary if, for any nonnegative integers a and b such that $a + b = n$, either a or b contains the digit 1 when written in base 10. Determine, with proof, the number of solitary integers less than 10^{2026} .

Solution 1

Lemma 1: $n = 2 \cdot 10^k - 1$ is solitary for all $k \in \mathbb{Z}_{\geq 0}$.

Suppose $a + b = n$. WLOG, $a \geq b$. Then, $2a \geq a + b = n \implies a \geq \frac{n}{2}$. Since a, n are integers, and n is odd, this implies $a \geq \frac{n+1}{2} = 10^k$. But, $a \leq n = 2 \cdot 10^k - 1$, so $10^k \leq a \leq 2 \cdot 10^k - 1$, so the leading digit of a is 1. Thus, n is solitary. $\square$

Lemma 2: $m = 2 \cdot 10^k + n, n < 10^k$ is solitary if and only if n is solitary.

First, suppose m is solitary and n is not. We write $n = a + b$ for some a, b with no 1's in their decimal representation. Then, let $a' = a + 2 \cdot 10^k$ and $b' = b$. It follows that

$$a' + b' = a + 2 \cdot 10^k + b = 2 \cdot 10^k + n = m$$

But a' and b' have no 1's in their decimal representation, since $10^k > n \geq a, b$, a contradiction. This proves the only if part.

Now, suppose n is solitary, and m is not. Write $m = a + b$ for some a, b with no 1's in their decimal representation.

Write $a = x \cdot 10^k + a', a' < 10^k$ (the leading digit of a is x). Similarly, $b = y \cdot 10^k + b', b' < 10^k$. By definition, $a, b \neq 1$. But

$$x = y = 0 \implies a + b < 2 \cdot 10^k < n \text{ and } x + y > 2 \implies a + b \geq 3 \cdot 10^k > m$$

Thus, one of x, y is 2, and the other is 0. WLOG $x = 2, y = 0$. Consider a' and b' . We have

$$a' + b' = m - x \cdot 10^k - y \cdot 10^k = m - 2 \cdot 10^k = n$$

But n is solitary, so either a' or b' has a 1 in their decimal representation. But, that implies a or b also has a 1, since $a', b' < 10^k$, a contradiction. This proves the if part. The lemma is complete. $\square$

Lemma 3: For any integer $m \in [10^k, 10^{k+1} - 1]$, if m is not of the form $2 \cdot 10^k - 1$ and not of the form $2 \cdot 10^k + n$ (where $n < 10^k$), then m is not solitary.

By Lemma 2, we can assume the leading digit of m is not 2. Now, take $a = 10^k - 1$ and $b = m - a$.

Let the decimal representations of a, b be $a_{k-1}a_{k-2}\dots a_0$ and $b_kb_{k-1}\dots b_0$, respectively. It follows that $a_i = 9$ for all i .

Also, $m \notin [2 \cdot 10^k - 1, 3 \cdot 10^k - 1] \implies b = m - a \notin [10^k, 2 \cdot 10^k]$, so $b_k \neq 1$.

Now, suppose $b_i = 1, i \in \{0, 1, \dots, k-1\}$. Increase b_i to 2 and lower a_i to 8. This increases b by 10^i while decreasing a by 10^i , so the sum stays the same. Repeating this for all i such that $b_i = 1$, we have a, b such that none of the digits are 1, $a + b = m$, so m is not solitary. $\square$

Now, we use the lemmas to finish. Let s_k be the number of solitary integers $< 10^k$, and let S_k be the set of these integers, so $|S_k| = s_k$. By the lemmas

$$S_{k+1} = \{2 \cdot 10^k - 1\} \cup \{2 \cdot 10^k + n : n \in S_k\} \cup S_k$$

Since no elements are shared, we have $s_{k+1} = 1 + s_k + s_k = 2s_k + 1$. Also, $S_1 = \{1\}$, so $s_1 = 1$. It follows by induction $s_k = 2^k - 1$. Thus, the answer is $\boxed{2^{2026} - 1}$.

Solution 2

Lemma 1: If x is even, then x can't be solitary.

Suppose x is even. Then take k such that $2k = x$. Also, let $k_1 = k$, and $k_2 = k$. Then, for each 1 digit in k , change that digit in k_1 to be 0, and change that digit in k_2 to be 2. This preserves the sum $k_1 + k_2 = x$, but now k_1 and k_2 don't have the digit 1, so x can't be solitary.

Lemma 2: If $10q + r$ is solitary, where r can be written as the sum of two non-1 digits (0 through 9) and $q > 0$, then q is solitary.

Suppose for the sake of contradiction that q isn't solitary. This means that we can take $a, b \geq 0$ such that $a + b = q$, and a and b don't contain any ones. Now we can also write r as $d_1 + d_2$ where $0 \leq d_1, d_2 \leq 9$, and neither d_1 or d_2 are 1. This gives us that

$$r = 10q + r = 10(a + b) + d_1 + d_2 = (10a + d_1) + (10b + d_2).$$

Note that $10a + d_1$ can't contain any 1's, and neither can $10b + d_2$, so therefore r isn't solitary, which is a contradiction.

Suppose x is solitary. Then its units digit must be 1 or 9. Write x as $10q + r$, where $0 \leq r < 10$. Then if $r \in \{3, 5, 7\}$, we can write x as $10q + a_1 + b_1$, or $10(q-1) + a_2 + b_2$, where a_1, a_2, b_1, b_2 are non-1 digits between 0 and 9 inclusively. Then both q and $q-1$ must be solitary by Lemma 2. However, either q or $q-1$ is even, which contradicts Lemma 1. Therefore, its unit digit must be either 1 or 9.

Claim: If $x > 11$ is solitary, then either $\frac{x-9}{10}$ is solitary or $\frac{x-11}{10}$ is solitary.

Suppose x is solitary. The first case is that x is 1 mod 10. Then we can write x as $10k + (5 + 6)$, where k is positive. By Lemma 2, k must also be solitary, so $k = \frac{x-11}{10}$ is solitary. The second case is that x is 9 mod 10. Then we can write x as $10k + (4 + 5)$ where $k > 0$, and by Lemma 2, k is solitary. Thus, the second case yields that $k = \frac{x-9}{10}$ is solitary.

Now I will show the other direction, that if k is solitary, then both $10k + 9$ and $10k + 11$ are solitary.

Suppose k is solitary. Then let's look at $10k + 9$, and suppose we can find non-negative integers a, b such that $a + b = 10k + 9$ and neither a nor b contains a 1.

The units digits of a and b must add to either 9 or 19, but the latter is impossible, so they add to 9. Therefore, we can write $a = 10a' + r_a$ and $b = 10b' + r_b$ where $r_a + r_b = 9$. The condition $a + b = 10k + 9$ turns into $10(a' + b') + 9 = 10k + 9$, which is equivalent to $a' + b' = k$. However, because k is solitary, one of a' and b' must contain the digit 1, which contradicts the fact that both a and b contain the digit 1.

The proof is identical for $10k + 11$, because if a, b add to $10k + 11$ with both a and b not containing a 1, then the units digits of a and b add to either 1 or 11, but they can't add to 1 because then the units digit of either a or b would be 1, so they must add to 11, and then we do the same thing as we did for $10k + 9$.

We have now established that $x > 11$ is solitary if and only if it can be written as $10k + 9$ or $10k + 11$ for some solitary k . This means for any solitary x , we can continually find smaller and smaller solitary integers by subtracting either 9 or 11 and dividing by 10 until we have a solitary integer less than 12. However, 1 is the only solitary integer less than 12, so therefore x is solitary if and only if it can be expressed by repeated application of $10x + 9$ or $10x + 11$, starting from 1.

Now let $S_1 = \{1\}$, and define

$$S_{k+1} = \{10x + 9 \mid x \in S_k\} \cup \{10x + 11 \mid x \in S_k\}.$$

Note that no elements are repeated, so we get that $|S_{k+1}| = 2|S_k|$, and $|S_k| = 2^{k-1}$.

Now an integer is solitary if and only if it is in S_k for some k . Now all that is left to prove is that S_k is precisely the set of k digit solitary integers. I will use induction to show that $\min(S_k) \geq 10^{k-1}$, and $\max(S_k) < 10^k - 2$.

Base case: $k = 1$. Then $1 \geq 10^0$, and $1 < 10 - 2$.

Inductive Hypothesis: Assuming that $\min(S_k) \geq 10^{k-1}$, and $\max(S_k) < 10^k - 2$. Then

$$\min(S_{k+1}) = 10\min(S_k) + 9 \geq 10^k + 9 \geq 10^k$$

and

$$\max(S_{k+1}) = 10\max(S_k) + 11 < (10^k - 2) \times 10 + 11 = 10^{k+1} - 9 < 10^{k+1} - 2.$$

Thus, by induction, S_k is precisely the set of k digit solitary integers, and the number of solitary integers less than 2026 is

$$\sum_{k=1}^{2026} |S_k| = \sum_{k=1}^{2026} 2^{k-1} = 2^{2026} - 1$$

Solution 3

Generalize to all positive integers. I claim all solitary numbers are of the form $2 \cdot 10^{a_n} + 2 \cdot 10^{a_{n-1}} + \dots + 2 \cdot 10^{a_1} \pm 1$ for some $1 \leq a_1 < a_2 < \dots < a_n$ and 1.

First, we prove these work. Proceed via contradiction; suppose we have some $b \geq c$ with $b + c = 2 \cdot 10^{a_n} + 2 \cdot 10^{a_{n-1}} + \dots + 2 \cdot 10^{a_1} \pm 1$ such that b, c do not contain the digit 1. It is not hard to see that the leading digit of b must be 2, ie $b = 2 \cdot 10^{a_n} + k$

for some natural k , which means that $c + k = 2 \cdot 10^{a_{n-1}} + 2 \cdot 10^{a_{n-2}} + \dots + 2 \cdot 10^{a_1} \pm 1$. Repeating this process recursively, each of $2 \cdot 10^{a_j}$ will be assigned to one of b or c , and adding or subtracting 1 at the end will make one of these two numbers have a 1 appear in their decimal representation; contradiction.

Call a number Ackerman if it is in the form $2 \cdot 10^{a_n} + 2 \cdot 10^{a_{n-1}} + \dots + 2 \cdot 10^{a_1} \pm 1$ (or 1), and for the sake of convenience let $2 \cdot 10^k + 2 \cdot 10^{k-1} + \dots + 2 \cdot 10^1 + 2 = S_k$.

It suffices to show that all non-Ackerman numbers are non-solitary.

Claim: If all non-Ackerman numbers $\leq S_k$ are non-solitary, then all non-Ackerman numbers $\leq S_{k+1}$ are non-solitary.

Proof: If a number x is non-solitary, then so are $10x + 2, 10x + 3, \dots, 10x + 11$, as if $x = y + z$, we can rewrite $10x + c$ for some $c \in \{2, 3, \dots, 11\}$ as $10y + l + 10z + p$ for some $l, p \in \{0, 2, 3, \dots, 9\}$. Consider the set $\{S_{n-1}, S_{n-1} + 1, \dots, S_n\}$. It follows that all numbers in the set $\{10S_{n-1} + 2, 10S_{n-1} + 3, \dots, 10S_n + 11\} \equiv \{S_n, S_n + 1, \dots, S_{n+1} + 9\}$ are non-solitary, set aside the ones in the form $10x + c$ for some Ackerman x and some $c \in \{2, 3, \dots, 11\}$. It follows that all of these numbers can be rewritten in the form $\sum_{k=1}^n 2 \cdot 10^{b_k} + m$ for some $2 \leq b_1 < b_2 < \dots < b_n$, and some $m \in \{-8, -7, \dots, 1, 12, 13, \dots, 21\}$. From here we can deduce that the only values of m which yield solitary numbers here is $m = -1, 1, 19, 21$, which all yield Ackerman numbers, finishing the inductive hypothesis. $\square$

As such, when constructing a solitary number $< 10^{2026}$, we effectively have 2026 digit slots and in each one we must place a 0 or 2, and at the end we must choose to add or subtract 1 from this number. Now, the units digit must be 0, as it cannot be 2, which means we have $2^{2025} \cdot 2 = 2^{2026}$ possible solitary numbers, but we must subtract 1 to account for the case where all the digits are 0 and we subtract 1, ie the number of solitary numbers $< 10^{2026}$ is $2^{2026} - 1$, and we are done. $\blacksquare$

Solution 4

Lemma 1: n must contain 1, otherwise it is not solitary.

Proof of Lemma 1:

Definition of solitary, choose $a = 0$ and $b = n$ and the result shows n is not solitary.

Lemma 2:: All such n has the property all digits to the right of "1" are "9".

Proof of Lemma 2:

Consider vertical addition. Let a_n and b_n be the n th place value of such numbers a and b . Then, $a_0 + b_0 = 9$. One can form such numbers using any orientation, either $a_n = 1$ and $b_n = 8$, vice versa, or some non-1 pair. Assume the worst case, where we construct sums of 9 that do not include 1. Then, if the leading digit itself is 1, then notice that one may "truncate" the 19 at the end of the string $1999 \dots 9$. Because 9 is solitary, any single combination of a_0, b_0 result in one value being greater than 10, hence we must carry a value. Specifically, that implies the length of $a < b$ or vice versa. Hence, if the length $a < b$ is obtained, then notice that then one can let, at the end of the truncation, b start with 9 and a begin with 10, giving the 19 that was truncated at the beginning. This fulfills solitary properties.

Lemma 3: All such n has the property all digits to the left of "1" are either "0" or "2" or a combination of both.

Proof of Lemma 3:

We continue from where we left off in Lemma 2. We realize that b must start with a 0 then move through with 9's. Afterwards, a and b are uniquely determined toward

the left side of a 's 1 and b 's 0. Consider forming a sum of 11 instead of 1, and we prove that a number can be formed that is solitary. More importantly, if a_n and b_n do not contain 1 at all yet still sum to 11, then that 1 carries over. Now, we still fulfill the properties of **Lemma 2**, but the number is not solitary! But if we were to carry over that extra 1, then a_{n+1} or b_{n+1} can have either 1 or 9, from which we terminate, or continue analogously. However, when we end, one of a or b has 1, hence the number n is solitary.

2026 USAJMO/Q5:

Given the conditions, n is *solitary* if and only if

- n contains 1
- All numbers to the right of the 1 in n are 9
- All numbers to the left of the 1 in n are "2" or "0"

The last part is to count all such n . From any number, the digit 1 can be placed at an index m from the left ($1 \leq m \leq 2026$).

There are exactly $2026 - m$ positions to the right of 1 that are all 9. Otherwise, there are exactly $m - 1$ positions to the left of 1. Those have two choices each, giving 2^{m-1} cases. So, on any given integer with m digits, we may form 2^{m-1} solitary integers.

We sum over all $1 \leq m \leq 2026$, or

$$\sum_{m=1}^{2026} 2^{m-1} = 1 + 2 + \cdots + 2^{2025} = \boxed{2^{2026} - 1} \quad \square$$

Remark REALLY annoying number theory problem

6. (*USAJMO 2026*) Emily has a red sheet of paper. She draws 2026 circles (not necessarily of equal size) on the piece of paper. She chooses a circle to color black, then cuts the paper around the circumference of all 2026 circles. She then separates the pieces of paper, into at least 2 black pieces and some number of red pieces. Is it possible that all black pieces are congruent?

Solution

Consider convexity. A circle is convex all throughout the perimeter, so any cut it makes is convex for half of the cut (the part closer to the circle's center) and is concave for the other part. So the main claim is that there always exists a purely convex black part as well as a part with both concave and convex sides. The prior can be proved by induction.

We start off with a single circle which is convex. Take that as the base case. Now assume by induction we have a piece that is completely convex. A extra cut through that piece can separate it into 2 pieces, a piece with a hole through it or 3 pieces. If it is 2 pieces, it is easy to see that the piece closer to the center has a convex side added, which means the shape is purely convex. If there is a piece with a hole, the "hole" itself as a circle is purely convex. If 3 pieces are created, the middle piece has 2 extra convex sides, which maintains the purely-convexity. So by induction there exists a piece that is purely convex.

Next we can prove that there also exists a concave shape. There are at least 2 black pieces, so there must be at least one cut of a circle through a circle which creates

at least one concave piece. Then, no number of convex cuts can make the concave arc "no longer concave". So the concave arc always exists, meaning there is always a concave shape.

A concave shape is definitely not congruent to a convex shape, so it is impossible for all pieces to be congruent.

Solution 2 (Concordance, Another version of Convexity)

No, it is not possible.

The extrema case is when two pieces are formed. Showing that the minima does not work proves that no matter how the minima increases there will always be a way to evaluate the diagram like the minima, and thus no cases will be prevalent.

Lemma 1:

It must be the case that a Black and Red circle overlap.

Proof of Lemma 1:

There must exist two black pieces. Consider overlapping the red and black circles. Then, cutting the circumference of the red circle yields a black piece. Cutting the circumference of what remains of the black and red circle overlap yields another black piece. If the circles are disjoint, then there would only exist red pieces, contradicting the problem statement, hence proven.

Lemma 2 (Cross Sections, Theorem of Concordance: 2D Geometry):

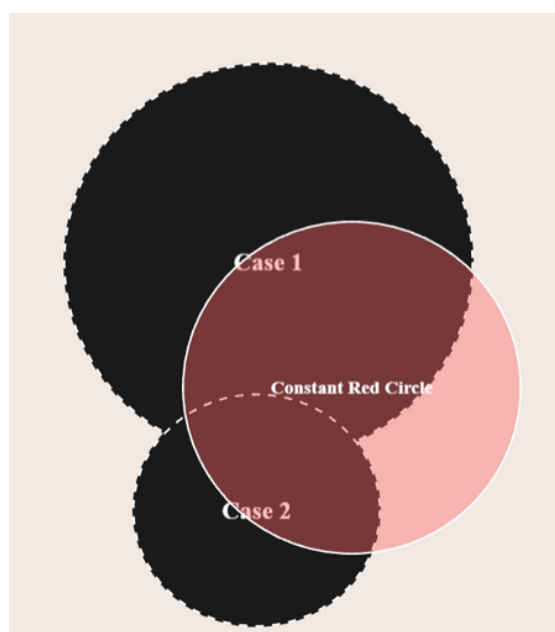
There will exist at least one black piece that is never congruent to all following black pieces.

Proof of Lemma 2

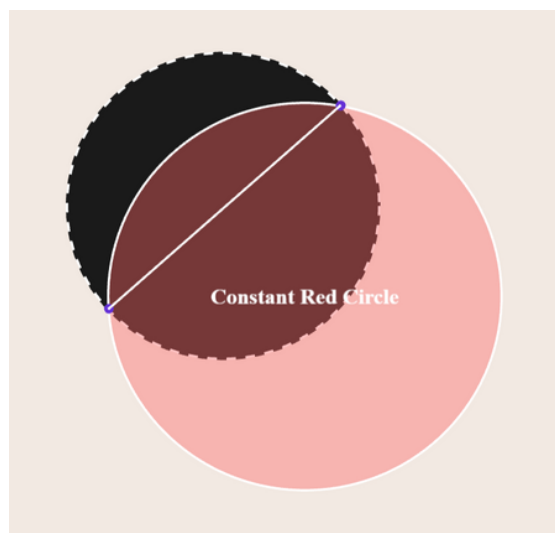
Through **Lemma 1** the circles must overlap.

Two figures are congruent if and only if they share the same shape and size. By proving that a figure would always yield a different shape or size, we show that no such configuration is possible.

Consider the diagram below, and assume the contrary.



If a black circle is larger or smaller than its overlapping red circle. Here, consider the arc with measure θ that is linked to the centre of the constant red circle Ω . It follows that if θ is the angle escribed within the two intersection points of the case 1 circle and Ω is arc measure (degrees) 2θ , then the contrary arc is $(360 - 2\theta)$ degrees. The curvature of the two black pieces do not yield the same shape, and thus are not congruent if $\theta \neq 90^\circ$. Then, consider the curvature of the red arc in terms of the black circle centre. The angle measure between the two points of intersection and the centre of the black circle is 180° . However, two intersection points of a circle, due and proven by the circle's curvature, if they truly do lie diametrically opposite, a small bound of one curvature will always lie below the diameter, as shown below.



Thus the curvature of the bounded figure between the intersection of the black circle and red circle will never be the same size as the red circle, as that arc imposed by the red circle forces one black shape to look like a "crescent moon", whereas the other looks like a bounded region, hence contradicting congruency.

2026 USAJMO Q6:

The case never holds if two circles, one red and one black, lie on the plane overlapping. Then, if the number of circles increase to 2026, I can always choose a red circle and a black circle, and two red circles with a bounded black union, and through the direct result of **Lemma 2** imply that no such configuration exists.

Therefore, no matter how we arrange the circles on the plane, it will never be possible.